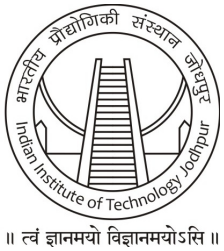
Content Based Analysis and Retrieval of Architectural Floor plans

A Thesis submitted by
Divya Sharma

in partial fulfillment of the requirements for the award of the degree of
Doctor of Philosophy

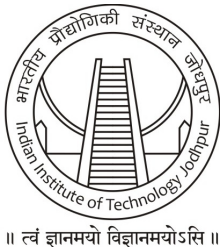


Indian Institute of Technology Jodhpur
Department of Computer Science and Engineering
August 2018

基于内容的建筑平面图分析与检索

迪维娅·夏尔马
提交的学位论文

为部分满足哲学博士学位授予要求



Indian Institute of Technology Jodhpur
Department of Computer Science and Engineering
August 2018

Declaration

I hereby declare that the work presented in this Thesis titled *Content Based Analysis and Retrieval of Architectural Floor Plans* submitted to the Indian Institute of Technology Jodhpur in partial fulfillment of the requirements for the award of the degree of Doctor of Philosophy, is a bonafide record of the research work carried out under the supervision of Dr. Chiranjoy Chattopadhyay. The contents of this thesis in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Divya Sharma
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兹证明，由*Divya Sharma (P15VSS002)*向印度理工学院焦特布尔分校提交、题为*Content Based Analysis and Retrieval of Architectural Floor Plans*的学位论文，系其在我指导下完成的研究工作的真实记录。据我所知，该论文的全部或部分内容未曾向任何其他院校或大学提交，以申请任何学位或文凭。

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List of Figures

Figure	*Title*	*Page*
1.1	Illustration of various subareas of document image analysis (DIA) on a sample scanned document [Sharma *et al.*, 2017].	2
1.2	An illustration of a basic Content Based Image Retrieval system.	3
1.3	Illustration of various query modes along with their corresponding retrieval results	4
1.4	Motivation of the problem.	5
1.5	Floor plans, showing 2D cross sections of buildings and displaying heterogeneity in design.	6
1.6	Retrieval in floor plans by extracting meaningful content and looking for similarity across floor plan databases.	7
1.7	Analysing floor plans for novel content aimed at retrieving similar layouts from a large repository.	8
3.1	Framework diagram for our structural retrieval based on topology	22
3.2	Various stages of structural and semantic analysis on an image from SESYD dataset [Delalandre *et al.*, 2010a]: (a) unsegmented layout, results of (b) morphological operation and wall, boundary extraction, (c) Detection of doors and windows (d) Closing gaps in walls and (e) labelling and segmenting the room structure	24
3.3	Detecting doors and windows using Harris corner detector	25
3.4	Topological graph for the segmented room layout	26
3.5	Example showing how a sink symbol in the furniture set is assigned a unique signature.	27
3.6	Room layout and corresponding graph consisting of both room adjacencies as well as room decor representation	28
3.7	Embedding features from the topological graphs into spectral space. Adjacency matrices with varying sizes are reduced to fixed length vectors in the pattern space. Kindly note, in the right hand side, similar vectors are clustered closely in the feature space.	29
3.8	Topological feature extraction from a room layout in a floor plan	30
3.9	Framework for Room Decor Matching	32
3.10	Top 5 retrieved results corresponding to a query from the SESYD dataset.	32
3.11	Retrieval results for two different query architectural floor plans taken from the SESYD dataset. The top row shows the query image, the corresponding room layout and the semantic graph generated from the floor plan image. For the given query images only the top six retrieval results are shown.	33
3.12	Retrieval results for two different query architectural floor plans taken from the SESYD dataset and the ROBIN dataset. 1. (a) and (b) correspond to the query given from SESYD dataset and the subsequent rank-ordered results. 2. (a) and (b) correspond to the query given from ROBIN dataset	34
3.13	Quantitative result comparing the RLM and the RDM approach on the ROBIN dataset	34
3.14	Plot of intra-class and inter-class distances between layouts using our feature matching technique on the SESYD datset.	36
3.15	Layout matching from floor plans taken from SESYD dataset. Two such results from Category 8 and 4 are highlighted with a red bounding box.	37
3.16	Execution time for the intermediate stages of processing of the framework for different categories of floor plans on the SESYD dataset. Values shown in the bar chart denote the ratio of time taken in seconds for each step to the total time taken for execution.	38

图表目录

Figure	*Title*	页面
1.1	Illustration of various subareas of document image analysis (DIA) on a sample scanned document [Sharma *et al.*, 2017].	2
1.2	An illustration of a basic Content Based Image Retrieval system.	3
1.3	Illustration of various query modes along with their corresponding retrieval results	4
1.4	Motivation of the problem.	5
1.5	Floor plans, showing 2D cross sections of buildings and displaying heterogeneity in design.	6
1.6	Retrieval in floor plans by extracting meaningful content and looking for similarity across floor plan databases.	7
1.7	Analysing floor plans for novel content aimed at retrieving similar layouts from a large repository.	8
3.1	Framework diagram for our structural retrieval based on topology	22
3.2	Various stages of structural and semantic analysis on an image from SESYD dataset [Delalandre *et al.*, 2010a]: (a) unsegmented layout, results of (b) morphological operation and wall, boundary extraction, (c) Detection of doors and windows (d) Closing gaps in walls and (e) labelling and segmenting the room structure	24
3.3	Detecting doors and windows using Harris corner detector	25
3.4	Topological graph for the segmented room layout	26
3.5	Example showing how a sink symbol in the furniture set is assigned a unique signature.	27
3.6	Room layout and corresponding graph consisting of both room adjacencies as well as room decor representation	28
3.7	Embedding features from the topological graphs into spectral space. Adjacency matrices with varying sizes are reduced to fixed length vectors in the pattern space. Kindly note, in the right hand side, similar vectors are clustered closely in the feature space.	29
3.8	Topological feature extraction from a room layout in a floor plan	30
3.9	Framework for Room Decor Matching	32
3.10	Top 5 retrieved results corresponding to a query from the SESYD dataset.	32
3.11	Retrieval results for two different query architectural floor plans taken from the SESYD dataset. The top row shows the query image, the corresponding room layout and the semantic graph generated from the floor plan image. For the given query images only the top six retrieval results are shown.	33
3.12	Retrieval results for two different query architectural floor plans taken from the SESYD dataset and the ROBIN dataset. 1. (a) and (b) correspond to the query given from SESYD dataset and the subsequent rank-ordered results. 2. (a) and (b) correspond to the query given from ROBIN dataset	34
3.13	Quantitative result comparing the RLM and the RDM approach on the ROBIN dataset	34
3.14	Plot of intra-class and inter-class distances between layouts using our feature matching technique on the SESYD datset.	36
3.15	Layout matching from floor plans taken from SESYD dataset. Two such results from Category 8 and 4 are highlighted with a red bounding box.	37
3.16	Execution time for the intermediate stages of processing of the framework for different categories of floor plans on the SESYD dataset. Values shown in the bar chart denote the ratio of time taken in seconds for each step to the total time taken for execution.	38

3.17 Execution time for the intermediate stages of processing of the framework averaged over all 51 categories of floor plans in the ROBIN dataset. Values shown in the bar chart denote the time taken in seconds. 38
3.18 Segmentation results on two images from CVC-FP [de las Heras *et al.*, 2015] dataset (a) Input floor plan, (b) Technique proposed in [de las Heras *et al.*, 2014], (c) Our approach 39

4.1 Deep Learning Model based Framework for Floor Plan Retrieval 42
4.2 Architecture of feed-forward neural network, with the input layer, a hidden layer and the output layer 44
4.3 Architecture of a Convolutional Neural Network Model for classification of an image [LeCun *et al.*, 2015] 45
4.4 Example of how convolution operation is performed on an image matrix. 45
4.5 Example of how a ReLU Operation works. 46
4.6 Example of how a Max-Pooling Operation behaves on an image matrix. 47
4.7 Sequence of layers in AlexNet Framework [Krizhevsky *et al.*, 2012a] 48
4.8 Architecture of the Convolutional Neural Network (CNN) model used for feature extraction from the floor plan images. 49
4.9 Rank ordered retrieval result of the proposed framework for five different query floor plans from the ROBIN dataset. Here, top five rank ordered floor plans are shown for each query. 51
4.10 Mean Average Precision values, comparing result of the framework on ROBIN dataset, using features from all the hidden layers of the CNN. 53
4.11 Precision and Recall (PR) plot, comparing state-of-art methods with the proposed framework on ROBIN dataset. 54

5.1 Q depicts the query floor plan and D1 depicts a resultant retrieved floor plan from the database of floor plans. The objects enclosed in red boxes in D1 are furniture components with similar shape as the components in Q, but different types. 57
5.2 Basic framework depicting extraction of high-level semantic features and weighted fusion for retrieval of floor plans. 59
5.3 Room level segmentation results on a pair of floor plan images from ROBIN dataset. 60
5.4 Figure depicting the ordered labeling of rooms after segmentation. (a) Distance is calculated from a leftmost origin point up to all room centroids. Room with the lowest distance from the origin is ordered number 1 and the other rooms are labeled in similar fashion. (b) Subsequent room ordering results. 61
5.5 Example of Room Adjacency String (RAS) feature: (a) Segmented rooms (b) Topological Graph (c) Adjacency matrix (Adjacency string corresponding to Room 2 is highlighted). 62
5.6 Carpet Area Ratio (CAR) feature: (a), (c) Segmented, ordered layout, while (b), (d) CAR feature corresponding to Example 1 and 2 respectively. 63
5.7 Furniture detection and categorization in each room. 63
5.8 Furniture detection and categorization in each room. Illustration underneath each room gives the array $\mathscr{T}$ containing the type of furniture in each room. 65
5.9 Distance score corresponding to each room of the query layout (Q) and the database layouts (S_1 and S_2). Attributed to the difference in features, lower the score, more similar the floor plans are. Hance, finally the score is subtracted from 1 to make the final greater Matching score to represent the floor plans that match more. Color codes are used to highlight the room correspondence. 68
5.10 Retrieval results for four different query architectural floor plans taken from the *ROBIN* dataset. Note that the floor plans highlighted in red are incorrectly retrieved as global layout shape differs from query. 69
5.11 Precision and Recall (PR) plot corresponding to a) Cumulative sum of all the features with equal weights b) Preferring one feature more than the others while retrieval. 70

3.17 Execution time for the intermediate stages of processing of the framework averaged over all 51 categories of floor plans in the ROBIN dataset. Values shown in the bar chart denote the time taken in seconds. 38
3.18 Segmentation results on two images from CVC-FP [de las Heras *et al.*, 2015] dataset (a) Input floor plan, (b) Technique proposed in [de las Heras *et al.*, 2014], (c) Our approach 39

4.1 Deep Learning Model based Framework for Floor Plan Retrieval 42
4.2 Architecture of feed-forward neural network, with the input layer, a hidden layer and the output layer 44
4.3 Architecture of a Convolutional Neural Network Model for classification of an image [LeCun *et al.*, 2015] 45
4.4 Example of how convolution operation is performed on an image matrix. 45
4.5 Example of how a ReLU Operation works. 46
4.6 Example of how a Max-Pooling Operation behaves on an image matrix. 47
4.7 Sequence of layers in AlexNet Framework [Krizhevsky *et al.*, 2012a] 48
4.8 Architecture of the Convolutional Neural Network (CNN) model used for feature extraction from the floor plan images. 49
4.9 Rank ordered retrieval result of the proposed framework for five different query floor plans from the ROBIN dataset. Here, top five rank ordered floor plans are shown for each query. 51
4.10 Mean Average Precision values, comparing result of the framework on ROBIN dataset, using features from all the hidden layers of the CNN. 53
4.11 Precision and Recall (PR) plot, comparing state-of-art methods with the proposed framework on ROBIN dataset. 54

5.1 Q depicts the query floor plan and D1 depicts a resultant retrieved floor plan from the database of floor plans. The objects enclosed in red boxes in D1 are furniture components with similar shape as the components in Q, but different types. 57
5.2 Basic framework depicting extraction of high-level semantic features and weighted fusion for retrieval of floor plans. 59
5.3 Room level segmentation results on a pair of floor plan images from ROBIN dataset. 60
5.4 Figure depicting the ordered labeling of rooms after segmentation. (a) Distance is calculated from a leftmost origin point up to all room centroids. Room with the lowest distance from the origin is ordered number 1 and the other rooms are labeled in similar fashion. (b) Subsequent room ordering results. 61
5.5 Example of Room Adjacency String (RAS) feature: (a) Segmented rooms (b) Topological Graph (c) Adjacency matrix (Adjacency string corresponding to Room 2 is highlighted). 62
5.6 Carpet Area Ratio (CAR) feature: (a), (c) Segmented, ordered layout, while (b), (d) CAR feature corresponding to Example 1 and 2 respectively. 63
5.7 Furniture detection and categorization in each room. 63
5.8 Furniture detection and categorization in each room. Illustration underneath each room gives the array $\mathscr{T}$ containing the type of furniture in each room. 65
5.9 Distance score corresponding to each room of the query layout (Q) and the database layouts (S_1 and S_2). Attributed to the difference in features, lower the score, more similar the floor plans are. Hance, finally the score is subtracted from 1 to make the final greater Matching score to represent the floor plans that match more. Color codes are used to highlight the room correspondence. 68
5.10 Retrieval results for four different query architectural floor plans taken from the *ROBIN* dataset. Note that the floor plans highlighted in red are incorrectly retrieved as global layout shape differs from query. 69
5.11 Precision and Recall (PR) plot corresponding to a) Cumulative sum of all the features with equal weights b) Preferring one feature more than the others while retrieval. 70

6.1 Flowchart comparing the different sub-components in two approaches for sketch based retrieval. (a) Approach 1: Basic framework using cyclic GAN for domain mapping and sketch based retrieval (b) Approach 2: An improved unified framework using cyclic GAN and autoencoders in conjunction for floor plan retrieval. 77
6.2 Framework for sketch based retrieval of floor plans. Domain adaptation is facilitated by a Cyclic GAN. 78
6.3 Comparing the original sketches/images with outputs generated from Cyclic GAN. (a) Original floor image from ROBIN dataset. (b) Generated floor plan image using Cyclic GAN. (c) Original floor plan sketch from S-ROBIN dataset. (d) Generated floor plan sketch using Cyclic GAN. Kindly note the quality of generated sketches is better than that of generated images. 79
6.4 Cyclic GAN framework, corresponding to the proposed approach with converting the sketches from the S-ROBIN dataset into GAN generated images (constituting GI-ROBIN dataset) closely similar in features to the to-be retrieved images and images from the ROBIN dataset to sketches (constituting GS-ROBIN dataset). 79
6.5 Network architecture for the Cyclic GAN framework, representing the inputs from ROBIN and S-ROBIN dataset. The outputs being generated constitute the GS-ROBIN dataset. 80
6.6 Schematic structure of the Cyclic GAN network used in the proposed framework for image floor plan to synthesized sketched floor plan mapping, and vice-versa. 81
6.7 Unified framework for multimodal retrieval. 84
6.8 Layerwise depiction of the Convolutional Neural Network (CNN) for query classification 85
6.9 Layerwise depiction of the autoencoder network. 86
6.10 A diagram depicting stages of sketch synthesis from an image through our Cyclic GAN framework. Note how, the discriminator and generator improve with increasing the number of training iterations. 90
6.11 Top five rank ordered retrieval results for the two proposed approaches for three different hand-drawn query floor plans. 92
6.12 Top five rank ordered retrieval result of proposed framework for three different query floor plan images using Approach 2 proposed in the Chapter. 93
6.13 Two examples showing results for querying the system for partial matches. (a) Query Sketch, (b),(c),(d) Retrieved results. Colour Coding establishes room correspondence between the query and the results. 94
6.14 (a) Effect of converting query sketch into image through Cyclic GAN and performing retrieval (CNN A (trained on ROBIN) and CNN B (trained on S-ROBIN)) as opposed to converting the retrieval database into sketches (CNN C). (b) Effect of training CNNs on ROBIN (CNN_1), S-ROBIN (CNN_2) and GS-ROBIN dataset (CNN_3) 95
6.15 Precision and Recall (PR) plot, comparing result of the framework, using all the hidden layers CNN features. 96
6.16 Precision and Recall (PR) plot, comparing result of this framework on autoencoder trained with ROBIN dataset (N1) and S-ROBIN dataset (N2). 96
6.17 PR plot, comparing the two approaches. Approach 1 extracts features and performs retrieval taking minimum of the precision values obtained by each layer (Approach 1 (min)) and taking the best layer (normalization 1 layer) giving the highest MAP value (Approach 1 (norm1)). Whereas, Approach 2 has two autoencoder networks, N1 trained on ROBIN and GI-ROBIN and N2 trained on S-ROBIN and GS-ROBIN. 97
6.18 PR plot, comparing the best of the proposed two approaches (Approach 1 (norm1)) and (Approach 2 (N2)) with SOA techniques. 98

7.1 Comparison of MAP values obtained by techniques mentioned in Chapters 3, 4, 5 and 6. 103

A.1 Symbol set used as furniture in ROBIN and S-ROBIN dataset. 105
A.2 3-Room layouts in the ROBIN and S-ROBIN dataset. 106

6.1 Flowchart comparing the different sub-components in two approaches for sketch based retrieval. (a) Approach 1: Basic framework using cyclic GAN for domain mapping and sketch based retrieval (b) Approach 2: An improved unified framework using cyclic GAN and autoencoders in conjunction for floor plan retrieval. 77
6.2 Framework for sketch based retrieval of floor plans. Domain adaptation is facilitated by a Cyclic GAN. 78
6.3 Comparing the original sketches/images with outputs generated from Cyclic GAN. (a) Original floor image from ROBIN dataset. (b) Generated floor plan image using Cyclic GAN. (c) Original floor plan sketch from S-ROBIN dataset. (d) Generated floor plan sketch using Cyclic GAN. Kindly note the quality of generated sketches is better than that of generated images. 79
6.4 Cyclic GAN framework, corresponding to the proposed approach with converting the sketches from the S-ROBIN dataset into GAN generated images (constituting GI-ROBIN dataset) closely similar in features to the to-be retrieved images and images from the ROBIN dataset to sketches (constituting GS-ROBIN dataset). 79
6.5 Network architecture for the Cyclic GAN framework, representing the inputs from ROBIN and S-ROBIN dataset. The outputs being generated constitute the GS-ROBIN dataset. 80
6.6 Schematic structure of the Cyclic GAN network used in the proposed framework for image floor plan to synthesized sketched floor plan mapping, and vice-versa. 81
6.7 Unified framework for multimodal retrieval. 84
6.8 Layerwise depiction of the Convolutional Neural Network (CNN) for query classification 85
6.9 Layerwise depiction of the autoencoder network. 86
6.10 A diagram depicting stages of sketch synthesis from an image through our Cyclic GAN framework. Note how, the discriminator and generator improve with increasing the number of training iterations. 90
6.11 Top five rank ordered retrieval results for the two proposed approaches for three different hand-drawn query floor plans. 92
6.12 Top five rank ordered retrieval result of proposed framework for three different query floor plan images using Approach 2 proposed in the Chapter. 93
6.13 Two examples showing results for querying the system for partial matches. (a) Query Sketch, (b),(c),(d) Retrieved results. Colour Coding establishes room correspondence between the query and the results. 94
6.14 (a) Effect of converting query sketch into image through Cyclic GAN and performing retrieval (CNN A (trained on ROBIN) and CNN B (trained on S-ROBIN)) as opposed to converting the retrieval database into sketches (CNN C). (b) Effect of training CNNs on ROBIN (CNN_1), S-ROBIN (CNN_2) and GS-ROBIN dataset (CNN_3) 95
6.15 Precision and Recall (PR) plot, comparing result of the framework, using all the hidden layers CNN features. 96
6.16 Precision and Recall (PR) plot, comparing result of this framework on autoencoder trained with ROBIN dataset (N1) and S-ROBIN dataset (N2). 96
6.17 PR plot, comparing the two approaches. Approach 1 extracts features and performs retrieval taking minimum of the precision values obtained by each layer (Approach 1 (min)) and taking the best layer (normalization 1 layer) giving the highest MAP value (Approach 1 (norm1)). Whereas, Approach 2 has two autoencoder networks, N1 trained on ROBIN and GI-ROBIN and N2 trained on S-ROBIN and GS-ROBIN. 97
6.18 PR plot, comparing the best of the proposed two approaches (Approach 1 (norm1)) and (Approach 2 (N2)) with SOA techniques. 98

7.1 Comparison of MAP values obtained by techniques mentioned in Chapters 3, 4, 5 and 6. 103

A.1 Symbol set used as furniture in ROBIN and S-ROBIN dataset. 105
A.2 3-Room layouts in the ROBIN and S-ROBIN dataset. 106

A.3 4-Room layouts in the ROBIN and S-ROBIN dataset. 107
A.4 5-Room layouts in the ROBIN and S-ROBIN dataset. 108

B.1 Architecture of the developed software for floor plan retrieval. 112
B.2 Sequence diagram depicting the overall flow of the software. 113
B.3 Screen prompting the user to select a floor plan of his/her choice. 113
B.4 Segmentation of the selected plan into labeled rooms 114
B.5 Graph embedding of the room level structure of the floor plan. 114
B.6 Selecting how many results to be displayed while retrieval. 115
B.7 User's required floor plans on display as per their similarity from the query floor plan. 116

A.3 4-Room layouts in the ROBIN and S-ROBIN dataset. 107
A.4 5-Room layouts in the ROBIN and S-ROBIN dataset. 108

B.1 Architecture of the developed software for floor plan retrieval. 112
B.2 Sequence diagram depicting the overall flow of the software. 113
B.3 Screen prompting the user to select a floor plan of his/her choice. 113
B.4 Segmentation of the selected plan into labeled rooms 114
B.5 Graph embedding of the room level structure of the floor plan. 114
B.6 Selecting how many results to be displayed while retrieval. 115
B.7 User's required floor plans on display as per their similarity from the query floor plan. 116

List of Tables

Table	*Title*	*Page*
2.1	Reviews of approaches for floor plan analysis	12
2.2	Reviews of approaches for symbol spotting	14
2.3	Details of publicly available existing floor plan datasets	18
4.1	Network Parameters for the proposed framework	49
4.2	Performance Comparison (MAP) of CNN layers used for floor plan retrieval task (on ROBIN dataset).	53
4.3	Performance Comparison in terms of MAP on *ROBIN* dataset.	55
5.1	Percentage Accuracy of the extracted high level semantic features	64
5.2	Comparison of techniques on ROBIN and SESYD Dataset	72
6.1	Network Parameters for the proposed framework	90
6.2	Parameter Values used while implementing Cyclic GAN	90
6.3	Parameter Values for Encoder Network	91
6.4	Mean Average Precision values for the proposed framework considering different distance metrics.	96

表格目录

Table	*Title*	*Page*
2.1	Reviews of approaches for floor plan analysis	12
2.2	Reviews of approaches for symbol spotting	14
2.3	Details of publicly available existing floor plan datasets	18
4.1	Network Parameters for the proposed framework	49
4.2	Performance Comparison (MAP) of CNN layers used for floor plan retrieval task (on ROBIN dataset).	53
4.3	Performance Comparison in terms of MAP on *ROBIN* dataset.	55
5.1	Percentage Accuracy of the extracted high level semantic features	64
5.2	Comparison of techniques on ROBIN and SESYD Dataset	72
6.1	Network Parameters for the proposed framework	90
6.2	Parameter Values used while implementing Cyclic GAN	90
6.3	Parameter Values for Encoder Network	91
6.4	Mean Average Precision values for the proposed framework considering different distance metrics.	96

List of Symbols

N	Total number of samples in a dataset
$\mathscr{I}$	Input floor plan image
$\oplus$	Dilation
$\ominus$	Erosion
S_a	Structuring Element
$\mathscr{I}_w$	Input floor plan image without walls
F	Furniture template signature
$\mathscr{C}$	Furniture count
$\mathscr{T}$	Furniture type set
$\mathscr{B}$	Blobs obtained through morphological fill operation
$\mathbf{CC()}$	Connected Component Operator
$\lvert\cdot\rvert$	Cardinality
C	Connected components of a blob
$\mathscr{A}$	Set containing areas of the connected components in a blob
$S()$	Signature
G_k	Topology graph representing adjacencies in k^{th} floor plan
V_k	Vertex set of a graph representing k^{th} floor plan
E_k	Edge set of a graph representing k^{th} floor plan
λ_k	Eigen values
$\vec{\phi}_k$	Eigen vectors
Adj_k	Adjacency matrix of the k^{th} graph
$\vec{F}_k$	Feature vector
Cv	Covariance Matrix
$cost_{qm}$	Match cost between two floor plan samples q and m
$\mathscr{C}^u$	Unique furniture inside each floor plan
Ar	Carpet Area of a room
L_k	k^{th} layout
$\lvert.\rvert$	Absolute value of a function
$A(\cdot)$	Carpet Area Ratio of a room corresponding to a layout
$e(\cdot,\cdot)$	Edit Distance
$\rho(\cdot)$	Room adjacency string matching score
$\psi(\cdot)$	Carpet Area Ratio matching score
$\phi(\cdot)$	Difference in number of furniture
$\theta(\cdot)$	Difference in type of furniture
F^+	Cumulative match score of a feature be it θ, ψ, ϕ, ρ
η	Weight Coefficients
I	Image domain
S	Sketch domain
f_I^i	i^{th} floor plan image sample
f_S^j	j^{th} floor plan sketch sample from S-ROBIN
$\mathbf{F(.)}$	Mapping function mapping images to sketches
$\mathbf{G(.)}$	Mapping function mapping sketches to images
D_S	Adversarial Discriminator for sketches
D_I	Adversarial Discriminator for images

List of Symbols

N 数据集中样本的总数 $\mathscr{I}$ 输入的平面布局图像 $\oplus$ 膨胀 $\ominus$ 腐蚀 S_a 结构元素 $\mathscr{I}_w$ 不含墙体的输入平面布局图像 F 家具模板特征 $\mathscr{C}$ 家具数量 $\mathscr{T}$ 家具类型集合 $\mathscr{B}$ 通过形态学填充操作 CC() 获取的连通区域 $|\cdot|$ 基数 C 连通区域的连通分量 $\mathscr{A}$ 包含连通区域内各连通分量面积的集合 $S()$ 特征 G_k 表示平面布局中邻接关系的拓扑图 k^{th} 平面布局 V_k 表示平面布局的图的顶点集 k^{th} 平面布局 E_k 表示平面布局的图的边集 k^{th} 平面布局 λ_k 特征值 $\vec{\phi}_k$ 特征向量 Adj_k 图的邻接矩阵 k^{th} 特征向量 $\vec{F}_k$ 协方差矩阵 Cv 两个平面布局样本 q 和 m 之间的匹配代价 $cost_{qm}$ 每个平面布局中独有的家具 $\mathscr{C}^u$ 房间的地毯面积 Ar 布局 L_k k^{th} 函数的绝对值 $|.|$ 对应某一布局的房间地毯面积比 $A(\cdot)$ 编辑距离 $e(\cdot,\cdot)$ 房间邻接字符串匹配得分 $\rho(\cdot)$ 地毯面积比匹配得分 $\psi(\cdot)$ 家具数量差异 $\phi(\cdot)$ 家具类型差异 $\theta(\cdot)$ 某一特征的累积匹配得分 F^+ 权重系数 θ,ψ,ϕ,ρ η 图像域 I 草图域 S 平面布局图像样本 f_I^i i^{th} S-ROBIN 中的平面布局草图样本 f_S^j j^{th} F(.) 将图像映射为草图的映射函数 G(.) 将草图映射为图像的映射函数 D_S 草图的对抗判别器 D_I 图像的对抗判别器

$f_S \rightarrow p(f_S)$	Probability distribution of a sample $f_S \in S$
$f_I \rightarrow p(f_I)$	Probability distribution of a sample $f_I \in I$
$S2I_A(.)$	Adversarial loss function for mapping a floor plan sketch to an image
$I2S_A(.)$	Adversarial loss function for mapping a floor plan image to a sketch
$CCL(.)$	Cycle consistency loss
$X(.)$	Final objective function in Cycle GAN
α	Hyperparameter in Cycle GAN network
$\mathscr{E}$	Encoder
$\mathscr{D}$	Decoder

$f_S \to p(f_S)$　　样本$f_S \in S$、$f_I \to p(f_I)$的概率分布 样本$f_I \in I$、S、$I_A(.)$的概率分布 将平面草图映射为图像的对抗损失函数 I、$S_A(.)$ 将平面图像映射为草图的对抗损失函数 $CCL(.)$ 循环一致性损失 $X(.)$ Cycle GAN中的最终目标函数 α Cycle GAN网络中的超参数 $\mathscr{E}$ 编码器 $\mathscr{D}$ 解码器

List of Abbreviations

ARD	Adjacent Room Detection
ARG	Attributed Relational Graphs
CAD	Computer Aided Design
CAR	Carpet Area Ratio
CBIR	Content Based Image Retrieval
CMS	Cumulative Match Score
CNN	Convolutional Neural Network
CVC-FP	Computer Vision Center- FLoor Plan
DAR	Document Analysis and Retrieval
DIA	Document Image Analysis
DNN	Deep Neural Networks
D-SIFT	Dense Scale Invariant Feature Transform
FCR	Furniture Composition Record
GAN	Generative Adversarial Network
GI	Generated Images
GS	Generated Sketches
GUI	Graphical User Interface
HOG	Histogram of Gradients
MAP	Mean Average Precision
OASIS	Online Algorithm for Scalable Image Similarity
PR	Precision Recall
RAS	Room Adjacency String
RDM	Room Decor Matching
ReLU	Rectified Linear Unit
RLH	Run Length Histogram
RLM	Room Layout Matching
ROBIN	Repository Of BuildIng plaNs
SESYD	Systems Evaluation SYnthetic Documents
SIFT	Scale Invariant Feature Transform
SOA	State of the Art
S-ROBIN	Sketched Repository Of BuildIng plaNs

List of Abbreviations

ARD 相邻房间检测 ARG 属性关系图 CAD 计算机辅助设计 CAR 地毯面积比 CBIR 基于内容的图像检索 CMS 累积匹配得分 CNN 卷积神经网络 CVC-FP 计算机视觉中心—楼层平面图 DAR 文档分析与检索 DIA 文档图像分析 DNN 深度神经网络 D-SIFT 密集尺度不变特征变换 FCR 家具构成记录 GAN 生成对抗网络 GI 生成图像 GS 生成草图 GUI 图形用户界面 HOG 梯度直方图 MAP 平均精度均值 OASIS 可扩展图像相似性在线算法 PR 查准率 RAS 房间相邻关系字符串 RDM 房间装饰匹配 ReLU 修正线性单元 RLH 游程长度直方图 RLM 房间布局匹配 ROBIN 建筑平面图资源库 SESYD 系统评估 SYnthetic Documents SIFT 尺度不变特征变换 SOA 最先进技术 S-ROBIN 草图建筑平面图资源库

Contents

	page
Abstract	*v*
Acknowledgements	*vii*
Contents	*ix*
List of Figures	*xi*
List of Tables	*xv*
List of Symbols	*xvii*
List of Abbreviations	*xix*
Chapter 1: Introduction	1
1.1 Motivation of the Problem	4
1.2 Problem Statement	6
1.2.1 Brief description of the Work Done	*7*
1.3 Research Issues	8
1.3.1 Organization of the Thesis	*9*
Chapter 2: Background and Related Work	11
2.1 Introduction	11
2.2 State of the art in Floor plan Analysis	11
2.2.1 3D reconstruction of floor plans	*11*
2.2.2 Semantic analysis of floor plans	*13*
2.2.3 Structural Analysis	*14*
2.3 State of the art in Symbol spotting	15
2.3.1 Region based symbol spotting	*15*
2.3.2 Segmentation and recognition of isolated symbols for symbol spotting	*15*
2.4 State of the art in generic features for retrieval	16
2.5 State of the art in Deep learning frameworks for image and sketch retrieval	17
2.6 Floor plan Datasets	18
2.7 Summary	19
Chapter 3: Topological features for representation and retrieval of similar floor plans	21
3.1 Brief Overview	22
3.2 Layout segmentation	23
3.3 Adjacent room detection	25
3.4 Furniture detection and categorization	26
3.5 Graph Spectral Embedding	28
3.5.1 Spectral Feature representation	*28*
3.5.2 Spectral feature embedding	*29*
3.5.3 Feature Matching and Retrieval	*30*
3.6 Experiments and Results	31
3.6.1 Qualitative Results	*32*
3.6.2 Quantitative Results	*35*
3.7 Discussion	36
3.8 Summary	39
Chapter 4: Deep convolutional neural networks for Floor plan Analysis and Retrieval	41
4.1 Brief Overview	42
4.2 Feature Representation	43
4.2.1 Multilayer feed-forward neural network Architecture	*43*
4.2.2 Proposed CNN architecture	*45*
4.3 Matching and Retrieval	50
4.4 Experiments and Results	51
4.4.1 Qualitative Results	*52*
4.4.2 Quantitative Results	*53*
4.5 Summary	55
Chapter 5: Requirement driven feature integration for floor plan retrieval	57
5.1 Brief Overview	58
5.2 High Level Semantic features	59

Contents

page
摘要 v
致谢 vii
内容 ix
图表目录 xi
表格目录 xv
符号列表 xvii
缩略语列表 xix

第一章：引言 1
1.1 问题的提出背景 4
1.2 问题陈述 6
1.2.1 工作概述 7
1.3 研究问题 8
1.3.1 论文的组织结构 9

第2章：背景与相关工作 11
2.1 引言 11
2.2 平面图分析的最新研究进展 11
2.2.1 平面图的三维重建 11
2.2.2 平面图的语义分析 13
2.2.3 结构分析 14
2.3 符号检测领域的研究现状 15
2.3.1 基于区域的符号检测 15
2.3.2 用于符号检测的孤立符号分割与识别 15
2.4 用于检索的通用特征研究现状 16
2.5 用于图像与草图检索的深度学习框架研究现状 17
2.6 平面布局数据集 18
2.7 总结 19

第三章：用于相似平面图表示与检索的拓扑特征 21
3.1 简要概述 22
3.2 版面分割 23
3.3 相邻房间检测 25
3.4 家具检测与分类 26
3.5 图谱嵌入 28
3.5.1 光谱特征表示 28
3.5.2 光谱特征嵌入 29
3.5.3 特征匹配与检索 30
3.6 实验与结果 31
3.6.1 定性结果 32
3.6.2 定量结果 35
3.7 讨论 36
3.8 总结 39

第4章：用于平面图分析与检索的深度卷积神经网络 41
4.1 简要概述 42
4.2 特征表示 43
4.2.1 多层前馈神经网络架构 43
4.2.2 提出的卷积神经网络架构 45
4.3 匹配与检索 50
4.4 实验与结果 51
4.4.1 定性结果 52
4.4.2 定量结果 53
4.5 总结 55

第5章：面向平面图检索的需求驱动特征集成 57
5.1 简要概述 58
5.2 高层次语义特征 59

5.2.1 Feature 1: Room Adjacency String (RAS) 60
5.2.2 Feature 2: Carpet Area Ratio (CAR) 61
5.2.3 Feature 3: Furniture Composition Record (FCR) 62
5.3 Fine-grained Matching and Retrieval 63
5.3.1 RAS Matching Score 64
5.3.2 CAR Matching Score 65
5.3.3 Furniture Composition Record Matching Score 66
5.3.4 Cumulative Match Score (CMS) 67
5.3.5 Total Match Score 67
5.4 Experiments and Results 68
5.4.1 Qualitative results 69
5.4.2 Quantitative results 70
5.5 Discussion 71
5.6 Summary 72

Chapter 6: Domain invariant feature transform for interactive floor plan retrieval 75
6.1 Brief Overview 75
6.2 Approach 1: Cyclic GAN for domain adaptation and sketch based floor plan retrieval 76
6.2.1 Domain Mapping through Cyclic GAN 78
6.2.2 Feature Representation through CNN 82
6.2.3 Matching and Retrieval 83
6.3 Approach 2: Unified framework for retrieval using autoencoder and Cyclic GAN 83
6.3.1 CNN for query classification 85
6.3.2 Autoencoder for domain mapping 85
6.3.3 Matching and Retrieval 87
6.4 Experiments and Results 88
6.4.1 Dataset Creation 88
6.4.2 Implementation Details 88
6.4.3 Parameter Setting 89
6.4.4 Qualitative Results 91
6.4.5 Quantitative Results 94
6.5 Summary 98

Chapter 7: Conclusions 101
7.1 Contributions 101
7.2 Discussion on the relative performance of the techniques proposed 102
7.3 Scope of Future Work 104

Chapter A: Floor plan datasets ROBIN and S-ROBIN 105
A.1 Number of Rooms 106
A.2 Global shape of the house 106
A.3 Connectivity within a house 107
A.4 Placement of furniture 107
A.5 Dataset creation process 108
A.6 Statistics of the dataset 109
A.7 Conclusion 109

Annexure B: GUI for floor plan analysis and retrieval 111
B.1 System Architecture 111
B.2 Software functionality 112
B.3 Screen-shots of the floor plan retrieval framework 112
B.4 Conclusion 116

References 117

5.2.1 特征1：房间邻接字符串（RAS） 60
5.2.2 特征2：地毯面积比（CAR） 61
5.2.3 特征3：家具构成记录（FCR） 62
5.3 细粒度匹配与检索 63
5.3.1 RAS匹配得分 64
5.3.2 CAR匹配得分 65
5.3.3 家具成分记录匹配得分 66
5.3.4 累积匹配得分（CMS） 67
5.3.5 总匹配得分 67
5.4 实验与结果 68
5.4.1 定性结果 69
5.4.2 定量结果 70
5.5 讨论 71
5.6 总结 72

第6章：用于交互式平面图检索的领域不变特征变换 75
6.1 简要概述 75
6.2 方法一：基于循环生成对抗网络的域适应与基于草图的平面布局检索 76
6.2.1 通过循环生成对抗网络进行领域映射 78
6.2.2 通过卷积神经网络进行特征表示 82
6.2.3 匹配与检索 83
6.3 方法二：基于自编码器与循环生成对抗网络的统一检索框架 83
6.3.1 用于查询分类的卷积神经网络 85
6.3.2 用于领域映射的自编码器 85
6.3.3 匹配与检索 87
6.4 实验与结果 88
6.4.1 数据集构建 88
6.4.2 实施细节 88
6.4.3 参数设置 89
6.4.4 定性结果 91
6.4.5 定量结果 94
6.5 总结 98

第七章：结论 101
7.1 贡献 101
7.2 关于所提出技术相对性能的讨论 102
7.3 未来工作范围 104

第A章：楼层平面图数据集ROBIN与S-ROBIN 105
A.1 房间数量 106
A.2 房屋的整体形状 106
A.3 房屋内部的连通性 107
A.4 家具的摆放 107
A.5 数据集构建过程 108
A.6 数据集的统计信息 109
A.7 结论 109

附录B：用于楼层平面分析与检索的图形用户界面 评估 111
B.1 系统架构 111
B.2 软件功能 112
B.3 平面图检索框架的截图 112
B.4 结论 116

参考文献 117

Abstract

Architects often refer to existing layouts, while designing new projects. This process aids in providing insight into how such similar architectural situations were handled in the past. By studying one or several previous reference projects the architect tries to derive a solution for the current problem. The manual look-up process for such similar projects through the layouts can be rather cumbersome. With several architectural projects archived in digital form, the research and development of fast automatic retrieval techniques in floor plans is the need of the hour.

Floor plan analysis is a special case of document image understanding. It aims at extracting semantic and structural details of an architectural layout by analysis of the 2D image of the floor plan. Symbol spotting and retrieval in architectural layouts have been solved as individual problems in the past. Moreover, both the image and sketch have been used as modalities for the symbol spotting task. Thus, upon performing a keen analysis related to the existing work in the area of architectural floor plans, it was concluded that retrieval in this particular domain is a challenging yet less researched area and has a variety of applications in today's digital scenario. Specific requirements by buyers during property rent/sale can be met using a composite, automated framework that takes into account semantics, as well as the content inside a floor plan for retrieval. Existing floor plan datasets like Systems Evaluation SYnthetic Documents (SESYD) and Computer Vision Center- Floor Plan (CVC-FP) dataset are not fit for a retrieval task as they don't suffice in number and variation in the floor plan samples for the task of floor plan retrieval. Keeping in mind the current status of floor plan analysis research, two publicly available benchmark datasets Repository Of BuildIng plaNs (ROBIN) and Sketched-Repository Of BuildIng plaNs (S-ROBIN) are proposed in this thesis that will aid research for the community in the area of floor plan analysis and retrieval.

In this thesis, techniques to analyse architectural floor plans and extraction of different features to aid in content based retrieval in floor plans is proposed. In one such attempt, retrieval in floor plans under the query by example paradigm, based on similar overall layout designs as well as the arrangement of the room decor present in the layouts, is proposed. Here query is taken as a floor plan image. A room layout segmentation and adjacent room detection algorithm is presented to represent layouts as an undirected graph. The vertices of the graph represent the rooms, while the edges represent the connectivity between them. Also a novel graph spectral embedding feature is proposed to uniquely represent the layout of the architectural floor plan. This helps in effective and efficient matching of the room layouts. To match the semantic similarity between a pair of floor plans, a two stage matching technique is proposed and high retrieval accuracy is obtained. An interactive graphical user interface to aid users to select, analyse and retrieve similar floor plans is also proposed in this thesis.

In another attempt a Convolutional Neural Network (CNN) framework to extract both low and high level semantic features is proposed for floor plan retrieval. Experiments were conducted on publicly available datasets as well as ROBIN. The key contributions in the proposed approach are, a novel deep learning framework to retrieve similar floor plan layouts from repository. Also, in this part of the work the effect of the individual deep convolutional neural network layers for floor plan retrieval task is analysed.

Experiments have shown that the deep learning frameworks work very well when the target image itself has a lot of features. Examples include natural images, textual documents, etc. On the other hand for images, which are not that feature rich, the off-the-shelf CNNs are not that effective.

Abstract

建筑师在设计新项目时，常常参考现有的平面布局。这一过程有助于深入了解以往类似建筑情境的处理方式。通过研究一个或多个既有的参考项目，建筑师试图为当前问题找到解决方案。然而，通过平面图手动查找此类相似项目的流程往往较为繁琐。随着大量建筑项目以数字化形式存档，研发用于快速自动检索建筑平面图的技术已成为当务之急。

平面图分析是文档图像理解的一个特例，其目标是通过分析平面图的二维图像来提取建筑布局的语义与结构信息。以往，建筑布局中的符号检测与检索通常被作为独立问题加以解决；此外，在符号检测任务中，既采用图像也采用草图作为数据模态。因此，通过对建筑平面图领域现有研究的深入分析，可以得出结论：在这一特定领域中，检索任务虽极具挑战性，却鲜有深入研究，且在当今数字化环境中具有广泛的应用前景。借助一个综合性的自动化框架，结合平面图中的语义信息及其内容特征进行检索，即可满足购房者在租赁或出售房产时的具体需求。然而，现有的平面图数据集，如Systems Evaluation SYnthetic Documents（SESYD）和Computer Vision Center- Floor Plan（CVC-FP）数据集，并不适用于检索任务，因为其样本数量及多样性均不足以支持平面图检索这一特定任务。鉴于当前平面图分析研究的现状，本论文提出了两个公开可用的基准数据集——Repository Of BuildIng plaNs（ROBIN）和Sketched-Repository Of BuildIng plaNs（S-ROBIN），以助力该领域的研究工作。

本论文提出了一种用于分析建筑平面图并提取不同特征以支持基于内容的平面图检索的技术。在其中一项研究中，针对“以例搜图”范式下的平面图检索，提出了既考虑整体布局设计相似性，又兼顾布局中房间装饰布置的检索方法。在此，查询对象为一张平面图图像。本文还提出了一种房间布局分割与相邻房间检测算法，用于将平面布局表示为无向图；图中的顶点代表各个房间，边则表征房间之间的连通关系。此外，本文还提出了一种新颖的图谱嵌入特征，用以唯一地表征建筑平面图的布局结构，从而实现房间布局的有效且高效匹配。为度量两幅平面图之间的语义相似性，本文进一步提出了一种两阶段匹配技术，并获得了较高的检索精度。最后，本论文还设计了一种交互式图形用户界面，以帮助用户选择、分析并检索相似的平面图。

在另一项研究中，提出了一种基于卷积神经网络（CNN）的框架，用于同时提取低级和高级语义特征，以实现平面布局检索。实验在公开可用的数据集以及ROBIN数据集上进行。所提方法的主要贡献在于：构建了一个新颖的深度学习框架，能够从知识库中检索出相似的平面布局；此外，在本研究的这一部分，还分析了各层深度卷积神经网络对平面布局检索任务的影响。

实验表明，当目标图像本身包含丰富的特征时，深度学习框架的性能表现非常优异，例如自然图像、文本文档等。然而，对于那些特征较为稀疏的图像，现成的卷积神经网络则效果欠佳。

Moreover, deep features mostly capture the global similarity in an image. It was envisaged that effective combination of domain specific features may give superior results. The effect of combining various extracted features in a weighted manner to aid in giving preference to a certain feature while retrieval is also proposed. A novel end-to-end framework for extracting high level semantic features like area and room-wise decor arrangement for the task of fine grained retrieval is proposed. Further, a technique to perform feature fusion to aggregate high-level semantic features extracted is also proposed. Weighted feature fusion helps in setting preferences to particular characteristics of the floor plan while retrieval and satisfying specific user demands.

In an attempt to explore other query modes than query by example in the form of floor plan image, sketch based retrieval is proposed. Sketch based retrieval comes with its own set of challenges in terms of both representation and recognition. However, this mode of query can aid in better correspondence while capturing the user's intent in a query. A composite network comprising of Cyclic Generative Adversarial Networks (Cycle-GAN) along with CNN is proposed to bridge the gap between sketch and image domains while retrieval. An improved approach using autencoders in conjunction with Cyclic Generative Adversarial Networks is proposed, which outperforms all other state-of-the-art techniques for sketch based floor plan retrieval by using an efficient domain mapping approach.

此外，深度特征主要捕捉图像的全局相似性。人们预期，通过有效融合领域特定特征，可以获得更优的性能。本文还提出了一种加权组合多种提取特征的方法，以在检索过程中对某一特征赋予更高的优先级。此外，本文提出了一种用于细粒度检索任务的新型端到端框架，该框架能够提取诸如面积及房间级装饰布局等高层次语义特征。进一步地，本文还提出了一种用于聚合所提取的高层次语义特征的特征融合技术。加权特征融合有助于在检索过程中对平面图的特定属性进行偏好设置，从而满足用户的个性化需求。

为了探索除基于平面图图像的示例查询之外的其他查询模式，本文提出了基于草图的检索方法。基于草图的检索在表示与识别两个方面均面临独特的挑战。然而，这种查询模式能够在捕捉用户查询意图的同时，实现更精确的匹配。为此，本文提出了一种由循环生成对抗网络（Cycle-GAN）与卷积神经网络（CNN）组成的复合网络架构，以在检索过程中弥合草图域与图像域之间的差异。此外，本文还提出了一种结合自编码器与循环生成对抗网络的改进方法，该方法通过采用高效的领域映射策略，在基于草图的平面图检索任务中优于现有的各类先进方法。

1
Introduction

In today's time, despite the increasing trend in the creation and processing of digital data, handwritten or paper based storage is still active and growing [Hilbert and López, 2011]. This has in turn led to a growth in the research and applications in Document Image Analysis (DIA) systems. The primary purpose of any DIA system is, automated storage, extraction, and processing of the information present in documents. The research in DIA lies at a very interesting intersection of individual fields like pattern recognition, image processing, machine learning, artificial intelligence, storage systems and linguistics and encompasses several sub research areas, like document categorization and understanding, hand-written and typed text recognition, graphics recognition and graphics retrieval. Figure 1.1 depicts a typical example and various types of tasks [Kasturi *et al.*, 2002] that fall under the umbrella of DIA.

Graphics recognition is a sub-domain in Document Image Analysis which aims at analyzing the content of graphical documents. A graphical document can be defined as a virtual or a physical object created to convey a piece of information. This piece can manifest itself in a picture or a diagrammatic form, conforming to a graphical language. Examples of graphical documents can be engineering drawings, flowcharts, maps, diagrams, architectural floor plans, etc. Graphical languages are human-made communication systems enabling the expression of complex semantic concepts via graphical information. In graphical languages, the vocabulary comprises of a symbol set, and the grammatical rules are context based relations between the symbols in the symbol set which provides the complete meaning to the given graphics in the document. Thus, the understanding of a graphical document comprises of both parsing of, the unique, isolated symbols as well as the relationship between their context. Still, graphical languages, have variations in terms of, the nature of reader, the difficulty level in the comprehension of the enclosed information, design and creativity matters, etc. Apart from recognition of graphical symbols in a graphical document, the next important thing is retrieval of such documents based on the semantic content derived from the symbols identified in those documents.

The amount of data available in digital format has grown exponentially in recent years. Data, specifically in the form of images are generated in large quantities and stored every day, and this results in a massive, distributed, and in most cases unstructured multimedia information. There are mainly two ways using which one can access images from a database; they are:

1. Text based retrieval: In this case, all the images are tagged using manual or semi-automatic tools during the cataloging stage. Indexes are formed over the database based on these tags. After that a suitable data structure is created that suits the retrieval task. During the retrieval stage, traditional text based retrieval techniques are applied to get the result based on the user query.

2. Content-based retrieval: In this type of retrieval method the search is performed by specifying the actual content of the image data in terms of query image, sketch etc. For example, a Content Based Retrieval System, which is described as the process of retrieving desired images from a large collection on the basis of syntactical image features. Such a system uses

1
Introduction

在当今时代，尽管数字数据的生成与处理呈日益增长的趋势，手写或纸质存储方式依然活跃并持续发展[Hilbert和López，2011]。这进而推动了文档图像分析（DIA）系统相关研究与应用的蓬勃发展。任何DIA系统的首要目标在于实现文档中信息的自动化存储、提取与处理。DIA领域的研究处于模式识别、图像处理、机器学习、人工智能、存储系统及语言学等多学科交叉的极具吸引力的交汇点，并涵盖多个子研究方向，如文档分类与理解、手写与印刷文本识别、图形识别以及图形检索。图1.1展示了一个典型示例，以及属于DIA范畴的各类任务[Kasturi等，2002]。

图形识别是文档图像分析的一个子领域，旨在分析图形化文档的内容。图形化文档可定义为一种用于传达信息的虚拟或实体对象，其信息以图像或图示形式呈现，并遵循特定的图形语言。图形化文档的典型示例包括工程图纸、流程图、地图、示意图、建筑平面图等。图形语言是人类构建的通信系统，通过图形信息表达复杂的语义概念。在图形语言中，词汇由一组符号构成，而语法规则则是符号集内各符号之间的上下文关系，这些关系共同决定了文档中图形的整体语义。因此，对图形化文档的理解既包括对单个孤立符号的解析，也包括对其上下文关系的把握。然而，不同的图形语言在读者群体、信息理解的难易程度、设计与创造性等方面仍存在差异。除了识别图形化文档中的图形符号之外，另一项重要任务是根据从这些符号中提取的语义内容实现此类文档的检索。

近年来，以数字形式存储的数据量呈指数级增长。其中，图像数据的生成与存储尤为频繁，每日产生的海量信息往往呈现出分布广泛、且在大多数情况下为非结构化的多媒体特征。从数据库中获取图像的主要方式主要有两种：

1. 基于文本的检索：在这种情况下，所有图像在编目阶段均通过人工或半自动工具进行标注。基于这些标注在数据库上构建索引，随后根据检索任务的需求建立相应的数据结构。在检索阶段，采用传统的基于文本的检索技术，根据用户查询获取检索结果。 2. 基于内容的检索：在这种检索方法中，通过以查询图像、草图等形式指定图像数据的实际内容来进行搜索。例如，一种基于内容的检索系统被定义为：根据图像的句法特征从大规模图像集合中检索出所需图像的过程。此类系统使用

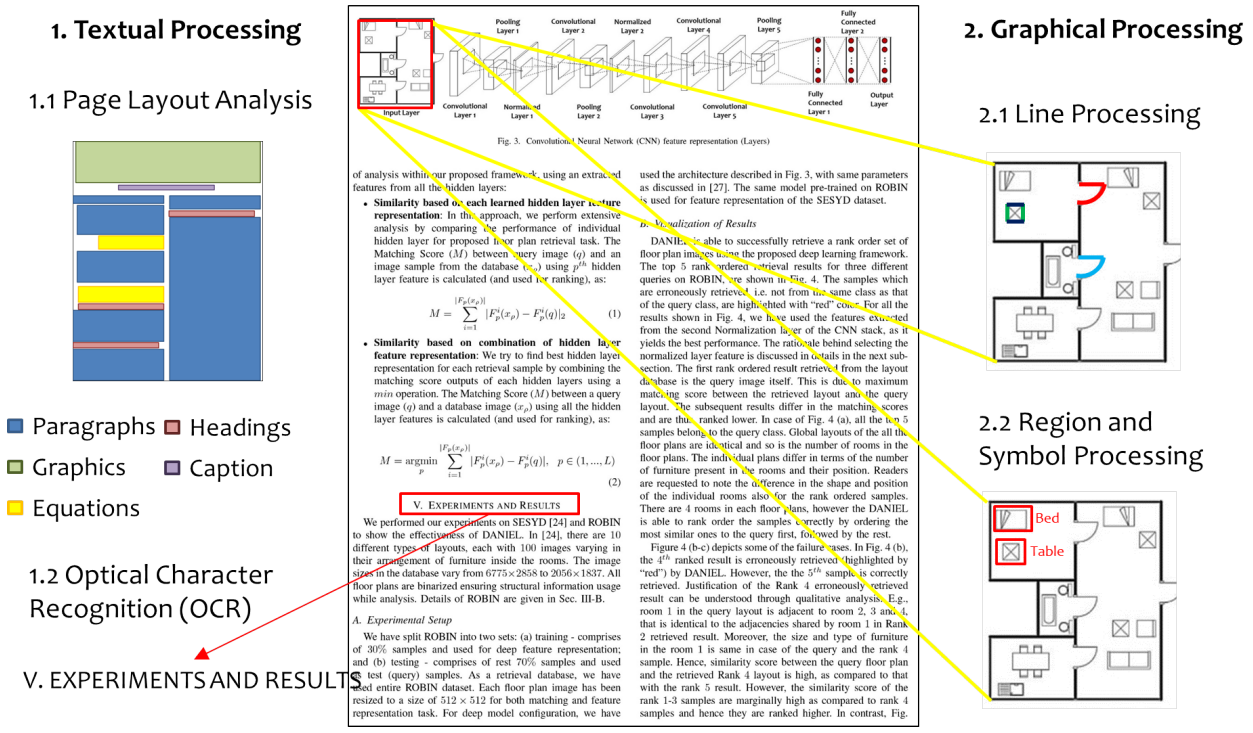


Figure 1.1. : Illustration of various subareas of document image analysis (DIA) on a sample scanned document [Sharma *et al.*, 2017].

different modes of query and assesses a wide variety of of content from the given query for retrieval purposes.

Design of a Content Based Image Retrieval (CBIR) system (refer Fig. 1.2) is a complex task and includes concepts from various academic disciplines such as image processing, computer vision, pattern recognition, machine learning, visual perception etc. Content-based means that the search analyzes the contents of the image rather than the metadata such as keywords, tags, or descriptions associated with the image. The term "content" in this context might refer to colors, shapes, textures, or any other information that can be derived from the image itself. CBIR is desirable because searches that rely purely on metadata are dependent on annotation quality and completeness. Having humans manually annotate images by entering keywords or metadata in a large database can be time consuming and may not capture the keywords desired to describe the image. The process of CBIR involves retrieving similar images in response to user's search queries, using the inherent properties (features) extracted from the image itself. These features are mapped onto a feature space and the images with the most closely placed features in the feature space (depends on a certain metric) are deemed similar and retrieved from the database. An illustration of the same is shown in Fig. 1.2, where the context is chosen to be natural images. One more key aspect that plays an important role in a CBIR system is the modality of the query to the system. As illustrated in Fig. 1.3, query can be in the form of text, image, sketch, colour layout which depicts the colour combination that a user wants to look in the retrieved images or a concept layout where images that contain multiple related objects as specified in the query needs to be retrieved. However, the quality of the retrieval result of any CBIR system depends on the quality of content representation in the feature database. In a content based retrieval system, given a query and a database to search, the following questions need to be addressed:

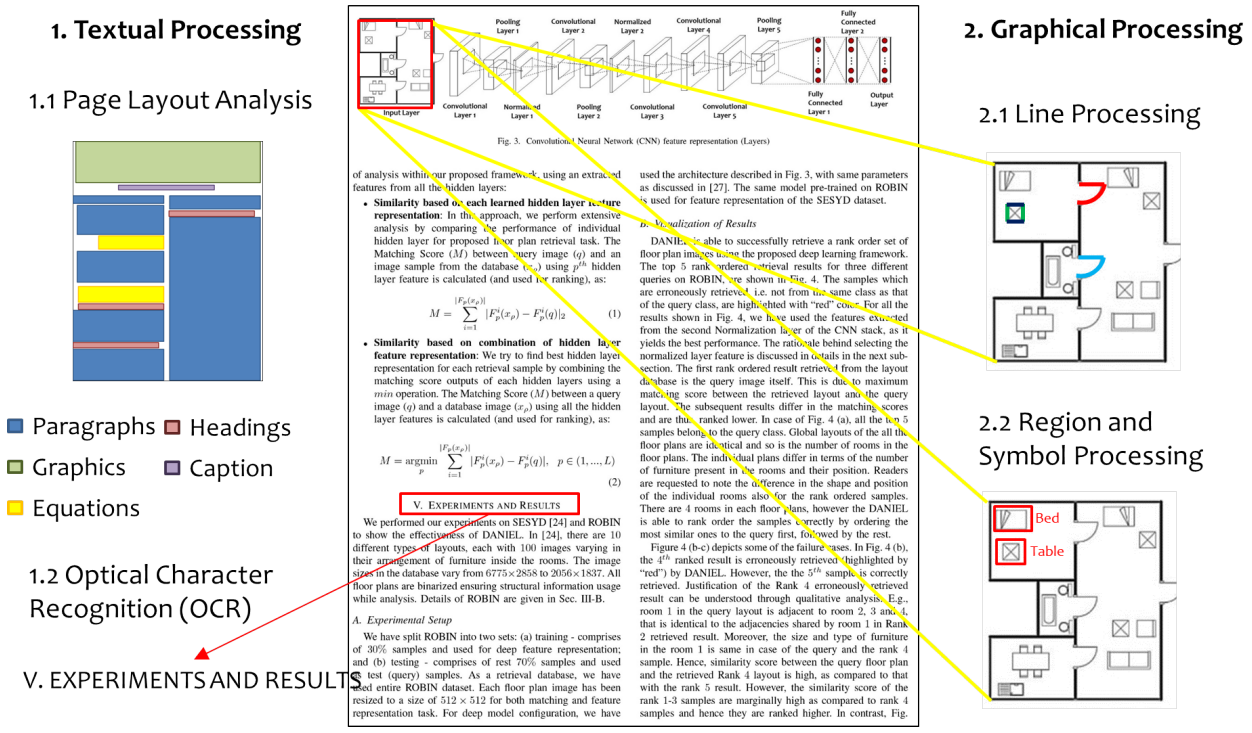


图1.1：在一张扫描文档样本上展示的文档图像分析（DIA）的各个子领域[Sharma等，2017]。

该系统支持多种查询模式，并针对给定查询评估多种类型的内容，以用于信息检索。

基于内容的图像检索（CBIR）系统的设计（参见图1.2）是一项复杂的任务，涉及图像处理、计算机视觉、模式识别、机器学习、视觉感知等多个学科领域的概念。所谓“基于内容”，是指检索过程分析的是图像本身的内在内容，而非与其关联的元数据，如关键词、标签或描述等。在此背景下，“内容”可指颜色、形状、纹理或其他可从图像本身提取的信息。采用CBIR技术具有优势，因为仅依赖元数据的检索方式会受到标注质量和完整性的影响。在大型数据库中由人工通过输入关键词或元数据对图像进行标注，不仅耗时，而且可能无法准确捕捉用于描述该图像的所需关键词。CBIR的过程是：根据用户提出的检索查询，利用从图像本身提取的固有属性（特征）来检索相似图像。这些特征被映射到一个特征空间中，并按照某种度量标准，将特征空间中距离最近的图像判定为相似图像，从而从数据库中予以检索。图1.2对此进行了说明，其中所选场景为自然图像。在CBIR系统中，另一个起关键作用的因素是向系统提交查询的方式。如图1.3所示，查询形式可以是文本、图像、草图，也可以是色彩布局——用于指定用户希望在检索结果中出现的颜色组合；或者采用概念布局，以检索包含查询中所指定的多个相关对象的图像。然而，任何CBIR系统的检索结果质量均取决于特征数据库中内容表示的质量。在基于内容的检索系统中，给定查询和待检索的数据库后，需要解决以下问题：

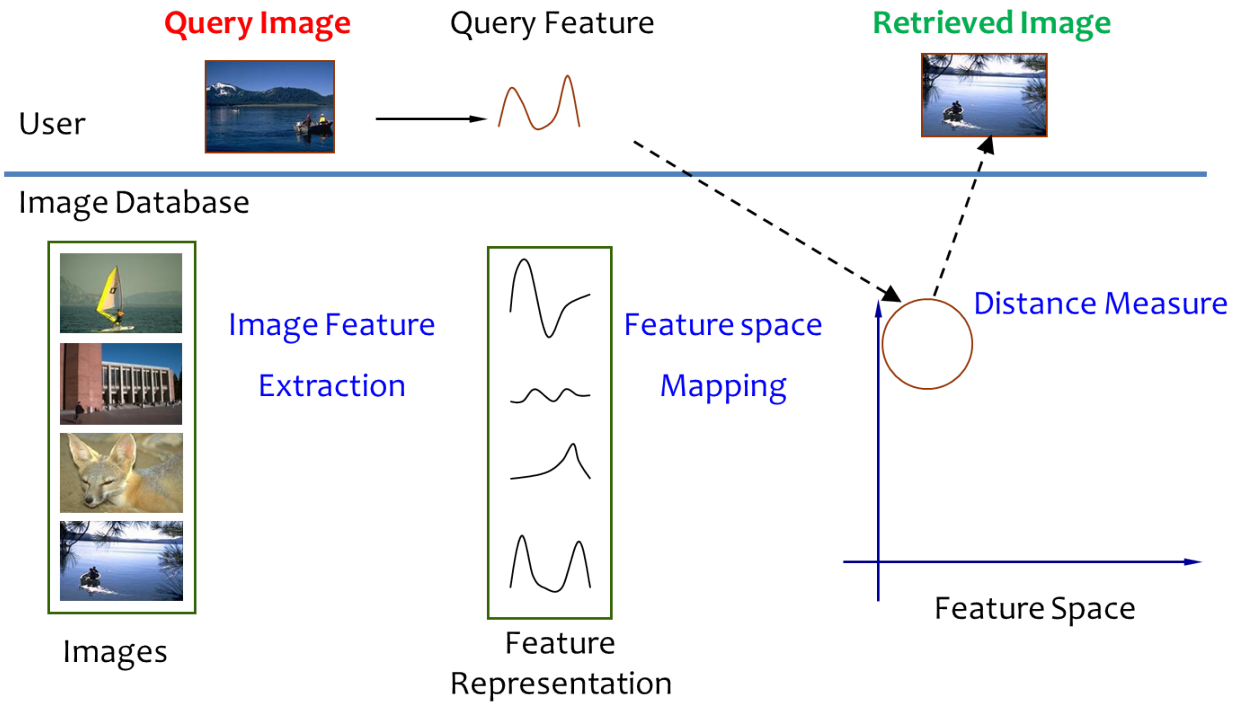


Figure 1.2. : An illustration of a basic Content Based Image Retrieval system.

1. What are the kinds of databases? This pertains to what kind of images are there in the database and what are the categories that the images have been classified into in the database.

2. What are the kind of queries? As illustrated in Fig. 1.3, the mode of the query can be an image, sketch, text, colour layout or a concept layout. Since each query mode has its own challenge, choice of modality is a critical research issue.

3. In which category does this image belong? What constitutes a category in the database and how one generates the feature set which represents each category present in the database distinctively.

4. In the given category, what are the "most similar" images in the database, with respect to the given query? Development of an efficient matching scheme, that helps evaluate the feature level similarity between the query as well as the categorized database images efficiently.

5. How does one make such searches efficient? What indexing mechanisms and efficient matching schemes can be used to make searching across databases fast and accurate.

Existing CBIR systems have been used for retrieval and have found applications in Art collections, Crime prevention, Geographical information, Remote sensing systems, Medical diagnosis, Retail catalogs, Face Finding, etc. Document image retrieval is a branch of CBIR that deals with extracting meaningful content out of document images such as, bank forms, cheques, scanned documents and then retrieving them for various applications. These days a great number of documents are scanned and archived in the form of digital images in digital libraries to make them available and accessible to the web. Therefore, a potential application of document image retrieval can be to make an online digital library where users would be able to do a search on a set of keywords, and get a list of relevant articles across the web, for viewing or printing. Architectural

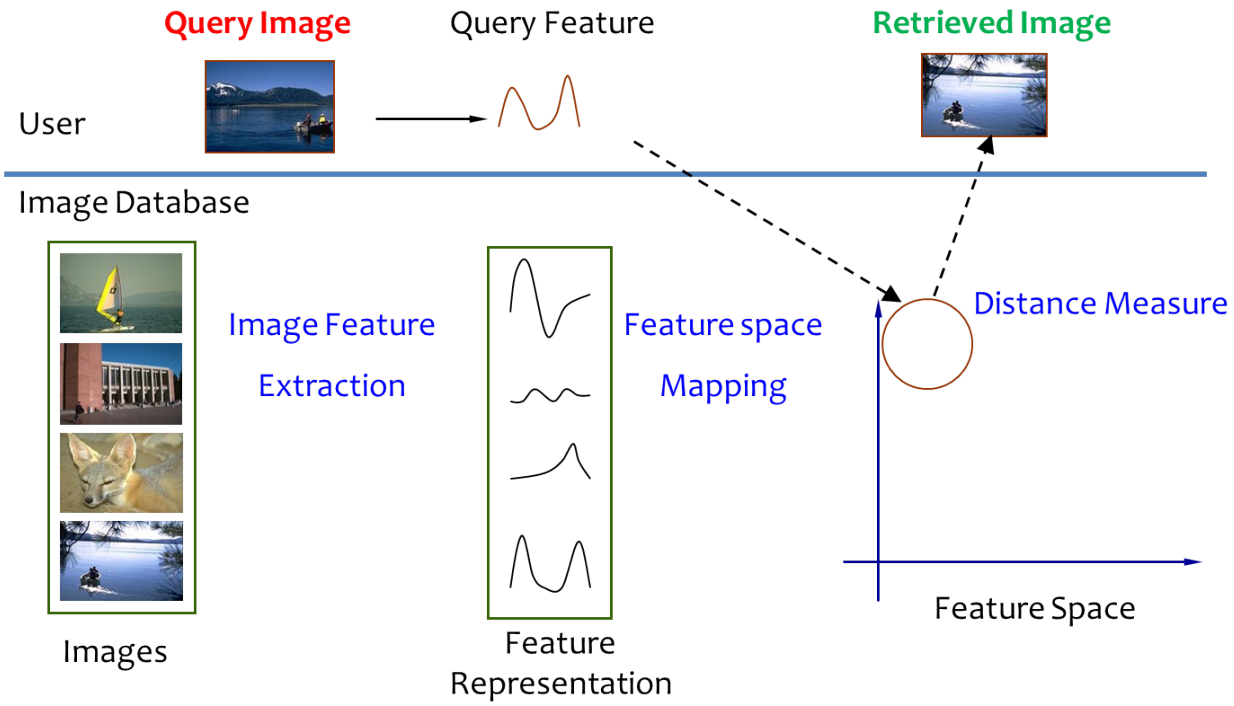


图1.2：一个基本的基于内容的图像检索系统的示意图。

1. 数据库有哪些类型？这涉及数据库中包含哪些类型的图像，以及这些图像在数据库中被划分为哪些类别。

2. 查询的类型有哪些？如图1.3所示，查询模式可以是图像、草图、文本、颜色布局或概念布局。由于每种查询模式都具有其独特的挑战，因此查询模态的选择是一个至关重要的研究问题。

3. 该图像属于哪一类别？数据库中的“类别”如何界定，以及如何构建能够区分性地表征数据库中各类别特征的特征集？

4. 在给定类别中，与给定查询图像在特征层面最为相似的数据库图像有哪些？需开发一种高效的匹配方案，以实现对查询图像与已分类数据库图像之间特征级相似度的高效评估。

5. 如何使此类检索高效进行？可以采用哪些索引机制和高效的匹配方案，以确保跨数据库检索的快速性和准确性？

现有的基于内容的图像检索系统已被广泛应用于各类场景，包括艺术收藏、犯罪预防、地理信息系统、遥感系统、医学诊断、零售目录以及人脸检测等。文档图像检索是基于内容的图像检索的一个分支，其核心任务是从诸如银行表格、支票、扫描文档等文档图像中提取语义信息，并针对不同应用进行检索。如今，大量文档被扫描并以数字图像的形式存档于数字图书馆中，以便通过网络实现共享与访问。因此，文档图像检索的一项潜在应用是构建一个在线数字图书馆，用户可通过输入关键词进行搜索，获取全网范围内相关的文献列表，以供浏览或打印。架构

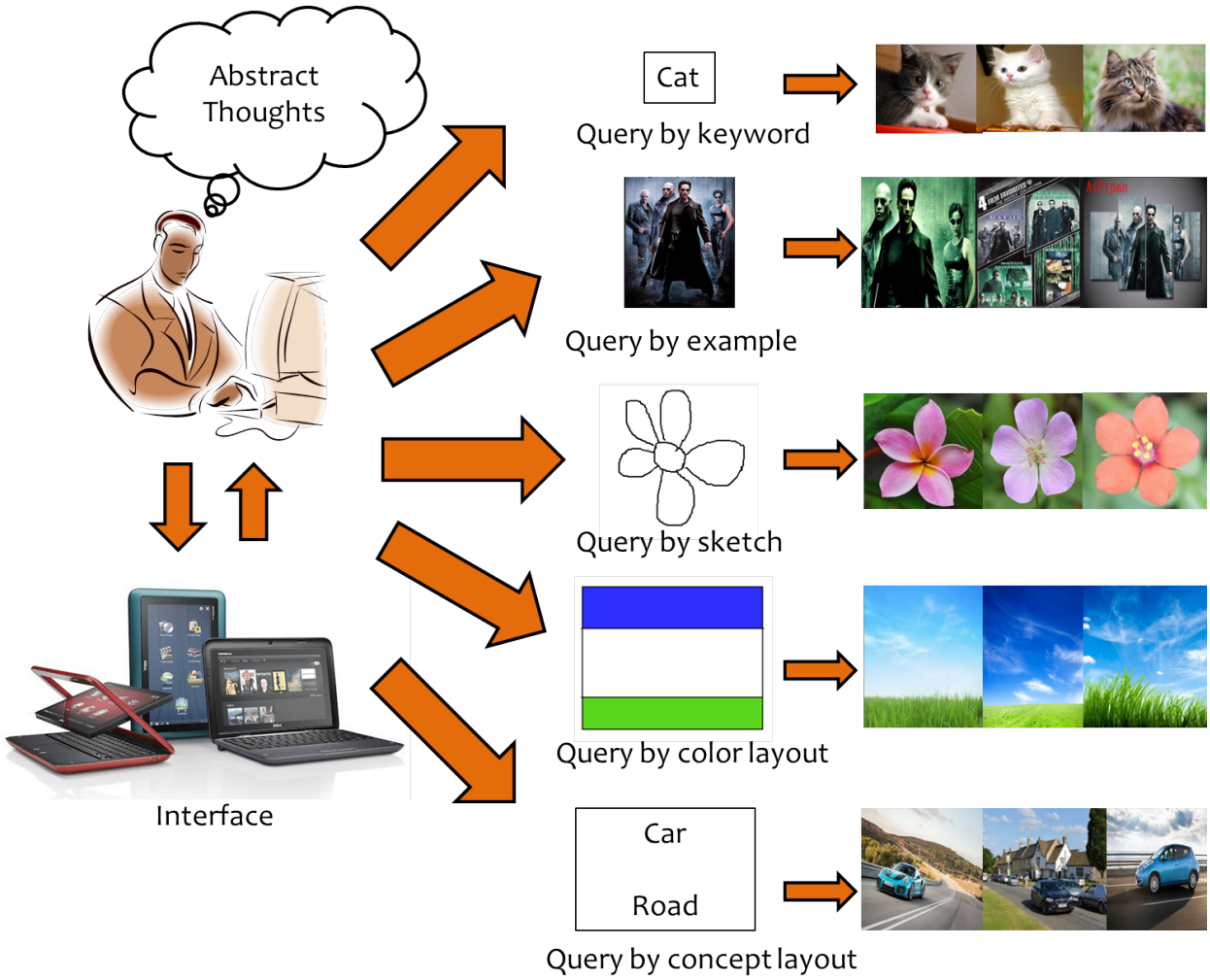


Figure 1.3. : Illustration of various query modes along with their corresponding retrieval results

floor plans are also a special type of documents, which have yet not been analysed with respect to its content and then retrieved based on similarity. Representing the inherent content present in a floor plan image/sketch automatically and identifying similar images from the database poses one of the fundamental challenge for the retrieval task. How to overcome the inherent challenges and come up with an efficient content based retrieval system specifically designed to represent content in architectural floor plans and retrieve thereafter, is a point which this thesis will reinforce.

1.1 MOTIVATION OF THE PROBLEM

The motivation behind the problem of retrieving similar floor plans by analysing meaningful content originated from the current needs and demands of the real-estate industry. To build a dream house in reality, customers visit architects, and specify the requirement "in words". An architect manually studies previous reference projects and suggests the best house and also searches for inspiration or a solution without having the final design at hand. This is an adhoc and tedious process. With the increase in number of online platforms, fast and automatic techniques for retrieval of similar projects are need of the hour. However, "*How to specify the requirements*

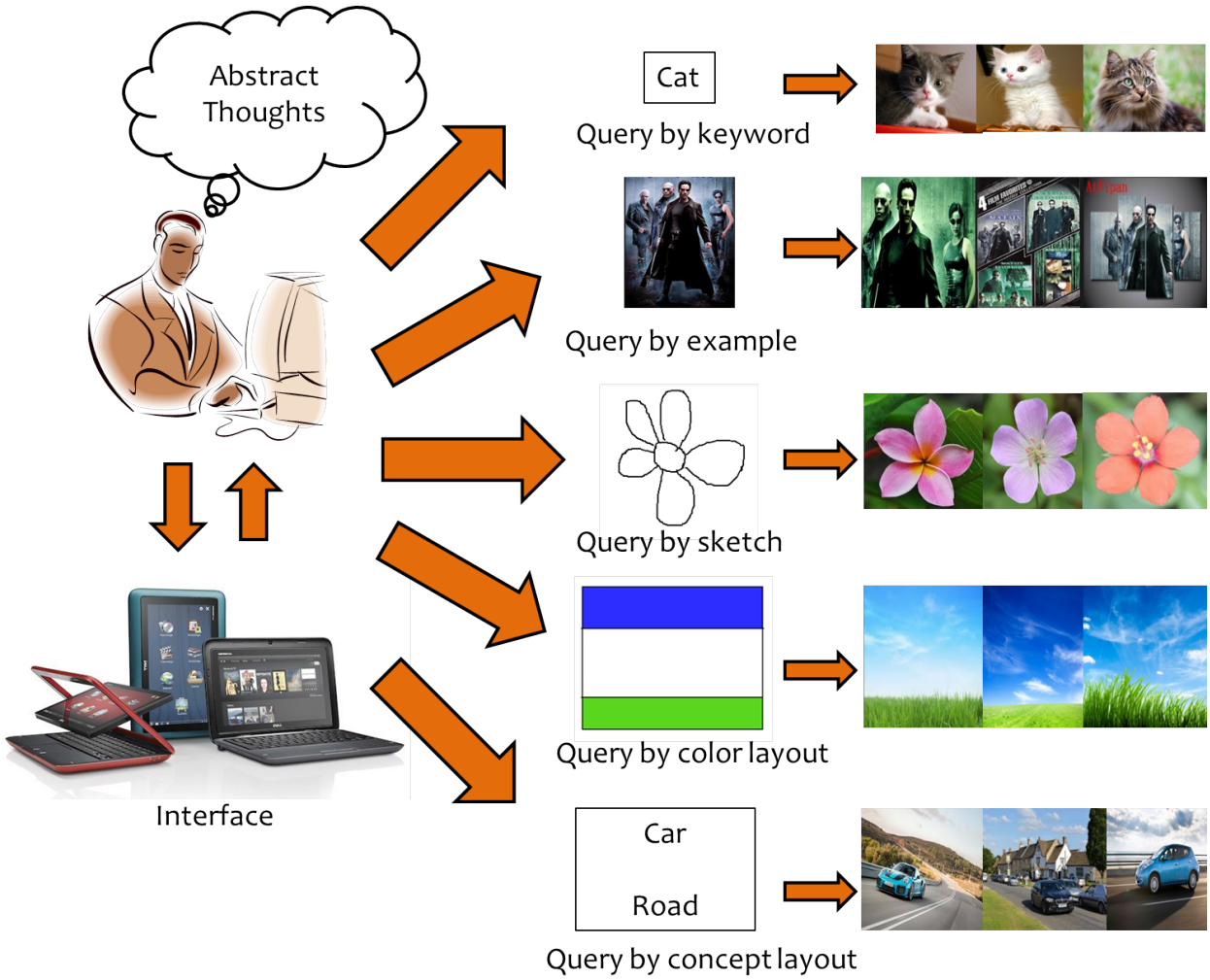


图1.3：各类查询模式及其相应检索结果的示意图

平面图亦是一类特殊的文档，迄今尚未对其内容进行分析并基于相似性实现检索。如何自动表征平面图图像或草图中所蕴含的内在内容，并从数据库中识别出相似图像，是检索任务面临的一项基本挑战。本论文将着重探讨如何克服这些固有难题，进而构建一套专门用于表征建筑平面图内容并实现高效检索的内容驱动型检索系统。

1.1 问题的提出

通过分析语义内容来检索相似平面图这一问题的提出，源于当前房地产行业的实际需求。为了在现实中建造梦想之家，客户会拜访建筑师，并以“口头描述”的方式提出需求。建筑师需手动研读以往的参考项目，从而推荐最合适的方案，并在尚未获得最终设计方案的情况下寻找灵感或解决方案。这一过程既临时性又繁琐。随着在线平台数量的不断增加，亟需开发快速且自动化的相似项目检索技术。然而，“*How to specify the requirements*

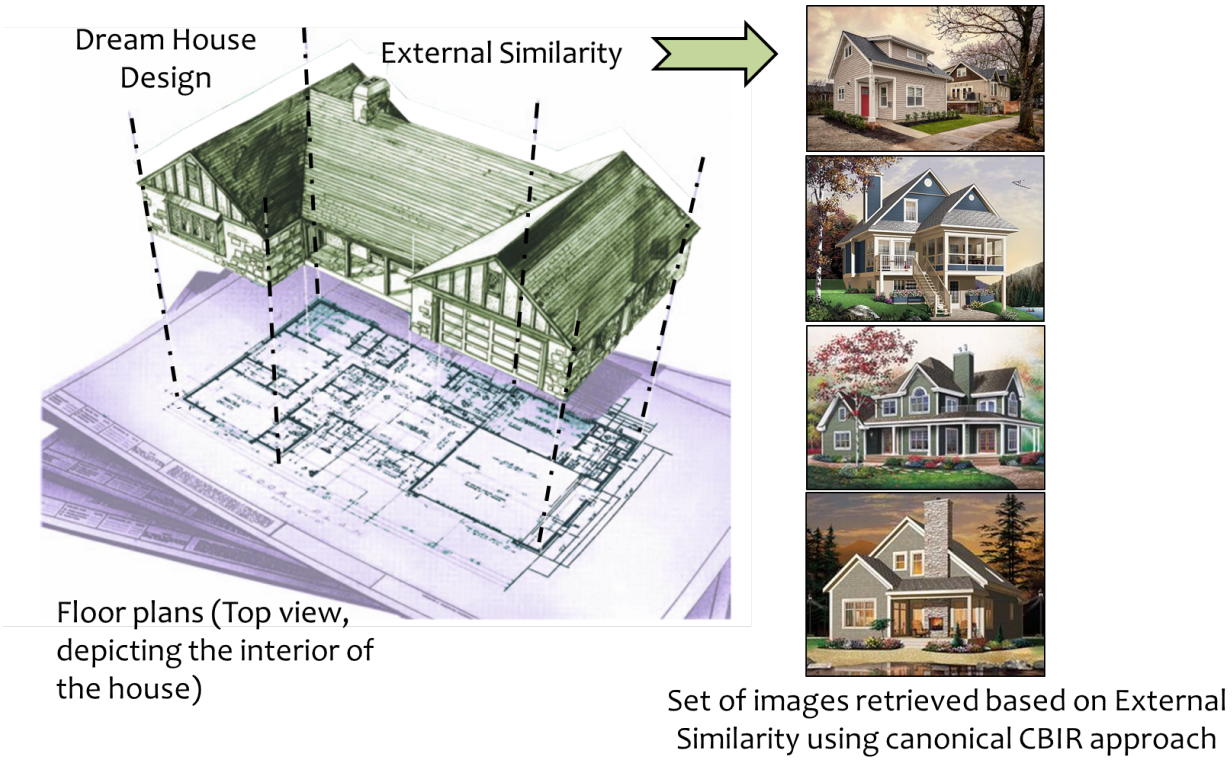


Figure 1.4. : Motivation of the problem.

of a dream house?" is still unclear. A naive solution is to follow the Content Based Image Retrieval (CBIR) approach to cater external similarity. Figure 1.4 shows such a scenario. It can be observed that the retrieved images of houses from the database are externally similar to that of the query image. However, there is no guarantee that these houses would match in their internal similarity as highlighted by the projection line and the floor plan.

However, the requirements of a user are usually pointing towards the internal structure of the home and the decors to be placed within each room. Consider a typical example where the requirement is specified as, "*the house must have* 4 *rooms, with* 2 *bedrooms with attached bathroom,* 1 *drawing room and* 1 *kitchen. The bedrooms must face towards west,...*", then interior similarities along with external structure similarities are also to be matched. Instead of using textual (subjective) query, a pictorial representation is always a better option. A floor plan is a fundamental form of architectural diagram, that gives a view of a house equivalent to a 2D horizontal cross section along with details of the interiors. Figure 1.5 depicts some examples of floor plan images corresponding to various types of houses.

Even though, floor plans are "*graphical language of communication*", however, investigation is required to answer "*how to match a pair of floor plans?*". In literature, researchers have carried out floor plan analysis in terms of symbol spotting and semantic segmentation. Symbol spotting in architectural floor plans deals with retrieving symbols from a database of floor plans, given a query symbol. Here the symbol set contains images of decor items like table, chair, bed, etc. It also finds the specific locations where the query image of the symbol is likely to be found. The other fields researched are floor plan analysis where existing works range from simple techniques for interpreting a hand-sketched floor plan for creating CAD models and automatically interpreting map drawings to analysing differentiation between thick, medium, and thin lines for segmenting

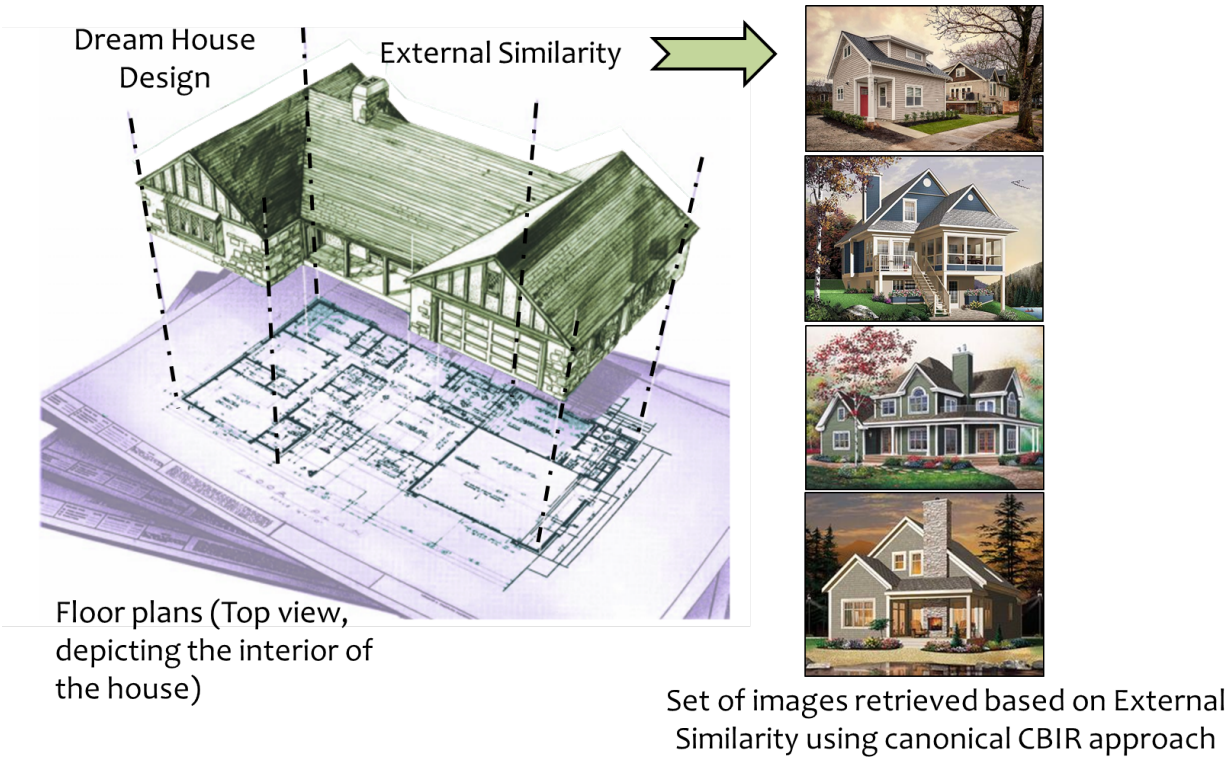


图1.4：问题的提出背景。

*of a dream house?*目前仍不明确。一种朴素的解决方案是沿用基于内容的图像检索（CBIR）方法来处理外部相似性。图1.4展示了此类情形。可以观察到，从数据库中检索出的房屋图像在外观上与查询图像较为相似。然而，如投影线和平面图所示，这些房屋在内部结构上的相似性并不能得到保证。

然而，用户的需求通常指向住宅的内部结构以及各房间内拟摆放的装饰。以一个典型示例为例，若需求被具体描述为“*the house must have* 4 *rooms, with* 2 *bedrooms with attached bathroom,* 1 *drawing room and* 1 *kitchen. The bedrooms must face towards west,...*”，则不仅需要匹配室内布局的相似性，还需同时考虑外部结构的相似性。与采用文本式（主观性较强）的查询方式相比，图像表达往往更为优越。平面图是一种基本的建筑图纸形式，它通过二维水平剖面展示房屋的整体布局，并附带详细的室内信息。图1.5展示了若干不同户型房屋的平面图示例。

尽管如此，平面图是“*graphical language of communication*”，但仍然需要通过调查来解答“*how to match a pair of floor plans?*”。在相关文献中，研究人员已从符号检测和语义分割两个方面开展了平面图分析研究。建筑平面图中的符号检测任务是在给定查询符号的情况下，从平面图数据库中检索出相应的符号；其中，符号集包含诸如桌子、椅子、床等装饰物品的图像，并进一步确定查询符号图像可能存在的具体位置。此外，还有其他研究领域涉及平面图分析，现有工作涵盖了从用于创建CAD模型的简单手绘平面图解读技术，到自动解析地图图纸，以及通过分析粗线、中线和细线之间的差异来进行分割等方法。

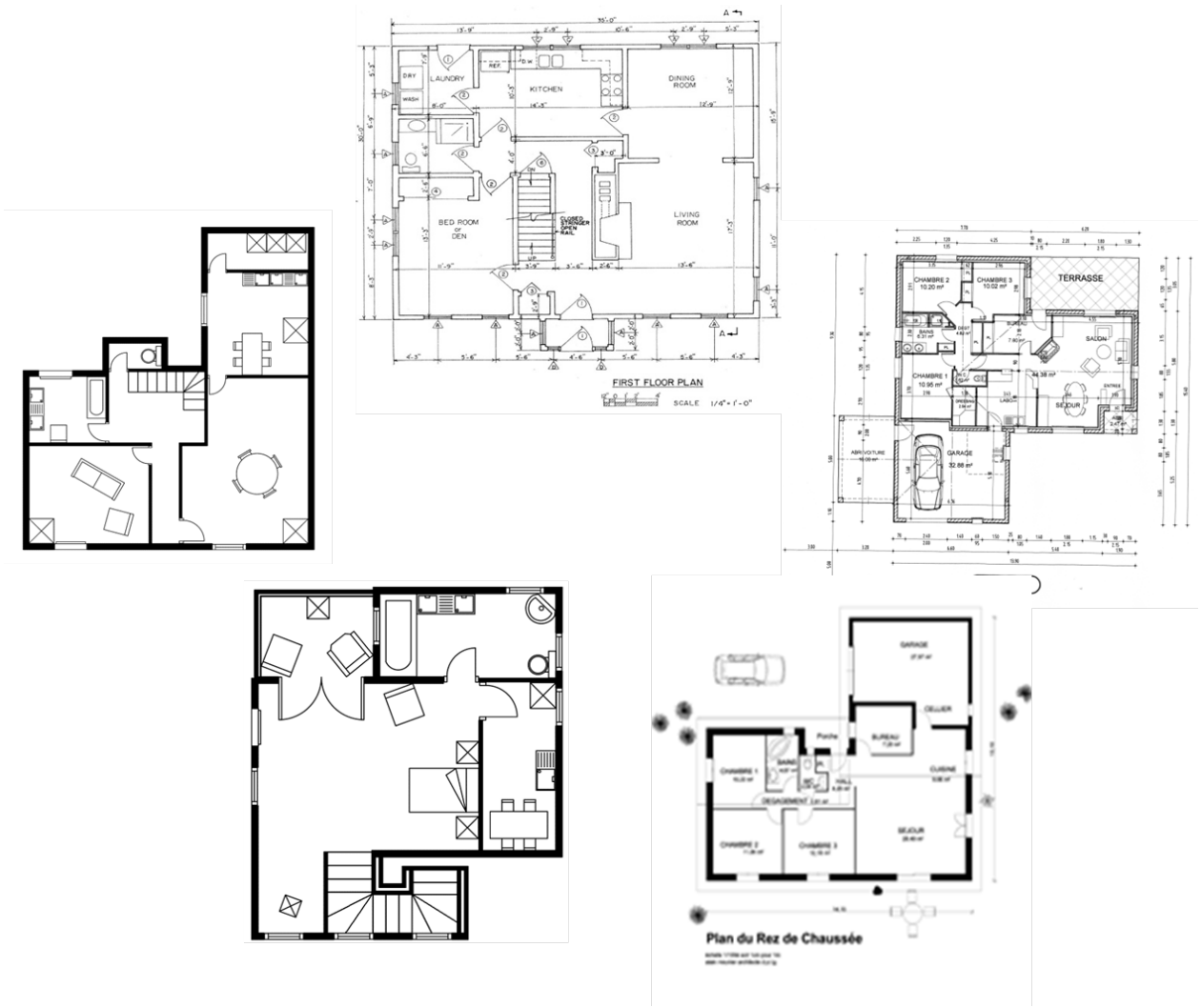


Figure 1.5. : Floor plans, showing 2D cross sections of buildings and displaying heterogeneity in design.

the room layouts in an efficient manner. Thus, upon doing a keen analysis of the existing work in the area of architectural floor plans, it was observed that retrieval in this particular domain is a challenging yet less researched area and has a lot of applications in today's digital scenario.

1.2 PROBLEM STATEMENT

The goal of the work proposed in this thesis is to design a content based retrieval system to perform the following task:

Given a query image/sketch of architectural floor plans retrieve alike floor plans from the database, rank-ordered by similarity in content (refer Fig. 1.6).

As proposed in this thesis, the input to the system would be a floor plan sketch or a floor plan image and the expected output would be similar floor plans as retrieved according to the specified content by the user. Content specification and appropriate feature set to represent the mentioned content plays a pivotal role in success of the retrieval system. Efficient techniques for

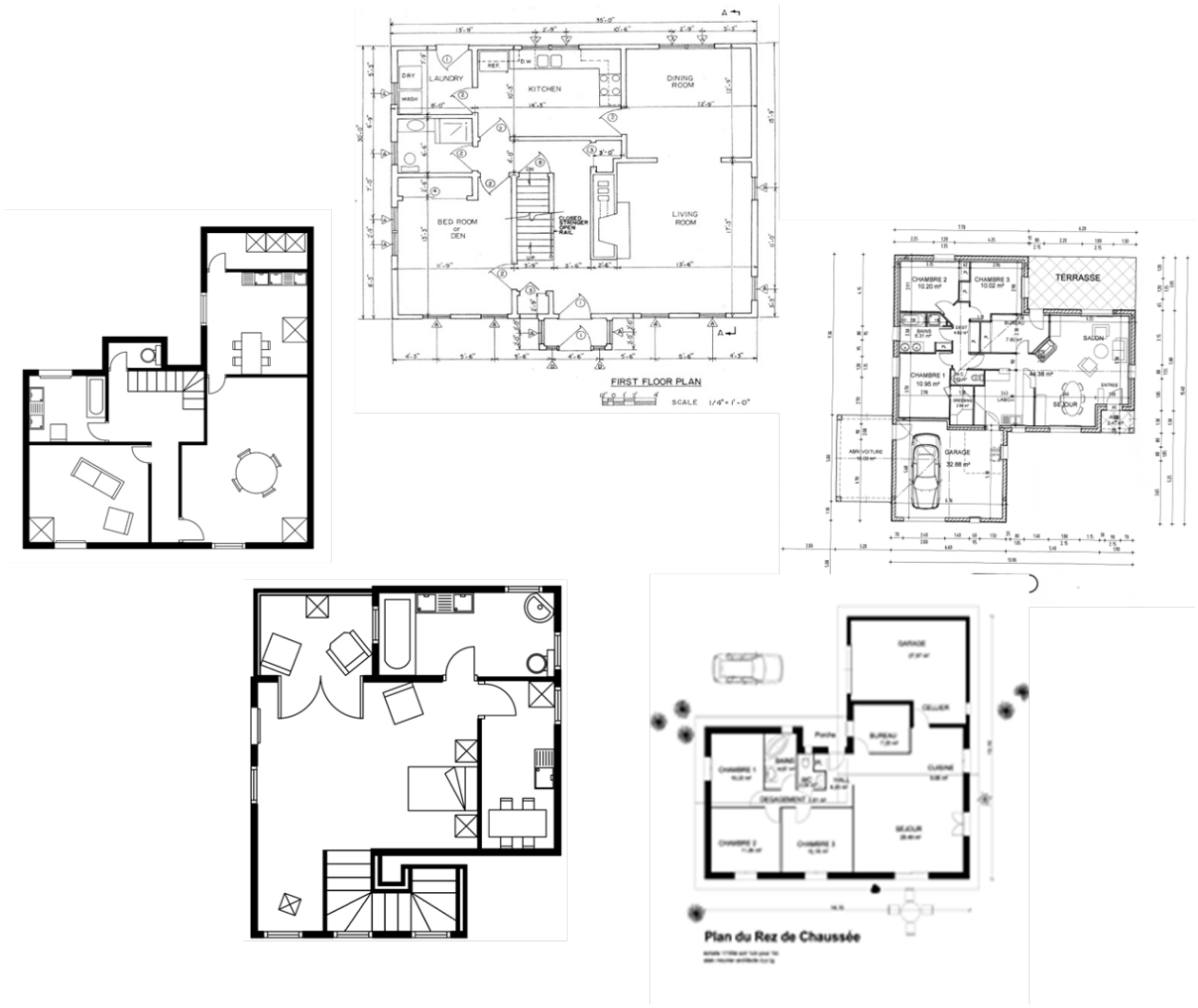


图1.5：平面布置图，展示了建筑物的二维剖面，并体现了设计上的异质性。

以高效的方式组织房间布局。因此，通过对建筑平面图领域现有研究的深入分析发现，在该特定领域中进行检索既具有挑战性，又鲜有深入探讨，并且在当今数字化环境中具有广泛的应用前景。

1.2 问题陈述

本论文所提出工作的目标是设计一个基于内容的检索系统，以实现以下任务：

给定一张建筑平面图的查询图像或草图，从数据库中检索出相似的平面图，并按内容相似度进行排序（参见图1.6）。

正如本论文所提出的，系统的输入将是一份平面草图或一张平面图图像，而预期的输出则是根据用户指定的内容检索到的相似平面图。内容的精确描述以及用于表征该内容的恰当特征集，在检索系统的成功中起着至关重要的作用。高效的检索技术能够……

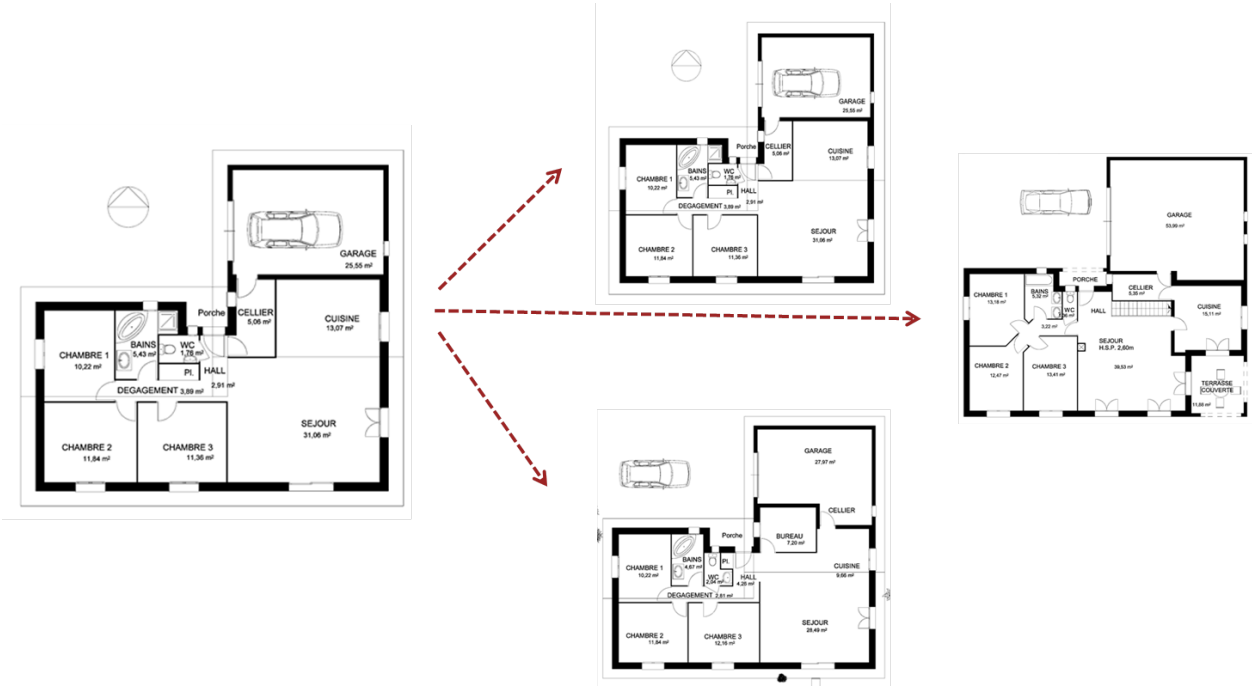


Figure 1.6. : Retrieval in floor plans by extracting meaningful content and looking for similarity across floor plan databases.

matching and retrieval are also proposed alongside capturing appropriate content from the floor plans.

1.2.1 Brief description of the Work Done

In Fig. 1.7, the problem of floor plan retrieval is represented alongwith its various possible solutions depicted in the form of a cube, containing a modality-axis, a feature-axis and an approach-axis. As shown in Fig. 1.7, the modality-axis of the cube corresponds to the type of modalities while querying. A content based retrieval system can have different modalities of representing the input query floor plan, for e.g. query can be an image (I) which represents floor plan images, a sketch (S), which can prove to be a convenient choice. Using pen and paper to scribble down ideas has always been a conventional way to query. The user looking for floor plans can easily draw an abstract floor plan with some interior details on a hand-held device and the sketched floor plan obtained thereafter can be used for querying. The third method can be a hybrid combination (H) of sketch and image representing a unified framework to capture the buyer's intent perfectly. Given a sketch or an image as query the framework should be able to retrieve similar floor plan images from the database. The term hybrid query mode is coined because in this mode the user can make online changes to the query, i.e. image or sketch, by adding or removing components in an existing floor plan. This mode currently serves as a future scope for this thesis. The feature-axis in Fig. 1.7 corresponds to the type of features extracted in the framework. The query and the existing layouts in the repository can be analysed for structural features as well as semantic features. Structural features correspond to the layout features that pertain to the room level analysis being used as content while extraction. These features use the topology and placement of rooms as matching criteria. Whereas, semantic features represent high-level features representing interior room details, sizes of the rooms, room-wise decor placement etc. It is to be noted that both of these features can be hand-crafted, which are typically

图1.6：通过提取有意义的内容并在不同楼层平面数据库之间寻找相似性，实现楼层平面的检索。

在从平面图中提取适当内容的同时，还提出了匹配与检索方法。

1.2.1 工作概述

在图1.7中，平面布局检索问题及其多种可能的解决方案以一个立方体的形式呈现，该立方体包含模态轴、特征轴和方法轴。s.

如图1.7所示，立方体的模态轴对应于查询时所采用的模态类型。基于内容的检索系统可以采用多种模态来表示输入的平面布局查询，例如，查询可以是一张图像（I），用于表示平面布局图像；也可以是一幅草图（S），这往往是一种更为便捷的选择。长期以来，利用笔和纸随手勾勒构思一直是常见的查询方式。寻求平面布局的用户可以在手持设备上轻松绘制一幅包含部分室内细节的抽象平面图，随后将所得的草图用于查询。第三种方式则为草图与图像的混合模式（H），通过构建统一的框架，能够更准确地捕捉用户的意图。给定草图或图像作为查询，该框架应能从数据库中检索出相似的平面布局图像。“混合查询模式”这一术语的提出，是因为在该模式下，用户可以通过增删现有平面布局中的构件，对查询图像或草图进行在线修改。目前，该模式被视为本论文的一个未来研究方向。图1.7中的特征轴则对应于框架中所提取的特征类型。查询以及存储库中已有的布局均可被分析其结构特征和语义特征。其中，结构特征是指在内容提取过程中用于匹配的、与房间层级分析相关的布局特征，其匹配依据为房间的拓扑结构及其空间位置；而语义特征则指更高层次的特征，包括室内细节、各房间的尺寸以及按房间划分的装饰布置等。需要指出的是，上述两类特征均可通过人工设计的方式进行提取，通常情况下……

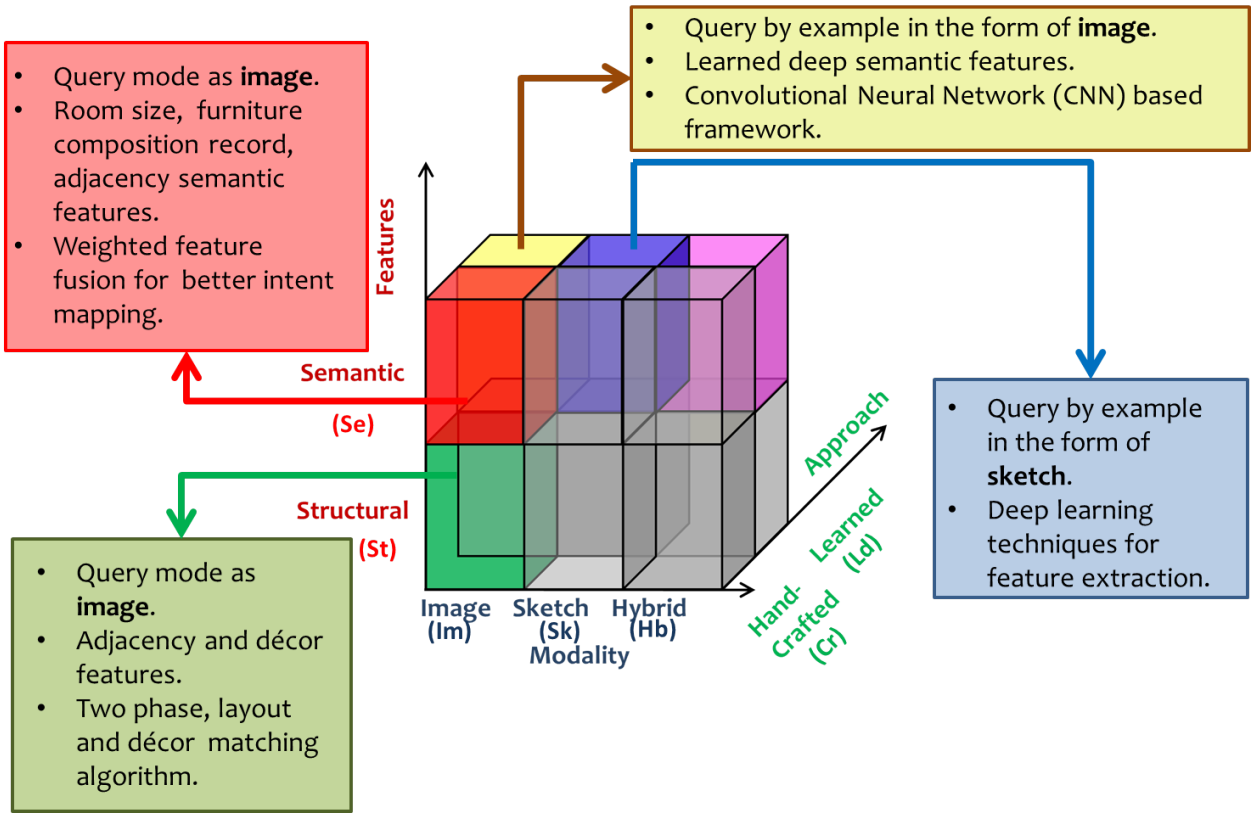


Figure 1.7. : Analysing floor plans for novel content aimed at retrieving similar layouts from a large repository.

chosen to represent the content in a floor plan, as well as, these features can be learned using a sophisticated learning framework, where explicit mentioning of the features is not necessary. The third axis of the cube in Fig. 1.7, represents such a scenario, where these two different approaches, i.e. the learned/ hand-crafted feature approaches are mapped. Thus, it can be inferred that such an amalgamation of various attributes while retrieval gives a great array of challenging problems to be dealt with while representation, analysis and finally matching and retrieval of floor plans. In this thesis, attempts were made to propose solutions to problems involving the following combinations (Im-St-Cr), (Im-Se-Ld), (Im-Se-Cr), (Sk-Se-Ld) as shown in Fig. 1.7.

1.3 RESEARCH ISSUES

Some of the key challenges encountered in the task of Content based retrieval and specifically floor plan analysis and retrieval are highlighted below:

1. In most cases, images/sketches require several stages of processing such as segmentation, smoothing, thresholding. The choice of an appropriate feature to correctly represent the content is in itself a big challenge.

2. As floor plans are 2D cross-sections of buildings, they are generally black and white and carry pixel information in a binary form. Moreover, there exists significant amount of whitespaces in the floor plan. Thus, throughout the image the information content is sparse in nature

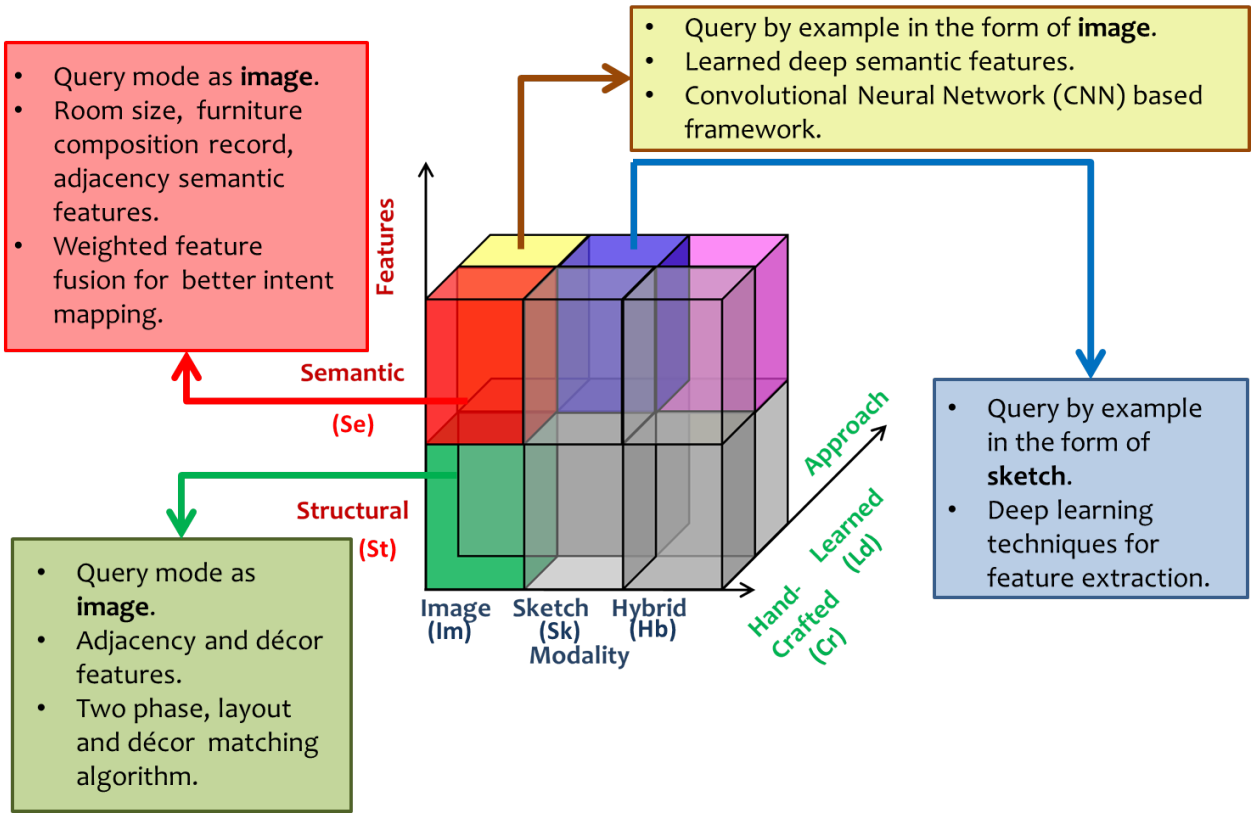


图1.7：通过分析平面布局，识别新颖内容，从而从大型数据库中检索相似的布局方案。

这些特征被选用来表征平面图的内容，并且可以通过一个复杂的学习框架进行自动学习，而无需显式地指定具体特征。图1.7中的立方体第三轴即刻画了这一情形，其中将两种不同的特征提取方法——即基于学习的特征提取与手工设计的特征提取——进行了映射。由此可见，在检索过程中综合运用多种属性，会在平面图的表示、分析以及最终的匹配与检索环节中带来一系列极具挑战性的难题。在本论文中，针对如图1.7所示的以下组合问题（图像–形状–颜色）、（图像–语义–布局）、（图像–语义–颜色）、（草图–语义–布局），均尝试提出了相应的解决方案。

1.3 研究问题

在基于内容的检索任务中，尤其是楼层平面图的分析与检索方面，所面临的一些关键挑战如下：

1. 在大多数情况下，图像或草图需要经过诸如分割、平滑和阈值化等多阶段的处理。如何选择恰当的特征以准确表征其内容本身就是一个重大挑战。2. 由于平面图是建筑物的二维剖面图，通常为黑白图像，并以二值形式存储像素信息。此外，平面图中存在大量空白区域，因此整幅图像的信息密度普遍较低。

which makes feature engineering a challenging task.

3. There is a dearth of publicly available datasets as architects refrain from sharing their designs. Thus, for implementation of an algorithm there is lack of publicly available sufficient data for the task of floor plan analysis and retrieval.

4. There is a huge-variation in the shapes and sizes of floor plans. This makes training a network, for instance, a deep learning framework, pretty challenging as resizing varied shaped floor plans leads to loss in information.

5. Sketch based query mode is in itself challenging to work with. Sketches incorporate a lot of user-variability and noise and require careful pre-processsing before analysis.

6. Representing image/sketch content using features demands a careful development of both hand-crafted and deep-learning features. The representation should ensure high intra-class similarity, at the same time high interclass dissimilarity. The representation should also be able to distinguish between overlapping classes. Moreover, the similarity criteria used for computing the match cost between the query and the model feature representation should also be unbiased and efficient.

Assumptions

The problem of content based retrieval in floor plans is solved under the following assumptions:

1. All the images are binarized.

2. Sketched drawings are scanned into image form for processing.

3. The floor plan images are digitally synthesised, i.e. they are not scanned and thus, free from any transformation like rotation, shear, scaling etc.

The focus of this thesis is on representation of floor plans and design of efficient similarity measures, under realistic situations. At the end of the thesis, an attempt has been made to bridge the gap between the two query modes, floor plan sketches and images using domain adaptation approach.

1.3.1 Organization of the Thesis

In this chapter, the importance of the problem of CBIR is introduced. Along with it a brief overview of the motivation behind a Content based retrieval for architectural floor plans is highlighted. The inherent challenges while proposing an efficient solution for this problem are listed. A brief outline of the approach to solve the problem is provided and the objectives that are trying to be achieved are listed.

In Chapter 2, a review of the existing work in the area of floor plan analysis is given, from which the motivation for this work is derived. Various aspects of architectural floor plan analysis dealing with symbol spotting, room level analysis and sketch based retrieval in floor plans are discussed. In the same Chapter, the datasets available in the floor plan analysis community are discussed. Also, basic image feature extraction strategies are reviewed. Some of the deep learning methodologies for feature extraction and domain adaptation are also discussed in Chapter 2, which serve as the motivation for the proposed multi-modal deep learning network.

这使得特征工程成为一项具有挑战性的任务。

3. 由于架构师不愿共享其设计，公开可用的数据集十分匮乏。因此，在算法的实现过程中，用于楼层平面图分析与检索任务的公开数据严重不足。

4. 平面图的形状和尺寸存在极大的差异性，这使得训练神经网络（例如深度学习模型）变得相当困难，因为对形状各异的平面图进行尺寸调整会导致信息丢失。

5. 基于草图的查询模式本身即具有较高的技术难度。草图中包含大量用户个体差异与噪声，在进行分析之前需要进行细致的预处理。

6. 利用特征表示图像/草图内容，需要精心设计手工特征与深度学习特征。该表示应确保类内高度相似、类间高度相异，并能够区分存在重叠的类别。此外，用于计算查询特征与模型特征表示之间匹配代价的相似性度量标准也应具备无偏性和高效性。

假设

在以下假设条件下，解决了建筑平面图的基于内容检索问题：

1. 所有图像均被二值化处理。

2. 草图被扫描为图像形式以供处理。

3. 平面图图像采用数字合成方式生成，即未经过扫描处理，因此不存在旋转、剪切、缩放等几何变换。

本论文的研究重点在于在实际应用场景下，对建筑平面图的表示方法及高效相似性度量的设计。在论文的最后，我们尝试通过领域自适应方法，弥合建筑平面草图与图像这两种查询模式之间的差距。

1.3.1 论文的组织结构

在本章中，首先介绍了基于内容的图像检索（CBIR）问题的重要性，并简要概述了针对建筑平面图开展基于内容检索的研究动机。随后，列出了在提出该问题的有效解决方案时所面临的固有挑战。最后，给出了问题求解方法的概要，并明确了拟达成的具体目标。

在第2章中，对平面图分析领域的现有研究进行了综述，并由此明确了本研究的动机。该章讨论了建筑平面图分析的多个方面，包括符号识别、房间级分析以及基于草图的平面图检索。此外，还介绍了平面图分析领域现有的数据集，并对基本的图像特征提取方法进行了回顾。第2章还探讨了一些用于特征提取与领域自适应的深度学习方法，这些内容为所提出的多模态深度学习网络提供了理论依据。

Chapter 3 proposes a framework for the matching and retrieval of similar architectural floor plans under the query by example paradigm. Here query is taken as a floor plan image. A room layout segmentation and adjacent room detection algorithm is presented to represent layouts as an undirected graph. Also a novel graph spectral embedding feature is proposed to uniquely represent the layout of the architectural floor plan. This graphical approach helps in effective and efficient matching of the room layouts. Room semantics in terms of both the room structures and room decor is used to retrieve similar floor plans from the repository. To match the semantic similarity between a pair of floor plans, a two stage matching technique is proposed and high retrieval accuracy is obtained.

In Chapter 4, a deep learning paradigm to extract both low and high level semantic features from a layout image is proposed. The key contributions in the proposed approach are, a novel deep learning framework to retrieve similar floor plan layouts from repository and analysing the effect of individual deep convolutional neural network layers for floor plan retrieval task.

In Chapter 5, a novel end-to-end framework for extracting high level semantic features like area and room-wise decor arrangement for the task of fine grained retrieval is proposed. Further, a technique to perform feature fusion to aggregate high-level semantic features extracted is also proposed. Weighted feature fusion helps in setting preferences to particular features while retrieval and satisfying specific user demands.

In Chapter 6, a novel system which uses sketch based query mode to retrieve corresponding similar floor plans from the repository is proposed. Two approaches are discussed, one with a system using Cyclic Generative Adversarial Networks (Cyclic GAN) for mapping between sketch and image domain. The key contributions here being a novel sketch based floor plan retrieval framework using an intuitive and convenient sketch query mode and conjunction of Cyclic GANs and Convolution Neural Networks (CNNs) for the task of hand-drawn floor plan image retrieval. The second approach being the usage of autoencoders along with Cyclic GANs for improving the performance of the retrieval framework and helping in better domain mapping between sketch and image domains.

Finally, in Chapter 8, the contributions of this thesis in the field of content based retrieval in architectural floor plans are summarized and a few possible extensions and future prospects for the work are identified.

Two appendices are also provided at the end of this thesis, where, Appendix 1 details out the creation of the floor plan datasets, Repository Of BuildIng floor plaNs (ROBIN) and Sketched-Repository Of BuildIng floor plaNs (S-ROBIN) that have been specifically designed to aid the analysis and retrieval in floor plans. Appendix 2 contains the description of a graphical user interface, designed for the task of interactive floor plan retrieval.

第三章在以图搜图的范式下，提出了一种用于相似建筑平面图匹配与检索的框架。在此框架中，查询对象被定义为一张平面图图像。本文提出了一种房间布局分割与相邻房间检测算法，将建筑布局表示为无向图；同时，还提出了一种新颖的图谱嵌入特征，用以唯一表征建筑平面图的布局结构。这种基于图的方法有助于实现房间布局的有效且高效的匹配。此外，通过综合考虑房间结构与室内装饰两方面的语义信息，从数据库中检索出相似的平面图。为了度量两幅平面图之间的语义相似性，本文进一步提出了一个两阶段的匹配方法，并获得了较高的检索精度。

在第4章中，提出了一种基于深度学习的范式，用于从布局图像中提取低层和高层语义特征。所提方法的主要贡献在于：构建了一个新颖的深度学习框架，用于从知识库中检索相似的平面布局，并分析了各深度卷积神经网络层对平面布局检索任务的影响。

在第5章中，提出了一种用于细粒度检索任务的新型端到端框架，该框架能够提取诸如区域及房间级装饰布局等高层次语义特征。此外，还提出了一种用于聚合所提取高层次语义特征的特征融合技术。加权特征融合有助于在检索过程中对特定特征进行优先级设置，从而满足用户的个性化需求。

在第6章中，提出了一种新颖的系统，该系统采用基于草图的查询模式，从知识库中检索相应的相似平面图。文中讨论了两种方法：其一，利用循环生成对抗网络（Cyclic GAN）实现草图域与图像域之间的映射；其主要贡献在于构建了一个基于草图的平面图检索框架，该框架采用直观便捷的草图查询模式，并将循环生成对抗网络与卷积神经网络（CNN）相结合，用于手绘平面图图像的检索任务。其二，通过结合自编码器与循环生成对抗网络，进一步提升检索框架的性能，并有助于实现草图域与图像域之间更优的域间映射。

最后，在第八章中，本文总结了在基于内容的建筑平面图检索领域所作出的贡献，并提出了若干可能的研究拓展方向及未来展望。

本论文末尾还附有两份附录：附录1详细介绍了用于辅助平面图分析与检索的两个数据集——建筑平面图资源库（Repository Of BuildIng floor plaNs，简称ROBIN）和草图版建筑平面图资源库（Sketched-Repository Of BuildIng floor plaNs，简称S-ROBIN）的构建过程；附录2则描述了一款专为交互式平面图检索任务设计的图形用户界面。

2
Background and Related Work

2.1 INTRODUCTION

In this chapter, the existing literature regarding the research areas discussed in this thesis is presented, followed by a brief review of the existing datasets in the area of floor plan analysis. Document Image Analysis (DIA) tasks involving Graphics Recognition specifically have evolved mainly due to the growing trend of not only storing digital documents these days, but also due to extraction, classification, and indexing of them according to the information conveyed by the documents. This gained popularity after the emergence of scanners that increased the managing and handling of digitized documents. Application of Graphics Recognition to domains like engineering drawings, electrical diagrams, flowcharts, maps, mathematical and chemical formulae, and architectural plans has been done from the very start of the evolution of this field. After one recognizes the graphics in a particular document the next task is to derive something meaningful out of it. As this thesis pertains with engineering drawings, specifically floor plans therefore, in the next section the existing work related to symbol spotting, floor plan analysis and retrieval in floor plans, the commonly used datasets in floor plan analysis and generic feature representation and deep learning retrieval methods is discussed.

2.2 STATE OF THE ART IN FLOOR PLAN ANALYSIS

Floor plan analysis is one of the most trendy applications in graphical document understanding due to the actual demand of real world applications. Thus, several systems dealing with heterogeneous architectural inputs such as sketch drawings, machine generated documents, and CAD files have been presented in the recent years. Nevertheless, floor plan interpretation problem has not yet been solved for two main reasons: the inherent difficulty of the problem i.e. the visual language may differ from plan to plan, and the lack of reusability and improvement of the existing approaches. In the following paragraphs, the most recent techniques for floor plan analysis are described in detail. In Tab. 2.1, some key approaches are listed with their salient features.

2.2.1 3D reconstruction of floor plans

The work proposed in [Ah-Soon and Tombre, 1997], [Dosch and Masini, 1999], and [Dosch *et al.*, 2000] deals with floor plan 3D reconstruction. In these works, the input is scanned floor plans. First, a preprocessing step is followed to separate both text and graphics information. Then the thick and thin lines present in the graphical layer are segregated and vectorized. Thick lines depict walls whereas, the thin ones depict the rest of the symbols, including windows and doors. In this process, two variations of walls are considered: first being the walls represented by thick parallel lines and the second case being walls represented by a single thick line. Arcs in the symbols represent doors, windows are detected using small closed loops, and rooms are generally

2

背景与相关工作

2.1 引言

在本章中，首先概述了与本论文所探讨的研究领域相关的现有文献，随后简要回顾了楼层平面图分析领域的现有数据集。涉及图形识别的文档图像分析（DIA）任务之所以得以发展，主要得益于当今不仅对数字文档进行存储的需求日益增长，而且还需要根据文档所承载的信息对其进行提取、分类和索引。这一趋势在扫描仪出现并推动数字化文档管理与处理效率提升之后愈发受到关注。自该领域发展之初，图形识别技术便已应用于工程图纸、电气原理图、流程图、地图、数学与化学公式以及建筑平面图等各类场景。在识别出特定文档中的图形元素之后，下一步便是从中提取出具有实际意义的信息。鉴于本论文聚焦于工程图纸，尤其是楼层平面图，因此在下一节中将重点讨论与符号检测、楼层平面图分析与检索相关的工作，以及楼层平面图分析领域常用的数据集、通用特征表示方法和基于深度学习的检索技术。

2.2 平面图分析领域的研究现状

由于实际应用需求，楼层平面图分析已成为图形文档理解领域中最热门的应用之一。因此，近年来已提出了若干能够处理多种异构建筑输入（如手绘草图、机器生成文档及CAD文件）的系统。然而，楼层平面图的语义理解问题至今仍未得到彻底解决，主要原因有二：其一，该问题本身具有固有的复杂性——不同平面图所采用的视觉语言可能存在差异；其二，现有方法在可复用性和可改进性方面仍显不足。在下文中，我们将详细阐述用于楼层平面图分析的最新技术。表2.1列出了若干关键方法及其主要特征。

2.2.1 平面图的三维重建

[Ah-Soon 和 Tombre，1997]、[Dosch 和 Masini，1999] 以及 [Dosch $et\ al.$，2000] 中提出的研究工作涉及平面图的三维重建。在这些研究中，输入为扫描得到的平面图。首先进行预处理步骤，以分离文本信息与图形信息。随后，对图形层中的粗线和细线进行区分并进行矢量化处理。其中，粗线用于表示墙体，而细线则用于表示其他符号，包括门窗等。在此过程中，考虑了两种类型的墙体：第一种是由两条平行的粗线表示的墙体，第二种是由单条粗线表示的墙体。符号中的弧线代表门，窗户则通过识别小型闭合环来检测，而房间通常……

Table 2.1. : Reviews of approaches for floor plan analysis

Paper Title	**Technique**	**References**
A system to understand hand-drawn floor plans using subgraph isomorphism and Hough transform	Attributed graph structure, Straight line Hough transform for speeding matching	Lladós *et al.* [1997]
A Prototype System for Interpreting Hand-Sketched Floor Plans	Automatically converts hand-sketched floor-plans into the CAD formation, Template matching of closed regions	Aoki *et al.* [1996]
A complete system for the analysis of architectural drawings	Converts the drawing into a description in terms of basic architectural entities, Reconstructing in 3D	Dosch *et al.* [2000]
Robust and Accurate Vectorization of Line Drawings	Vectorizing the graphical parts of line drawings, Compute feasibility domains for accurate vectorization	Hilaire and Tombre [2006]
The Room Connectivity Graph: Shape Retrieval in the Architectural Domain	Characterize 3D architectural models on underlying arrangement of their rooms, Retrieval of 3D models.	Wessel *et al.* [2008]
A System to Detect Rooms in Architectural Floor Plan Images	Hough transform, Decomposing rooms to near convex regions	Macé *et al.* [2010]

a composition of bigger connected components. Finally, a single level in a floor plan or several floors in the same building are identified and their 3D reconstruction is performed [Ah-Soon and Tombre, 1997] by finding special symbols as beams, staircases, etc. [Dosch and Masini, 1999]. Both in [Dosch and Masini, 1999] and [Dosch *et al.*, 2000], emphasis is made on the necessity of human feedback while working with complex plans. Or *et al.* [2005] focus on 3D model generation from a 2D plan. Rendek *et al.* [2004] preprocess the image by firstly segregating the text and graphics and then converting the graphical layer to a vector form. They further, manually delete other noisy lines and symbols to finally detect the plan structure. Once the remaining structure consists of only lines belonging to walls, doors, and windows, a set of polygons using each polyline of the vectorized image is generated. At the end, blocks are represented by polygons; thick lines represent outer walls, rectangles represent windows inside the walls, and arcs represent doors. This system is able to generate the 3D model corresponding to single story buildings. Again, this approach is plan specific and slight change in the floor plan leads to modification of the technique. Some works such as proposed by Lu *et al.* [2007] for 3D reconstruction contain a CAD file as input which consists of lines of the architectural design and real non-distorted original polylines. First, T, L and X shapes are identified by extracting parallel overlapped lines. Then, the 3D reconstruction of the plan is

表2.1：平面布局分析方法综述

Paper Title	**Technique**	**References**
A system to understand hand-drawn floor plans using subgraph isomorphism and Hough transform	Attributed graph structure, Straight line Hough transform for speeding matching	Lladós *et al.* [1997]
A Prototype System for Interpreting Hand-Sketched Floor Plans	Automatically converts hand-sketched floor-plans into the CAD formation, Template matching of closed regions	Aoki *et al.* [1996]
A complete system for the analysis of architectural drawings	Converts the drawing into a description in terms of basic architectural entities, Reconstructing in 3D	Dosch *et al.* [2000]
Robust and Accurate Vectorization of Line Drawings	Vectorizing the graphical parts of line drawings, Compute feasibility domains for accurate vectorization	Hilaire and Tombre [2006]
The Room Connectivity Graph: Shape Retrieval in the Architectural Domain	Characterize 3D architectural models on underlying arrangement of their rooms, Retrieval of 3D models.	Wessel *et al.* [2008]
A System to Detect Rooms in Architectural Floor Plan Images	Hough transform, Decomposing rooms to near convex regions	Macé *et al.* [2010]

由较大的连通组件构成的组合。最后，通过识别梁、楼梯等特定符号，对同一建筑中的一层平面或若干楼层进行识别，并完成其三维重建[Ah-Soon和Tombre，1997] [Dosch和Masini，1999]。在[Dosch和Masini，1999]以及[Dosch *et al.*，2000]中，均强调了在处理复杂平面图时人工交互的必要性。Or *et al.* [2005]则专注于从二维平面图生成三维模型。Rendek *et al.* [2004]首先将图像中的文本与图形分离，再将图形层转换为矢量形式；随后，人工去除其他噪声线条和符号，最终提取出平面结构。当剩余结构仅包含墙体、门和窗的线条时，利用矢量化图像中的每一条多段线生成一组多边形。最终，各功能区块以多边形表示：粗线代表外墙，矩形代表墙内的窗户，弧线代表门。该系统能够为单层建筑生成相应的三维模型。然而，该方法具有较强的平面依赖性，平面布局的细微变化即需相应调整技术流程。此外，一些研究，例如Lu *et al.* [2007]提出的三维重建方案，以CAD文件作为输入，其中包含建筑设计的线条及未经畸变的原始多段线。首先通过提取平行且重叠的线条来识别T形、L形和X形结构，进而进行平面的三维重建。

retrieved by identifying connections between the lines. After that, segmentation of the graphical symbols such as furniture and staircases is done by deleting the structural lines. This method is primarily based on symbol recognition using geometrical features extracted from the image.

2.2.2 Semantic analysis of floor plans

Cherneff *et al.* [1992] present a knowledge-based interpretation method for architectural drawings. Their aim is to extract the complete structure of the plan, which comprises of the recognition of walls, doors, windows, rooms, and the relations between them. The input is a floor plan vectorized in order to obtain unique symbols for doors. Then, the system is composed of two models: the semantic model and the structural model. The semantic model represents the contextual composition of the architectural elements. The structural model entails the geometrical configuration of the plan, including the 2D spatial indexing of primitives. Walls are defined as parallel segments incidenting to doors or windows at their ending points. This fact places a constraint on the possibilities of interpretation, since it is based on the assumption that walls are always straight, however in real world scenario walls might be curved or even not be modeled using parallel lines. The work presented by Cherneff *et al.* [1992] focuses on room segmentation. A proximity map based semi-automatic approach is used for searching for regions in machine printed floor plan images. This method serves as an extension of the area-filling approach which mainly deals with splitting rooms when there is a lack of physical separation. However, the drawback here is that this method retrieves many false positives in the form of all those objects that are modeled as closed regions, for example, doors, tables, and staircases. [Macé *et al.*, 2010] focus on the extraction of the building structure from scanned images. Similarly to [Dosch *et al.*, 2000], [Dosch and Masini, 1999] they firstly separate textual from graphical information. Then, on the graphical layer, they recognize walls by detecting parallel lines at the contours of the thicker lines. Thick lines are earlier separated from the thin ones morphologically. Finally, rooms are recognized by a recursive decomposition of the images into convex regions. In this method, the wall detector strongly depends on the wall notation, which encompasses a redefinition in order to be able to deal with a variety of floor plan models. Work proposed by Ahmed *et al.* [2012] also starts with the classical text/graphics separation. After that, they morphologically separate the graphical components into thin, medium, and thick lines. Medium and thick lines are considered to be walls whereas small lines model the rest of the architectural elements. Then, doors and windows are detected by means of a patch-based description of the image using SURF [Bay *et al.*, 2006]. Rooms are recognized by finding closed areas enclosed by walls, doors, and windows. At the end, the textual labelling inside the room areas are used to verify and correct the final segmentation. This whole method focuses on a single graphical notation, specially in the process of wall extraction. Also, the work of Zhi *et al.* [2003] takes input in the form of a CAD file. It then automatically extracts the topological and geometrical information from a 2D architectural drawing and builds a simulator that helps in building evacuation in case of crisis situations. Firstly, filtering of redundant information such as dimensions, text, and furniture, is carried out in a semi-automatic fashion and only the remaining necessary objects like doors, windows and walls are kept. Then, the plan is transformed into an attributed graph and loops are searched for in the floor plan. Further, according to their respective attributes, loops are classified into different categories such as rooms and corridors are represented by spatial loops, walls and columns through physical loops, window loops, door loops and unrecognized loops. Finally, the compartments in the floor plans are identified and the system is integrated in a simulation model that aids in difficult evacuation strategies by connecting rooms to their nearest exits/staircases in complex buildings.

通过识别线条之间的连接关系来实现提取。随后，通过删除结构线条对家具、楼梯等图形符号进行分割。该方法主要基于从图像中提取的几何特征进行符号识别。

2.2.2 平面图的语义分析

Cherneff *et al.* [1992] 提出了一种基于知识的建筑图纸解析方法。其目标是提取平面图的完整结构，包括墙体、门、窗、房间及其相互关系的识别。输入为经过矢量化的楼层平面图，以便为门获取唯一的符号表示。系统由两个模型组成：语义模型和结构模型。语义模型用于表征建筑元素的上下文构成；结构模型则描述平面的几何构型，包括基本图元的二维空间索引。墙体被定义为在其端点与门或窗相交的平行线段。这一定义对解释的可能性施加了约束，因为它假定墙体始终为直线，而在实际场景中，墙体可能是曲线，甚至未必采用平行线来建模。Cherneff *et al.* [1992] 的工作侧重于房间分割。他们采用一种基于邻近度图的半自动方法，在机器打印的楼层平面图像中搜索区域。该方法可视为面积填充法的扩展，主要用于在缺乏物理分隔时对房间进行分割。然而，其缺点在于会产生大量误检，即所有被建模为封闭区域的对象，如门、桌椅和楼梯等，均会被错误地识别为房间。[Macé *et al.*, 2010] 则致力于从扫描图像中提取建筑结构。与 [Dosch *et al.*, 2000] 及 [Dosch 和 Masini, 1999] 类似，他们首先将文本信息与图形信息分离。随后，在图形层面上，通过检测较粗线条轮廓处的平行线来识别墙体；较粗线条会先通过形态学方法与细线条分离。最后，通过对图像进行递归分解为凸区域来识别房间。在此方法中，墙体检测器高度依赖于墙体的表示方式，而这种表示需要重新定义，以适应多种类型的楼层平面模型。Ahmed *et al.* [2012] 的工作同样始于经典的文本/图形分离。之后，他们通过形态学方法将图形组件划分为细线、中线和粗线。其中，中线和粗线被视为墙体，而细线则用于建模其他建筑元素。接着，利用基于SURF特征的图像块描述方法（Bay *et al.*, 2006）检测门和窗；房间则通过寻找由墙体、门和窗围合的封闭区域来识别。最后，借助房间区域内标注的文本信息对最终分割结果进行验证与修正。整个方法主要针对单一的图形表示形式，尤其是在墙体提取环节。此外，Zhi *et al.* [2003] 的输入为CAD文件，该方法能够自动从二维建筑图纸中提取拓扑与几何信息，并构建一个模拟系统，以支持危机情况下的人员疏散。首先，以半自动方式过滤掉尺寸、文字及家具等冗余信息，仅保留门、窗和墙体等必要对象。随后，将平面图转换为带属性的图结构，并在平面中搜索环路。进一步地，根据各环路的属性将其划分为不同类别：空间环路用于表示房间和走廊，物理环路用于表示墙体和柱子，另有窗环、门环以及未识别环路。最终，确定平面中的各个功能分区，并将系统集成到仿真模型中，通过在复杂建筑中将房间与其最近的出口或楼梯相连，从而辅助制定困难条件下的疏散策略。

2.2.3 Structural Analysis

Works as proposed by Aoki *et al.* [1996] and Lladós *et al.* [1997] analyze hand-sketched floor plans. On the one hand, [Aoki *et al.*, 1996] transforms the sketches into CAD files. In this particular work the lines that model the building structure are extracted, which are basically handwritten on a preprinted paper with a grid of lines. This method again describes line elements in the form of windows and walls, and closed region elements here represent doors. On the other hand, [Lladós *et al.*, 1997] recognizes building elements and their respective topological properties using Hough transform and subgraph isomorphism. Subgraph isomorphism aids in symbol recognition whereas Hough transform helps in detecting walls modeled using hatched patterns. It should be noted that in both, [Aoki *et al.*, 1996] and [Lladós *et al.*, 1997]], the conventions to be used while drawing are prescribed beforehand. Floor plan structural retrieval has gathered the attention of architects recently. The works of [Ahmed *et al.*, 2014], [Weber *et al.*, 2010] and [Wessel *et al.*, 2008] are three examples of this application. In the case of the work proposed by Weber *et al.* [2010], the query is specified in the form of a sketch drawn online by a user. Their system allows the user to sketch an abstract schematic representation of a floor plan and uses this representation to search for structurally similar documents. The sketch is modelled as a graph, which encloses the structure of the plan, and it is compared to the repository representations using subgraph matching algorithms. As proposed by Wessel *et al.* [2008], the specified input is in the form of a polygon representing a 3D floor plan, so they do not need to vectorize the plan. They construct a graph where attributed nodes are rooms and attributed edges are connections between them: doors or windows. Based on this connectivity graph, fast and efficient shape retrieval is achieved.

Table 2.2. : Reviews of approaches for symbol spotting

Paper Title	**Technique**	**References**
Statistical Grouping For Segmenting Symbols Parts From Line Drawings, With Application To Symbol Spotting	Groups line drawings into shapes, Identifying ROI and aiding matching process	Nayef and Breuel [2011]
Symbol Spotting in Line Drawings Through Graph Paths Hashing	Hashing the shape descriptors of graph paths, Local sensitivity hashing, Spatial voting scheme	Dutta *et al.* [2011]
Combination of Product Graph and Random Walk Kernel for Symbol Spotting in Graphical Documents	Path similarity using product graph, Random Walk Kernel for calculating distance measures	Dutta *et al.* [2012]
An integer linear program for substitution-tolerant subgraph isomorphism and its use for symbol spotting in technical drawings	Integer linear formulation, Substitution-tolerant subgraph isomorphism.	Le Bodic *et al.* [2012]
A symbol spotting approach in graphical documents by hashing serialized graphs	Hashing the shape descriptors of graph paths, Local sensitivity hashing, Spatial voting scheme	Dutta *et al.* [2013]
Object Recognition in Floor Plans by Graphs of White Connected Components	Region Adjacency Graph, White connected components as nodes	Barducci and Marinai [2012]

2.2.3 结构分析

Aoki（1996）和Lladós（1997）所提出的方法均用于分析手绘的平面图。一方面，Aoki（1996）将这些草图转换为CAD文件；在该研究中，主要提取的是建筑物结构的线条，这些线条通常是手绘在带有网格线的预印纸上的。该方法同样以线段元素来表示窗和墙，而封闭区域元素则用于表征门。另一方面，Lladós（1997）利用霍夫变换与子图同构技术识别建筑构件及其相应的拓扑属性；其中，子图同构用于符号识别，而霍夫变换则有助于检测采用阴影线模式绘制的墙体。值得注意的是，在Aoki（1996）和Lladós（1997）的研究中，绘图时均预先规定了应遵循的制图规范。近年来，平面图结构检索已引起建筑师的广泛关注。Ahmed（2014）、Weber（2010）以及Wessel（2008）的相关工作便是这一应用的三个典型实例。在Weber（2010）提出的方案中，查询以用户在线绘制的草图形式给出；其系统允许用户勾勒出平面图的抽象示意图，并以此为基础检索结构相似的文档。该草图被建模为一张图，其中包含平面的结构信息，并通过子图匹配算法与数据库中的表示进行比对。而根据Wessel（2008）的方案，输入则为表示三维平面的多边形，因此无需对平面进行矢量化处理。他们构建了一张图，其中带属性的节点代表房间，带属性的边代表房间之间的连接——即门或窗。基于该连通性图，实现了快速且高效的形状检索。

表2.2：符号检测方法综述

Paper Title	**Technique**	**References**
Statistical Grouping For Segmenting Symbols Parts From Line Drawings, With Application To Symbol Spotting	Groups line drawings into shapes, Identifying ROI and aiding matching process	Nayef and Breuel [2011]
Symbol Spotting in Line Drawings Through Graph Paths Hashing	Hashing the shape descriptors of graph paths, Local sensitivity hashing, Spatial voting scheme	Dutta *et al.* [2011]
Combination of Product Graph and Random Walk Kernel for Symbol Spotting in Graphical Documents	Path similarity using product graph, Random Walk Kernel for calculating distance measures	Dutta *et al.* [2012]
An integer linear program for substitution-tolerant subgraph isomorphism and its use for symbol spotting in technical drawings	Integer linear formulation, Substitution-tolerant subgraph isomorphism.	Le Bodic *et al.* [2012]
A symbol spotting approach in graphical documents by hashing serialized graphs	Hashing the shape descriptors of graph paths, Local sensitivity hashing, Spatial voting scheme	Dutta *et al.* [2013]
Object Recognition in Floor Plans by Graphs of White Connected Components	Region Adjacency Graph, White connected components as nodes	Barducci and Marinai [2012]

2.3 STATE OF THE ART IN SYMBOL SPOTTING

Symbol spotting in floor plans deals with given an image of a query symbol, location of the query symbol and associated documents are retrieved. There are two approaches to symbol spotting. The first pertains to a region-based approach, which holds its basis on searching for regions of interest inside drawings, using various descriptors to describe them, and finally using a query symbol to index them. The second approach for spotting deals with isolating the symbols using segmentation to get the symbols segregated from their background, and then performing their recognition. Generally the spotting methods mentioned in the literature can be categorized into the region-based approach. However, this approach has the following inherent problems: (1) It is difficult to locate regions of interest in a scaled down drawing; (2) Rotation invariance in such type of drawings is hard to achieve; (3) As the patterns in line segments are simplistic in nature, special descriptors need to be developed to represent line drawings, instead of using the regular textual based descriptors and, (4) Indexing techniques like hashing fail to scale well when one brings large databases into the picture. Due to these problems, the methods implementing the region based approach result in low recall and precision rates. Inspite of the drawbacks, a primary reason to follow this approach, rather than the segmentation-based approach, is the incapability of the current segmentation techniques to peform while dealing with complicated technical drawings. Table 2.2 lists some important works related to symbol spotting.

2.3.1 Region based symbol spotting

In this spotting approach, various methods for detection and description of regions of interest within drawings are employed by the state-of-the-art methods. Authors in [Rusinol and Lladós, 2007] and [Rusiñol *et al.*, 2010], propose the use of different spotting methods working with detecting closed regions to further identify the regions of interest. In [Rusinol and Lladós, 2007], a bipolar coordinate system is used to depict a region of interest and for indexing purposes a hash table is used. In [Rusiñol *et al.*, 2010], attributed strings are used to represent regions of interest. A polygonal approximation of the contour of the closed regions is computed and association of adjacent segments is done in the form of polylines. These polylines are encoded as attributed strings, where attributes consist of, the length of the segment and the angle between the current and the previous segment. Similar or alike strings are then clustered together in a look-up table and a set of median strings represent the indexing keys to index into the table. String matching is used for comparison of the query to regions in the image. Finally, a Hough voting scheme is used for the verification of the initially found matches. In another work as proposed in [Ah-Soon and Tombre, 2001], overlapping sliding windows are used to detect regions. The regions here are described as feature vectors depicting the relations between the connections of the region segments, and graphs are used to describe the descriptions. The symbol search is then ensured by calculating the similarity of the relationship between the segments in the network.

2.3.2 Segmentation and recognition of isolated symbols for symbol spotting

Graph representations carry their fair share of popularity in isolated symbol recognition methods for symbol spotting. In [Qureshi *et al.*, 2007], authors propose the use of attributed relational graphs (ARG) for the representation of line primitives in the form of vectors, to represent floor plan drawings. The relationship between the identified vectorial primitives paves way for the detection of regions inside the floor plans. Matching of the query to the regions is done using a graph description and matching method proposed by Qureshi *et al.* [2006]. A combination of graph parsing and matching for representing line drawings was proposed by Llados and Sanchez

2.3 符号检测领域的研究现状

在平面图中进行符号识别的任务，是指给定一个查询符号的图像后，检索出该符号在图纸中的位置及其相关文档。目前主要有两种符号识别方法：第一种是基于区域的方法，其核心是在图纸中搜索感兴趣区域，利用多种描述子对其进行表征，最后通过查询符号建立索引；第二种方法则通过图像分割将符号与其背景分离，进而实现符号的识别。通常，文献中提到的符号识别方法均可归为基于区域的方法。然而，这种方法存在以下固有问题：（1）在缩放后的图纸中难以准确定位感兴趣区域；（2）此类图纸的旋转不变性难以实现；（3）由于线段图案较为简单，需要开发专门的描述子来表征线条图，而不能直接采用常规的文本型描述子；（4）当处理大规模数据库时，诸如哈希等索引技术的可扩展性较差。鉴于上述问题，基于区域的方法往往导致较低的召回率和精确率。尽管存在这些不足，但选择该方法而非基于分割的方法的主要原因在于，现有的分割技术在处理复杂的技术图纸时仍显力不从心。表2.2列出了与符号识别相关的一些重要研究成果。

2.3.1 基于区域的符号检测

在这一检测方法中，当前最先进的技术采用了多种用于识别和描述图纸中感兴趣区域的手段。Rusinol 和 Lladós（2007）以及 Rusiñol（2010）提出使用不同的检测方法，通过识别封闭区域来进一步确定感兴趣区域。在 Rusinol 和 Lladós（2007）的工作中，采用双极坐标系来表征感兴趣区域，并借助哈希表实现索引。而在 Rusiñol（2010）的研究中，则利用带属性的字符串来表示感兴趣区域。首先对封闭区域的轮廓进行多边形近似，并以折线的形式建立相邻线段之间的关联；这些折线被编码为带属性的字符串，其中属性包括线段的长度以及当前线段与前一线段之间的夹角。随后，将相似或相近的字符串聚类存储于查找表中，由一组中值字符串作为索引键，用于对该表进行索引检索。查询与图像中各区域的比较则采用字符串匹配算法。最后，通过霍夫投票机制对初步匹配结果进行验证。此外，在 Ah-Soon 和 Tombre（2001）提出的另一项工作中，采用重叠滑动窗口来检测区域；这里的区域被描述为特征向量，用以刻画区域各线段之间连接关系的模式，并借助图结构进行描述。符号搜索则通过计算网络中各线段间关系的相似性来实现。

2.3.2 分割　　以及对孤立符号的识别　　油斑

图表示在用于符号定位的孤立符号识别方法中也颇受欢迎。在文献[Qureshi v2, 2007]中，作者提出采用带属性的关联图（ARG）以向量形式表示线段基元，从而对平面图进行建模。通过这些已识别的向量化基元之间的关系，可以进一步实现对平面图内部区域的检测。查询与这些区域的匹配则采用Qureshi v3于2006年提出的图描述与匹配方法。Llados和Sanchez还提出了一种结合图解析与图匹配来表示线画的方法。

[2004]. In this paper, the authors present a mechanism for graph indexing to spot known symbols in the drawings. Locteau *et al.* [2007] again used a graph based approach for representation of geometrical shapes which collectively represent geometrical primitives in line drawings. Detection of the regions is done based on open or closed curves. Subsequent query matching to the regions is performed using edit distance metric. Tabbone *et al.* [2003] presented a spotting method that uses the segmentation-based spotting approach. In this particular approach authors propose the need of a segmentation method to segregate text and graphics along with user feedback to further perform symbol segmentation in simplistic line drawings. The descriptor used here is a discrete version of the F-signature which is the histogram of forces. For matching the query to the segmented symbols, the sum of absolute differences is used.

It can be safely stated for the spotting methods following the region-based approach, that if one is able to accurately and precisely identify the regions in the line drawings, then the subsequent steps of matching and description can achieve good results. On the other hand, methods employing isolated symbol spotting approach, rely on efficient prior segmentation, which makes this approach a bit challenging. It is observed that there is an increase in the number of spotting methods which project good resultant recall rates, but usually have a drawback, that they give low precision rates.

The online process for symbol spotting is faster and efficient and usually deals with the following steps: (1) locating regions of interest, (2) clustering (3) creating an efficient indexing data structure from the obtained clusters. During the time of retrieval, the query description is compared with the entries already stored apriori in the data structure. Approaches mentioned in [Nguyen *et al.*, 2009], [Luqman *et al.*, 2010], [Kong *et al.*, 2011] and [Rusiñol *et al.*, 2010] follow this particular technique for symbol spotting. To summarize, the issue with using these methods is that they are lacking in projection of efficient steps for content analysis, which further results in a not-so-good representation of the floor plan drawings, and thus, inadvertently affects the recall and precision rates of retrieval, reducing them to low values.

2.4 STATE OF THE ART IN GENERIC FEATURES FOR RETRIEVAL

So far the existing techniques involved in floor plan analysis and retrieval are discussed. In the latter chapters of the thesis, comparisons with generic retrieval based approaches are done. This section presents a review of features used for retrieval tasks. Several book chapters and papers in the literature present a thorough review of the advances in the feature descriptors [Doermann *et al.*, 2014; Zhang and Lu, 2004]. The primary goals of these description techniques can be listed down as: (1) minimizing the intra-class distances and maximizing the inter-class distances, (2) minimizing the spatial dimensionality to enhance the time complexity and the classification efficiency, and (3) dealing with noise, transformations and distortions in the images. Some popular techniques involve adapting existing feature descriptors from popular research fields like pattern recognition, e.g SIFT [Lowe, 2004], SURF [Bay *et al.*, 2006], and HOG [Dalal and Triggs, 2005]. In HOG [Dalal and Triggs, 2005], the idea is that the shape and appearance of an object within an image can be described by a histogram of intensity gradients. This descriptor is built by dividing the image into small regions, and computing a histogram of the edge orientations for each region. The combination of these histograms form the descriptor for the image. This method has been used for human detection in images. SIFT [Lowe, 2004] has been one of the most widely used feature descriptor in object recognition. Lowe [2004] first proposed this method of representing an image as a collection of local feature vectors which are invariant to translation, scaling, and rotation. These features are initially detected using a Harris-Laplace corner detector in scale space. Once corners are detected, the scale space at which they were detected are stored. Then for each of these keypoints, the dominant gradient is computed over a neighborhood of 16×16. This is also stored

[2004]。在该文中，作者提出了一种用于图索引的机制，以识别图纸中的已知符号。Locteau *et al.* [2007] 再次采用基于图的方法来表示几何图形，这些图形整体上代表了线画中的几何基元。区域的检测是基于开放或闭合曲线进行的，随后通过编辑距离度量对这些区域进行查询匹配。Tabbone *et al.* [2003] 提出了一种基于分割的识别方法。在该方法中，作者建议采用一种分割算法，将文本与图形分开，并结合用户反馈进一步实现对简单线画中符号的分割。此处使用的描述子是力直方图的离散版本，即F-特征。对于查询与分割后符号之间的匹配，则采用绝对差之和作为度量。

对于采用基于区域的方法进行符号检测的算法而言，可以明确指出：若能准确且精确地识别线条图中的各个区域，则后续的匹配与描述步骤便能取得良好效果。另一方面，采用孤立符号检测方法的算法则依赖于高效的预分割，这使得该方法在实施上具有一定难度。观察发现，近年来涌现出大量能够获得较高召回率的符号检测方法，但其普遍存在一个缺陷，即往往导致较低的精确率。

用于符号检测的在线处理流程速度快、效率高，通常包括以下步骤：（1）定位感兴趣区域；（2）聚类；（3）基于所得聚类构建高效的索引数据结构。在检索过程中，将查询描述与预先存储于该数据结构中的条目进行比对。文献[Nguyen *et al.*, 2009]、[Luqman *et al.*, 2010]、[Kong *et al.*, 2011]及[Rusiñol *et al.*, 2010]中所提及的方法均采用这一技术来实现符号检测。综上所述，这些方法的主要问题在于缺乏针对内容分析的有效步骤设计，从而导致对平面图的表征效果欠佳，并间接影响检索的召回率和精确率，使其降至较低水平。

2.4 用于信息检索的通用特征研究现状

迄今为止，已讨论了用于平面图分析与检索的现有技术。在论文的后续章节中，将与基于通用检索的方法进行对比。本节对用于检索任务的特征进行了综述。文献中的若干书籍章节和论文对特征描述子领域的研究进展进行了详尽的回顾[Doermann等，2014；Zhang和Lu，2004]。这些描述技术的主要目标可归纳为：（1）最小化类内距离并最大化类间距离；（2）降低空间维度以提升时间复杂度并提高分类效率；以及（3）有效应对图像中的噪声、变换与畸变。一些常用方法是从模式识别等热门研究领域直接移植现有的特征描述子，例如SIFT[Lowe，2004]、SURF[Bay等，2006]和HOG[Dalal和Triggs，2005]。在HOG[Dalal和Triggs，2005]中，其核心思想是通过梯度方向直方图来刻画图像中目标的形状与外观。该描述子的构建过程是将图像划分为若干小区域，并为每个区域计算边缘方向的直方图，最终由这些直方图的组合构成整幅图像的描述子。该方法曾被用于图像中的人体检测。SIFT[Lowe，2004]则是目标识别领域应用最为广泛的特征描述子之一。Lowe[2004]首次提出将图像表示为一组局部特征向量的集合，这些特征对平移、缩放和旋转具有不变性。这些特征首先通过尺度空间中的Harris-Laplace角点检测器进行检测，并记录其检测时所在的尺度空间。随后，针对每一个关键点，在一个16×16的邻域范围内计算其主梯度方向，并将其一并存储。

and after which, the descriptor for this keypoint is computed. This keypoint descriptor is described by a 128 dimensional vector. This vector is obtained by selecting a 16×16 neighborhood around the keypoint. This is then partitioned into 4×4 regions with each region containing 4×4 pixels. For each region, a histogram with 8 bins is computed from gradient magnitudes in that region. Thus, the 128-dimensional vector is formed. SIFT has also many variants. The first of which is SURF or Speed Up Robust Features [Bay *et al.*, 2006]. The key difference is that this uses integral images to speed up convolutions. Other variants include RGB-SIFT, HSV-SIFT, and Opponent-SIFT. These mainly use SIFT on different color spaces since SIFT was primarily made for grayscale images. RGB-SIFT operates on the RGB channels, HSV-SIFT on HSV channels, and C-SIFT on opponent channels. One final variant is called dense SIFT (or D-SIFT) where descriptors are computed for every pixel. In certain cases, describing the symbol characteristics requires strategies which are highly domain specific, e.g. zoning [Escalera *et al.*, 2009], geometric moments [El Rube *et al.*, 2006], and histogram-oriented [Yang, 2005] techniques. Finally, classification techniques on extracting the features are highly benefited by mathematical techniques such as the computation of sums, products and distances. Thus, several effective algorithms for classification tasks, such as K-nearest neighbour, SVM, boosting and neural network learning have been widely employed in the literature. Another technique used for extracting features is Run Length Histograms (RLH) [de las Heras *et al.*, 2013] approach where, the style of an architectural drawing is characterized by the perception of lines, shapes and texture. Authors propose runlength histograms extracted in vertical, horizontal and diagonal directions as a characterization of line and space properties in floor plans that further estimate a description of walls and room structure. Another efficient technique is an Online Algorithm for Scalable Image Similarity learning (OASIS) [Chechik *et al.*, 2010], which learns a bilinear similarity measure over sparse representations. OASIS is an online dual approach using the passive-aggressive family of learning algorithms that employ an efficient hinge loss cost.

2.5 STATE OF THE ART IN DEEP LEARNING FRAMEWORKS FOR IMAGE AND SKETCH RETRIEVAL

With the extensive research on deep neural networks (DNN) [Krizhevsky and Hinton, 2011; Szegedy *et al.*, 2015; Simonyan and Zisserman, 2014], the success and efficiency of the learning-based features have been observed in multiple areas. High-level abstractions providing an approximation to the human cognition system can be learned [Bengio *et al.*, 2009] using deep architectures. As a result, DNN can be used to extract semantically oriented features using the activation in different layers in the networks. In the work proposed by Hörster and Lienhart [2008], feature extraction is carried out using local patches employing a deep restricted Boltzmann machine. Convolutional neural networks (CNN) form a sub-component of the DNNs. Authors in [Krizhevsky *et al.*, 2012b] have demonstrated a highly superior performance in various tasks on image recognition and retrieval [Razavian *et al.*, 2014]. In the paper proposed by Wan *et al.* [2014], detailed studies are conducted on the capability of learned visual features using deep CNN for a variety of applications which include content based image retrieval as well. Razavian *et al.* [2014] study the VGG-Net and Alex-Net and further try to tap the response of the last convolutional layers along with max pooling for image representation for the final task of image retrieval [Razavian *et al.*, 2016]. In the work of Zheng *et al.* [2015], the sixth layer of the Alex-Net [Krizhevsky *et al.*, 2012b] is activated and the activations are taken out as a DNN feature for each image, which is integrated in the image similarity match score level along side traditional visual features that include HSV histogram, SIFT-based BoW feature, and GIST. The state-of-the-art sketch based deep retrieval models [Sangkloy *et al.*, 2016] are deep models that aim to close the domain gap by learning a joint feature embedding for the two domains. Multibranch deep convolutional neural networks (CNNs) are employed where each branch corresponds to one domain and the final shared layer defines the

随后，计算该关键点的描述子。该关键点描述子由一个128维向量表示。该向量是通过在关键点周围选取一个16×16的邻域区域获得的，然后将该区域划分为4×4个子区域，每个子区域包含4×4个像素。对于每个子区域，基于该区域内梯度幅值计算一个具有8个 bin 的直方图，从而构成最终的128维向量。SIFT算法还存在多种变体，其中最早出现的是SURF，即加速稳健特征算法[Bay，2006]。其主要区别在于采用积分图像来加速卷积运算。其他变体还包括RGB-SIFT、HSV-SIFT和Opponent-SIFT，这些方法主要是在不同的颜色空间上应用SIFT，因为SIFT最初是为灰度图像设计的。RGB-SIFT作用于RGB通道，HSV-SIFT作用于HSV通道，而C-SIFT则作用于对立色通道。最后一个变体称为密集SIFT（或D-SIFT），即对图像中的每个像素都计算描述子。在某些情况下，刻画符号特征需要采用高度领域特定的策略，例如分区法[Escalera，2009]、几何矩[El Rube，2006]以及方向直方图技术[Yang，2005]。此外，在特征提取后的分类过程中，数学方法如求和、乘积及距离计算等能够显著提升分类效果。因此，K近邻、支持向量机、提升算法以及神经网络学习等多种有效的分类算法已在相关文献中得到广泛应用。另一种用于特征提取的技术是游程长度直方图（RLH）方法[de las Heras，2013]，该方法通过线条、形状和纹理的感知来表征建筑图纸的风格。作者提出，分别沿垂直、水平和对角方向提取的游程长度直方图可用于刻画平面图中线条与空间的属性，进而推断墙体及房间结构的描述。此外，一种高效的在线可扩展图像相似性学习算法（OASIS）[Chechik，2010]也值得关注，该算法基于稀疏表示学习双线性相似性度量。OASIS是一种在线双重优化方法，采用被动-攻击型学习算法族，并使用高效的铰链损失函数。

2.5 用于图像与草图检索的深度学习框架研究现状

随着对深度神经网络（DNN）的广泛研究[Krizhevsky 和 Hinton，2011；Szegedy 等，2015；Simonyan 和 Zisserman，2014]，基于学习的特征在多个领域已展现出显著的成功与高效性。借助深度架构，可以学习到能够近似人类认知系统的高层次抽象表示[Bengio，2009]。因此，深度神经网络可通过其不同层的激活来提取语义导向的特征。在Hörster和Lienhart[2008]提出的方案中，特征提取采用局部块，并利用深度受限玻尔兹曼机实现。卷积神经网络（CNN）是深度神经网络的一个子类。Krizhevsky[2012b]的研究表明，CNN在图像识别与检索等各类任务中均表现出极为优异的性能[Razavian 等，2014]。Wan[2014]的论文详细探讨了基于深度CNN所学视觉特征在多种应用中的能力，其中包括基于内容的图像检索。Razavian[2014]对VGG-Net和Alex-Net进行了研究，并进一步尝试利用最后一层卷积层的响应及最大池化结果进行图像表征，以用于最终的图像检索任务[Razavian，2016]。Zheng[2015]的工作则激活了Alex-Net的第六层[Krizhevsky，2012b]，并将各层的激活作为每幅图像的深度神经网络特征，在图像相似性匹配得分层面与传统视觉特征（如HSV直方图、基于SIFT的词袋特征以及GIST）相结合。最先进的基于草图的深度检索模型[Sangkloy等，2016]是一类旨在通过学习两个域的联合特征嵌入来缩小域间差异的深度模型。该类模型采用多分支深度卷积神经网络，其中每个分支对应一个域，而最终的共享层则定义了

embedding space which is subject to various discriminative losses such as pairwise contrastive loss or triplet ranking loss. These losses are designed to pull matching pairs of photos and sketches close and push mismatched pairs away. These models thus indirectly align the two domains. However, with limited training data and by focusing only on discriminative losses, these models struggle to capture all the domain-invariant information and thus generalise poorly to test data where the domain discrepancies and misalignments could be different from those in the training data. Alternatively, Generative Adversarial Networks (GANs) have also been employed for generic sketch retrieval [Creswell and Bharath, 2016] and [Chen and Hays, 2018] but have not yet been explored for application in floor plan retrieval. GANs combine the discriminative model with a generative component. One of the typical cases is to use source images, noise vectors or both to generate simulated samples that are similar to the target samples and preserve the annotation information of the source domain. One more approach that is used is the reconstruction-based domain adaptation approach, which assumes that the data reconstruction of the source or target samples can be helpful for improving the performance of domain adaptation. The reconstructor can ensure both specificity of intra-domain representations and indistinguishability of inter-domain representations by using stacked autoencoders (SAEs), that combine the encoder network for representation learning with a decoder network for data reconstruction [Ghifary *et al.*, 2016],[Zhuang *et al.*, 2015].

2.6 FLOOR PLAN DATASETS

Table 2.3. : Details of publicly available existing floor plan datasets

Floor plan Datasets		
Dataset	Count	Remarks
CVC-FP [de las Heras *et al.*, 2015]	122	4 Sub-Categories, varying in wall textures, to study graphical notations in floor plans
FPLAN-POLY [Rusiñol *et al.*, 2010]	42	Used for floor plan analysis and room analysis
SESYD [Delalandre *et al.*, 2007]	1000	100 layouts/ class, differ in arrangement of symbols, used for symbol spotting tasks

Performance evaluation of symbol spotting tasks and layout analysis on floor plans requires datasets that are publicly available to aid the research in this area. Table 2.3 contains the details of some publicly available floor plan datasets. One of the widely popular databases for symbol recognition and spotting tasks is the Systems Evaluation SYnthetic Documents (SESYD) database [Delalandre *et al.*, 2007]. It is a collection of labelled synthetic images. It consists of electrical and architectural drawings, designed specifically for symbol spotting, recognition and retrieval. The whole dataset consists of 5 document collections containing $11,100$ images with $128,700$ symbols. It contains 1000 floor plans divided into 10 broad categories, each containing 100 samples. Inter-class similarity is very low as layouts across each category have a huge difference. Intra-class similarity is very high as floor plans within a class only differ in the furniture/symbol arrangement and orientation in each plan.

Another floor plan dataset FPLAN-POLY proposed in [Rusiñol *et al.*, 2010] contains 42 floor plan vectorial images in dxf format which can be viewed with the common CAD softwares. The main goal of this database also is to provide a framework for the evaluation of different symbol spotting methods in vectorized graphic documents. CVC-FP dataset proposed in [de las Heras *et al.*, 2015] is annotated for the architectural objects and their structural relations. It contains 122 floor plans spread over 4 different layout categories. This dataset aims at aiding in learning and

嵌入空间会受到多种判别性损失的约束，例如成对对比损失或三元组排序损失。这些损失旨在使匹配的图像与素描样本彼此靠近，同时将不匹配的样本相互推开。因此，这类模型能够间接实现两个域之间的对齐。然而，在训练数据有限且仅依赖判别性损失的情况下，这些模型难以充分捕捉所有域不变信息，从而在测试数据上的泛化性能较差；而测试数据中的域差异与对齐偏差可能与训练数据存在显著不同。此外，生成对抗网络（GAN）也被用于通用素描检索任务[Creswell和Bharath，2016]以及[Chen和Hays，2018]，但尚未在楼层平面图检索中得到应用。GAN将判别模型与生成组件相结合。一种典型的应用是利用源域图像、噪声向量或两者的组合来生成与目标样本相似且保留源域标注信息的模拟样本。另一种方法是基于重建的域适应方法，该方法假定通过对源域或目标域样本进行数据重建，有助于提升域适应性能。通过使用堆叠自编码器（SAE），即结合用于表征学习的编码器网络与用于数据重建的解码器网络，重建器能够在确保域内表示特异性的同时，实现域间表示的不可区分性[Ghifary等，2016；Zhuang等，2015]。

2.6 平面布局数据集

表2.3：公开可用的现有平面图数据集详情

Floor plan Datasets		
Dataset	Count	Remarks
CVC-FP [de las Heras *et al.*, 2015]	122	4 Sub-Categories, varying in wall textures, to study graphical notations in floor plans
FPLAN-POLY [Rusiñol *et al.*, 2010]	42	Used for floor plan analysis and room analysis
SESYD [Delalandre *et al.*, 2007]	1000	100 layouts/ class, differ in arrangement of symbols, used for symbol spotting tasks

对平面图中的符号检测任务和版面分析进行性能评估，需要公开可用的数据集以支持该领域的研究。表2.3列出了部分公开的平面图数据集的详细信息。在符号识别与检测任务中，广为使用的数据库之一是系统评估合成文档（SESYD）数据库[Delalandre v3, 2007]。该数据库是一组带有标注的合成图像，包含电气图和建筑图，专为符号检测、识别及检索任务而设计。整个数据集由5个文档集合组成，共包含11,100张图像，其中包含128,700个符号。数据集中包含1,000张平面图，分为10个大类，每类包含100个样本。不同类别之间的相似性很低，因为各分类的版面布局差异显著；同一类别内的相似性则很高，因为同一类别的平面图仅在家具或符号的布置及其朝向上存在差异。

另一份由Rusiñol等人于2010年提出的平面图数据集FPLAN-POLY包含42幅DXF格式的矢量化平面图图像，可使用常见的CAD软件进行查看。该数据库的主要目标同样是为评估矢量化图形文档中的各类符号检测方法提供一个框架。由de las Heras等人于2015年提出的CVC-FP数据集则针对建筑对象及其结构关系进行了标注，共包含122幅平面图，并涵盖4种不同的布局类别。该数据集旨在辅助学习和

better interpretation about how the elements are structurally arranged in a floor plan.

2.7 SUMMARY

In this Chapter, various research areas such as floor plan analysis, symbol spotting, retrieval, feature extraction and deep learning models for analysis are discussed. Although the literature in the above mentioned areas is quite rich, there does not exist a composite framework in the area of floor plan analysis that takes meaningful information extracted out of floor plans to perform content based retrieval. Moreover, the publicly available datasets in the area of floor plans are mainly suited for symbol detection and structural analysis in floor plans. Neither are the existing datasets diverse enough to offer varied content for retrieval task nor are the samples in them adequate in number to implement deep learning approaches for efficient analysis and retrieval. Thus, in the consequent chapters of this thesis techniques for content based retrieval in floor plans are proposed to aid users/architects to help find their dream property according to their specific requirements.

更准确地理解各元素在平面布局中的结构排列方式。

2.7 摘要

在本章中，讨论了平面图分析、符号识别、检索、特征提取以及用于分析的深度学习模型等多个研究领域。尽管上述各领域的文献已相当丰富，但在平面图分析领域仍缺乏一个综合性框架，能够利用从平面图中提取的语义信息实现基于内容的检索。此外，现有公开的平面图数据集主要适用于平面图中的符号检测与结构分析任务；这些数据集在内容多样性方面不足以支持多样化的检索需求，且样本数量也难以满足实施高效分析与检索的深度学习方法的要求。因此，在本论文的后续章节中，将提出基于内容的平面图检索技术，以帮助用户或建筑师根据其特定需求找到理想的房产。

3

Topological features for representation and retrieval of similar floor plans

As observed by the detailed literature survey given in Chapter 2, researchers in the area of analysis of architectural floor plans have mainly explored areas like:

- *Symbol spotting*: Symbols are described as a set of logically related graphical parts. These parts are generally visual primitives such as vectors, lines, and arcs or a certain combination of all of these. These primitives in a certain context denote a symbol instance. In the task of symbol spotting thus, all the decor symbols present inside a floor plan are localized and recognized.

- *Floor plan Analysis*: This pertains to structural and semantic analysis of floor plans. Examples of various categories of problems addressed in this type of task are: detecting the various type of walls in a floor plan, segmentation of various rooms in a floor plan, understanding the relationship between the rooms, identifying various functionalities of various parts of a floor plan.

From the literature it can be inferred that symbol spotting and analysis in architectural floor plans have been active research areas in the past. However, a composite framework comprising of analysing a floor plan structurally as well as semantically, and further retrieving floor plans based on certain content or categories is a relatively novel problem. Upon closer observation it was found that a floor plan has various types of meaningful information (content) that can be explored while retrieval. At the same time, inter-class dissimilarity and intra-class similarity among the samples of a floor plan dataset are also to be taken into consideration. Some examples for a meaningful content are, layout information, inner components or decor information, accessibility or approachability of each room from the entry or exit or from each other, aesthetics information or proper placement of the rooms inside the layout, and so on.

In this chapter, the primary focus is on the first two types of content namely, the layout information and the decor information to distinguish between a pair of floor plans. A baseline global framework comprising of match scores computed based on both the topological and decor features is proposed, which paves way for a novel framework for the introduction of retrieval in the area of architectural floor plans. In the subsequent chapters of this thesis, improvement over this baseline is discussed. The Chapter organization is as follows: Sec. 3.1 gives a brief insight into the proposed approach. Subsequent sections detail out the methodology of the proposed approach. In Sec. 3.6 the experimental setup, qualitative and quantitative results of the proposed technique are presented. Section 3.7 provides an analysis of the results obtained and finally Sec. 3.8 concludes the Chapter.

3

用于相似平面图表示与检索的拓扑特征

正如第2章所进行的详细文献调研所示，建筑平面图分析领域的研究者主要探讨了以下方面：

- *Symbol spotting*符号被定义为一组逻辑相关的图形元素，这些元素通常是矢量、直线、弧线等基本视觉图元，或它们的某种组合。在特定上下文中，这些图元共同构成一个符号实例。因此，在符号检测任务中，需要对建筑平面图中的所有装饰性符号进行定位与识别。

- *Floor plan Analysis*这涉及对建筑平面图的结构与语义分析。此类任务所处理的问题类别包括：识别平面图中的各类墙体、分割平面图中的各个房间、理解各房间之间的关系，以及确定平面图中各部分的具体功能。

从现有文献可知，建筑平面图中的符号识别与分析在过去一直是活跃的研究领域。然而，构建一个综合框架，既从结构层面又从语义层面分析建筑平面图，并在此基础上按特定内容或类别进行检索，仍是一个相对新颖的问题。通过深入观察发现，建筑平面图中蕴含着多种可被检索利用的有意义信息（内容）。与此同时，在建筑平面图数据集中，还需考虑样本间的类间差异性和类内相似性。一些具有实际意义的内容示例包括：布局信息、内部构件或装饰信息、各房间相对于出入口或其他房间的可达性或接近程度、美学信息以及房间在布局中的合理位置等。

在本章中，主要关注前两类内容，即布局信息和装饰信息，以区分一对建筑平面图。我们提出了一种基于拓扑特征与装饰特征综合计算匹配得分的基准全局框架，为在建筑平面图检索领域引入新型检索框架奠定了基础。在本论文的后续章节中，将讨论对该基准框架的改进。本章的组织结构如下：第3.1节简要介绍所提出的方案；随后各节详细阐述该方案的具体方法；第3.6节展示实验设置以及所提方法的定性和定量结果；第3.7节对所得结果进行分析；最后，第3.8节对本章进行总结。

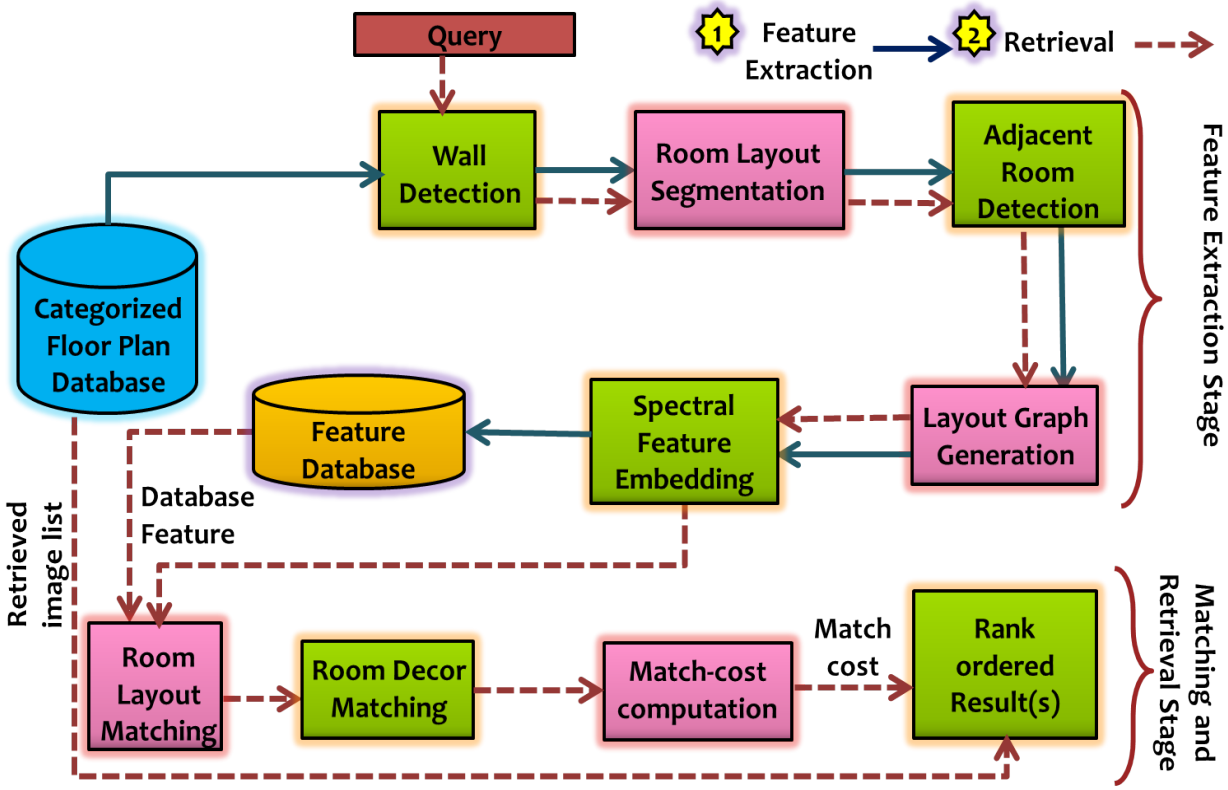


Figure 3.1. : Framework diagram for our structural retrieval based on topology

3.1 BRIEF OVERVIEW

A query by example paradigm has been followed to retrieve similar floor plan images. Figure 3.1 depicts the complete framework of the proposed technique of semantic based architectural floor plan retrieval. The framework has two phases : A) Feature extraction, and B) Matching and Retrieval phase. The solid arrows in the framework diagram (Fig. 3.1) correspond to the feature extraction phase, whereas, the dashed arrows correspond to the retrieval phase. The various components of the framework, in order of operation, are:

1. Floor plan repository, which in this case are the architectural floor plan databases SESYD [Delalandre *et al.*, 2010a] and ROBIN [Sharma *et al.*, 2017].

2. The Wall Detection block, which focuses on delineating a floor plan and detecting room boundaries.

3. Room Layout Segmentation block, which segments the layout into meaningful rooms discarding all the inner decor components present inside the rooms.

4. Adjacent Room Detection (ARD) block wherein the relationship between the segmented rooms is established. Here, the relationship pertains to the topological information of the layout.

5. The output of the ARD block is sent to the Layout Graph generation step where, the topological information of a layout is modelled into a graph for feature representation.

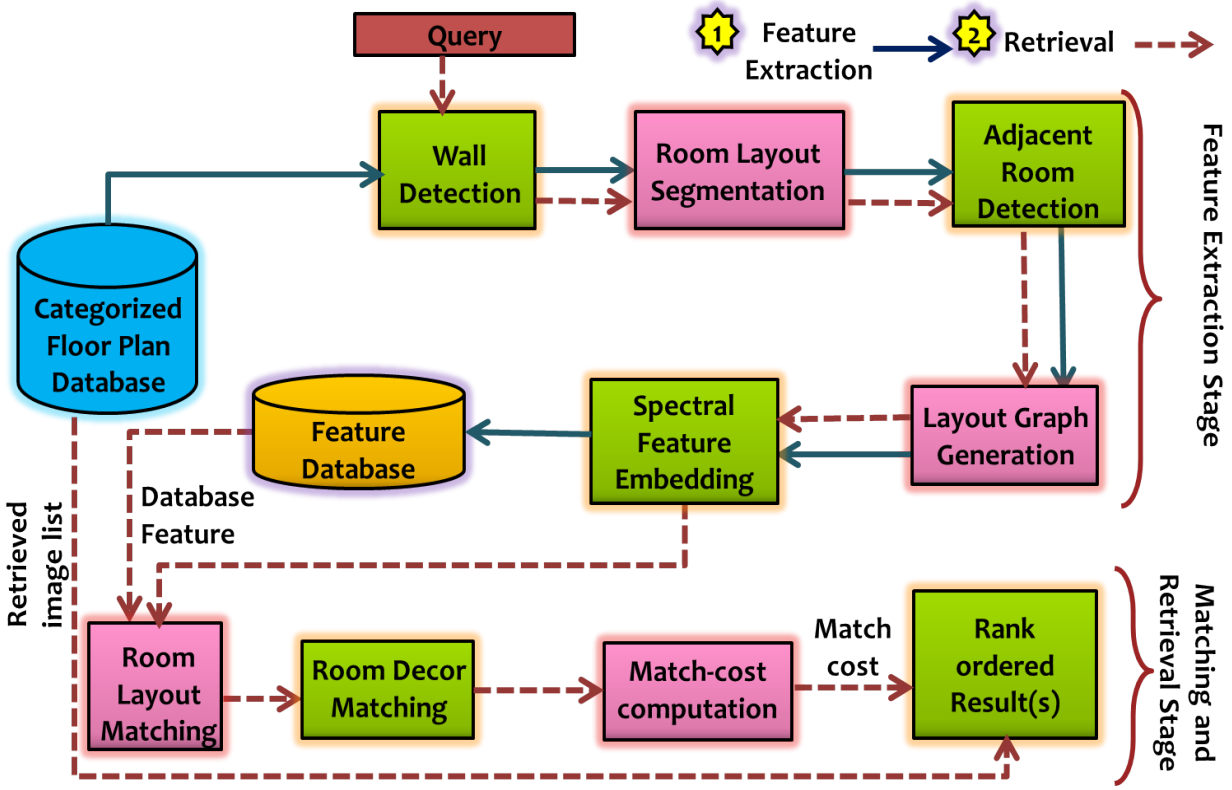


图3.1：基于拓扑结构的结构检索框架示意图

3.1 简要概述

采用基于示例的查询范式来检索相似的平面图图像。图3.1展示了所提出的基于语义的建筑平面图检索技术的完整框架。该框架分为两个阶段：A）特征提取阶段，以及B）匹配与检索阶段。框架图（图3.1）中的实线箭头对应于特征提取阶段，而虚线箭头则对应于检索阶段。按照执行顺序，该框架的各个组成部分依次为：

1. 平面图知识库，本研究中采用的为建筑平面图数据库 SESYD [Delalandre *et al.*, 2010a] 和 ROBIN [Sharma *et al.*, 2017]。 2. 墙体检测模块，用于勾勒平面图并识别房间边界。 3. 房间布局分割模块，将布局划分为具有语义意义的房间，并剔除房间内部的所有装饰性构件。 4. 相邻房间检测（ARD）模块，在此模块中建立各分割后房间之间的关系，该关系反映布局的拓扑信息。 5. ARD 模块的输出被送入布局图生成步骤，其中布局的拓扑信息被建模为图结构，以用于特征表示。

Considering the adjacencies a graph of the floor plan is generated using the topology graph idea proposed in [da Fonseca, 2004].

6. The next block is the Spectral Feature Embedding block where, the topology (neighbourhood) information is embedded into a pattern space as a feature vector. The adjacency matrix obtained from this layout graph is used to extract the spectral features.

7. The next component, the feature database, stores all the feature vectors extracted from the floor plans and serves as a feature repository. This procedure is carried out apriori for all the images present in the floor plan database.

The blocks mentioned above constitute the Feature Extraction Stage. As soon as the query is fired, first the features are extracted from the floor plans using the Stage 1 blocks and then the Matching and Retrieval Stage (Stage 2) comes into picture. The main blocks in Stage 2 are:

1. Room Layout Matching (RLM) block, where, the dissimilarity between the query feature vector and each of the model feature vectors is computed using Euclidean distance metric. The model corresponding to the minimum distance of its feature vector from the query feature vector is chosen as the best match.

2. Room Decor Matching (RDM) block, where the layout from the database to which the query layout matches the best is then chosen and further in the images of that matching layout category, the arrangement of furniture is analyzed. A match cost is computed by matching the room decor of each image of the layout category with the query image and this match cost is then used to rank order the results for obtaining the best matching floor plan image to the query floor plan image.

3. Match-cost Computation block helps in calculating the cumulative match cost obtained from both the Room Layout Matching block and the Room Decor Matching block.

4. Finally, using this cumulative match cost floor plans are fetched from the repository and rank-ordered on the basis of their similarity with the query floor plan.

The main contributions towards the first attempt for designing a baseline framework for floor plan retrieval are as follows:

1. A novel approach to perform floor plan retrieval incorporating both symbol spotting and room level analysis.

2. Performing two level matching for floor plans based on semantics such as layout and decor.

3. Graph spectral embedding of layouts into a pattern space for efficient and fast matching.

All these steps finally complete the retrieval framework for similar floor plans. The details of each step mentioned in the framework diagram are further discussed in the following sections.

3.2 LAYOUT SEGMENTATION

The first step in the proposed framework is layout image segmentation, which consists of structural and semantic analysis to segment the floor plans into respective rooms. Figure 3.2 depicts the various stages of layout segmentation and analysis. Boundary extraction is performed where

考虑到邻接关系，利用文献[da Fonseca, 2004]中提出的拓扑图思想，生成了楼层平面的图表示。6. 下一个模块是谱特征嵌入模块，在该模块中，拓扑（邻域）信息被作为特征向量嵌入到模式空间中。从该布局图得到的邻接矩阵用于提取谱特征。7. 接下来的组件——特征数据库，存储了从所有楼层平面图中提取的特征向量，用作特征库。这一过程预先对楼层平面数据库中的所有图像进行处理。

上述各模块构成了特征提取阶段。一旦发出查询请求，首先通过第一阶段的模块从平面图中提取特征，随后进入匹配与检索阶段（第二阶段）。第二阶段的主要模块包括:

1. 房间布局匹配（RLM）模块，其中，查询特征向量与每个模型特征向量之间的差异性采用欧氏距离度量进行计算。将与其特征向量到查询特征向量的距离最小的模型作为最佳匹配结果。

2. 房间装饰匹配（RDM）模块：首先从数据库中选择与查询布局匹配度最高的布局；随后，在该匹配布局类别的图像集中，对家具的摆放方式进行分析。通过将该布局类别中每张图像的房间装饰与查询图像进行匹配，计算出匹配代价，并以此匹配代价对结果进行排序，从而获得与查询平面图最为匹配的平面图图像。

3. 匹配代价计算模块用于计算由房间布局匹配模块和房间装饰匹配模块共同得到的累积匹配代价。

4. 最后，基于该累积匹配代价，从知识库中检索平面布局，并根据其与查询平面布局的相似度进行排序。

首次尝试构建用于平面图检索的基准框架的主要贡献如下:

1. 提出一种新颖的方法，通过融合符号检测与房间级分析来实现建筑平面图检索。2. 基于布局与装饰等语义特征，对建筑平面图进行双层匹配。3. 将布局结构进行图谱嵌入，映射至模式空间，以实现高效、快速的匹配。

所有这些步骤最终共同构成了相似平面图的检索框架。框架图中所提及的各步骤的具体内容将在后续章节中进一步讨论。

3.2 布局分割

所提出的框架的第一步是布局图像分割，该步骤包括结构分析与语义分析，以将平面图分割为各个房间。图3.2展示了布局分割与分析的各个阶段。其中，边界提取是在……处进行的。

Figure 3.2. : Various stages of structural and semantic analysis on an image from SESYD dataset [Delalandre *et al.*, 2010a]: (a) unsegmented layout, results of (b) morphological operation and wall, boundary extraction, (c) Detection of doors and windows (d) Closing gaps in walls and (e) labelling and segmenting the room structure

the boundary extraction algorithm and morphological operations are carried out detecting the walls in the layout. Let the input floor image be denoted by $\mathscr{I}$. This floor plan image is first dilated ($\oplus$) by a disk type structuring element (S_a) of unit radius followed by erosion operation ($\ominus$) by the same element, according to the following equation:

$$\mathscr{I} = \mathscr{I} \oplus S_a \;\; ; \;\; \mathscr{I} = \mathscr{I} \ominus S_a \tag{3.1}$$

These operations lead to segregation of the layout boundary from small unnecessary designs present inside the layout (Fig. 3.2(b)). This detects the walls in the layout to represent the room structures and thus, delineates the rooms within the layout from each other. Next gap closing is performed at the location of doors and window in the floor plan. Floor plans contain doors and windows that connect each room, therefore, are important for semantic analysis. However, the presence of these doors and windows doesn't influence the overall structure of the building. Hence, these two components are removed for structural analysis. The doors and windows inside the layout are present in various orientations and their elimination leads to production of gaps in the walls at their location (refer Fig. 3.2 (c)). The first task therefore, is to detect the doors and windows in the floor plan and close the gaps at their locations so as to construct definite boundaries for the rooms (Fig. 3.2(d)). To detect the doors and windows Harris corner detector [Harris and Stephens, 1988] is employed.

It is observed that the door symbol has an arc like structure (triangular structure) and thus, Harris corner detector detects three corners at the three triangle edges for the door symbol (refer Fig. 3.3(a)). It is to be noted that only the door symbol out of all the decor symbols in the symbol library has this type of a unique triangle-bound 3-corner structure. Through this unique

图3.2：对SESYD数据集中的图像进行结构与语义分析的各个阶段[Delalandre等，2010a]：（a）未分割的版面布局；（b）形态学操作及墙体、边界提取的结果；（c）门窗检测；（d）墙体缝隙填充；（e）房间结构的标注与分割。

通过边界提取算法和形态学运算对布局中的墙体进行检测。设输入的楼层图像为$\mathcal{I}$。首先对该楼层平面图像进行膨胀处理（得到$\oplus$），所使用的结构元素为单位半径的圆形结构元素（记为S_a）；随后再对该图像进行腐蚀处理（得到$\ominus$），所用结构元素与前述相同，具体过程如下式所示：

$$\mathcal{I} = \mathcal{I} \oplus S_a \ ; \ \mathcal{I} = \mathcal{I} \ominus S_a \tag{3.1}$$

这些操作能够将版图边界与版图内部存在的小型非必要结构分离（图3.2(b)）。在此过程中，通过检测版图中的墙体以表征房间结构，从而在版图中划分出各个独立的房间。接下来，在平面图中对门和窗的位置进行间隙填补。由于平面图中存在连接各房间的门和窗，它们对于语义分析具有重要意义；然而，这些门和窗的存在并不影响建筑的整体结构，因此在进行结构分析时需将其移除。版图中的门和窗往往呈现不同的朝向，其被移除后会在相应位置的墙体上形成间隙（参见图3.2(c)）。因此，首要任务是在平面图中检测门和窗，并在其所在位置填补间隙，以构建各房间的明确边界（图3.2(d)）。为实现这一目标，采用Harris角点检测器[Harris和Stephens，1988]进行门和窗的检测。

观察发现，门符号具有类似弧形的结构（三角形结构），因此，Harris角点检测器在该门符号的三条三角形边缘处分别检测到三个角点（参见图3.3(a)）。需要指出的是，在符号库中的所有装饰符号中，唯有门符号具备这种独特的由三角形边界构成的三角结构。通过这一独特特性

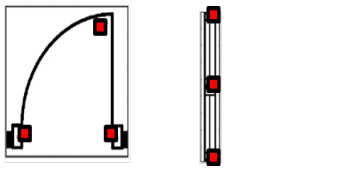


Figure 3.3. : Detecting doors and windows using Harris corner detector

structure the doors stand out as compared to the rest of the symbols in the symbol library. To detect whether the obtained corners form a triangle or not, the three corners identified through Harris corner detector for the door symbol are joined with their adjacent corner point, creating a closed triangle. Further, length of each side (a,b,c) of this triangle is computed and using the triangular inequality : "sum of any two sides should be greater than the third side ($a+b>c$, $a+c>b$, $b+c>a$)", the closed structure is determined to be a triangle. Thus, using this approach doors are detected. The next symbol that needs to be detected in a floor plan for gap closing, is the window. The windows are observed to have collinear corners (refer Fig. 3.3(b)) and thus, get detected through the detection algorithm as the only symbol with collinear corners. Again to identify whether the obtained corners are collinear or not, the co-ordinates of the corner points are noted (x_1, y_1, x_2, y_2 and x_3, y_3). Further, slope of the line between each pair of points is calculated as:

$$slope = \frac{y_1 - y_2}{x_1 - x_2}; \qquad \frac{y_2 - y_3}{x_2 - x_3}; \qquad \frac{y_3 - y_1}{x_3 - x_1} \tag{3.2}$$

If the slope is the same between any pair of corner points, then they are identified to be collinear, hence belonging to the window structure. Thus, uptil now the door and the window locations have been identified.

Upon identifying and eliminating the doors and windows from the floor plan, the next task is to close the gaps in the walls (as shown in Fig. 3.2 (c)). Closing of the gaps is achieved by joining the detected co-ordinates of the doors (in this case the co-ordinates forming 0° angle with each other) and windows setting the values of the pixel location between them as 0, with a thickness of about 40 pixels. Hence, creating a closed wall-structure as in Fig. 3.2 (d). Next, connected component analysis using 8-neighbourhood criterion is performed, to label each room in the layout separately. By this step, the rooms are segmented and are surrounded with walls separating each one of them 3.2(e)). The segmented room layout is sent to the Adjacent Room Detection block for adjacency graph creation.

3.3 ADJACENT ROOM DETECTION

In this step the segmented image is passed through an Adjacent Room Detection (ARD) module to determine the adjacencies between the rooms in the layout. As the segmented rooms are still divided by walls surrounding them, a threshold corresponding to the wall width of about 40 pixels is taken along both vertical and horizontal directions to detect the differently labelled pixels in the image corresponding to the adjacent room. By determining the adjacencies the topology graph

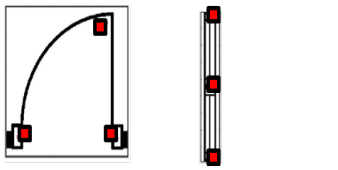


图3.3：利用Harris角点检测器检测门窗

在符号库中，门的结构与其他符号相比显得尤为突出。为判断所提取的角点是否构成三角形，首先将通过Harris角点检测器识别出的门符号的三个角点与其相邻的角点相连，从而形成一个闭合三角形。随后，计算该三角形每条边的长度（a、b、c），并利用三角不等式——“任意两边之和应大于第三边（$a+b>c$、$a+c>b$、$b+c>a$）”——来判定该闭合结构是否为三角形。由此，采用上述方法实现对门的检测。在用于填补空隙的平面图中，下一个需要检测的符号是窗。观察发现，窗的角点呈共线分布（参见图3.3(b)），因此可通过检测算法将其识别为唯一具有共线角点的符号。同样，为判断所提取的角点是否共线，需记录各角点的坐标（x_1, y_1、x_2, y_2和x_3, y_3）。进一步地，计算每对点之间连线的斜率：

$$slope = \frac{y_1 - y_2}{x_1 - x_2}; \qquad \frac{y_2 - y_3}{x_2 - x_3}; \qquad \frac{y_3 - y_1}{x_3 - x_1} \tag{3.2}$$

如果任意一对角点之间的斜率相同，则判定这些角点共线，从而归属于窗体结构。因此，截至目前，门和窗的位置均已得到识别。

在从平面图中识别并移除门窗后，下一步是封闭墙体中的缝隙（如图3.2(c)所示）。通过将检测到的门窗坐标（此处指彼此之间形成0°夹角的坐标）连接起来，并将其间的像素位置值设置为0，同时赋予约40像素的厚度，即可实现缝隙的封闭，从而构建出如图3.2(d)所示的封闭墙体结构。随后，采用8邻域准则进行连通组件分析，以对布局中的每个房间分别进行标记。至此，各房间已被分割，并由墙体相互分隔（如图3.2(e)所示）。分割后的房间布局被送入相邻房间检测模块，用于生成相邻关系图。

3.3 相邻房间检测

在该步骤中，将分割后的图像输入到邻室检测（ARD）模块，以确定布局中各房间之间的邻接关系。由于分割后的房间仍被墙体包围，因此沿垂直和水平方向分别设定一个对应于约40像素墙宽的阈值，用于识别图像中属于相邻房间的不同标记像素。通过确定这些邻接关系，即可构建拓扑图。

for the layout is created, according to the technique mentioned in [da Fonseca, 2004]. Figure 3.4(c) depicts an example of a topology graph created from the room adjacencies (Fig. 3.4(b)) determined from a given floor plan (Fig. 3.4(a)). The parent node (the one at the centre) represents the layout and the child nodes represent the rooms in the layout. The solid edges between the parent node and the child nodes represent inclusion and dashed edges between the child nodes represents adjacencies. This graph creation further, aids in constructing an adjacency matrix for the rooms detected in the layout which is then used for feature extraction.

However, topology graphs are not directly used for searching similar drawings, since graph matching is a NP-complete problem. Instead of that, the corresponding graph spectrum is used. For each topology graph to be indexed in a database, the descriptors are computed on the basis of its spectrum. In this way, the problem of isomorphism between topology graphs is reduced to computing distances between descriptors. The next section details out the steps to identify the decor components inside the floor plan which are also used for matching purposes.

3.4 FURNITURE DETECTION AND CATEGORIZATION

Algorithm 1 depicts the steps involved in furniture categorization. Firstly, the walls are detected in the floor plan image using a morphological operation as walls are thicker than the other objects inside a floor plan. Then both, the original floor plan image and the image containing only walls are subtracted to obtain only the furniture present inside the floor plan. The decor present inside the whole floor plan is counted on the basis of the application of morphological fill operation applied to the image. Blobs obtained are taken as the decor components inside the floor plan. Further, to categorize the obtained blobs, area-ratio of the individual connected components inside the blob is taken and a unique signature for each furniture component is established by considering the area-ratio of three largest components in each blob. Prior to this, such an area-ratio feature taking the three largest connected components in a furniture symbol is obtained for all the 12 symbols in the symbol library and stored. Signature obtained of all the components inside a floor plan image is finally compared with the signature of the components present in the symbol library and thus, the type of the furniture is assigned to each decor item in the floor plan. Figure 3.5 shows how a furniture in the furniture set is assigned a signature according to Algo. 1.

(a) Architectural Floor plan (b) Results of proposed room layout segmentation technique (c) Layout graph of (a)

Figure 3.4. : Topological graph for the segmented room layout

布局的生成遵循[da Fonseca, 2004]中所述的技术。图3.4(c)展示了一个由房间邻接关系构建的拓扑图示例（图3.4(b)），该邻接关系是从给定的平面图（图3.4(a)）中确定的。其中，父节点（位于中心）表示整个布局，子节点则表示布局中的各个房间。父节点与子节点之间的实线边表示包含关系，而子节点之间的虚线边则表示邻接关系。这一拓扑图的构建进一步有助于为布局中检测到的各房间构造邻接矩阵，该矩阵随后用于特征提取。

然而，拓扑图并不直接用于相似图纸的检索，因为图匹配问题属于NP完全问题。因此，转而采用相应的图谱特征。对于需在数据库中建立索引的每一张拓扑图，均基于其图谱计算特征描述子。由此，拓扑图之间的同构性判定问题被转化为特征描述子之间距离的计算问题。下一节将详细阐述用于识别平面图中装饰构件的具体步骤，这些步骤同样可用于匹配目的。

3.4 家具检测与分类

算法1展示了家具分类所涉及的步骤。首先，通过形态学操作检测平面图中的墙体，因为墙体在平面图中通常比其他物体更粗。随后，将原始平面图与仅包含墙体的图像相减，以提取平面图中所有的家具。对于整个平面图中的装饰元素，则基于对图像施加的形态学填充操作进行计数，所得的连通区域被视为平面图中的装饰组件。为进一步对这些连通区域进行分类，计算每个连通区域内各连通分量的面积比，并通过考虑每个连通区域内三个最大分量的面积比，为每种家具组件建立唯一的特征签名。在此之前，已针对符号库中的所有12种符号，分别提取了以其三个最大连通分量为基础的面积比特征，并予以存储。最后，将平面图中各组件所获得的特征签名与符号库中相应组件的特征签名进行比对，从而为平面图中的每一项装饰元素确定其家具类型。图3.5展示了根据算法1如何为家具集合中的某件家具分配特征签名。

(a) Architectural Floor plan　(b) Results of proposed room layout segmentation technique　(c) Layout graph of (a)

图3.4：分割后房间布局的拓扑图

Algorithm 1 Furniture detection and categorization

Input: Room image without outer walls ($\mathscr{I}_w$), Furniture Template Signatures (F)
Output: Furniture Count (C), Furniture Type Set ($\mathscr{T}$)

C=0, $\mathscr{T}$={}
$\mathscr{B}$=Morphological fill operation($\mathscr{I}_w$)
$C = |\mathscr{B}|$ ▷ $|\cdot|$: *Cardinality*
for $j = 1$ to C **do**
  $C = CC(\mathscr{B}_j)$ ▷ CC()=Connected Component
  for $k = 1$ to $|C|$ **do**
    $\mathscr{A}_k = Area(c_k)$, where $c_k \in C$
  end for
  $A = Sort_{desc}(\mathscr{A})$
  $S(\mathscr{B}_j) = \{(A_1/A_3), (A_2/A_3), 1)\}$ ▷ $S(\cdot)$: Signature
  for $k = 1$ to $|F|$ **do**
    if $S(\mathscr{B}_j) == F_k$ **then**
      $T(\mathscr{B}_j) = T(F_k)$ ▷ $T(\cdot)$: Type of Furniture
    end if
  end for
  $\mathscr{T} = \{\mathscr{T} \cup (T(\mathscr{B}_j)\}$
end for

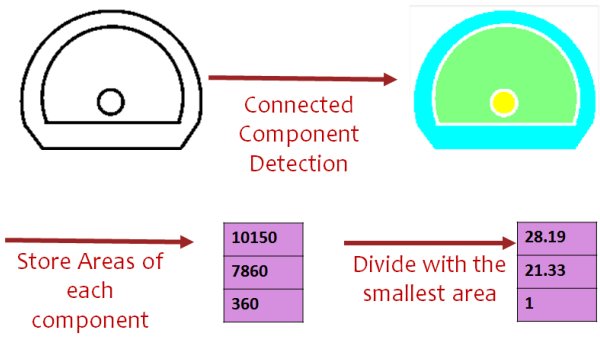


Figure 3.5. : Example showing how a sink symbol in the furniture set is assigned a unique signature.

Figure 3.6(b) depicts an example of the semantic layout graph, consisting of both room adjacencies as well as, room decor representation, corresponding to a layout shown in Fig. 3.6(a) from the SESYD [Delalandre *et al.*, 2010a] dataset. The uncoloured nodes labelled with numbers correspond to the number of rooms present in the layout. The adjacencies between these rooms are depicted by dashed edges. The coloured nodes in the graph correspond to the furniture inside each room. A 360° colour wheel is quantized into uniform bins of value 30° each. According to the orientation of the furniture inside the rooms, a corresponding colour value from the colour wheel is assigned to the furnitures. After obtaining the graph depicting the adjacencies between the rooms and also the decor components, the next task is graph spectral embedding that helps in mapping all graphs to pattern space for retrieval purposes.

Algorithm 1 Furniture detection and categorization

Input: Room image without outer walls ($\mathscr{I}_w$), Furniture Template Signatures (F)
Output: Furniture Count (C), Furniture Type Set ($\mathscr{T}$)

C=0, $\mathscr{T}$={}
$\mathscr{B}$=Morphological fill operation($\mathscr{I}_w$)
$C = |\mathscr{B}|$ ▷ $|\cdot|$: *Cardinality*
for $j = 1$ to C **do**
  $C = CC(\mathscr{B}_j)$ ▷ CC()=Connected Component
  for $k = 1$ to $|C|$ **do**
    $\mathscr{A}_k = Area(c_k)$, where $c_k \in C$
  end for
  $A = Sort_{desc}(\mathscr{A})$
  $S(\mathscr{B}_j) = \{(A_1/A_3), (A_2/A_3), 1)\}$ ▷ $S(\cdot)$: Signature
  for $k = 1$ to $|F|$ **do**
    if $S(\mathscr{B}_j) == F_k$ **then**
      $T(\mathscr{B}_j) = T(F_k)$ ▷ $T(\cdot)$: Type of Furniture
    end if
  end for
  $\mathscr{T} = \{\mathscr{T} \cup (T(\mathscr{B}_j)\}$
end for

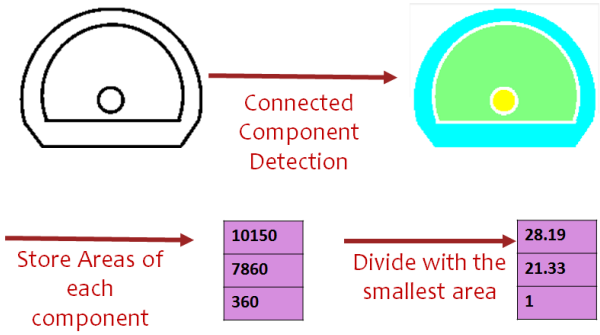
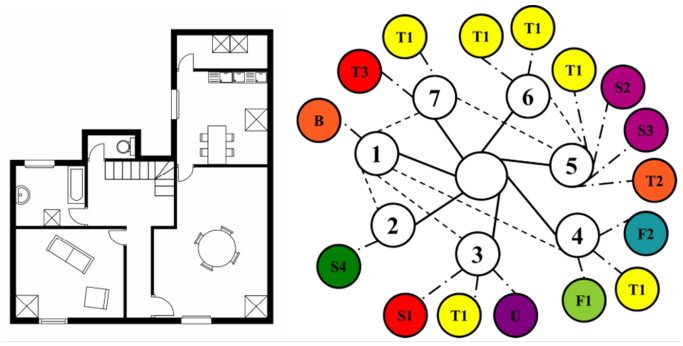


图3.5：示例展示了家具集中如何为一个水槽符号分配唯一标识符。

图3.6(b)展示了一个语义布局图的示例，该图既包含了房间之间的邻接关系，也包含了房间的装饰信息，对应于图3.6(a)中所示的布局，该布局来自SESYD [Delalandre *et al.*, 2010a] 数据集。未着色的节点以数字标注，表示布局中包含的房间数量；这些房间之间的邻接关系则由虚线边表示。图中的彩色节点代表各房间内的家具。一个360°颜色环被等分为每个区间为30°的均匀子区间。根据房间内家具的朝向，从该颜色环中为每件家具分配相应的颜色值。在获得同时刻画房间邻接关系及装饰组件的图结构后，下一步是进行图谱嵌入，从而将所有图映射至模式空间，以支持检索任务。



(a) Architectural Floor plan (b) Semantic graph of (a)

Figure 3.6. : Room layout and corresponding graph consisting of both room adjacencies as well as room decor representation

3.5 GRAPH SPECTRAL EMBEDDING

Graph Spectral Embedding [Luo *et al.*, 2003] approach is adopted to represent the room layout graphs created in the previous stage in a pattern-space. Such a spectral embedding helps to map similar floor plans closer in the pattern space and thus while retrieval all similar plans are clustered together for efficient matching. Using this technique one can map graphs containing a number of nodes to a feature vector of uniform size for all the floor plans in the database. To overcome the problem of how to map the structure of a graph onto a vector of fixed length, the graph-spectral decomposition methods are explored. Leading Eigenvectors of the graph adjacency matrix are used to define Eigenmodes of the adjacency matrix. Thereafter, the spectral decomposition of the adjacency matrix is performed. The use of graph spectrum as an indexing method is beneficial as (1) it captures local topology, (2) it is invariant to subgraph re-order, and (3) it is stable, since small changes in the graph produce little changes in its spectrum. The graph obtained in the previous step is further represented as an adjacency matrix for spectral decomposition as shown in the next subsection.

3.5.1 Spectral Feature representation

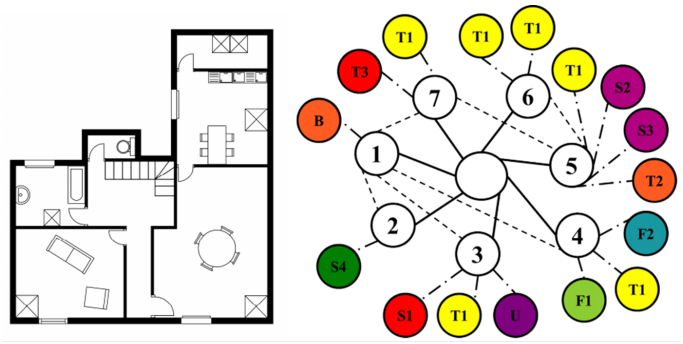
Given N images in the database, their graphs are represented as $G_1, G_2 ..., G_N$. The k^{th} graph is denoted as $G_k = (V_k, E_k)$, where V_k is the set of vertices and $E_k \in V_k \times V_k$ is the edge-set. For each graph G_k, an adjacency matrix A_k of size $|V_k| \times |V_k|$ is computed, where $|\cdot|$ denotes the size of the set.

From the calculated adjacency matrices $A_k, k = 1 \ldots N$, where, N is the total number of graphs for the images in the database, the Eigenvalues λ_k are computed by solving the equation

$$|A_k - \lambda_k I| = 0 \tag{3.3}$$

and the associated Eigenvectors $\vec{\phi}_k$ by solving the system of equations,

$$A_k \phi_k^i = \lambda_k^i \phi_k^i \tag{3.4}$$



(a) Architectural Floor plan (b) Semantic graph of (a)

图3.6：房间布局及其对应的图，该图同时包含房间的邻接关系和房间装饰的表示。

3.5 图谱嵌入

采用图谱嵌入方法[罗等，2003]，将前一阶段构建的房间布局图在模式空间中进行表示。这种谱嵌入有助于使相似的平面布局在模式空间中彼此靠近，从而在检索时将所有相似的平面布局聚类在一起，实现高效的匹配。借助该技术，可以将包含多个节点的图映射为具有统一维度的特征向量，适用于数据库中的所有平面布局。为解决如何将图的结构映射为固定长度向量的问题，本文探讨了图谱分解方法。利用图邻接矩阵的主导特征向量来定义邻接矩阵的本征模式，并在此基础上对邻接矩阵进行谱分解。将图谱用作索引方法具有以下优势：（1）能够刻画局部拓扑结构；（2）对子图的节点顺序变换具有不变性；（3）具有良好的稳定性，因为图的微小变化仅引起其谱的微小变化。在前一步骤中得到的图将进一步表示为邻接矩阵，以供下一小节所述的谱分解使用。

3.5.1 光谱特征表示

给定数据库中的N个图像，其对应的图表示为$G_1, G_2 ..., G_N$。该图记为k^{th}，其中V_k为顶点集，$E_k \in V_k \times V_k$为边集。对于每个图G_k，计算一个大小为$|V_k| \times |V_k|$的邻接矩阵A_k，其中$|\cdot|$表示该集合的大小。

从计算得到的邻接矩阵$A_k, k =$至$...N$中，其中N为数据库中图像的总图数，特征值λ_k通过求解该方程获得。

$$|A_k - \lambda_k I| = 0 \tag{3.3}$$

并通过求解该方程组得到相应的特征向量$\vec{\phi}_k$,

$$A_k \phi_k^i = \lambda_k^i \phi_k^i \tag{3.4}$$

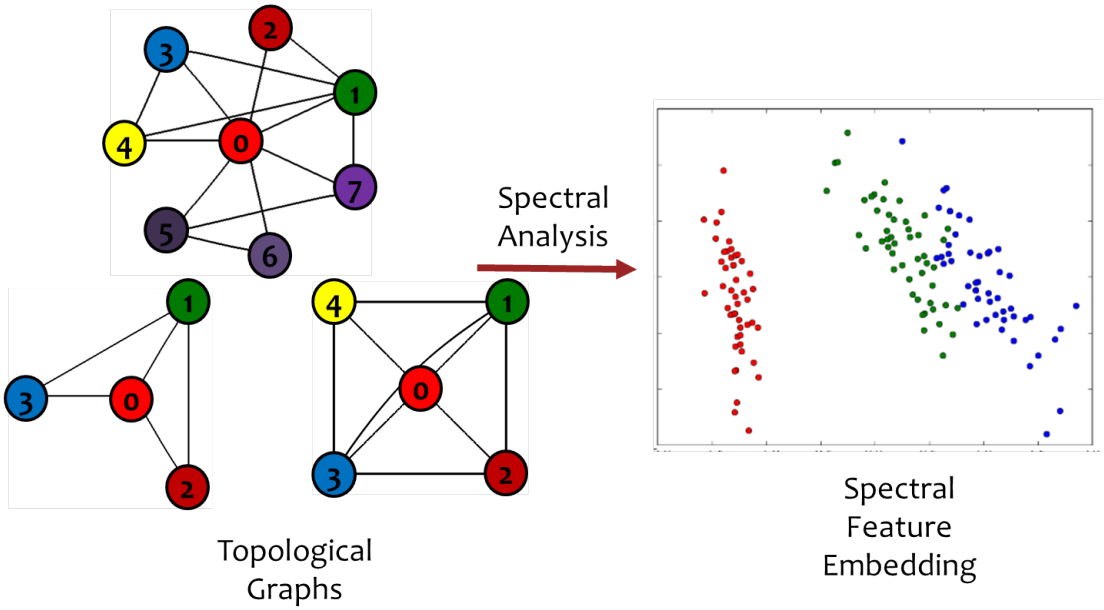


Figure 3.7. : Embedding features from the topological graphs into spectral space. Adjacency matrices with varying sizes are reduced to fixed length vectors in the pattern space. Kindly note, in the right hand side, similar vectors are clustered closely in the feature space.

, where, i is the Eigenmode index.

Spectral feature vector representing the spectrum of the graph G_k is constructed from the top n Eigenvalues of A_k taken in decreasing order. For the k^{th} graph, this vector is represented as :

$$\vec{F}_k = (\lambda_k^1, \lambda_k^2 .... \lambda_k^n)^T. \tag{3.5}$$

Figure 3.8 (a) represents a floor plan image and its corresponding segmentation into rooms. Figure 3.8 (b) represents the topological graph for this segmented layout depicting adjacencies and relationships between the rooms. Node colours correspond to the rooms in the floor plan. Figure 3.8 (c) is an adjacency matrix representation of this topological graph. This adjacency matrix is spectrally analysed and broken down into it's leading Eigen values arranged in a decreasing order as shown in Fig. 3.8 (d). This further forms the feature vector for each layout.

3.5.2 Spectral feature embedding

Once a feature vector is obtained as mentioned in Sec. 3.5.1 the next task is to analyse this feature vector and reduce it into a uniform size vector representing the layout information from each floor plan (refer Fig. 3.7, where due to feature extraction all graphs have been projected on a space of uniform dimension). Uniformity in the feature size is needed as floor plans in the dataset have different number of rooms leading to different sizes of adjacency matrices. Hence, to compare all the adjacency matrices for similarity there was a need to reduce them to a same size feature vector. To achieve this, principal components analysis is performed, following the parametric Eigenspace idea of [Murase and Nayar, 1994]. Rationale behind using this approach is to organise graphs into a pattern-space in which similar structures are close to one-another, and dissimilar structures are far apart. The graphs extracted from each image are vectorised in the way outlined in the Sec. 3.5.1. Principal components analysis (PCA) of the feature vector $\vec{F}_k$ is performed next. For the N

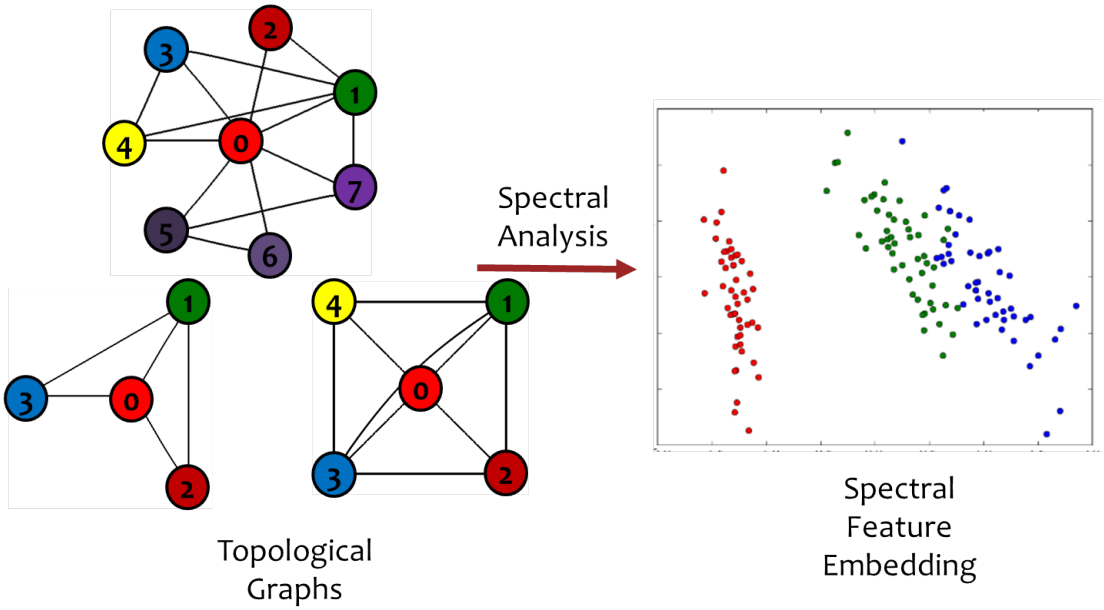


图3.7：将拓扑图的特征嵌入到谱空间中。不同尺寸的邻接矩阵被降维为模式空间中的固定长度向量。请注意，在右侧，相似的向量在特征空间中聚集成簇。

，其中，i 为特征模态序号。

用于表示图G_k谱的谱特征向量，由按降序排列的前n个特征值构成。对于图k^{th}，该向量可表示为：

$$\vec{F}_k = (\lambda_k^1, \lambda_k^2 .... \lambda_k^n)^T. \tag{3.5}$$

图3.8（a）展示了某楼层的平面布局及其对应的房间分割结果。图3.8（b）给出了该分割后布局的拓扑图，其中刻画了各房间之间的邻接关系与相互联系；节点颜色与平面布局中的房间一一对应。图3.8（c）为该拓扑图的邻接矩阵表示。对该邻接矩阵进行谱分析，并按降序排列其主导特征值，如图3.8（d）所示。由此进一步构建出每个布局的特征向量。

3.5.2 光谱特征嵌入

如第3.5.1节所述，一旦获得特征向量，下一步便是对该特征向量进行分析，并将其降维为一个固定尺寸的向量，以表征各楼层平面图的布局信息（参见图3.7；由于进行了特征提取，所有图均被投影到一个维度统一的空间中）。之所以需要使特征尺寸统一，是因为数据集中各楼层平面图的房间数量不同，从而导致邻接矩阵的大小各异。因此，为了比较所有邻接矩阵的相似性，有必要将其统一降维为相同尺寸的特征向量。为此，我们采用了主成分分析方法，并沿用了[Murase和Nayar，1994]提出的参数化特征空间思想。采用该方法的依据在于：将图结构组织到一个模式空间中，使得相似的结构彼此靠近，而不相似的结构则相距较远。从每幅图像中提取的图均按照第3.5.1节所述的方式进行向量化表示。接下来，对特征向量$\vec{F}_k$进行主成分分析（PCA）。对于N

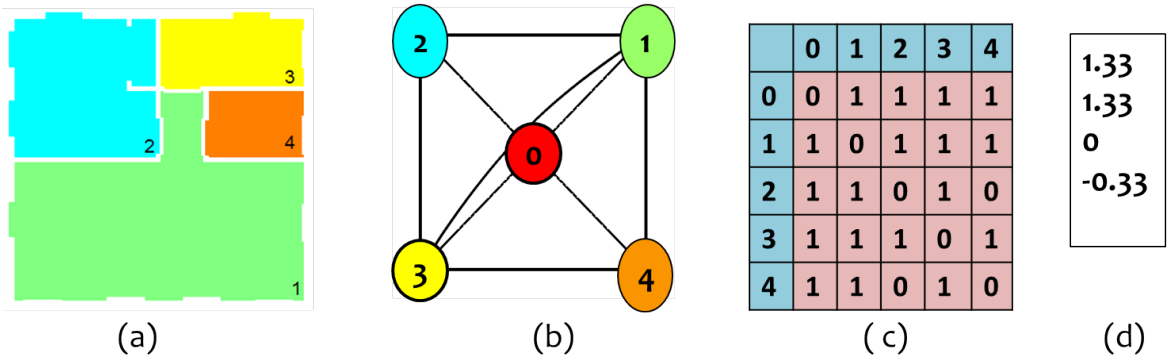


Figure 3.8. : Topological feature extraction from a room layout in a floor plan

different graph representations of the layouts in the database their spectra are arranged in a matrix as:

$$R = [\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_N]. \tag{3.6}$$

The Covariance matrix, Cv for R is computed as,

$$Cv = RR^T. \tag{3.7}$$

A spectral decomposition of Cv results in Eigenvalues ρ upon solving the Eigenvalue equation

$$|Cv - \rho I| = 0 \tag{3.8}$$

and the corresponding Eigenvectors $\vec{\psi}$ are found by solving the Eigenvector equation.

$$Cv\vec{\psi}_i = \rho_i \vec{\psi}_i. \tag{3.9}$$

The Principal components directions are obtained by using the first three leading Eigenvectors of C. The three leading Eigenvectors give a wholesome estimation of the adjacencies in the layout [Luo *et al.*, 2003]. Three orthogonal vectors span the co-ordinate system of the Eigenspace as $\Phi = (\vec{\psi}_1, \vec{\psi}_2, \vec{\psi}_3)$. This aids in projecting the individual graphs represented by the vectors $\vec{F}_k; k = 1, 2, \ldots, N$ on the pattern space as

$$\vec{x}_k = \Phi^T \vec{F}_k. \tag{3.10}$$

Hence, each graph obtained from the layouts is represented as a three-component vector $\vec{x}_k$ in the Eigenspace as $\vec{x}_k = (x'_k, x''_k, x'''_k)^T$. After the computation of the features, in the next section, the matching technique is described.

3.5.3 Feature Matching and Retrieval

A two phase matching and retrieval, Room Layout Matching (RLM) and Room Decor Matching (RDM) is proposed. Feature matching and retrieval of similar layouts (RLM) is performed by determining the proximities between all the layouts. The distance (d) between query graph's feature vector $\vec{x}_q = (x'_q, x''_q, x'''_q)^T$ and that of the model graph's feature vector $\vec{x}_m = (x'_m, x''_m, x'''_m)^T$ is calculated as:

$$d = \sqrt{(x'_q - x'_m)^2 + (x''_q - x''_m)^2 + (x'''_q - x'''_m)^2} \tag{3.11}$$

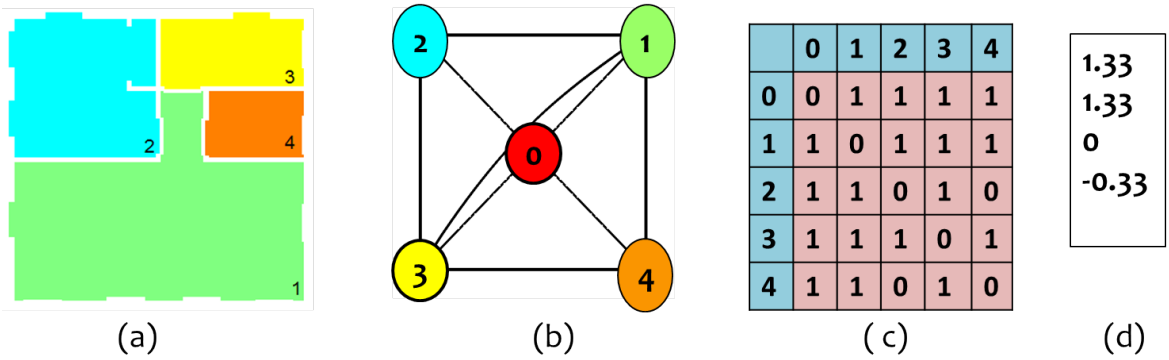


图3.8：从楼层平面图中提取房间布局的拓扑特征

数据库中布局的不同图表示形式，其谱被排列成一个矩阵，如下所示：

$$R = [\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_N]. \tag{3.6}$$

协方差矩阵 Cv 对于 R 的计算公式为：

$$Cv = RR^T. \tag{3.7}$$

对Cv进行谱分解，在求解特征值方程后得到特征值ρ。

$$|Cv - \rho I| = 0 \tag{3.8}$$

相应的特征向量$\vec{\psi}$可通过求解特征向量方程得到。

$$Cv\vec{\psi}_i = \rho_i \vec{\psi}_i. \tag{3.9}$$

主成分方向通过使用C的前三个主导特征向量获得。这三个主导特征向量能够对布局中的邻接关系给出较为全面的估计[Luo *et al.*, 2003]。三个两两正交的向量构成了特征空间的坐标系，记为$\Phi = (\vec{\psi}_1, \vec{\psi}_2, \vec{\psi}_3)$。这有助于将由向量$\vec{F}_k$、$k$ =、1,、2,...,N所表示的各个图投影到模式空间中，即

$$\vec{x}_k = \Phi^T \vec{F}_k. \tag{3.10}$$

因此，从布局中获得的每个图在特征空间中均表示为一个三元向量$\vec{x}_k$，即$\vec{x}_k = (x'_k, x''_k, x'''_k)^T$。在完成特征计算后，下一节将介绍匹配技术。

3.5.3 特征匹配与检索

提出了一种两阶段的匹配与检索方法，包括房间布局匹配（RLM）和房间装饰匹配（RDM）。通过计算所有布局之间的相似度来实现相似布局的特征匹配与检索（RLM）。查询图的特征向量$\vec{x}_q = (x'_q, x''_q, x'''_q)^T$与模型图的特征向量$\vec{x}_m = (x'_m, x''_m, x'''_m)^T$之间的距离d按如下公式计算：

$$d = \sqrt{(x'_q - x'_m)^2 + (x''_q - x''_m)^2 + (x'''_q - x'''_m)^2} \tag{3.11}$$

.

The smaller the distance more similar the graphs are. Thus, this similarity measure helps us to order the architectural layouts represented as graphs. Rank ordering of the images helps in generating the most similar images to the query input and can efficiently retrieve exactly matched or approximately similar layouts from the database to suit the application.

After the first level matching of the layouts the next level match to find the semantic difference between the layouts (RDM) is done. An algorithm to compare the number and type of the objects inside the room (see Alg.2) is proposed, where $\mathscr{T}$ represents type of furniture, $\mathscr{C}^u$ represents number of unique furniture in the whole floor plan. After determining the closest layout, the unique furnitures present in each room are identified. To compute the semantic difference, between a pair of layouts, the number and type of the objects present inside the rooms of the query layout is compared with the model layout. Further a matching cost ($cost_{qm}$) is assigned which increments if there is an object level matching between the layouts and decrements otherwise. The final $cost_{qm}$ obtained from Alg.2 is used to rank order retrieved results based on semantic difference. A flowchart for Room Decor Matching is also provided in Fig. 3.9 to depict the various steps involved in the RDM framework.

Algorithm 2 Calculate semantic difference between layouts

Input: Query Image ($\mathscr{I}_q$), Model Image ($\mathscr{I}_m$)
Output: Matching Cost ($cost_{qm}$)

$\mathscr{T}_q = \{\mathscr{T}_q^i\}; 1 \leq i \leq |\mathscr{C}_q^u|$ ▷ $\mathscr{C}^u$: No. of unique items
$\mathscr{T}_m = \{\mathscr{T}_m^j\}; 1 \leq j \leq |\mathscr{C}_m^u|$ ▷ $\mathscr{T}$: type of furniture
$cost_{qm} = 0$
if $(|\mathscr{C}_q| \neq |\mathscr{C}_m|)$ **then** ▷ $\mathscr{C}$: No. of items
    $cost_{qm} = -1$
else
    for all i, j **do**
        if $\mathscr{T}_q^i = \mathscr{T}_m^j$ **then**
            $cost_{qm} \leftarrow cost_{qm} + 1$
        else
            $cost_{qm} \leftarrow cost_{qm} - 1$
        end if
    end for
end if
    return $cost_{qm}$

3.6 EXPERIMENTS AND RESULTS

The experiments are performed on the SESYD dataset [Delalandre *et al.*, 2010a] and the ROBIN dataset [Sharma *et al.*, 2017]. The SESYD dataset has 10 different types of layouts, each with 100 images varying in their arrangement of furnitures inside the rooms. The image sizes in the database vary from 6775×2858 to 2056×1837. On the other hand, the ROBIN dataset has about 510 floor plans each divided into 3 broad categories based on the number of rooms present inside them and varying in the global structure. All floor plans are binarized to ensure that only structural information of the floor plans is used for the analysis.

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距离越小，图的相似度越高。因此，该相似性度量有助于对以图表示的建筑布局进行排序。通过对图像进行等级排序，可以生成与查询输入最为相似的图像，并从数据库中高效检索出完全匹配或近似相似的布局，以满足应用需求。

在对布局进行第一层匹配之后，接下来进行第二层匹配，以识别两个布局之间的语义差异（RDM）。本文提出了一种用于比较房间内对象数量和类型的算法（见算法2），其中$\mathscr{T}$表示家具类型，$\mathscr{C}^u$表示整层平面图中不同家具的种类数。确定最接近的布局后，进一步识别每个房间中存在的独特家具。为计算一对布局之间的语义差异，将查询布局中各房间内的对象数量和类型与模型布局进行对比。此外，还引入一个匹配代价$cost_{qm}$：若布局间存在对象级别的匹配，则该代价增加；否则，该代价减少。由算法2得到的最终值$cost_{qm}$用于根据语义差异对检索结果进行排序。图3.9还给出了房间装饰匹配的流程图，以直观展示RDM框架中的各个步骤。

Algorithm 2 Calculate semantic difference between layouts

Input: Query Image ($\mathscr{I}_q$), Model Image ($\mathscr{I}_m$)
Output: Matching Cost ($cost_{qm}$)

$\mathscr{T}_q = \{\mathscr{T}_q^i\}; 1 \le i \le |\mathscr{C}_q^u|$ ▷ $\mathscr{C}^u$: No. of unique items
$\mathscr{T}_m = \{\mathscr{T}_m^j\}; 1 \le j \le |\mathscr{C}_m^u|$ ▷ $\mathscr{T}$: type of furniture
$cost_{qm} = 0$
if $(|\mathscr{C}_q| \neq |\mathscr{C}_m|)$ **then** ▷ $\mathscr{C}$: No. of items
    $cost_{qm} = -1$
else
    for all i, j **do**
        if $\mathscr{T}_q^i = \mathscr{T}_m^j$ **then**
            $cost_{qm} \leftarrow cost_{qm} + 1$
        else
            $cost_{qm} \leftarrow cost_{qm} - 1$
        end if
    end for
end if
    return $cost_{qm}$

3.6 实验与结果

实验在SESYD数据集[Delalandre，2010a]和ROBIN数据集[Sharma，2017]上进行。SESYD数据集包含10种不同类型的布局，每种布局有100张图像，这些图像中房间内的家具布置各不相同。该数据库中图像的尺寸范围为6775×2858至2056×1837。另一方面，ROBIN数据集约有510个平面图，根据其内部房间数量被划分为三大类，且整体结构存在差异。所有平面图均被二值化处理，以确保分析仅基于平面图的结构信息。

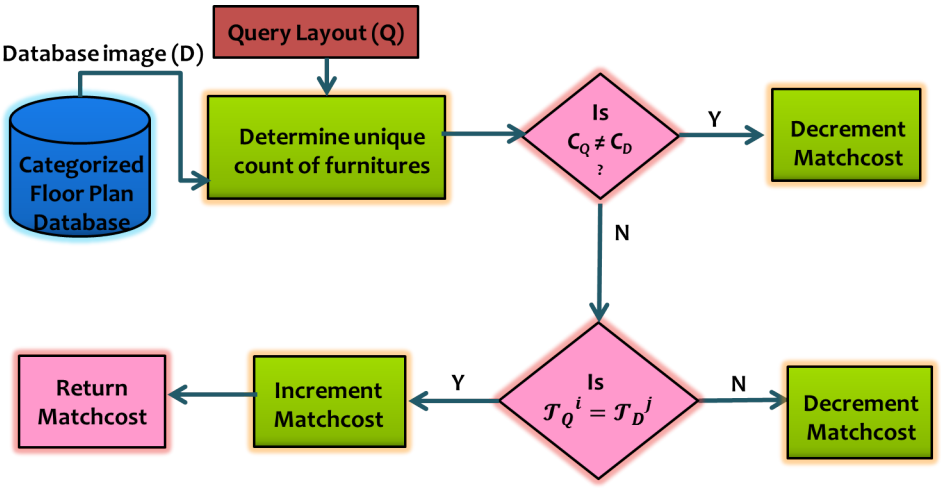


Figure 3.9. : Framework for Room Decor Matching

Figure 3.10. : Top 5 retrieved results corresponding to a query from the SESYD dataset.

3.6.1 Qualitative Results

The framework starts off with segmenting the architectural layouts into rooms followed by, detecting the various furnitures present in each room. A semantic representation of the layout graph (see Sec. 3.3) is obtained, by connecting the furniture nodes with the room node that they belong to using dash-dot edges.

The proposed RLM and RDM techniques, helped in rank ordering of the matched layouts based on the semantic difference i.e. differences in the arrangement and number of furniture inside the rooms. Figure 3.10, shows qualitative retrieval result corresponding to a query from the SESYD dataset. The red bounding boxes depict a change of decor between the query layout and the retrieved layouts, thus leading to dissimilar layouts being retrieved later than the similar ones

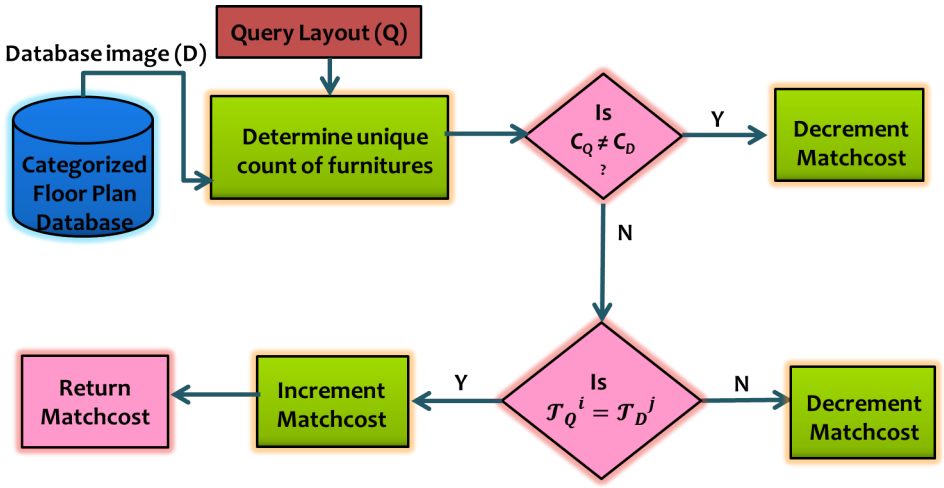


图3.9：房间装饰匹配框架

图3.10：来自SESYD数据集的查询所对应的前5个检索结果。

3.6.1 定性结果

该框架首先将建筑布局划分为各个房间，随后检测每个房间中存在的各类家具。通过使用虚点线将家具节点与其所属的房间节点相连，从而获得布局图的语义表示（参见第3.3节）。

所提出的RLM和RDM技术有助于根据语义差异，即房间内家具的布置方式及数量差异，对匹配的布局进行排序。图3.10展示了来自SESYD数据集的一条查询所对应的定性检索结果。红色边界框标示出查询布局与检索到的布局之间在装饰风格上的差异，因此导致与查询相似度较低的布局在检索结果中排在相似度较高的布局之后。

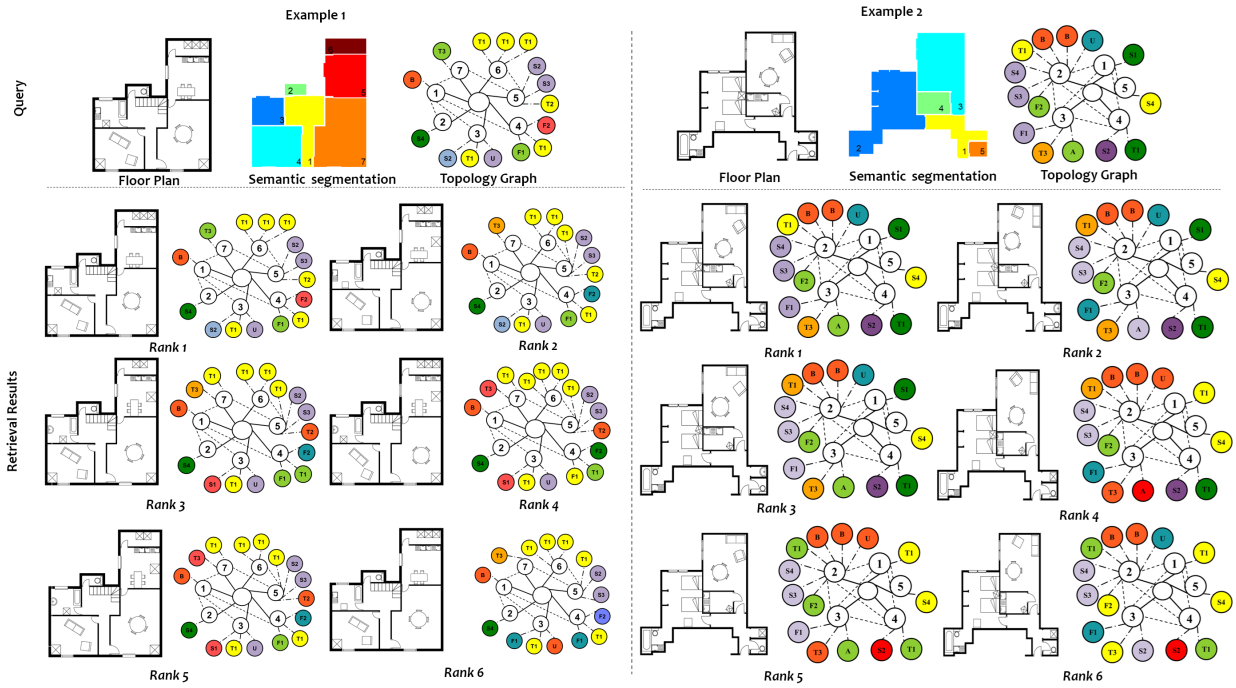


Figure 3.11. : Retrieval results for two different query architectural floor plans taken from the SESYD dataset. The top row shows the query image, the corresponding room layout and the semantic graph generated from the floor plan image. For the given query images only the top six retrieval results are shown.

during matching. As the result is demonstrated on SESYD dataset and it is observed that within a category in SESYD dataset only the decor arrangement differs, and the layout remains the same. Hence, the first rank ordered result is the query layout itself. The subsequent results differ in the placement, position, type and number of the furniture present inside the floor plan. For example, the Rank 5 result differs from the query floor plan in terms of a table's orientation as well as lacks a small table in comparison to the query. Hence, is retrieved later. To elaborate further, the rank ordered retrieval results along with their corresponding semantic graphs for two such queries, are shown in Fig. 3.11. The first row in Fig. 3.11 represents the query image, its room level segmentation, and the corresponding semantic graph. The following rows correspond to the top six rank ordered results along with their semantic graphs. It can be observed that the layout category in the rank ordered results is the same as that of the query and thus, ensures high retrieval accuracy. The first rank ordered result obtained from the layout database is the query image itself due to minimum distance from the query. The subsequent results differ in the type, number and orientation of furnitures inside each room.

After explanation of how the retrieval results differ in their layout/ decor arrangement, further, more qualitative results on both SESYD dataset and ROBIN dataset are shown in Fig. 3.12. In Fig. 3.12, the subfigures 1. (a) and (b) correspond to the query given from SESYD dataset and the subsequent rank-ordered results. While, 2. (a) and (b) correspond to the query given from ROBIN dataset. On closer observation it can be seen that retrieval framework performs well while layout matching and all layouts from the same category are retrieved as can be seen form the same global shapes of the layouts in the query as well as the retrieved results.

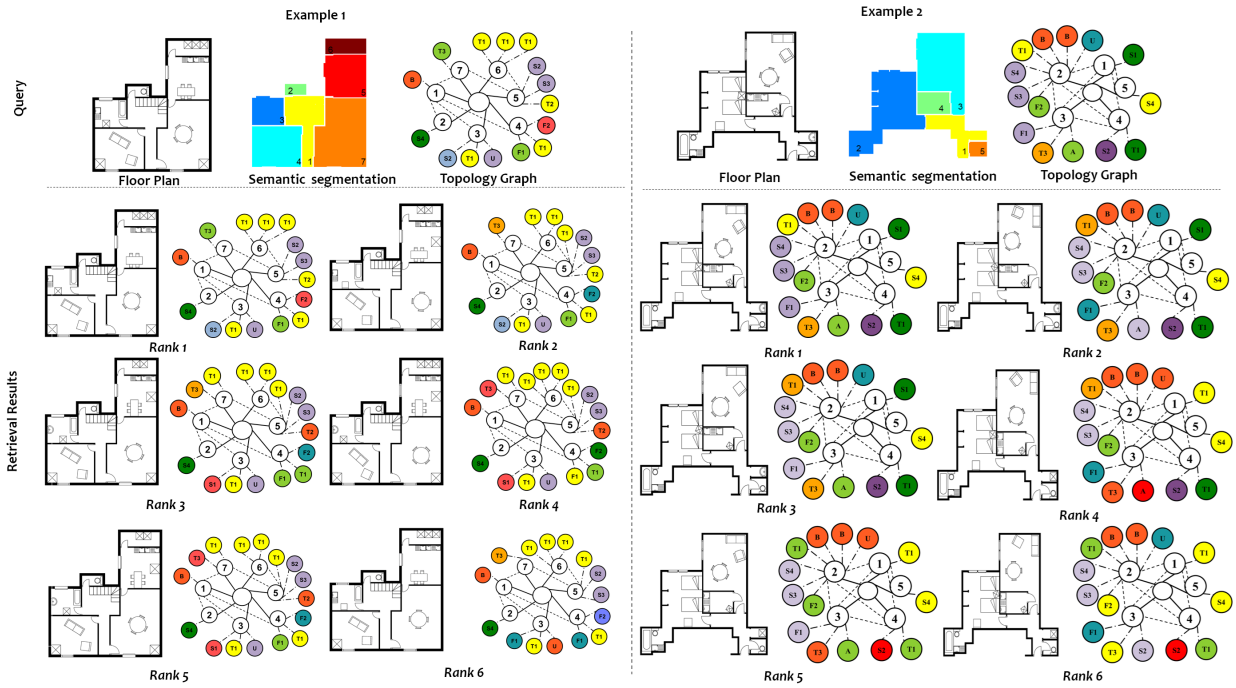


图3.11：基于SESYD数据集的两张不同查询建筑平面图的检索结果。第一行为查询图像、对应的房间布局以及由该平面图生成的语义图。对于给定的查询图像，仅展示前六名检索结果。

在匹配过程中，以SESYD数据集为例进行验证。观察发现，SESYD数据集中同一类别内的样本仅在装饰布置上存在差异，而布局结构保持一致。因此，排序第一的结果即为查询布局本身。后续结果则在平面图中家具的摆放位置、朝向、类型及数量等方面有所不同。例如，排名第五的结果在餐桌的朝向上与查询平面图存在差异，并且缺少一张小桌，因而被排在较后的位置。为进一步说明，图3.11展示了两组此类查询的排序检索结果及其对应的语义图。图3.11的第一行分别展示了查询图像、其房间级分割结果以及相应的语义图；随后的各行为前六名排序结果及其语义图。可以看出，排序结果中的布局类别与查询一致，从而保证了较高的检索精度。从布局数据库中获得的排序第一的结果即为查询图像自身，这是由于其与查询之间的距离最小。后续结果则在各房间内家具的类型、数量及朝向上有所差异。

在阐述了检索结果在布局与装饰配置上的差异之后，图3.12进一步展示了基于SESYD数据集和ROBIN数据集的更多定性分析结果。在图3.12中，子图1.(a)和(b)对应于来自SESYD数据集的查询及其后续的排序结果；而子图2.(a)和(b)则对应于来自ROBIN数据集的查询。仔细观察可以发现，该检索框架在布局匹配方面表现良好，且同一类别的所有布局均被检索出来，这从查询与检索结果在整体形态上的一致性即可看出。

Figure 3.12. : Retrieval results for two different query architectural floor plans taken from the SESYD dataset and the ROBIN dataset. 1. (a) and (b) correspond to the query given from SESYD dataset and the subsequent rank-ordered results. 2. (a) and (b) correspond to the query given from ROBIN dataset

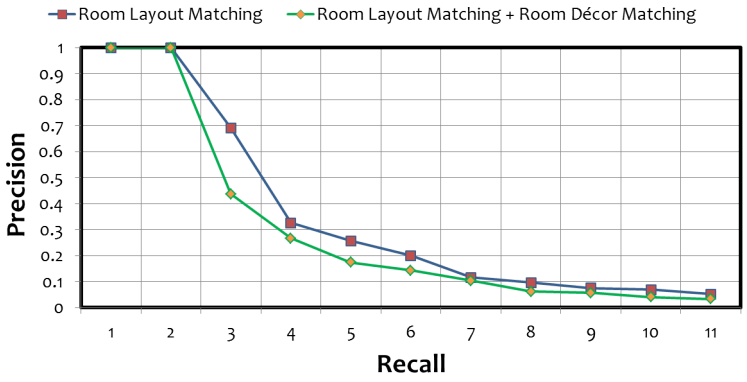


Figure 3.13. : Quantitative result comparing the RLM and the RDM approach on the ROBIN dataset

图3.12：分别基于SESYD数据集和ROBIN数据集中两张不同的查询建筑平面图所得到的检索结果。1.（a）和（b）对应于SESYD数据集中的查询及其后续的排序结果。2.（a）和（b）对应于ROBIN数据集中的查询。

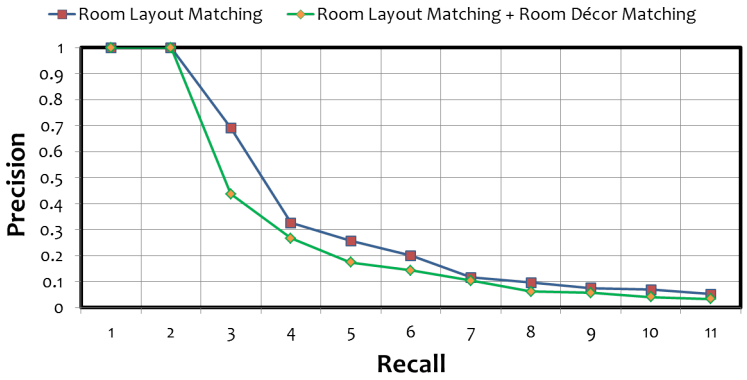


图3.13：定量　　比较RLM与RDM方法的对比结果　　在 ROBIN 数据集上 h

3.6.2 Quantitative Results

Quantitative analysis of the proposed algorithm was also done to observe the results. Precision (P) and Recall (R) are calculated using the equations mentioned below:

$$Precision(P) = \frac{|\{\text{relevant floor plan images}\} \cap \{\text{retrieved floor plan images}\}|}{|\{\text{retrieved floor plan images}\}|} \tag{3.12}$$

$$Recall(R) = \frac{|\{\text{relevant floor plan images}\} \cap \{\text{retrieved floor plan images}\}|}{|\{\text{relevant floor plan images}\}|} \tag{3.13}$$

Figure 3.13 shows the PR plot obtained by taking query from the ROBIN dataset and retrieving based on layout matching and a combination of layout and decor matching separately. In Fig. 3.13, the blue line marked with red squares denotes the PR plot obtained through layout matching (RLM) and the green line with yellow diamond markers depicts PR plot obtained with combination of both RLM and RDM. This study helped in observing how well the algorithm behaves in case of both matching strategies taken exclusively. The initial Recall value was obtained as 1 due to the first rank ordered retrieved result to be the same as the query. The subsequent results had some differences in the layout and decor. It was observed that taking the global layout, i.e. the placement of rooms inside a floor plan was a broader way of classifying the layouts, hence, the area under the PR plot with layout matching was obtained to be greater. This also owes to the fact that as more detailed decor characterization and matching is looked into, it leads to a slightly poor categorization of the layouts because of the specificity while retrieval.

Similarly, quantitative analysis of retrieval performance on the SESYD dataset was also performed. As discussed before, SESYD dataset has large differences between the layouts across its 10 categories but inside a category, all the layout shapes are the same, only differing in their decor arrangement. The values of the distance metric depicting the similarity between layouts from SESYD dataset are shown in Figure 3.14. In Fig. 3.14, the x-axis corresponds to the ten different categories of layouts present in the database, whereas, the y-axis corresponds to the distance value between two layout categories in terms of the distance between the feature vectors representing them. The nodes depict the distance of each category from the 10 categories in the database. For example, as shown in Fig. 3.14, yellow line with triangles corresponds to the distances with respect to layout category 3 in the SESYD database. Note that the distance is 0 with respect to category 3 and highest with respect to category 9. This is due to the very large difference in the layout characteristics between category 3 and 9. It is evident from the plot that our proposed RLM technique (see Sec. 3.5.3) yields high intra-category similarity and low inter-category similarity between layouts, as expected. To illustrate further, one can see in Fig. 3.15 for layout 8 and 4 that the first rank retrieved result is layout 8 itself due to exact matching between the query and the Rank 1 result. Further, Rank 2 result has similar number of rooms and adjacency information as the query. Other retrieved results follow suit. Due to high inter-class dissimilarity between floor plans in the SESYD dataset, a perfect Precision value was obtained for all the recall values while retrieval.

3.6.2 定量结果

还对所提出的算法进行了定量分析以观察其结果。精确率（P）和召回率（R）按照如下公式计算:

$$Precision(P) = \frac{|\{\text{relevant floor plan images}\} \cap \{\text{retrieved floor plan images}\}|}{|\{\text{retrieved floor plan images}\}|} \tag{3.12}$$

$$Recall(R) = \frac{|\{\text{relevant floor plan images}\} \cap \{\text{retrieved floor plan images}\}|}{|\{\text{relevant floor plan images}\}|} \tag{3.13}$$

图3.13展示了分别基于布局匹配以及布局与装饰匹配相结合的检索结果所得到的PR曲线。在图3.13中，以红色方块标记的蓝线表示仅通过布局匹配（RLM）获得的PR曲线，而以黄色菱形标记的绿线则表示同时采用RLM和装饰匹配（RDM）时的PR曲线。本研究有助于观察该算法在两种匹配策略单独使用时的表现情况。由于首个检索结果与查询完全一致，初始召回率为1。后续检索结果在布局与装饰方面存在一定差异。研究发现，以整体布局，即平面图中各房间的布置方式作为分类依据，是一种更为宽泛的布局划分方法，因此，仅基于布局匹配的PR曲线下的面积更大。此外，随着装饰特征刻画与匹配的精细化程度提高，在检索过程中由于其特异性，会导致布局分类效果略有下降。

同样地，我们也对SESYD数据集上的检索性能进行了定量分析。如前所述，SESYD数据集的10个类别之间在布局形式上存在显著差异，而在同一类别内部，所有布局的形状均相同，仅在装饰布置上有所区别。图3.14展示了用于表征SESYD数据集中布局间相似性的距离度量值。在该图中，横轴对应数据库中的10个不同布局类别，纵轴则表示以特征向量间距离为度量标准的两类布局之间的距离值。各节点代表某一类别与数据库中其余9个类别之间的距离。例如，如图3.14所示，带有三角形标记的黄色线条对应于SESYD数据库中第3类布局与其他各类别之间的距离；可见，其与第3类的距 离为0，而与第9类的距离最大，这是由于第3类与第9类在布局特征上存在显著差异所致。从图中可以看出，我们提出的RLM技术（参见第3.5.3节）能够实现布局间的高内类相似性和低类间相似性，符合预期。为进一步说明，如图3.15所示，对于布局8和布局4，由于查询与排名第一的检索结果完全匹配，因此第一排名的检索结果即为布局8本身；此外，第二排名的结果在房间数量及邻接关系等方面也与查询高度相似，其余检索结果亦与此相符。鉴于SESYD数据集中各平面布局之间的类间差异较大，在检索过程中针对所有召回率均获得了完美的精确率值。

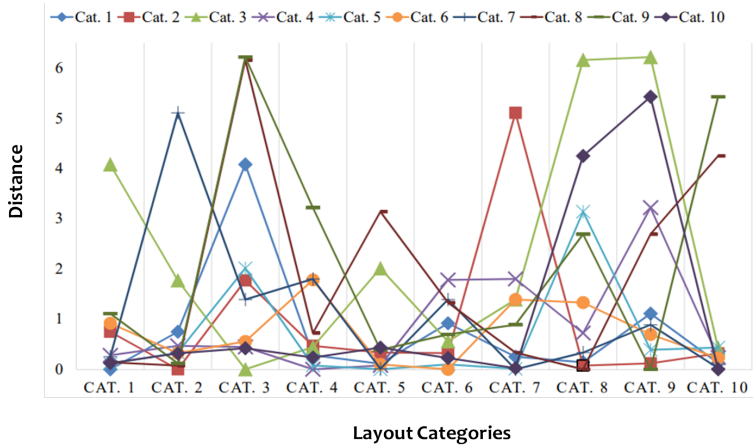


Figure 3.14. : Plot of intra-class and inter-class distances between layouts using our feature matching technique on the SESYD datset.

3.7 DISCUSSION

The framework discussed in this Chapter efficiently performs the various processing steps starting from, wall detection to room segmentation, graph creation of the layouts, feature computation, graph matching between the layouts, object level matching between the layouts and finally rank ordering the results. Another important aspect to be analyzed here is the search time. To compute the search time of the proposed approach time taken in seconds is computed at every step for the SESYD dataset. This time is averaged over 1000 samples belonging to a single class in the SESYD dataset. Figure 3.16 shows the class-wise computation time-ratio for the layouts in the dataset. Colour coding signifies various components of the proposed framework and helps establishing correspondence to understand which step performs computationally efficient in the proposed framework.

It can be observed that the wall detection step took an average time of 4.481 secs. for all the layout categories. Segmentation into rooms was a little computationally expensive step owing to a number of intermediate processing steps. The average time taken by the segmentation step was 18.047 secs. Graph generation and spectral feature embedding of the graphs took an average time of 5.513 secs. and 5.292 secs. respectively. Figure 3.16 shows this analysis, where, values shown in the bar chart denote the ratio of time taken in seconds for each step to the total time taken for execution.

As discussed in Sec. 3.3, graph matching is a computationally expensive process. To avoid this spectral graph embedding is proposed in this Chapter, which creates uniform three component vectors for representing all the graphs, which helps in clustering similar graphs closer in the pattern space. Therefore, this technique considerably reduced the computation time while matching the layouts. The average time taken at this step is 7.569 secs.

Analysis revealed that the furniture level matching inside the layouts was the most computationally expensive step in the proposed approach. This step took the maximum average time of 154.036 secs for every layout because of 1) iteratively searching for the arrangement of furniture inside each layout, and 2) large dimension (6775×2858) of the layout images. An efficient indexing method or hashing technique for the second level match could yield better results.

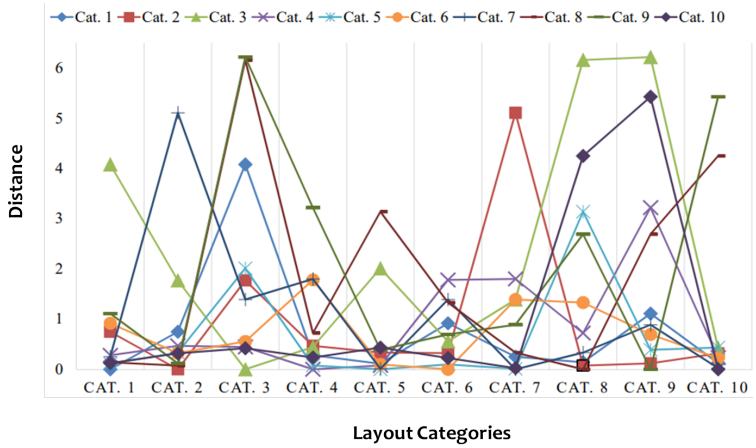


图3.14：基于SESYD数据集，采用我们的特征匹配技术计算得到的布局间类内距离与类间距离的分布图。

3.7 讨论

本章所讨论的框架能够高效地完成从墙体检测、房间分割，到布局图的图构建、特征计算、布局间的图匹配以及对象级匹配等一系列处理步骤，并最终对结果进行排序。另一个需要在此分析的重要方面是搜索时间。为评估所提出方法的搜索耗时，针对SESYD数据集，在每个步骤均以秒为单位记录了运行时间，并对该数据集中属于同一类别的1000个样本的运行时间取平均值。图3.16展示了该数据集中各布局按类别划分的计算时间比。通过颜色编码标识了所提框架的各个组成部分，有助于明确各步骤在计算效率上的表现。

可以观察到，对于所有布局类别，墙体检测步骤的平均耗时为4.481秒。由于涉及多个中间处理步骤，房间分割这一环节的计算开销略高，其平均耗时为18.047秒。图的生成以及图的谱特征嵌入分别平均耗时5.513秒和5.292秒。图3.16展示了该项分析；其中，柱状图中所示的数值表示各步骤所用时间（以秒为单位）占总执行时间的比例。

如第3.3节所述，图匹配是一项计算开销较高的过程。为避免这一问题，本章提出了谱图嵌入方法，该方法为所有图构造统一的三元向量表示，从而使得相似图在模式空间中更易于聚类。因此，该技术在布局匹配过程中显著降低了计算时间，此步骤的平均耗时为7.569秒。

分析表明，所提出方法中，布局内部的家具层级匹配是计算开销最大的步骤。由于该步骤需要：1）对每个布局内的家具排列进行迭代搜索；2）处理尺寸较大的布局图像（6775 × 2858像素），因此每幅布局平均耗时高达154,036秒。若能为第二层级匹配采用高效的索引方法或哈希技术，有望进一步提升性能。

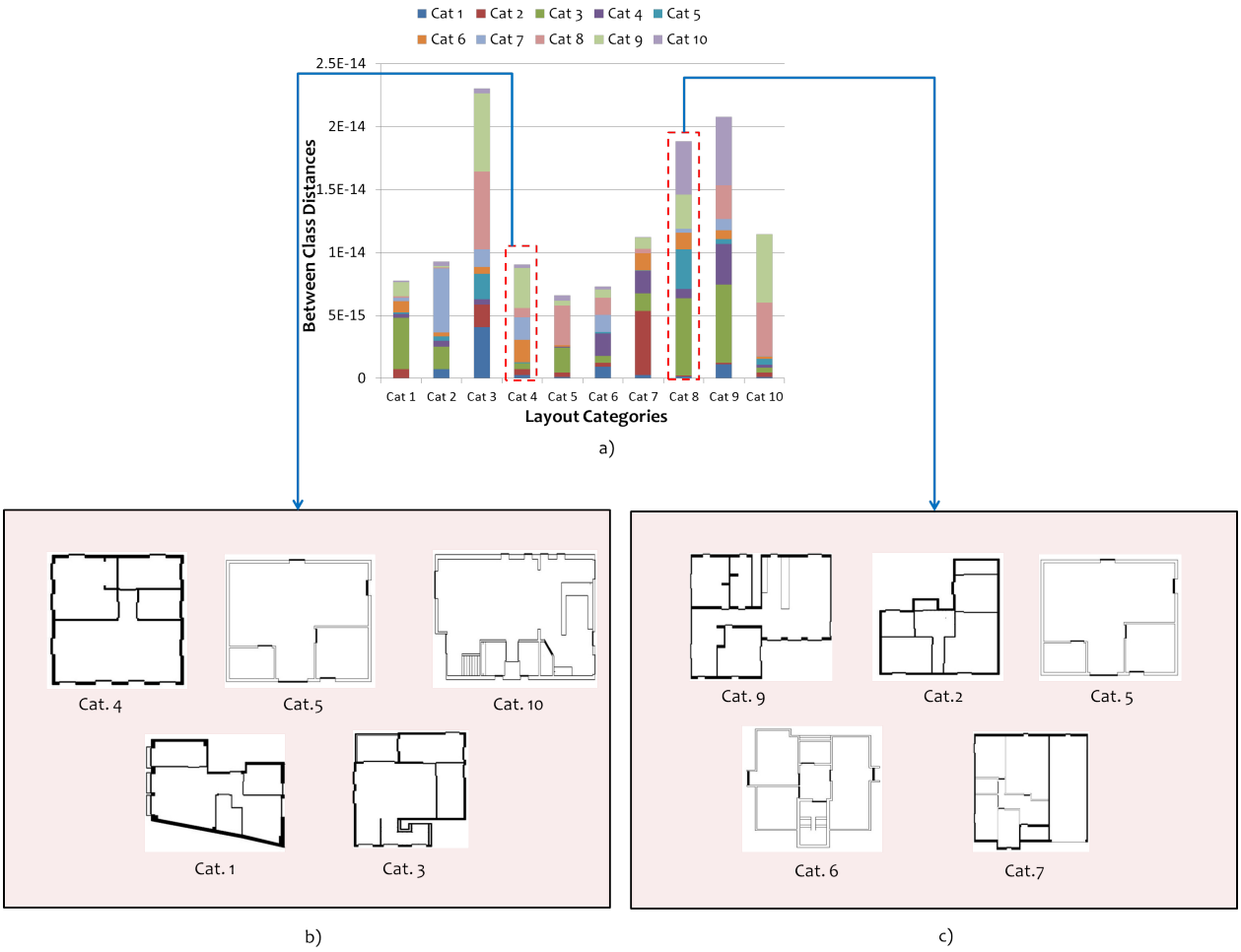


Figure 3.15. : Layout matching from floor plans taken from SESYD dataset. Two such results from Category 8 and 4 are highlighted with a red bounding box.

Similar analysis was also performed for the ROBIN dataset. As the classes in ROBIN dataset are 51 in number, therefore, it is not feasible to show class-wise search time graph for the ROBIN dataset. Alternatively, in Fig. 3.17 the search time taken in seconds averaged over all the 510 samples in the ROBIN dataset, at every step of the framework is represented as a bar-graph. It was observed that the most computationally step was the decor matching step again taking an average time of about 7.34 secs., which results from the fact that, iteratively searching for the arrangement of furniture inside each layout is computationally expensive. On the other hand, the most computationally efficient step is the Automatic Room Detection step.

As this was a novel baseline framework proposed in this Chapter, therefore, due to a lack of availability of the implementation of such a composite framework for retrieval and matching of floor plans in the literature, there is an inability to compare the results with other approaches. Although, in Fig. 3.18 the comparison of segmentation step to the one proposed in [de las Heras *et al.*, 2014] is shown. Where, segmentation results on two images from CVC-FP [de las Heras *et al.*, 2015] dataset are shown, with, the left hand side being (a) the input floor plan and (b) and (c) correspond to the technique proposed in [de las Heras *et al.*, 2014] and the proposed approach respectively. It can be observed that the proposed segmentation technique performs at par with the technique proposed in [de las Heras *et al.*, 2014].

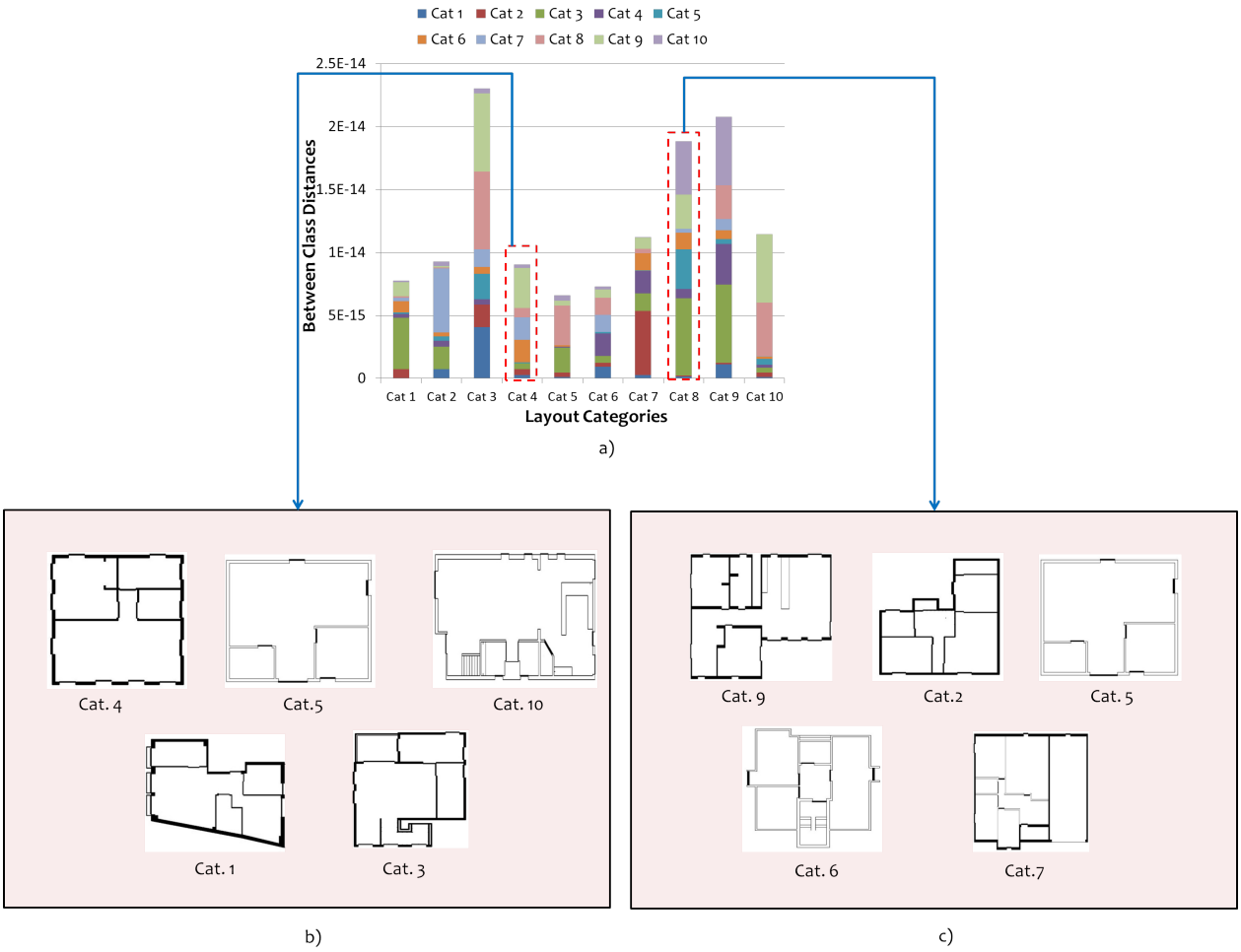


图3.15：基于SESYD数据集中的平面图进行的布局匹配。其中来自第8类和第4类的两个匹配结果以红色边界框突出显示。

类似的分析也针对ROBIN数据集进行了开展。由于ROBIN数据集包含51个类别，因此无法为该数据集绘制按类别划分的搜索时间曲线图。作为替代，在图3.17中，以柱状图的形式展示了框架在每一步骤中所耗费的搜索时间（以秒为单位），该时间是基于ROBIN数据集中全部510个样本的平均值。观察发现，计算开销最大的步骤仍然是装饰匹配步骤，其平均耗时约为7.34秒，这是因为在每个布局内部迭代地搜索家具排列方式需要较高的计算复杂度。另一方面，计算效率最高的步骤则是自动房间检测步骤。

由于本章提出了一种全新的基线框架，且现有文献中尚无用于平面图检索与匹配的此类综合框架的实现，因此无法将其结果与其他方法进行对比。不过，在图3.18中展示了所提分割步骤与[de las Heras，2014]中提出的方案之间的对比。其中，给出了CVC-FP [de las Heras，2015]数据集中的两张图像的分割结果，左侧分别为（a）输入的平面图，以及（b）和（c），分别对应于[de las Heras，2014]中提出的算法与本研究所提方法。可以观察到，所提分割技术的性能与[de las Heras，2014]中提出的算法相当。

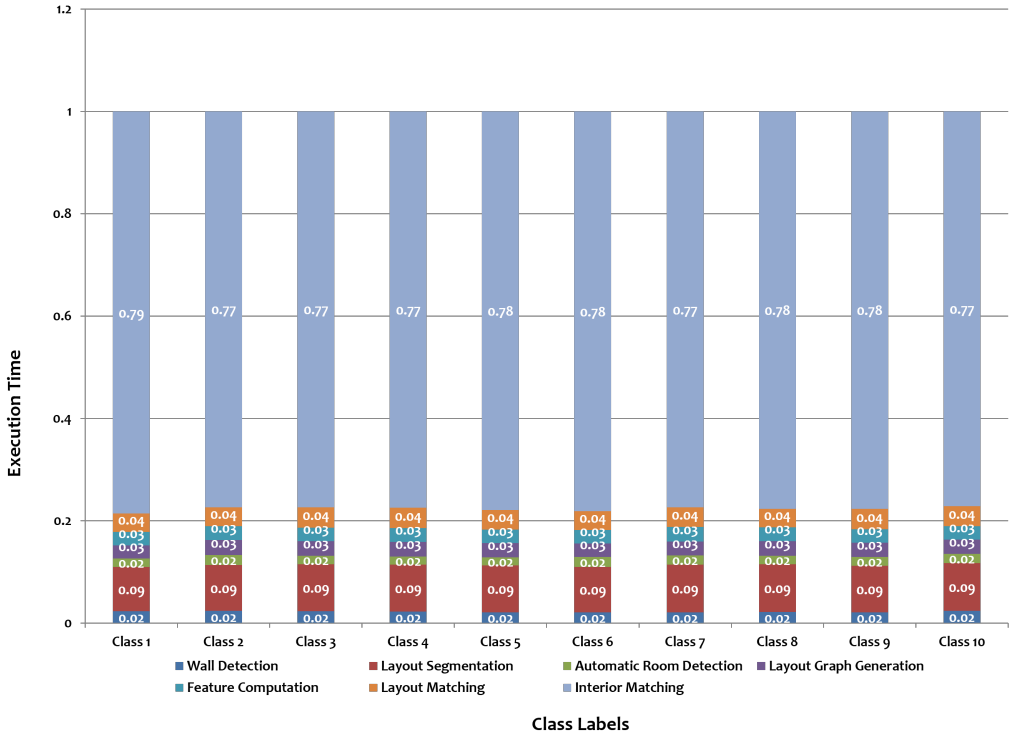


Figure 3.16. : Execution time for the intermediate stages of processing of the framework for different categories of floor plans on the SESYD dataset. Values shown in the bar chart denote the ratio of time taken in seconds for each step to the total time taken for execution.

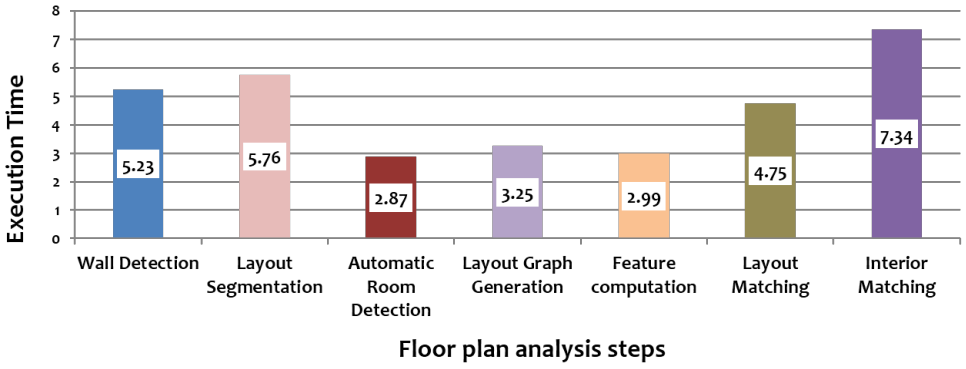


Figure 3.17. : Execution time for the intermediate stages of processing of the framework averaged over all 51 categories of floor plans in the ROBIN dataset. Values shown in the bar chart denote the time taken in seconds.

The approach proposed in [Weber *et al.*, 2010] taking sketch as a query to retrieve floor plans is partly related to the work proposed in this Chapter. However, the key difference is that the authors in [Weber *et al.*, 2010] represent query layouts as sketches using only primitive shapes

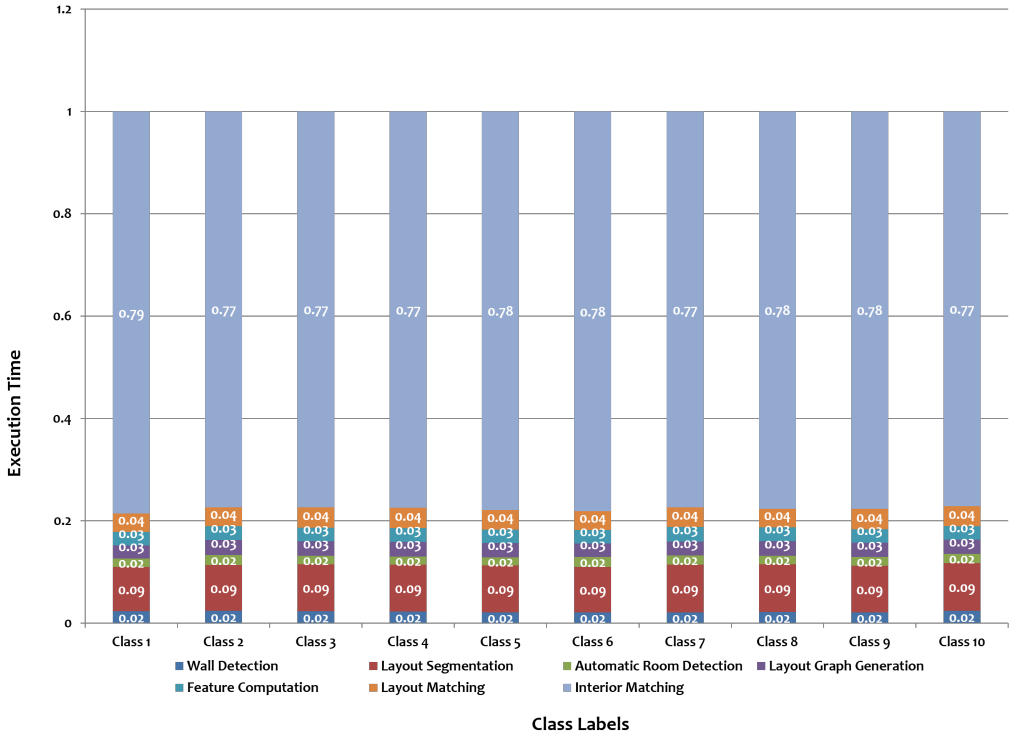


图3.16：在SESYD数据集上，针对不同类别平面图，框架各中间处理阶段的执行时间。柱状图中所示数值为各步骤所用时间（以秒计）占总执行时间的比值。

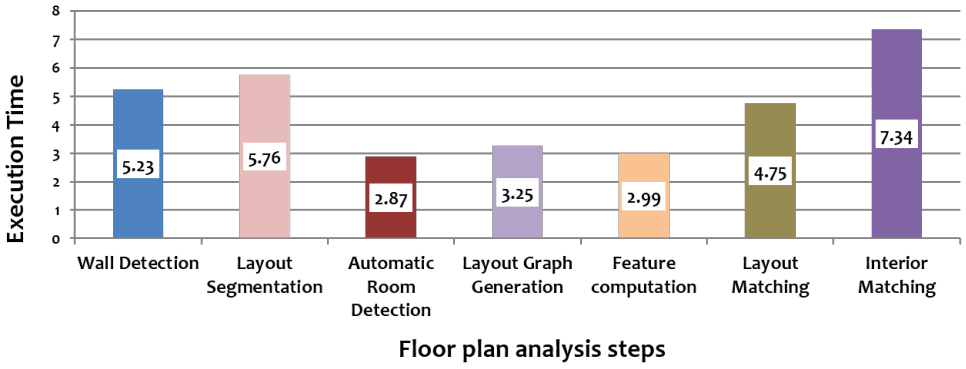


图3.17：在ROBIN数据集中，框架处理各中间阶段的平均执行时间，统计范围涵盖全部51类平面图。条形图中的数值以秒为单位。

文献[Weber v0, 2010]中提出的以草图作为查询来检索平面图的方法与本章所提出的工作在一定程度上相关。然而，关键区别在于，文献[Weber v1, 2010]的作者仅采用基本几何形状来表示查询布局的草图。

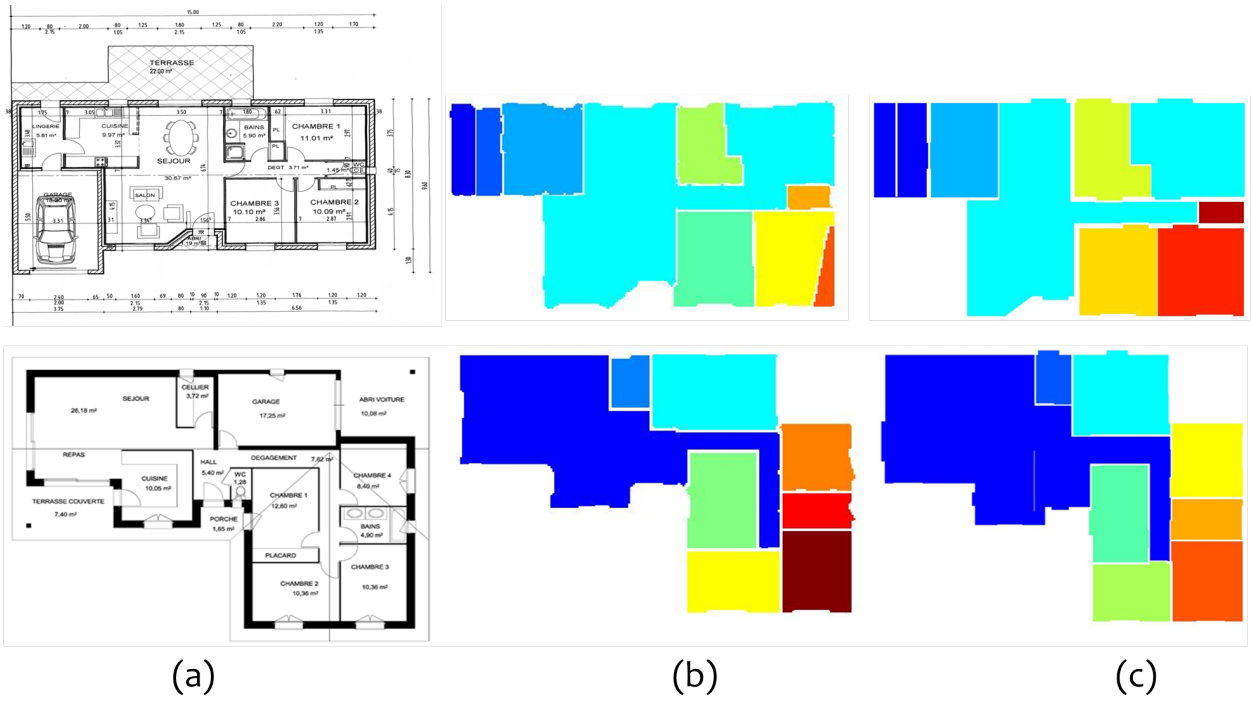


Figure 3.18. : Segmentation results on two images from CVC-FP [de las Heras *et al.*, 2015] dataset (a) Input floor plan, (b) Technique proposed in [de las Heras *et al.*, 2014], (c) Our approach

such as rectangles. In contrast, in this Chapter the room decor represented by complex geometric primitives is considered. Moreover, in [Weber *et al.*, 2010] the adjacencies between the rooms represented in the layouts, in sketch form were very simplistic in nature which is quite complex in our case. Furthermore, they have ignored the representation and matching of the room decor between the layouts.

3.8 SUMMARY

In this Chapter, a framework for searching floor plans by structural analysis, as well as, semantic similarity is proposed. Inclusion of the room decor arrangement during retrieval is a novel idea that can find application in scenarios where the user has specific choices in terms of the fully furnished floor plans. Furthermore, a spectral embedding approach to represent graphs obtained from the layout is proposed in the baseline retrieval framework proposed in this Chapter. This technique helped in reducing the computation time taken during the graph matching stage to a considerable level.

In this Chapter, handcrafted topological features were proposed to be extracted from the floor plans. Topology and room decor features although help in distinguishing two floor plans and retrieving similar ones, but inclusively are not a good enough measure to analyse a floor plan completely. It is very difficult to curate handcrafted features powerful enough to extract all the details from a floor plan. On the other hand, deep neural networks are able to analyse a floor plan in a minute manner extracting all the relevant features at once. In the literature, deep neural network based features have shown promising results for the retrieval of natural images. However their application to floor plan images has not been researched. Therefore, in the next chapter, deep

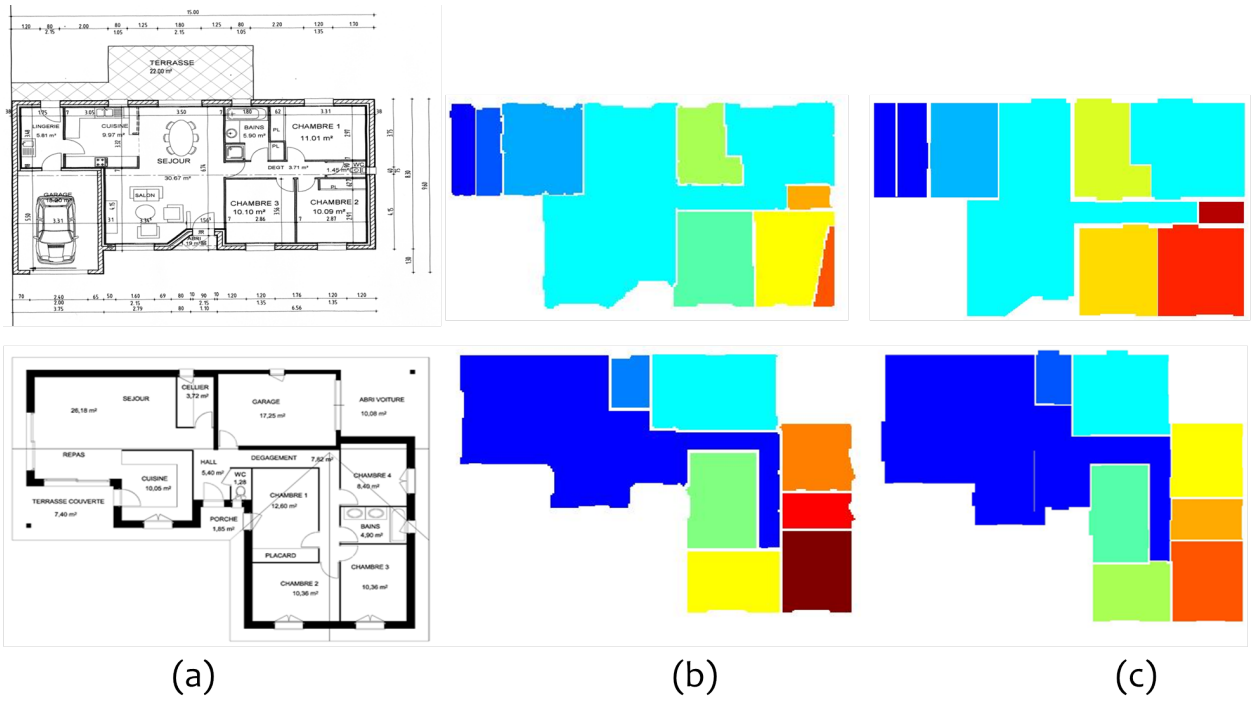


图3.18：在CVC-FP [de las Heras et al., 2015] 数据集中的两张图像上的分割结果。（a）输入的平面图，（b）[de las Heras et al., 2014] 中提出的方法，（c）我们的方法。

例如矩形。相比之下，本章考虑的是由复杂几何图元表示的室内装饰。此外，在[Weber *et al.*, 2010]中，以草图形式呈现的布局中各房间之间的邻接关系极为简化，而在本研究中则要复杂得多。更进一步地，他们在该研究中忽略了布局之间室内装饰的表示与匹配问题。

3.8 摘要

在本章中，提出了一种基于结构分析及语义相似性的平面图检索框架。在检索过程中纳入房间装饰布局这一新颖思路，可在用户对全装修平面图有特定偏好时得到应用。此外，在本章提出的基线检索框架中，还提出了一种用于表示由平面布局生成的图的谱嵌入方法；该技术显著降低了图匹配阶段的计算耗时。

在本章中，我们提出从平面布局图中提取手工构建的拓扑特征。尽管拓扑特征和房间装饰特征有助于区分不同的平面布局并检索相似的布局，但综合来看，它们仍不足以对平面布局进行全面分析。要设计出能够充分捕捉平面布局所有细节的手工特征极为困难。相比之下，深度神经网络能够在微观层面同时提取所有相关特征。在现有文献中，基于深度神经网络的特征已在自然图像检索任务中展现出良好的性能；然而，其在平面布局图像上的应用尚未得到深入研究。因此，在下一章中，我们将探讨深度

neural networks are explored to extract structural and semantic information from a floor plan for the retrieval task.

neural networks are explored to extract structural and semantic information from a floor plan for the retrieval task.

4

Deep convolutional neural networks for Floor plan Analysis and Retrieval

In Chapter 3, a baseline retrieval framework was explored by extracting topological features for understanding the layout in a floor plan. Additionally the decor features were extracted, to understand the room semantics better. During retrieval, these two features are combined and the floor plans are rank ordered based on the feature similarity. For efficient and accurate characterization of a given floor plan, as well as coming up with a feature representation that is invariant to various perturbation, feature engineering is a key component. Broadly the feature engineering approaches are classified into two categories, namely: hand-crafted features and machine learned features (deep features). The features discussed in Chapter 3, are hand-crafted. For hand-crafted features the pipeline for feature extraction is designed such that it is particularly suitable for the underlying application, and sufficient to meet the requirements. These features are robust and give satisfactory retrieval performance. However, the hand-crafted features have the following deficiencies:

1. The hand-crafted features mostly reflect the low-level identities of the objects present in the scene or the image under consideration. As a result these features are not directly suitable for representing the mid-level or even high-level depiction of the objects and require further processing or integration with additional knowledge.

2. During the feature extraction process the image level information is ignored, i.e. the features extracted are independent of the content present in the image.

This limits the matching algorithm to identify the similarities between two given floor plans. Recently, Deep Neural Networks (DNNs) have demonstrated their great power in computer vision and have achieved better performance for many tasks. For example, Convolutional Neural Networks (CNNs) have achieved good performance for ImageNet image classification [Russakovsky *et al.*, 2015]. The CNNs automatically learn the image representation from the images. The image labels are used to make the image representations separable. Such CNN learned features are termed as the machine learned features. These machine learned features capture the distribution information about the underlying data, and such information is especially important to the image representation. Hence, in this Chapter a deep learning framework is proposed for the analysis and retrieval of similar floor plans, with query by example paradigm, where the queries are floor plan images.

The rest of the chapter is organized as follows: Section 4.1, gives a brief description of the proposed technique. Section 4.2 and 4.3 give an insight into the proposed deep architecture for floor plan feature extraction and retrieval. Section 4.4 analyses the qualitative and quantitative results obtained through the extensive experiments that were carried out. Finally, Sec. 4.5 concludes the Chapter.

4

用于楼层平面分析与检索的深度卷积神经网络

在第3章中，通过提取拓扑特征来理解平面图的布局，从而构建了一个基准检索框架。此外，还提取了装饰特征，以更好地理解房间语义。在检索过程中，将这两类特征进行融合，并根据特征相似度对平面图进行排序。为了高效且准确地表征给定的平面图，并获得对各类扰动具有不变性的特征表示，特征工程是关键环节。广义上，特征工程方法可分为两类：手工设计特征和机器学习特征（深度特征）。第3章中讨论的特征均为手工设计特征。对于手工设计特征，其特征提取流程被专门针对具体应用场景进行设计，能够充分满足应用需求。这类特征具有较强的鲁棒性，并能取得令人满意的检索性能。然而，手工设计特征也存在以下不足：

1. 手工设计的特征主要反映了场景或待处理图像中对象的低层属性，因此这些特征并不直接适用于表征对象的中层甚至高层语义，需要进一步的处理或与额外的知识进行融合。2. 在特征提取过程中，图像层面的信息被忽略，即所提取的特征与图像中的具体内容无关。

这限制了匹配算法仅能识别两个给定平面图之间的相似性。近年来，深度神经网络（DNN）在计算机视觉领域展现出强大的能力，并在多项任务中取得了更优的性能。例如，卷积神经网络（CNN）在ImageNet图像分类任务中表现优异[参见Russakovsky等，2015]。CNN能够从图像中自动学习特征表示，而图像标签则用于使这些特征表示具有可区分性。此类由CNN学习得到的特征被称为机器学习特征。这些机器学习特征能够捕捉底层数据的分布信息，而这类信息对于图像表示尤为重要。因此，在本章中，我们提出了一种基于深度学习的框架，用于分析和检索相似的平面图，并采用以图搜图的查询范式，其中查询对象为平面图图像。

本章其余部分的组织安排如下：第4.1节简要介绍了所提出的算法；第4.2节和第4.3节深入阐述了用于平面布局特征提取与检索的深度网络架构；第4.4节对通过大量实验获得的定性和定量结果进行了分析；最后，第4.5节对本章进行总结。

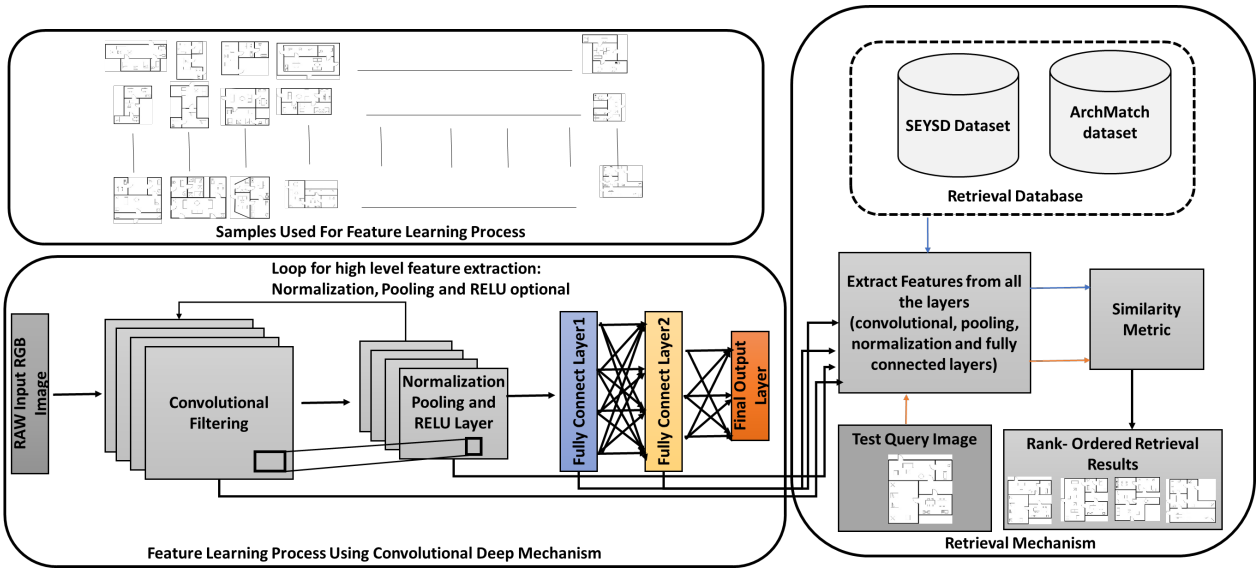


Figure 4.1. : Deep Learning Model based Framework for Floor Plan Retrieval

4.1 BRIEF OVERVIEW

Figure 4.1 depicts the complete framework of the proposed deep learning technique. A brief overview of the proposed framework for similar floor plan retrieval is as follows. The entire framework has been divided into two main phases. They are: (1) deep feature representation; and (2) matching and retrieval. In the deep feature representation task, deep neural network layers like Convolution, Normalization, Pooling and ReLU are used. In this work an additional Normalization layer is used which basically is useful when ReLU neurons are being dealt with. This is due to the fact that ReLU neurons have unbounded output and thus need to be normalized. If there is a normalization around the local neighborhood of the excited neuron, it becomes even more sensitive as compared to its neighbors. At the same time, it dampens the responses that are uniformly large in any given local neighborhood. If all the values are large, then normalizing those values diminishes all of them. Therefore, the normalization layer encourages some kind of inhibition and boosts the neurons with relatively larger activations. The output of all the layers mentioned above is passed through a network of fully connected layers to finally give the feature vector for an input image. The framework first learns the deep feature representation for floor plan retrieval task using a few samples from the ROBIN dataset. The learned deep representation helps to extract deep features from the samples of ROBIN, and stores it in the feature database. During the retrieval stage, the framework extracts the deep features from query image also using the same deep representation, where query image is both from the SESYD and the ROBIN dataset. The extracted deep features are then matched with the deep features in the database. A similarity score for a particular layout is thus calculated for all the query samples. To understand the effectiveness of the individual layers or feature representation of the deep learning network, the similarity is calculated using the features at the outputs of all the layers.

In this Chapter, the goal is to come up with new effective deep learning based feature representation and to investigate the effectiveness of learned features on floor plan retrieval tasks. In particular, two open issues are addressed:

- How to learn a new efficient deep learning feature representation for floor plan retrieval task?

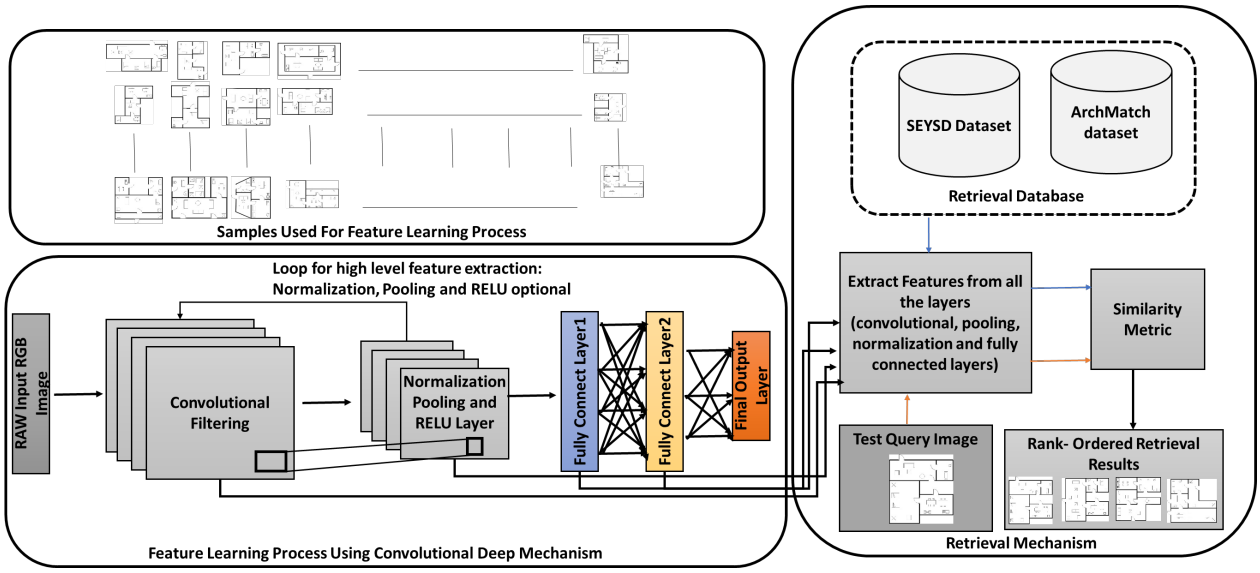


图4.1：基于深度学习的平面图检索框架

4.1 简要概述

图4.1展示了所提出的深度学习技术的完整框架。关于用于相似平面图检索的该框架的简要概述如下：整个框架被划分为两个主要阶段，分别为：（1）深度特征表示；以及（2）匹配与检索。在深度特征表示任务中，采用了卷积、归一化、池化和ReLU等深度神经网络层。在本工作中，还额外引入了一个归一化层，这在处理ReLU神经元时尤为有用。这是因为ReLU神经元的输出无界，因此需要进行归一化。如果在激活神经元的局部邻域周围实施归一化，则该神经元相对于其邻域中的其他神经元会变得更加敏感；同时，它也会抑制给定局部邻域内普遍较大的响应。若所有数值均较大，对其进行归一化则会使这些数值整体减小。因此，归一化层能够促进一定程度的抑制，并增强相对激活程度较高的神经元的响应。上述各层的输出经由全连接层网络传递，最终为输入图像生成特征向量。该框架首先利用ROBIN数据集中的少量样本，学习用于平面图检索任务的深度特征表示。所学得的深度表示有助于从ROBIN数据集的样本中提取深度特征，并将其存储于特征数据库中。在检索阶段，该框架同样基于该深度表示从查询图像中提取深度特征，其中查询图像既来自SESYD数据集，也来自ROBIN数据集。随后，将提取的深度特征与数据库中的深度特征进行匹配，从而为每个查询样本计算特定布局的相似度得分。为了评估深度学习网络中各层或特征表示的有效性，还基于各层输出的特征计算了相似度。

在本章中，我们的目标是提出一种新的、有效的基于深度学习的特征表示方法，并探究所学特征在楼层平面图检索任务中的有效性。具体而言，本章重点解决两个尚未解决的问题：

- 如何为楼层平面图检索任务学习一种新的高效深度学习特征表示?

- How the individual deep feature layers affect the performance of the retrieval system?

The proposed architecture has two main components, (1) Feature Representation and (2) Matching and Retrieval, which are discussed in detail in the subsequent subsections:

4.2 FEATURE REPRESENTATION

Traditionally, feature extractors are manually engineered and optimised through a laborious trial-and-error cycle involving re-learning the classifiers. In this work, the representation is learned using a CNN instead, by jointly optimising the performance of the features, as well as, of the classifiers. Existing literature also validates that deep features are a better way to represent an image. However, it is also shown that improper training of deep layers may lead to over-fitting or under-fitting of a model which can be a disadvantage.

Convolutional Neural Networks (CNNs) have shown promising results for various computer vision tasks like classification and detection. Convolutional neural networks are similar to feed-forward neural networks. They are made up of neurons that have learnable weights and biases. Each neuron receives some inputs, performs a dot product and generally follows it with a non-linearity. The entire network expresses a single differentiable score function: from the raw image pixels on one end to class scores at the other end. There is a loss function on the final fully-connected layer. The difference that convolutional neural networks have from feed-forward networks is that, these architectures make the explicit assumption that the inputs are images. This assumption then leads to a vast reduction in the number of parameters in the network. Convolutional neural networks have a variety of layers serving different purposes. Using the approach of CNNs, many classification related tasks have been done in the past. It was observed that researchers have not extensively explored the CNNs for floor plan retrieval tasks. The work proposed in this Chapter uses a combination of convolution, pooling, normalization, ReLU, and fully connected layers along with dropout regularization technique (see Fig. 4.1). Before going into the detailed description of the proposed architecture, in the next sub-section, for the sake of completeness, a brief description of multilayer feed forward neural network and its relation to CNNs is presented.

4.2.1 Multilayer feed-forward neural network Architecture

Multilayer feed-forward neural networks (MLFFNN) receive an input (a single vector), and transform it through a series of hidden layers. Figure 4.2 depicts an architecture of a typical MLFFNN. Each hidden layer has a number of neurons. Each neuron is connected to all neurons in the previous layer (Fig. 4.2(a)). Neurons in a single layer do not share any connections. The final fully-connected layer called the output layer represents the class scores. Let, the input feature vector $\bar{x}$ be of d dimension and there are k output classes (in a k-class classification problem). According to the Universal Approximation theorem, if $\Phi(.)$ is a non-constant, bounded, monotonically increasing continuous function, then there exists an integer J such that:

$$\tilde{f}(\bar{x}) = \sum_{j=1}^{J} \alpha_j \Phi_j \left(\sum_{i=1}^{d} w_{ij} x_i + w_{jo} \right) \tag{4.1}$$

is an approximate realization of the desired function $f(\bar{x})$. Here, J is the number of neurons in the hidden layer, and w_{ij}s are the synaptic weights between the pre-synaptic neuron i, and post-synaptic neuron j. The theorem guarantees that we will be able to design an artificial neural network with J

- How the individual deep feature layers affect the performance of the retrieval system?

所提出的架构包含两个主要组成部分：（1）特征表示与（2）匹配与检索，相关内容将在后续小节中详述。

4.2 特征表示

传统上，特征提取器是通过人工设计，并借助反复试错的迭代过程来优化，其间还需重新训练分类器。在本研究中，我们采用卷积神经网络（CNN）来学习特征表示，通过联合优化特征与分类器的性能实现这一目标。现有文献也证实，深度特征是更优的图像表示方式。然而，也有研究表明，深层网络的训练不当可能导致模型过拟合或欠拟合，从而带来不利影响。

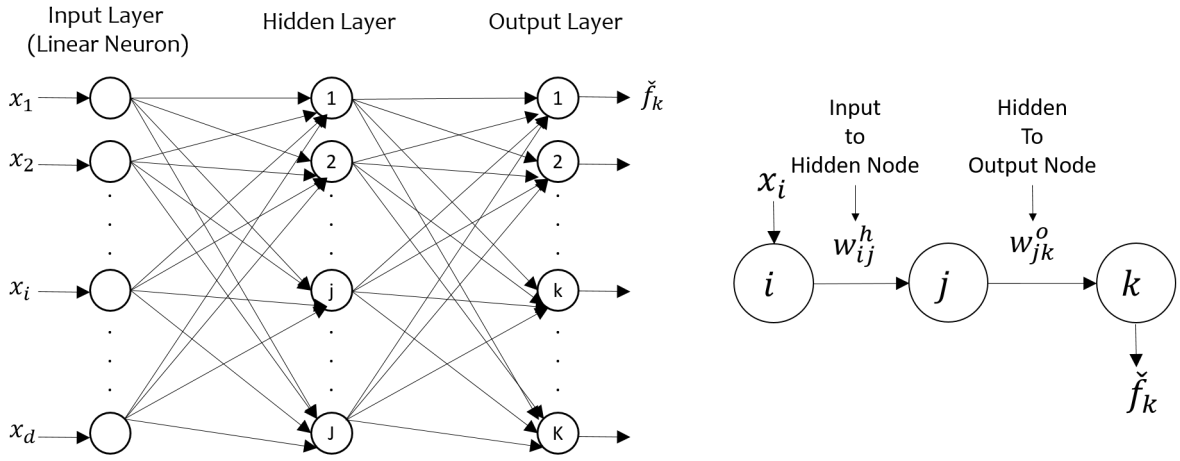
卷积神经网络（CNN）在分类与检测等多种计算机视觉任务中已展现出良好的性能。卷积神经网络与前馈神经网络类似，均由具有可学习权重和偏置的神经元组成。每个神经元接收若干输入，执行点积运算，并通常在其后引入非线性激活函数。整个网络定义了一个可微分的评分函数：从一端的原始图像像素映射到另一端的类别得分。在最后一个全连接层上定义了损失函数。卷积神经网络与前馈网络的主要区别在于，前者显式地假设输入为图像，这一假设显著减少了网络中的参数数量。卷积神经网络包含多种不同功能的层。借助卷积神经网络的方法，以往已成功解决了诸多分类相关任务。然而，研究发现，针对平面图检索任务，卷积神经网络的应用尚不充分。本章所提出的方案采用卷积、池化、归一化、ReLU以及全连接层的组合，并辅以Dropout正则化技术（参见图4.1）。在详细介绍所提架构之前，为便于理解，下一小节将简要阐述多层前馈神经网络及其与卷积神经网络的关系。

4.2.1 多层前馈神经网络架构

多层前馈神经网络（MLFFNN）接收一个输入（即一个单一的向量），并通过一系列隐藏层对其进行变换。图4.2展示了一个典型的MLFFNN架构。每个隐藏层包含若干个神经元，且每个神经元与前一层的所有神经元相连（见图4.2(a)）。同一层内的神经元之间不存在连接。最后一层为全连接层，称为输出层，用于表示各类别的得分。设输入特征向量$\bar{x}$的维度为d，且存在k个输出类别（在k-分类问题中）。根据通用逼近定理，若$\Phi(.)$为非常值、有界且单调递增的连续函数，则存在一个正整数J，使得：

$$\tilde{f}(\bar{x}) = \sum_{j=1}^{J} \alpha_j \Phi_j \left(\sum_{i=1}^{d} w_{ij} x_i + w_{jo} \right) \tag{4.1}$$

是对期望函数$f(\bar{x})$的一个近似实现。其中，J为隐藏层中的神经元数量，w_{ij}s为前突触神经元i与后突触神经元j之间的突触权重。该定理保证我们能够设计出一个具有J的神经网络。



(a) Illustration of a typical multilayer feed-forward neural network

(b) Illustration of a connection between pre- and post-synaptic neurons, interconnected with synaptic weights

Figure 4.2. : Architecture of feed-forward neural network, with the input layer, a hidden layer and the output layer

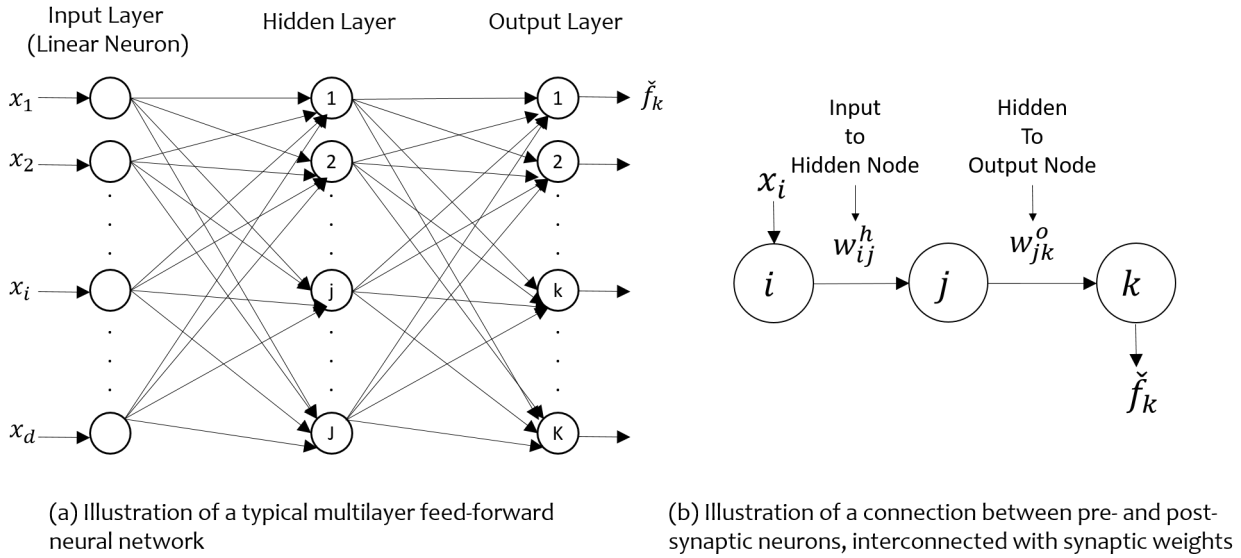
hidden nodes and synaptic connections, which will be able to represent the input feature vector to the best possible extent. In Fig. 4.2(b), the synaptic weights between input to hidden and hidden to output layer are explicitly shown. To determine the class label for a given test sample, "winner takes all" decision logic is followed.

In case of MLFFNN, the input vector dimension plays an important role in the scalability of the entire solution. An increase in the value of d leads to increase in the number of synopsis, and hence the synaptic weights. As a result, there are huge number of parameters to be learned. For example, a given image of size, say, $32 \times 32 \times 3$, requires 3072 weights for each neuron. For a bigger image that is 200×200, each neuron in the first layer would require 120000 parameters. This network with a huge number of parameters, would not only take long to train, but may also be prone to overfitting.

On the other hand, the convolutional neural network has its origin deeply rooted at the seminal work by Hubel and Wiesel (Nobel prize awarded in the 1980's). Hubel and Wiesel's "simple" and "complex" cell's computational properties were later well-described by linear models and rectifiers. The key components incorporated in convolutional neural networks, which are similar to the various components of the visual cortex are:

- CNNs are spatially organized, so that nearby cells act on nearby parts of the input image (similar to the Retinotopic map)
- CNNs use spatially localized linear filters, which are followed by thresholding (resembling the simple cells)
- CNNs use pooling units to incorporate invariance to shifts of the position of the feature (mimics the complex cells)

Some recent work has shown that artificial neurons in convolutional neural networks have similar hierarchical structure to the visual pathway. CNN (refer Fig. 4.3) takes advantage of the fact



(a) Illustration of a typical multilayer feed-forward neural network

(b) Illustration of a connection between pre- and post-synaptic neurons, interconnected with synaptic weights

图4.2：前馈神经网络的架构，包括输入层、隐藏层和输出层。

隐藏节点与突触连接能够以尽可能最佳的方式表征输入特征向量。在图4.2(b)中，输入层到隐藏层以及隐藏层到输出层之间的突触权重被明确展示。对于给定的测试样本，采用“胜者全得”的决策逻辑来确定其类别标签。

在多层前馈神经网络（MLFFNN）中，输入向量的维度对整个解决方案的可扩展性具有重要影响。当d的值增大时，概要的数量以及相应的突触权重也会随之增加，从而导致需要学习的参数数量急剧增多。例如，对于一幅尺寸为32×32×3的图像，每个神经元就需要3072个权重；而对于一幅更大的200×200的图像，第一层中的每个神经元则需要120000个参数。这种参数规模巨大的网络不仅训练时间较长，而且还容易出现过拟合现象。

另一方面，卷积神经网络的起源深深植根于Hubel和Wiesel的开创性工作（该研究于20世纪80年代荣获诺贝尔奖）。Hubel和Wiesel提出的“简单”细胞与“复杂”细胞的计算特性，后来被线性模型和整流函数很好地刻画。卷积神经网络中所包含的关键组件，与视觉皮层中的各类结构单元具有相似之处，具体包括：

- 卷积神经网络具有空间组织特性，因此相邻的神经元会对输入图像的邻近区域进行处理（类似于视网膜拓扑映射）。
- 卷积神经网络采用空间局部化的线性滤波器，其后通常接有阈值处理（类似于简单细胞的特性）。
- 卷积神经网络通过池化单元引入对特征位置平移的不变性（模拟了复杂细胞的功能）。

近期的一些研究表明，卷积神经网络中的人工神经元具有与视觉通路类似的层级结构。卷积神经网络（参见图4.3）利用了这一特性。

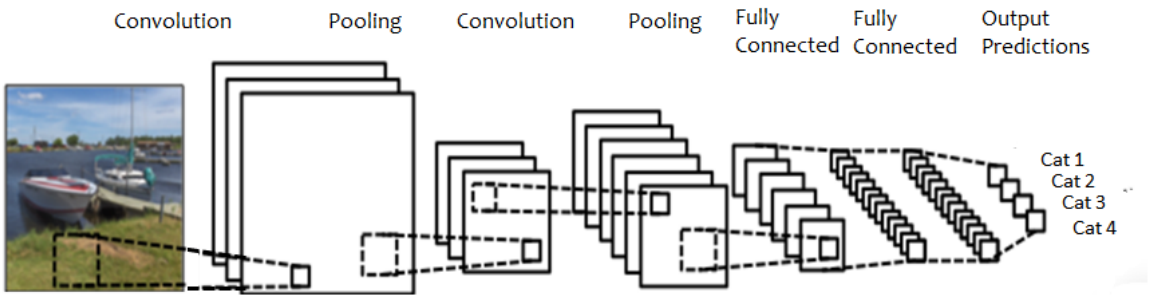


Figure 4.3. : Architecture of a Convolutional Neural Network Model for classification of an image [LeCun *et al.*, 2015]

that the input are images and they constrain the architecture. In particular, the layers of a CNN have neurons arranged in 3 dimensions: width, height and depth, where, depth refers to the third dimension of an activation volume.

4.2.2 Proposed CNN architecture

The Convolutional Neural Network (CNN) proposed in this Chapter for the classification of floor plans has an architecture comprising of an Input layer, a Convolution layer, a ReLU layer, a Pooling layer, a Normalization layer and a Fully Connected layer. Details of each layer are as follows:

- The Input layer holds the raw pixel values of the floor plan image.
- Convolutional layer computes the output of neurons that are connected to local regions in the input, each neuron computing a dot product (weighted sum) between its weights and a small region it is connected to in the input image. Figure 4.4 depicts an example of such a weighted sum between an input image and a filter. The convolutional layer defines a collection of filters (or activation maps), each with the same dimension as the input. Let, the input be of dimensionality (w,h,d), and the filter dimensionality be (w_f,h_f,d), typically $w_f < w$. The depths being equal means that the output of this convolution operation is 2D. The output slices of the convolution operations with each filter are composed together to form a $(w+w_f-1,h+h_f-1,n_f)$ tensor, where n_f is the number of filters. A convolutional layer

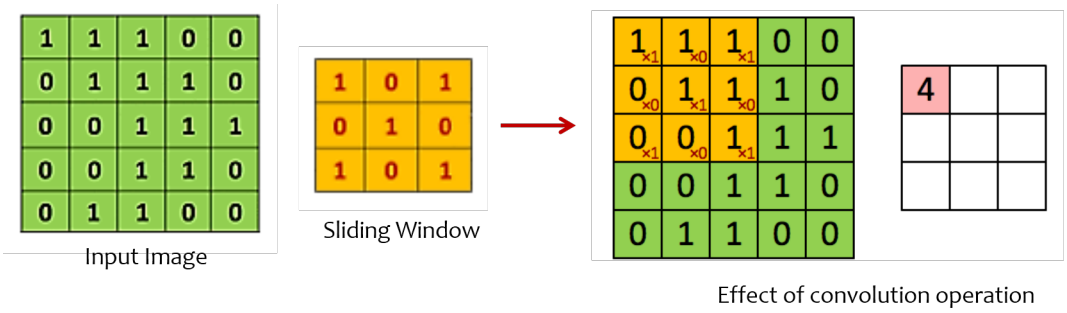


Figure 4.4. : Example of how convolution operation is performed on an image matrix.

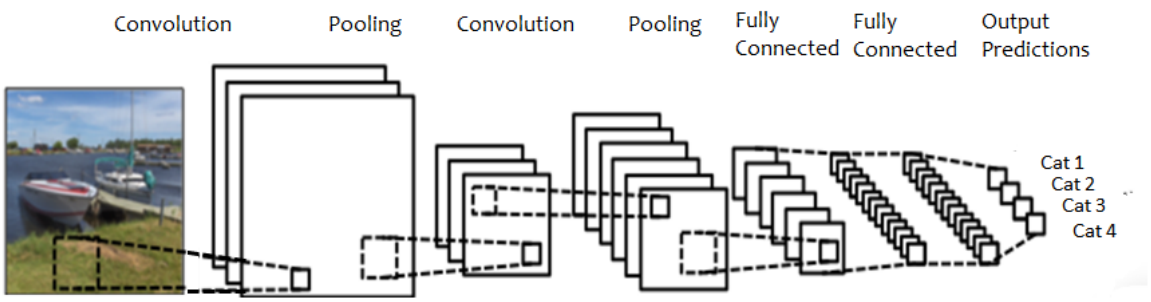


图4.3：用于图像分类的卷积神经网络模型架构 [LeCun等，2015]

输入为图像，并且它们对网络架构施加了约束。具体而言，卷积神经网络的各层中的神经元按三维排列：宽度、高度和深度，其中深度指激活张量的第三个维度。

4.2.2 提出的卷积神经网络架构

本章提出的用于平面图分类的卷积神经网络（CNN）采用如下架构：包括输入层、卷积层、ReLU激活层、池化层、归一化层以及全连接层。各层的具体细节如下：

- 输入层存储了平面图图像的原始像素值。
- 卷积层计算与输入中局部区域相连的神经元的输出，每个神经元对其权重与输入图像中与其相连的小区域之间进行点积（加权求和）。图4.4展示了输入图像与滤波器之间这种加权求和的一个示例。卷积层定义了一组滤波器（或激活图），每组滤波器的尺寸与输入相同。设输入的维度为(w,h,d)，滤波器的维度通常为(w_f,h_f,d)，典型值为$w_f < w$。当深度相等时，该卷积操作的输出为二维。将各滤波器对应的卷积操作输出切片组合在一起，形成一个$(w+w_f-$ $\times 1 \times ,n_f)$张量，其中n_f表示滤波器的数量。卷积层

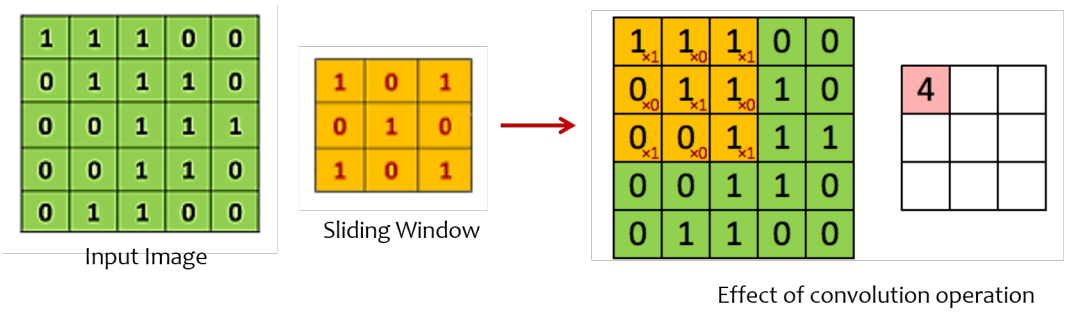


图4.4：对图像矩阵进行卷积运算的示例。

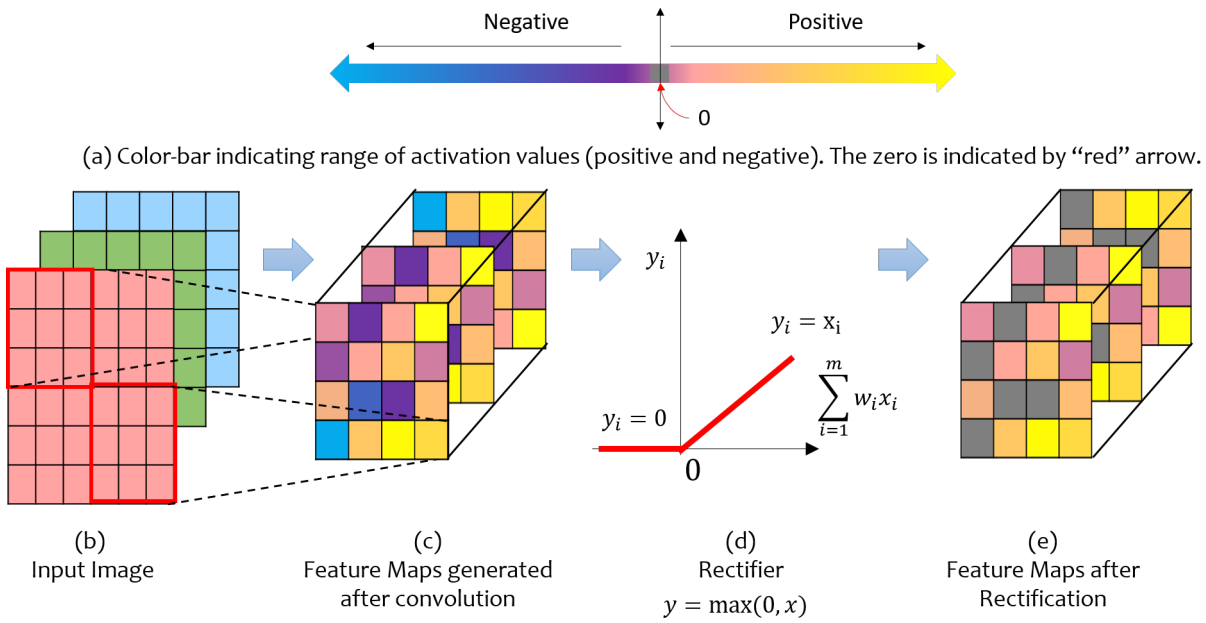


Figure 4.5. : Example of how a ReLU Operation works.

finally learns a set of n_f filters, $Filt = \{filt_1, filt_2 .... filt_{n_f}\}$ with input floor plan image I to produce a set of 2D feature maps:

$$z_k = filt_k \circledast I \tag{4.2}$$

where $\circledast$ is the convolution operator. Sharing of weights is done over the entire image which in turn results in the reduction of the number of parameters. Also, if there are m inputs and n outputs in a hidden layer, a MLFFNN would require $O(mn)$ operations to compute the output. Whereas, in CNN each output is connected to only $k = w_f h_f d$ inputs, and thus requires $O(kn)$ operations to compute the output, thereby reducing the computation time.

- In the network proposed in this Chapter, ReLU (Rectified Linear Unit) layer is used to induce non-linearity. The ReLU clips negative values to zero while keeping positve values unchanged. Figure 4.5 depicts how ReLU works. The popularity of ReLU is due to the fact that it highly enhances the acceleration of the convergence of stochastic gradient descent as opposed to the sigmoid/tanh functions due to its linear non-saturating form. It also avoids and rectifies vanishing gradient problem [Krizhevsky *et al.*, 2012b]. This is due to the fact that as seen in Eq. 4.3 the gradient maintains a constant value. An added advantage is that it is computationally less expensive. ReLU has the following mathematical form:

$$y = max(0, x) \tag{4.3}$$

It applies an element-wise activation function, in the form of $max(0,x)$ with thresholding at zero which leaves the size of the volume unchanged.

- In the proposed architecture, a pooling layer is used for reducing the size of the feature maps and to provide invariance between floor plan images with very little differences. Max pooling is usually preferred as it prevents blurring of the activations and gradients throughout the network because the gradient is placed in a single location during back propagation. It also avoids cancellation of negative elements (refer Fig. 4.6) [Cadène *et al.*, 2016; Yin *et al.*, 2016]. Pooling layer performs a down-sampling operation along the spatial dimensions.

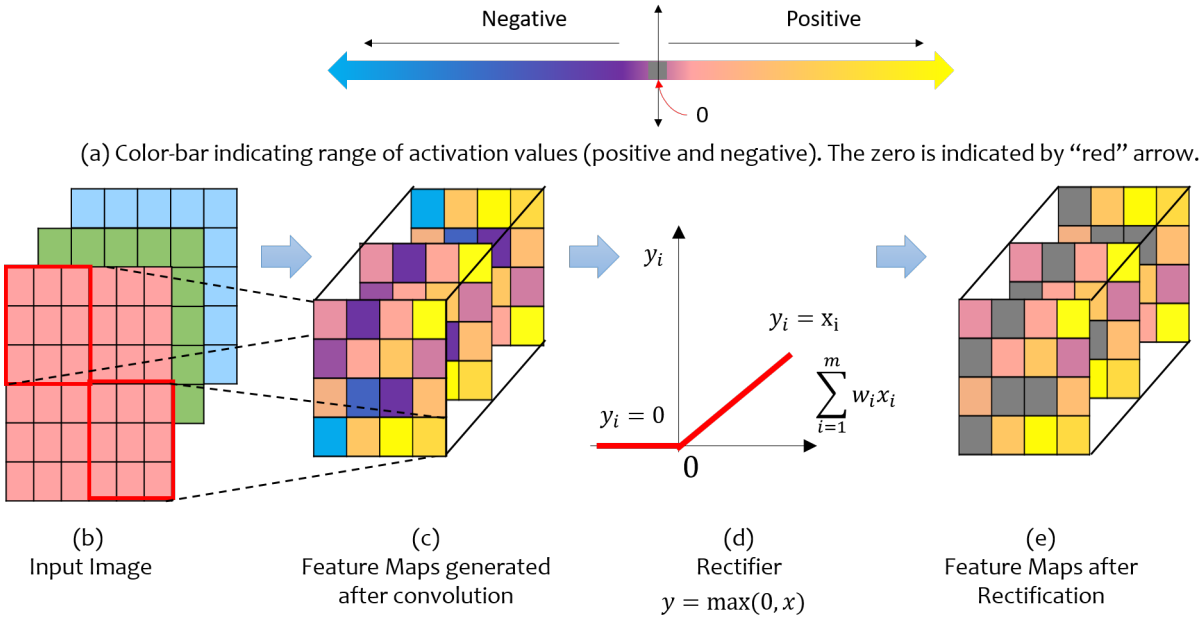


图4.5：ReLU运算的工作示例　　ks。

最终学习得到一组n_f个滤波器，这些滤波器以输入的平面布局图像I为输入，生成一组二维特征图：

$$z_k = filt_k \circledast I \tag{4.2}$$

其中，$\circledast$ 为卷积算子。在整个图像上共享权重，从而减少了参数数量。此外，若某隐层有 m 个输入和 n 个输出，则全连接前馈神经网络（MLFFNN）需要进行 $O(mn)$ 次运算才能得到输出。而在卷积神经网络（CNN）中，每个输出仅与 $k = w_f h_f d$ 个输入相连，因此只需 $O(kn)$ 次运算即可得到输出，从而缩短了计算时间。

- 在本章所提出的网络中，采用ReLU（修正线性单元）层以引入非线性。ReLU将负值截断为零，而对正值则保持不变。图4.5展示了ReLU的运作机制。ReLU之所以广受欢迎，是因为其线性且非饱和的特性，与Sigmoid/tanh函数相比，能够显著加速随机梯度下降的收敛过程；同时，它还能有效避免并缓解梯度消失问题[Krizhevsky *et al.*, 2012b]。这是由于如式4.3所示，其梯度始终保持恒定值。此外，ReLU在计算上也更为高效。ReLU的数学表达式如下：

$$y = max(0, x) \tag{4.3}$$

它应用了一个逐元素的激活函数，形式为$max(0,x)$，并在零处进行阈值处理，从而保持体积的大小不变。

- 在所提出的架构中，池化层用于降低特征图的尺寸，并使具有细微差异的平面布局图像之间具有不变性。通常优先采用最大池化，因为它能够在反向传播过程中将梯度集中于单一位置，从而避免激活值和梯度的模糊化。此外，它还能防止负值元素相互抵消（参见图4.6）[Cadène *et al.*, 2016; Yin *et al.*, 2016]。池化层沿空间维度执行下采样操作。

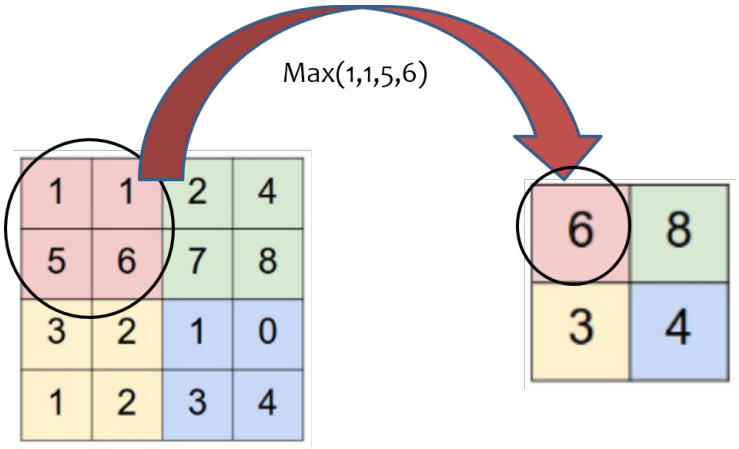


Figure 4.6. : Example of how a Max-Pooling Operation behaves on an image matrix.

- The next layer in the proposed architecture is the Batch normalization layer, which is widely popular as it converges faster [Ioffe and Szegedy, 2015]. The Batch normalization layer leads to an addition of a normalization step which helps in shifting the inputs to zero-mean and unit variance [Cadène *et al.*, 2016]. It allows activation functions to not get stuck in the saturation mode (gradient equal to 0).
- The fully-connected layer computes the class scores. Classification into categories is done by this layer. Neurons in a fully connected layer have connections to all activations in the previous layer.

In this way, the proposed Convolutional Neural Network transforms the original floor plan image, layer by layer from the original pixel values to the final class scores. In particular, the Convolutional/Fully Connected layers perform transformations that are a function of not only the activations in the input volume, but also of the parameters (the weights and biases of the neurons). On the other hand, the RELU/Pooling layers implement a fixed function. The parameters in the Convolutional/Fully Connected layers are trained with gradient descent so that the class scores that the Convolutional Neural Network computes are consistent with the labels in the training set for each image.

This proposed architecture takes inspiration from the methodology described in the work AlexNet [Krizhevsky *et al.*, 2012a], to obtain an effective feature representation from floor plan images (refer Fig. 4.7).

The basic highlights of the AlexNet framework [Krizhevsky *et al.*, 2012a] are:

- ReLU is used instead of Tanh to add non-linearity. This layer accelerates the speed by 6 times at the same accuracy for classification tasks.
- Dropout is used instead of regularisation to deal with overfitting.
- Pooling is overlapped to reduce the size of network. It reduces the top-1 and top-5 error rates by 0.4% and 0.3%, respectively.
  The AlexNet framework contains 5 convolutional layers and 3 fully connected layers. ReLU is applied after every convolutional and fully connected layer. Dropout is applied before the first and the second fully connected layer. Through Dropout, at each training stage, individual nodes are dropped out of the net to ensure that there is no overfitting in the network.

However, the model proposed in this Chapter is different from AlexNet in terms of two

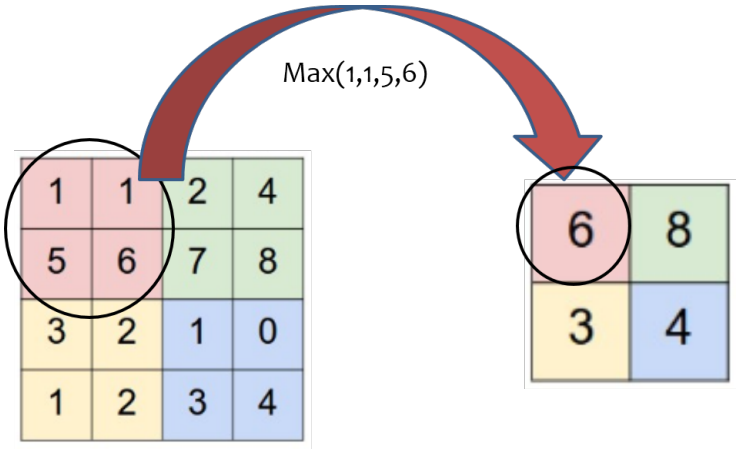


图4.6：最大池化操作在图像矩阵上的示例。

- 所提出的架构中的下一层是批量归一化层，该层因其收敛速度更快而广受欢迎[Ioffe和Szegedy，2015]。批量归一化层引入了一步归一化操作，有助于将输入的均值调整为零、方差调整为单位[加德纳等，2016]。这使得激活函数不会陷入饱和状态（梯度为零）。

- 全连接层用于计算各类别的得分。分类任务由该层完成。全连接层中的每个神经元与前一层的所有激活值均存在连接。

通过这种方式，所提出的卷积神经网络逐层地将原始平面图图像从像素值映射至最终的类别得分。具体而言，卷积层与全连接层执行的变换不仅取决于输入特征图中的激活值，还取决于其参数（即神经元的权重和偏置）。相比之下，ReLU层与池化层则实现固定的函数变换。卷积层与全连接层的参数通过梯度下降法进行优化训练，以使卷积神经网络为每幅图像计算出的类别得分与其在训练集中的标签保持一致。

所提出的架构借鉴了AlexNet [Krizhevsky *et al.*, 2012a] 中描述的方法，以从平面布局图中提取有效的特征表示（参见图4.7）。

AlexNet框架的基本要点如下[Krizhevsky *et al.*, 2012a]：

- ReLU被用于替代Tanh以引入非线性。在分类任务中，该层在保持相同精度的前提下可将计算速度提升6倍。
- 为应对过拟合问题，采用丢弃法（Dropout）而非正则化方法。
- 通过重叠池化来减小网络规模，从而分别将Top-1和Top-5错误率降低了0.4%和0.3%。AlexNet框架包含5个卷积层和3个全连接层；在每个卷积层和全连接层之后均应用ReLU激活函数；在第一个和第二个全连接层之前应用了Dropout。通过Dropout，在每个训练阶段随机丢弃网络中的部分节点，以防止网络出现过拟合。

然而，本章所提出的模型在两个方面与AlexNet存在差异。

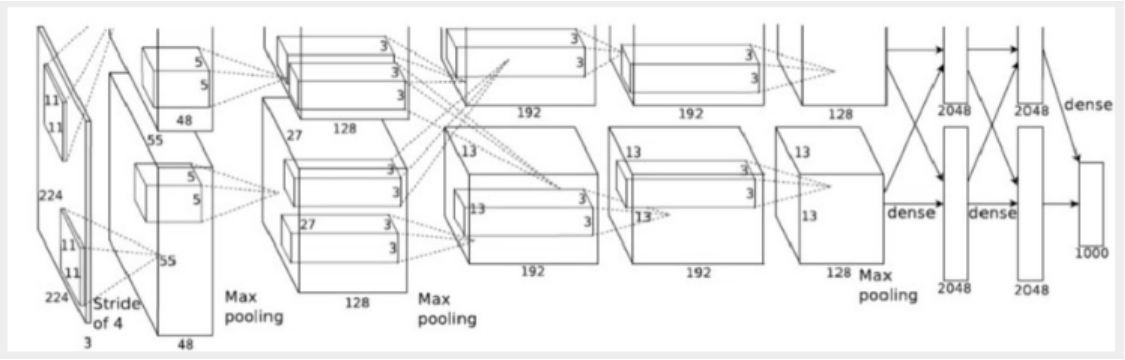


Figure 4.7. : Sequence of layers in AlexNet Framework [Krizhevsky *et al.*, 2012a]

properties:

- The network is not trained with data-augmentation.
- The order of pooling and normalization layers is switched (pooling is done before normalization) [Jia *et al.*, 2014].

The values of the parameters used in the framework such as, number of layers, type of layers, channels, filter sizes, convolution strides, pooling size, pooling stride and the padding size are listed out in Tab. 4.1. The parameters left blank in the table are the ones which do not have any role to play in that particular layer.

As an example taking the first layer, which is a combination of a convolution layer, a maxpool layer and a normalization layer. The size of the output image after passing through a convolution layer can be computed as:

Let, O_c = Size of the output image, I_c = Size of input image, K_c = Size of channels used in the convolution layer, S_c = Stride of the convolution operation and P_c = Padding Size. The size of the output image O_c is calculated using the equation below:

$$O_c = \frac{I_c - K_c + 2P_c}{S_c} + 1 \tag{4.4}$$

Now as, our input size I_c is 512×512. The first convolutional layer has 96 kernels of size 11×11. The stride is 4 and the padding is 0, therefore, the output image size is:

$$O_c = \frac{512 - 11 + 2 \times 0}{4} + 1 = 126 \tag{4.5}$$

The output image size is : $126 \times 126 \times 96$, corresponding to the number of channels. This image is then sent to the maxpool layer.

Let O_m be the size of the output image, I_m be the size of the input image to maxpool layer, S_m be the stride of the pooling operation and P_m be the pool size. The size of the output image is

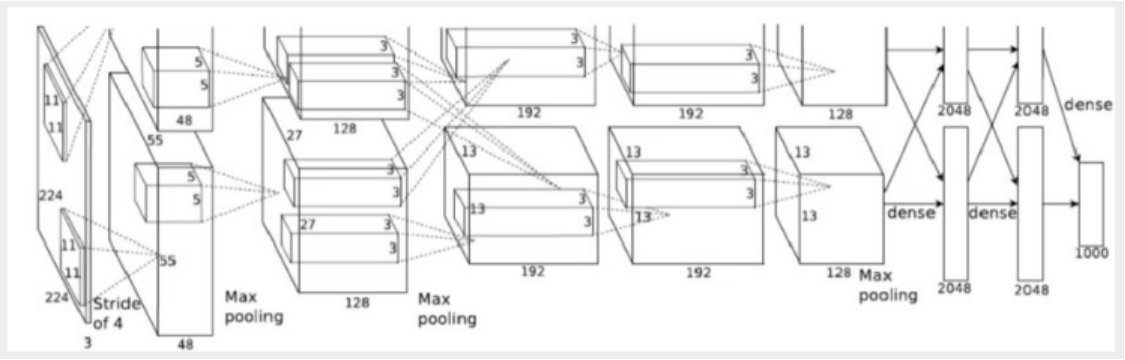


图4.7：AlexNet框架中的层序列[Krizhevsky等，2012a]

属性：

- 该网络未采用数据增强进行训练。
- 池化层与归一化层的顺序被互换（即先进行池化，再进行归一化）[Jia *et al.*, 2014]。

框架中所使用的各参数值，如层数、层类型、通道数、卷积核大小、卷积步长、池化窗口大小、池化步长以及填充大小，均列于表4.1中。表中未填写的参数为在该特定层中不发挥作用的参数。

以第一层为例，该层由卷积层、最大池化层和归一化层组成。经过卷积层处理后，输出图像的尺寸可按如下公式计算：

设O_c =为输出图像的尺寸，I_c =为输入图像的尺寸，K_c =为卷积层所使用的通道数，S_c =为卷积操作的步幅，P_c =为填充大小。输出图像的尺寸O_c可通过下式计算：

$$O_c = \frac{I_c - K_c + 2P_c}{S_c} + 1 \tag{4.4}$$

现在，我们的输入尺寸为512×512。第一层卷积层包含96个大小为11×11的卷积核，步幅为4，填充为0，因此输出图像的尺寸为：

$$O_c = \frac{512 - 11 + 2 \times 0}{4} + 1 = 126 \tag{4.5}$$

输出图像的尺寸为：126 × 126 × 96，其中最后一个维度对应通道数。随后，该图像被送入最大池化层。

设O_m为输出图像的尺寸，I_m为最大池化层的输入图像尺寸，S_m为池化操作的步幅，P_m为池化窗口的大小。输出图像的尺寸为

Layers	1	2	3	4	5	6
Type	conv1+ maxpool1+ norm1	conv2+ maxpool2+ norm2	conv3	conv4	conv5+ maxpool3	full (fc6,fc7)
Channels	96	256	384	384	256	4096
Filter Size	11×11	5×5	3×3	3×3	3×3	-
Convolution Stride	4×4	1×1	1×1	1×1	1×1	-
Pooling Size	3×3	3×3	-	-	3×3	-
Pooling Stride	2×2	2×2	-	-	2×2	-
Padding Size	-	1×1	1×1	1×1	1×1	-

Table 4.1. : Network Parameters for the proposed framework

calculated using the following equation:

$$O_m = \frac{I_m - P_m}{S_m} + 1 \tag{4.6}$$

As per this equation, the size of the output image after passing through the first maxpool layer is:

$$O_m = \frac{126 - 3}{2} + 1 = 62 \tag{4.7}$$

So, the output image is of size $62 \times 62 \times 96$. After passing through the normalization layer, the size of the output remains unchanged, i.e. $62 \times 62 \times 96$. In a similar manner according to the parameters, sizes of the output across each of the layers are calculated.

For learning a representation of CNN (deep) layers, a few samples from the proposed floor plan dataset are used. Unlike the networks mentioned in the literature, the activations of all the

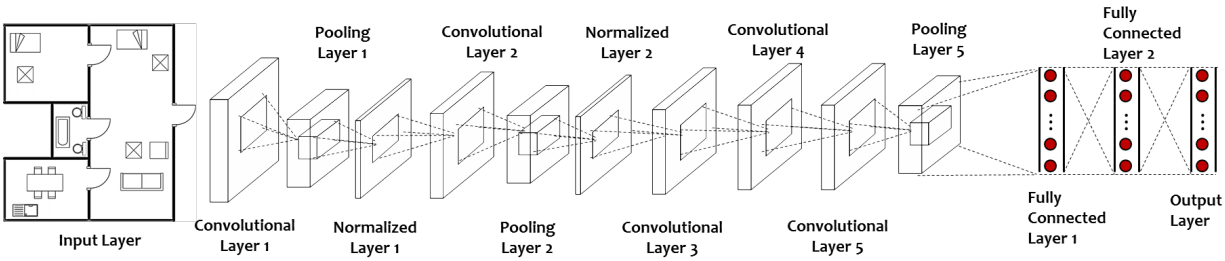


Figure 4.8. : Architecture of the Convolutional Neural Network (CNN) model used for feature extraction from the floor plan images.

Layers	1	2	3	4	5	6
Type	conv1+ maxpool1+ norm1	conv2+ maxpool2+ norm2	conv3	conv4	conv5+ maxpool3	full (fc6,fc7)
Channels	96	256	384	384	256	4096
Filter Size	11×11	5×5	3×3	3×3	3×3	-
Convolution Stride	4×4	1×1	1×1	1×1	1×1	-
Pooling Size	3×3	3×3	-	-	3×3	-
Pooling Stride	2×2	2×2	-	-	2×2	-
Padding Size	-	1×1	1×1	1×1	1×1	-

表4.1：所提框架的网络参数

calculated using the following equation:

$$O_m = \frac{I_m - P_m}{S_m} + 1 \tag{4.6}$$

根据该公式，经过第一个最大池化层后的输出图像尺寸为:

$$O_m = \frac{126 - 3}{2} + 1 = 62 \tag{4.7}$$

因此，输出图像的尺寸为62×62×96。经过归一化层后，输出尺寸保持不变，仍为62×62×96。按照相同的参数设置，可依次计算各层的输出尺寸。

为了学习卷积神经网络（深度）层的表示，我们使用了所提出的楼层平面图数据集中的少量样本。与文献中提到的网络不同，所有层的激活值

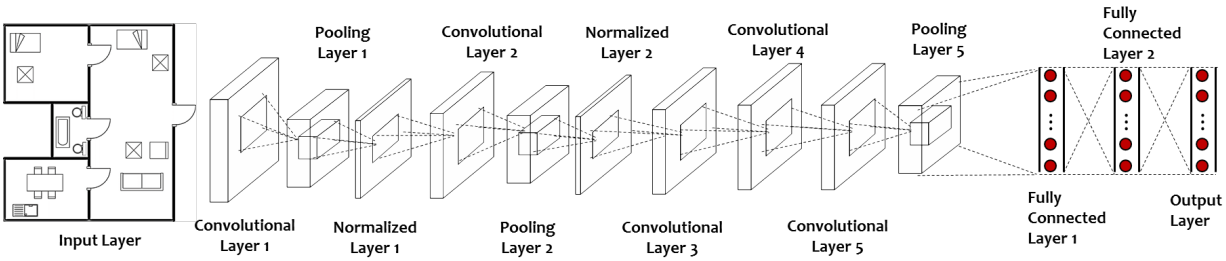


图4.8：用于从平面布局图像中提取特征的卷积神经网络（CNN）模型架构。

layers (convolution, pooling, norm, and fully connected) for the task of floor plan retrieval are taken to investigate the effectiveness of each individual layer (see Fig. 4.8). The major motivation behind using lower convolution neural network layers is to understand the effectiveness of low level features as compared to higher-level feature representation for floor plan retrieval tasks.

To obtain the feature representation from query as well as the database samples, the images are directly fed into the input layer of the learned CNN model, and then the activation values from all the layers are taken (see Fig. 4.8). For conv, norm, and pool layers, the activation output is vectorized. Since computation from the feed-forward network based on the matrix multiplication is only needed once, the whole scheme is highly time efficient. In the next stage, i.e. matching and retrieval, query samples are matched with database samples to effectively retrieve the rank-ordered samples from the database.

4.3 MATCHING AND RETRIEVAL

Floor plan layout matching using the extracted features is essential for retrieving similar layouts to the query layout. In the rest of this section, the procedure to calculate matching score M for a pair (query and database) of layouts is described, corresponding to learned deep features.

Let N be the total number of images in the form of (x_ρ, y_ρ), $\rho = 1,2...,N$ where, x_ρ is the observed variable and y_ρ is the corresponding class label in a retrieval database. Let $F_H(x_\rho) : \{F_{H1}(x_\rho), F_{H2}(x_\rho), .., F_{HL}(x_\rho)\}$ be the set of hidden layer features extracted for x_ρ sample, where $F_{Hp}(x_\rho)$ represents p^{th} hidden layer feature extracted from x_ρ sample in database and L represents the total number of hidden layers in deep feature hierarchy. For experimentation purpose, features are extracted from 12 hidden layers of CNN model (see Fig. 4.8). For each sample in the database, features from each hidden layer ($F_{Hp}(x_\rho)$) are extracted and stored in the feature database as discussed. Let $F_H(q) : \{F_{H1}(q), F_{H2}(q), .., F_{HL}(q)\}$ be the the set of hidden layer features extracted from a query image (q). In this Chapter, analysis of the proposed framework is done using features features extracted from all the hidden layers:

- **Similarity based on each learned hidden layer feature representation**: In this approach, extensive experimental analysis was performed by comparing the performance of individual hidden layer for proposed floor plan retrieval task. The Matching Score (M) between query image (q) and an image sample from the database (x_ρ) using p^{th} hidden layer feature is calculated (and used for ranking), as:

$$M = \sum_{i=1}^{|F_{Hp}(x_\rho)|} |F^i{}_{Hp}(x_\rho) - F^i{}_{Hp}(q)|_2 \tag{4.8}$$

- **Similarity based on combination of hidden layer feature representation**: On the other hand, in this approach, the focus is on identification of the best performing hidden layer representation for each retrieval sample. This was achieved by combining the matching score outputs of each hidden layers using a *min* operation. The Matching score (M) between a query image (q) and a database image (x_ρ) using all the hidden layer features is calculated (and

针对平面图检索任务，分别选取卷积层、池化层、归一化层以及全连接层，以探究各单一层的效能（见图4.8）。采用卷积神经网络底层的主要动机在于，相较于高层特征表示，明确低层特征在平面图检索任务中的有效性。

为了从查询图像以及数据库样本中获取特征表示，将图像直接输入已训练的卷积神经网络模型的输入层，并提取所有层的激活值（见图4.8）。对于卷积层、归一化层和池化层，其激活输出被转化为向量形式。由于基于矩阵乘法的前向传播计算只需进行一次，整个方案具有极高的时间效率。在下一阶段，即匹配与检索过程中，将查询样本与数据库样本进行匹配，从而有效地从数据库中检索出按相关性排序的样本。

4.3 匹配与检索

利用提取的特征进行平面布局匹配，对于检索与查询布局相似的布局至关重要。在本节余下部分，将描述针对查询布局与数据库中布局构成的一对布局，基于所学习的深度特征计算匹配得分M的具体流程。

设N为形如(x_ρ, y_ρ)的图像总数，其中$\rho = 1, 2..., N$，且x_ρ为观测变量，y_ρ为检索数据库中对应的类别标签。设$F_H(x_\rho) : \{F_{H1}(x_\rho), F_{H2}(x_\rho), .., F_{HL}(x_\rho)\}$为针对$x_\rho$个样本提取的隐藏层特征集合，其中$F_{Hp}(x_\rho)$表示从数据库中的第$x_\rho$个样本提取的第$p^{th}$个隐藏层特征，$L$表示深度特征层级结构中的隐藏层总数。为便于实验，我们从CNN模型的12个隐藏层中提取特征（见图4.8）。对于数据库中的每个样本，均按前述方法提取并存储各隐藏层的特征（$F_{Hp}(x_\rho)$）。设$F_H(q) : \{F_{H1}(q), F_{H2}(q), .., F_{HL}(q)\}$为从查询图像（$q$）提取的隐藏层特征集合。在本章中，所提出的框架分析基于从所有隐藏层提取的特征进行：

- 基于各层学习到的隐藏层特征表示的相似度：在该方法中，通过比较所提出的楼层平面检索任务中各隐藏层的性能，进行了详尽的实验分析。利用第p^{th}层的隐藏层特征，计算查询图像(q)与数据库中的样本图像(x_ρ)之间的匹配得分(M)（并用于排序），其计算公式为：

$$M = \sum_{i=1}^{|F_{Hp}(x_\rho)|} |F^i{}_{Hp}(x_\rho) - F^i{}_{Hp}(q)|_2 \tag{4.8}$$

- 基于隐藏层特征表示组合的相似度：另一方面，在该方法中，重点在于为每个检索样本确定性能最优的隐藏层表示。为此，通过采用min操作将各隐藏层的匹配得分输出进行组合。利用所有隐藏层特征计算查询图像（q）与数据库图像（x_ρ）之间的匹配得分（M），并

Figure 4.9. : Rank ordered retrieval result of the proposed framework for five different query floor plans from the ROBIN dataset. Here, top five rank ordered floor plans are shown for each query.

used for ranking), as:

$$M = argmin_p \sum_{i=1}^{|F_{Hp}(x_\rho)|} |F^i{}_{Hp}(x_\rho) - F^i{}_{Hp}(q)|_2, \quad p \in (1,...,L) \tag{4.9}$$

The next section describes the qualitative and quantitative assessment of the proposed framework.

4.4 EXPERIMENTS AND RESULTS

The experiments were performed on the SESYD [Delalandre *et al.*, 2010b] and the ROBIN dataset to show the effectiveness of the framework. As already illustrated the SESYD dataset [Delalandre *et al.*, 2010b], contains 10 different types of layouts, each with 100 images varying in their arrangement of furniture inside the rooms. The image sizes in the database vary from

图4.9：所提出框架在ROBIN数据集上对五个不同查询平面图的排序检索结果。此处展示了每个查询对应的前五名排序平面图。

用于排序），如下：

$$M = argmin_p \sum_{i=1}^{|F_{Hp}(x_\rho)|} |F^i{}_{Hp}(x_\rho) - F^i{}_{Hp}(q)|_2, \quad p \in (1,...,L) \tag{4.9}$$

下一节将对所提出的框架进行定性和定量评估。

4.4 实验与结果

实验在SESYD [Delalandre v1, 2010b] 和ROBIN数据集上进行，以验证该框架的有效性。如前所述，SESYD数据集 [Delalandre v2, 2010b] 包含10种不同类型的布局，每种布局包含100张图像，这些图像中房间内的家具布置各不相同。数据库中图像的尺寸范围为

6775×2858 to 2056×1837. All floor plans are binarized to ensure use of only structural information while analysis. On the other hand, ROBIN dataset contains 510 floor plans divided into 3 broad categories based on the global layout shape.

The ROBIN dataset is split into two sets: (a) training - comprises of 30% samples and used for deep feature representation; and (b) testing - comprises of rest 70% samples and used as test (query) samples. As a retrieval database, the entire ROBIN dataset was used. Each floor plan image has been resized to a size of 512×512 for both matching and feature representation task. For deep model configuration, the architecture described in Fig. 4.8 is used, with same parameters as discussed in [Krizhevsky *et al.*, 2012a]. The same model pretrained on the ROBIN dataset is used for feature representation of the SESYD dataset.

4.4.1 Qualitative Results

The proposed framework is able to successfully retrieve a rank order set of floor plan images using the proposed deep learning framework. The top 5 rank ordered retrieval results for five different queries on ROBIN, are shown in Fig. 4.9. The arrow underneath the diagram depicts the decreasing order of similarity as one goes from left to right in Fig. 4.9 with green colour signifying high similarity and red signifying low similarity. The samples which are erroneously retrieved, i.e. not from the same class as that of the query class, are highlighted with "red" color. For all the results shown in Fig. 4.9, features extracted from the second normalization layer of the CNN stack were used. The rationale behind this choice was that the second normalizaton layer yields the best performance. Details analysis of such a selection is discussed in details in the next sub-section.

The first rank ordered result retrieved from the layout database is the query image itself. This is due to maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower. In case of Fig. 4.9 (a), all the top 5 samples belong to the query class. Global layouts of the all the floor plans are identical and so is the number of rooms in the floor plans. The individual plans differ in terms of the number of furniture present in the rooms and their position. Readers are requested to note the difference in the shape and position of the individual rooms also for the rank ordered samples. There are 4 rooms in each floor plans, however the framework is able to rank order the samples correctly by ordering the most similar ones to the query first, followed by the rest.

Figure 4.9 (b-c) depict some of the failure cases. In Fig. 4.9 (b), the 4^{th} ranked result is erroneously retrieved (highlighted by "red") by the framework. However, the the 5^{th} sample is correctly retrieved. Justification of the rank 4 erroneously retrieved result can be understood through qualitative analysis. For example, room 1 in the query layout is adjacent to room 2, 3 and 4, that is identical to the adjacencies shared by room 1 in rank 2 retrieved result. Moreover, the size and type of furniture in the room 1 is same in case of the query and the rank 4 sample. Hence, similarity score between the query floor plan and the retrieved rank 4 layout is high, as compared to that with the rank 5 result. However, the similarity score of the rank 1-3 samples are marginally high as compared to rank 4 samples and hence they are ranked higher. In contrast, Fig. 4.9 (c) shows an instance where two out of top five retrieved results are incorrect. The rank 4 image has only three rooms, while the the rank 5 floor plan has four rooms in it. In this case, the framework is unable to capture the detailed features of the floor plans correctly. Even though, the number of rooms in a floor plan is different than the query, the framework ranks it higher.

6775 × 2858 至 2056 × 1837。所有平面图均被二值化处理，以确保在分析过程中仅使用结构信息。另一方面，ROBIN 数据集包含 510 张平面图，这些平面图根据整体布局形状被划分为三大类。

ROBIN数据集被划分为两部分：（a）训练集，包含30%的样本，用于深度特征表示；（b）测试集，包含剩余70%的样本，用作测试（查询）样本。作为检索数据库，使用了整个ROBIN数据集。为匹配任务和特征表示任务，每张平面图图像均被调整至512×512的尺寸。在深度模型配置方面，采用图4.8所示的架构，并沿用[Krizhevsky, 2012a]中所讨论的相同参数。对于SESYD数据集的特征表示，同样使用在ROBIN数据集上预训练的模型。

4.4.1 定性结果

所提出的框架能够利用所设计的深度学习模型成功地检索出一组按相似度排序的平面布局图像。图4.9展示了在ROBIN数据集上针对五个不同查询的前5位排序检索结果。图中下方的箭头表示从左至右相似度递减的顺序，其中绿色代表高相似度，红色代表低相似度。被错误检索的样本——即不属于与查询类别相同的类别——以“红色”突出显示。对于图4.9中所示的所有结果，均采用了从卷积神经网络堆栈的第二个归一化层提取的特征。选择该层的主要依据是其能够获得最佳性能。关于这一选择的详细分析将在下一小节中予以阐述。

从布局数据库中检索出的首个排序结果即为查询图像本身，这是由于该检索到的布局与查询布局之间的匹配得分最高所致。后续结果的匹配得分依次降低，因此排名靠后。以图4.9(a)为例，前5个样本均属于查询类别。所有平面图的整体布局完全相同，房间数量也一致；各平面图之间的差异主要体现在房间内家具的数量及其位置上。此外，读者还应注意不同样本中各房间的形状与位置差异。每个平面图均包含4个房间，但所提出的框架仍能正确地对样本进行排序，将与查询最为相似的样本排在最前面，其余样本则依次排列。

图4.9（b）和（c）展示了部分失败案例。在图4.9（b）中，框架错误地检索出了排名为第4位的结果（以红色标注），而排名第5位的样本则被正确检索。对该错误检索结果的合理性可通过定性分析加以解释：例如，查询布局中的房间1分别与房间2、3和4相邻，这一邻接关系与排名为第2位的结果中房间1的邻接关系完全一致；此外，房间1内的家具尺寸及类型在查询样本与排名为第4位的样本中也完全相同。因此，与排名为第5位的结果相比，查询平面图与排名为第4位的检索结果之间的相似度更高。然而，由于排名为第1至第3位的样本与排名为第4位的样本相比，其相似度仅略高，故这些样本被排在了更靠前的位置。相比之下，图4.9（c）展示了一个实例，其中前五名检索结果中有两个是错误的。排名为第4位的图像仅包含三个房间，而排名为第5位的平面图则包含四个房间。在这种情况下，该框架未能准确捕捉平面图的细节特征；尽管平面图中的房间数量与查询不一致，但框架仍将其排名靠前。

Layer	conv1	pool1	norm1	conv2	pool2	norm2	conv3	conv4	conv5	pool5	fc6	fc7
MAP	0.503	0.527	0.554	0.537	0.557	0.561	0.539	0.537	0.541	0.536	0.472	0.465

Table 4.2. : Performance Comparison (MAP) of CNN layers used for floor plan retrieval task (on ROBIN dataset).

4.4.2 Quantitative Results

To quantify the performance of the framework the Precision (P) and Recall (R) metric is used. The precision values are averaged over all the queries for the particular recall values. Given a query floor plan image, retrieved layouts should belong to the same sub-category of layouts as the query, keeping in mind the preference set by the property seeker during querying. Remember that the global shape of the layout is the criteria for a given floor plan to belong to a certain class. The PR plot is shown for only the proposed *ROBIN* Dataset, as for SESYD dataset, the proposed and some of the existing state-of-the-art techniques [Sharma *et al.*, 2016a,b; Dalal and Triggs, 2005] yielded a flat PR curve (Precision value 1 for all Recall values). The reason behind such a flat curve for SESYD dataset is the simplicity of the dataset.

To understand the effectiveness of the proposed CNN layers, the Mean Average Precision (MAP) values were calculated for floor plan retrieval task using each layer separately (see Fig. 4.10 and Tab. 4.2). From Fig. 4.10 and Tab. 4.2, it can be concluded that: (1) it is not always true that the final layer i.e. the fully connected layer will always work best; and (2) normalization and pooling layers are more powerful as compared to convolutional layer for floor plan retrieval task.

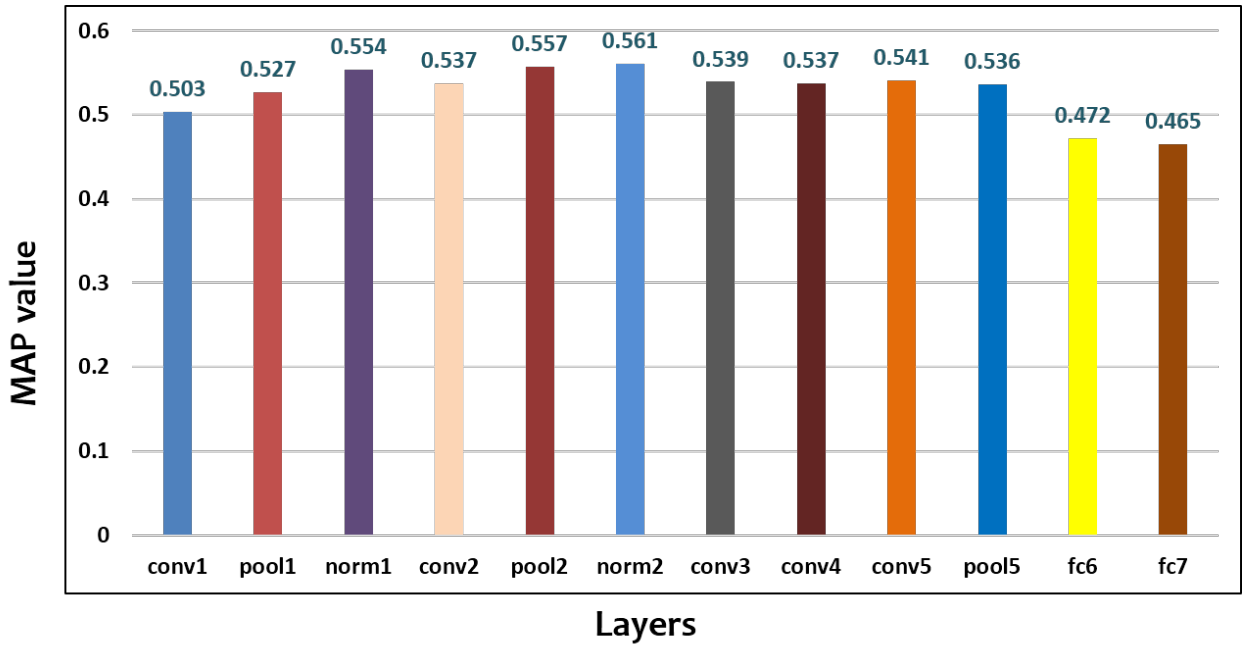


Figure 4.10. : Mean Average Precision values, comparing result of the framework on ROBIN dataset, using features from all the hidden layers of the CNN.

To show the effectiveness of the proposed framework, the technique proposed in this chapter

Layer	conv1	pool1	norm1	conv2	pool2	norm2	conv3	conv4	conv5	pool5	fc6	fc7
MAP	0.503	0.527	0.554	0.537	0.557	0.561	0.539	0.537	0.541	0.536	0.472	0.465

表4.2：用于楼层平面图检索任务的CNN各层性能比较（基于ROBIN数据集，以平均精度MAP为指标）。

4.4.2 定量结果

为量化该框架的性能，采用精确率（P）和召回率（R）指标。对于特定的召回率值，精确率取所有查询的平均值。给定一张查询的平面图，检索出的布局应与查询属于同一子类别，同时需考虑房产需求者在查询时设定的偏好。需要注意的是，布局的整体形状是判定某平面图归属某一类别的主要依据。PR曲线仅针对所提出的*ROBIN*数据集绘制；而对于SESYD数据集，所提出的方法以及部分现有的最先进方法[Sharma *et al.*, 2016a,b；Dalal和Triggs, 2005]均产生了平坦的PR曲线（即在所有召回率下精确率为1）。SESYD数据集呈现如此平坦曲线的原因在于其数据集本身的简单性。

为评估所提出的卷积神经网络各层的有效性，针对平面图检索任务，分别对每一层计算了平均精度均值（MAP）指标（见图4.10及表4.2）。由图4.10和表4.2可知：（1）并非总是最后一层，即全连接层的效果最佳；（2）与卷积层相比，归一化层和池化层在平面图检索任务中表现更为优越。

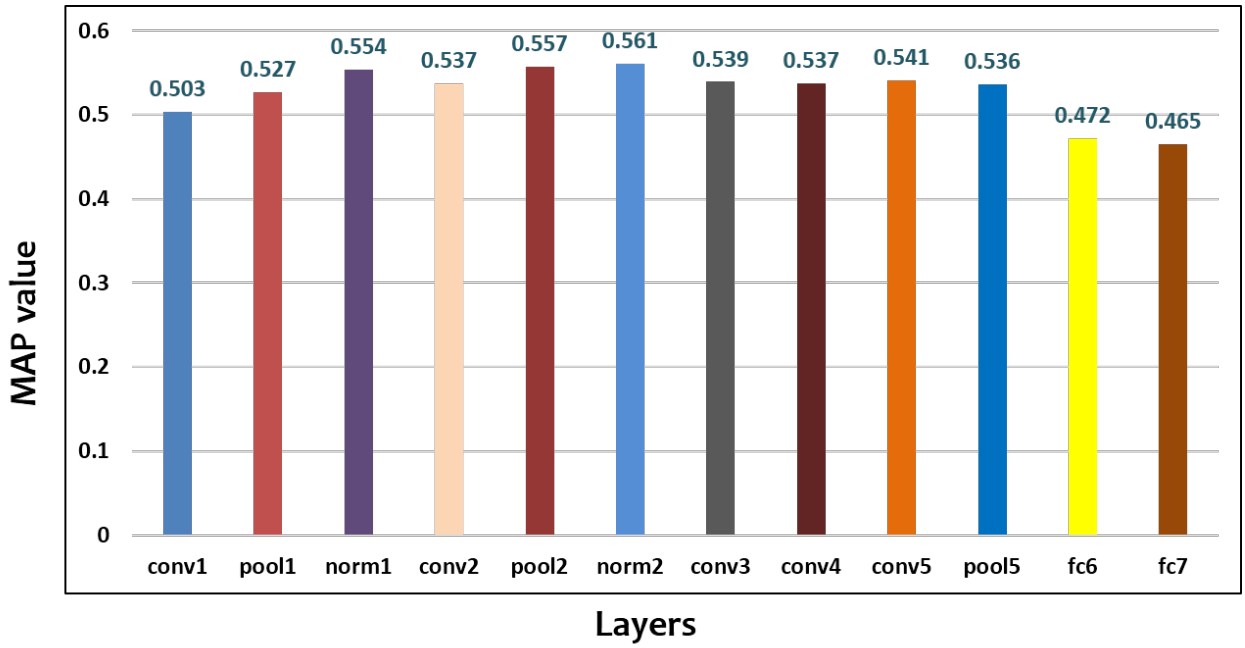


图4.10：平均精度均值（mAP）结果，比较了在ROBIN数据集上采用CNN所有隐藏层特征时该框架的性能。

为验证所提出框架的有效性，本章所提出的这一技术

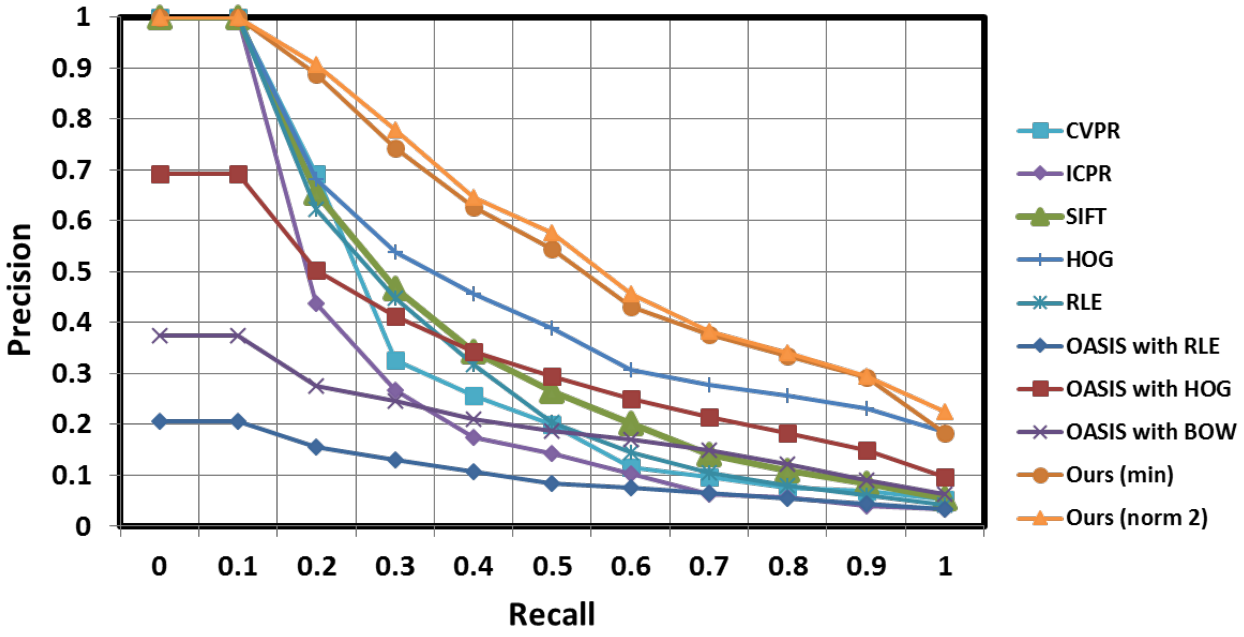


Figure 4.11. : Precision and Recall (PR) plot, comparing state-of-art methods with the proposed framework on ROBIN dataset.

is also compared with several state-of-the-art (SOA) techniques. Figure 4.11 depicts the PR plot comparing result of the framework on ROBIN dataset, with the other SOA techniques. It can be observed that the framework with features extracted from the Normalized layer 2 (see Fig. 4.10) is giving the best retrieval performance. It can also be noted from Fig. 4.11 that the proposed min based approach (which selects best layer for each retrieval sample, refer Eq. 4.9) is also outperforming all the state-of-the-art techniques. There is a significant improvement in the performance while using deep features as compared to other features or techniques.

As shown in Fig. 4.9 (a), given a query sample the rank 1 result is the query floor plan itself. This leads to the highest precision value of 1 during the initial recall. With further retrieval the average precision value decreases due to some incorrectly retrieved results belonging to other categories of layouts as compared to the query layout. It is to be noted that area under the PR curve is the highest for proposed deep learning based retrieval framework, as compared to other techniques applied for floor plan retrieval task as can be noted through the orange marker on the PR plot (Fig. 4.11).

The SOA techniques with which the proposed approach is compared with are standard image retrieval techniques that use hand-crafted features such as HOG [Dalal and Triggs, 2005], SIFT [Lazebnik *et al.*, 2006a], Run-Length Histogram and Online Algorithm for Scalable Image Similarity (OASIS) [Chechik *et al.*, 2009]) but are not able to capture well the complete content of the floor plan images. Also, reason behind the under performance of OASIS [Chechik *et al.*, 2009] is that, it focuses on the difference of similarity values between relevant and irrelevant image pairs, while ignoring the similarity value between highly similar images. Hence, when the same features (HOG, SIFT, RLH) are used under a canonical CBIR paradigm, better results are obtained in comparison to using them in conjuction with OASIS (see Tab. 4.3).

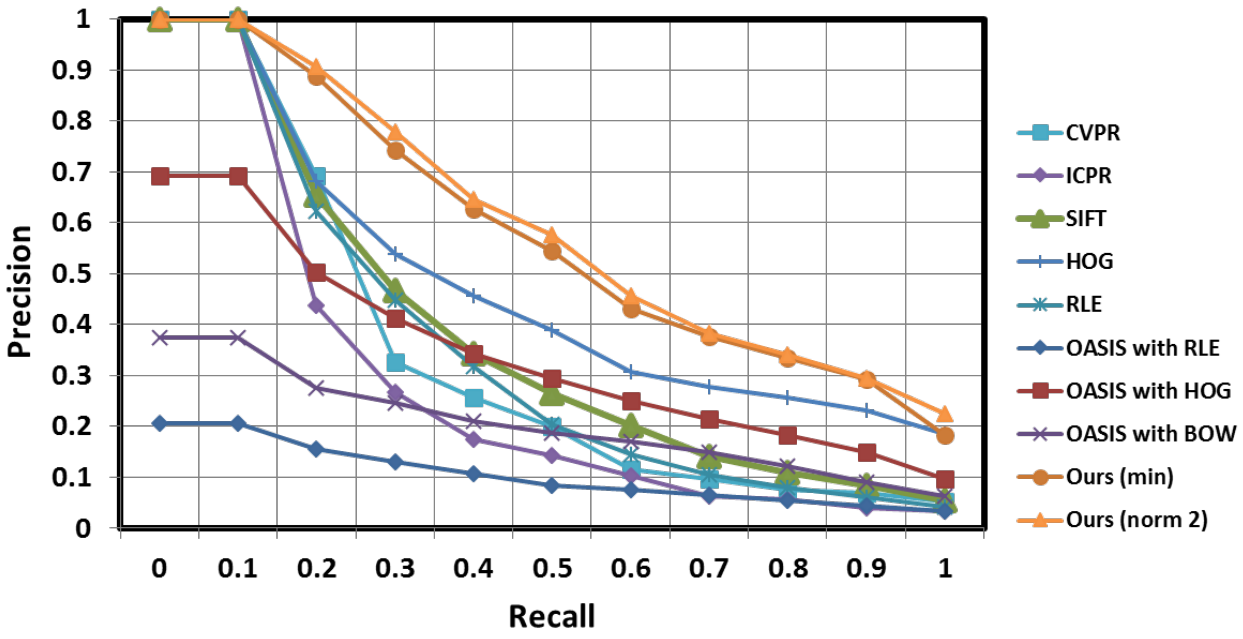


图4.11：精确率-召回率（PR）曲线，对比了ROBIN数据集上现有最先进方法与所提出的框架。

该方法还与若干当前最先进的（SOA）技术进行了对比。图4.11展示了在ROBIN数据集上，本框架与其他SOA技术的PR曲线比较结果。可以观察到，采用归一化层2所提取特征的框架取得了最佳的检索性能。此外，从图4.11还可以看出，所提出的基于最小值的策略（即为每个检索样本选择最优层，参见式4.9）同样优于所有当前最先进的技术。与使用其他特征或技术相比，采用深度特征显著提升了检索性能。

如图4.9(a)所示，对于给定的查询样本，排名为1的结果即为该查询的平面布局本身。这使得在初始召回阶段的精确率值达到最高。随着检索过程的深入，由于部分检索结果属于与查询布局不同的类别，平均精确率随之下降。值得注意的是，与用于平面布局检索任务的其他方法相比，所提出的基于深度学习的检索框架在PR曲线下面积最大，这一点可通过PR曲线图（图4.11）中的橙色标记加以验证。

与所提方法进行比较的SOA技术均为基于手工设计特征的标准图像检索方法，如HOG [Dalal和Triggs，2005]、SIFT [Lazebnik等，2006a]、游程长度直方图以及可扩展图像相似性在线算法（OASIS）[Chechik等，2009]，但这些方法难以充分捕捉平面布局图像的完整内容。此外，OASIS [Chechik等，2009]性能欠佳的原因在于，其仅关注相关与不相关图像对之间相似度值的差异，而忽略了高度相似图像之间的相似度本身。因此，在经典的基于内容的图像检索（CBIR）范式下使用相同的特征（HOG、SIFT、RLH），其效果优于将其与OASIS结合使用（参见表4.3）。

Techniques	**MAP**
CVPR [Sharma *et al.*, 2016b]	0.29
ICPR [Sharma *et al.*, 2016a]	0.23
HOG [Dalal and Triggs, 2005] based CBIR	0.40
SIFT [Lazebnik *et al.*, 2006a] based CBIR	0.33
RLH based CBIR	0.30
OASIS [Chechik *et al.*, 2009] with HOG	0.31
OASIS [Chechik *et al.*, 2009] with BOW	0.19
OASIS [Chechik *et al.*, 2009] with RLH	0.01
Ours (min)	0.54
Ours (norm2)	**0.56**

Table 4.3. : Performance Comparison in terms of MAP on *ROBIN* dataset.

4.5 SUMMARY

In this Chapter a deep learning framework is proposed to extract semantic features from floor plans and those features are used for matching and retrieval of similar floor plan images. Extensive experimentation was conducted and performance was compared with eight different state-of-the-art methods to demonstrate the edge that the deep learning framework has, as compared to the hand-crafted feature based retrieval. Although, deep learning is convenient and bypasses the need of curating features manually, the issue lies in the fact that the layers in the deep learning framework extract features implicitly. This implicit feature extraction at each layer does not give the user, the ability to give priorities to certain aspects or features while retrieval. For example, if a user wants to give higher preference to the feature "size of the rooms" while retrieval, then in the deep learning framework there is no such provision of giving additional weights to a particular feature. This, makes the application of the framework a little less intuitive. Hence, in Chapter 5, high level semantic analysis of floor plans and their subsequent retrieval is proposed. Additionally, a provision to adjust the weights of each feature while retrieval is given in the proposed technique. Such a provision can prove to be more practical to users while looking for similar floor plans sufficing their specific preferences.

Techniques	**MAP**
CVPR [Sharma *et al.*, 2016b]	0.29
ICPR [Sharma *et al.*, 2016a]	0.23
HOG [Dalal and Triggs, 2005] based CBIR	0.40
SIFT [Lazebnik *et al.*, 2006a] based CBIR	0.33
RLH based CBIR	0.30
OASIS [Chechik *et al.*, 2009] with HOG	0.31
OASIS [Chechik *et al.*, 2009] with BOW	0.19
OASIS [Chechik *et al.*, 2009] with RLH	0.01
Ours (min)	0.54
Ours (norm2)	**0.56**

表4.3：在ROBIN数据集上以平均精度（MAP）为指标的性能比较。

4.5 摘要

在本章中，我们提出了一种深度学习框架，用于从平面图中提取语义特征，并利用这些特征实现相似平面图图像的匹配与检索。我们开展了大量实验，并将该框架的性能与八种不同的最新方法进行了对比，以证明相较于基于手工设计特征的检索方法，该深度学习框架所具有的优势。尽管深度学习具有便捷性，且无需手动构建特征，但其问题在于：深度学习框架中的各层以隐式方式提取特征。这种逐层进行的隐式特征提取使得用户在检索时无法对特定方面或特征赋予优先级。例如，若用户希望在检索过程中对“房间尺寸”这一特征给予更高的权重，则在现有的深度学习框架中并不存在为某一特定特征分配额外权重的机制。这在一定程度上降低了该框架的应用直观性。因此，在第5章中，我们提出了面向平面图的高层次语义分析及其后续检索方法，并进一步在所提技术中增加了在检索过程中调整各特征权重的功能。这一功能在用户根据自身特定偏好搜索相似平面图时，将更具实用性。

5

Requirement driven feature integration for floor plan retrieval

In Chapter 4, a deep learning architecture for floor plan retrieval is proposed which was the first attempt of applying deep learning to the task of floor plan retrieval. A naive approach was introduced to learn a deep learning feature representation and study how the individual deep feature layers affect the performance of the system during retrieval. In the deep learning framework, a combination of convolutional, pooling, normalization, ReLU and fully connected layers in the Convolutional Neural Network (CNN) framework were used to obtain an effective feature representation from floor plan images. Quantitatively this framework gave encouraging results with a mean average precision of 0.56. However, certain key issues were observed during the retrieval process:

- The CNN framework is a simple network that mainly distinguishes by the global shape of a floor plan. It would fail while doing fine grain categorization of furniture components in the layouts. It would also not perform well if two layouts from different classes have almost similar shapes.

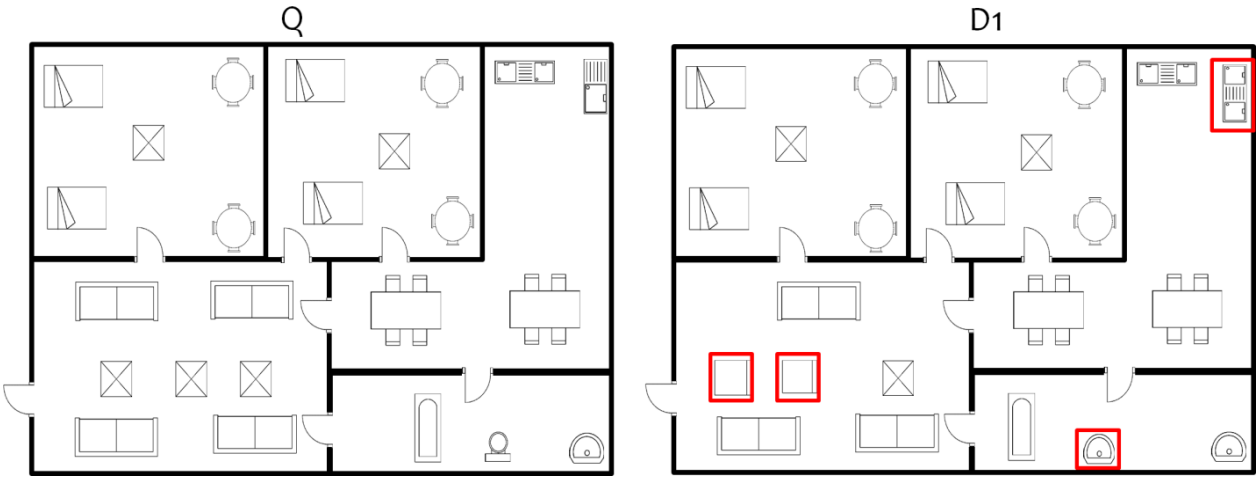


Figure 5.1. : Q depicts the query floor plan and D1 depicts a resultant retrieved floor plan from the database of floor plans. The objects enclosed in red boxes in D1 are furniture components with similar shape as the components in Q, but different types.

An example in support of the above argument is given in Fig. 5.1. Here, the shape of the square table and the small sofa (marked in red in D1) are quite similar, therefore, while retrieval through the deep framework, query floor plan Q and database floor plan D1 are deemed similar and D1 is ranked higher in the retrieved results, in spite of their dissimilarity. Moreover, the layout sub-categories in the ROBIN dataset are mainly proposed to be same in the global layout shape, which justifies the higher mean average precision value using DANIEL framework, but does not result into efficient qualitative results.

5

面向平面图检索的需求驱动特征集成

在第4章中，提出了一种用于平面图检索的深度学习架构，这是首次将深度学习应用于平面图检索任务的尝试。文中介绍了一种朴素的方法，用于学习深度特征表示，并研究各层深度特征对检索系统性能的影响。在该深度学习框架中，通过在卷积神经网络（CNN）中组合使用卷积层、池化层、归一化层、ReLU激活函数以及全连接层，从平面图图像中提取有效的特征表示。定量评估结果显示，该框架取得了令人鼓舞的性能，平均精度均值达到0.56。然而，在检索过程中也观察到了一些关键问题：

- 卷积神经网络框架是一种结构较为简单的网络，主要依据平面布局的整体形状进行区分。在对布局中的家具组件进行细粒度分类时，该框架会表现不佳；此外，当来自不同类别但形状几乎相似的两个布局出现时，其性能也会受到影响。

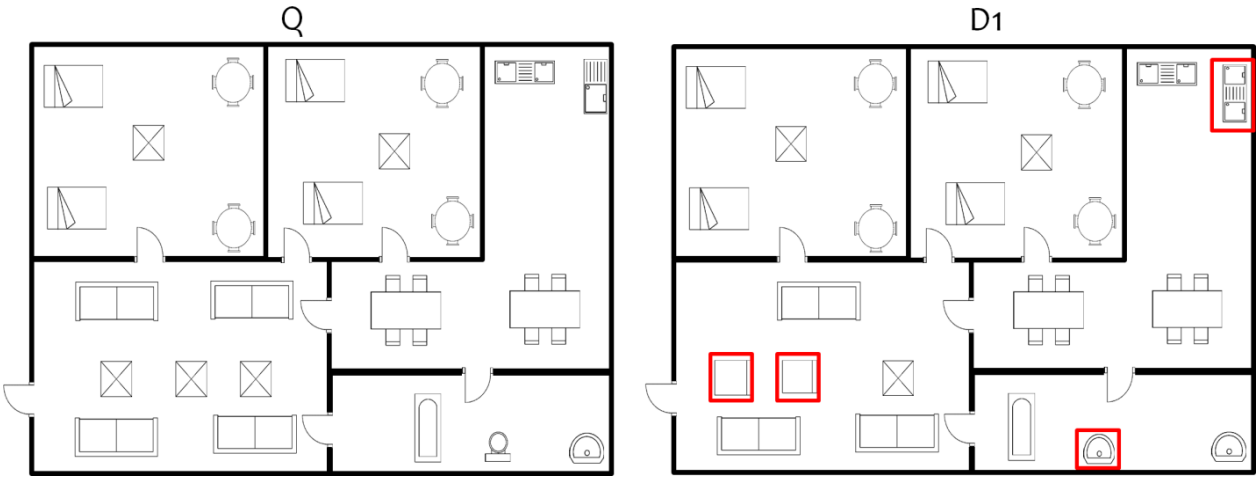


图5.1：Q表示查询平面图，D1表示从平面图数据库中检索出的相应平面图。D1中被红色框圈出的物体为家具构件，其形状与Q中的构件相似，但类型不同。

支持上述论点的一个示例如图5.1所示。在该示例中，方形餐桌与小型沙发的形状（在D1中以红色标注）颇为相似，因此，在基于深度学习框架的检索过程中，查询平面图Q与数据库平面图D1被判定为相似，导致D1在检索结果中排名靠前，尽管二者实际上存在显著差异。此外，ROBIN数据集中各布局子类别的划分主要依据全局布局形状的一致性，这虽然使得DANIEL框架的平均精度均值较高，却未能带来理想的定性检索效果。

The other trait where the deep framework lacks is that there is no provision of a weighted combination of the features, which does not let the user give preferences to individual characteristics, according to the requirement. Choices cannot be set using the proposed deep framework, as the layers extract and combine the features implicitly without the provision of manipulating the weights by the user.

The hand-crafted approach, with non-deep features proposed in Chapter 3 was the baseline, and first of its kind framework for retrieval in floor plans. However, it had limitations like (i) the matching was done in two phase, i.e. layout matching followed by decor matching, and thus not a unified/ combined approach, (ii) a few key features were overlooked that define a buyer's requirement, e.g., room dimensions were not considered. In this Chapter, a framework for "fine-grained" retrieval of floor plan images from the repository is proposed.

The key contributions of this work are:

1. A novel end-to-end framework for extracting high level semantic features like the area and room-wise decor arrangement for the task of fine-grained retrieval

2. A technique to perform feature fusion to aggregate high-level semantic features to retrieve floor plans.

3. Extensive experimentation on a large dataset to demonstrate robustness and scalability. The technique proposed in this Chapter can find application in online property sale/rent scenarios where the buyer has preferred features related to the room semantics.

The Chapter is organized as follows: Sec. 5.1 gives a brief overview of the proposed framework. The fine-grained feature extraction process is presented in Sec. 5.2. The matching algorithm is described in Sec. 5.3. Details of the experimental findings are discussed in Sec. 5.4, followed by critical observations in Sec. 5.5. Section 5.6 summarizes this Chapter and leads the thesis towards the final approach.

5.1 BRIEF OVERVIEW

Figure 5.2 depicts the complete framework of the proposed technique. The entire framework has two phases: (i) feature extraction (solid line) and (ii) retrieval (dotted line). Structural analysis of a floor plan is performed. For this purpose, the layouts are segmented into rooms by identifying the walls separating all the rooms. Then, in a floor plan, the high-level semantic features like adjacencies between rooms, carpet area of rooms [Bannister, 2010] and also the number and type of furniture inside each room, termed as the Furniture Composition Record are extracted. During the retrieval phase, these features are extracted from a query floor plan and a weighted sum is computed of all the features extracted to generate a matching score for a particular layout as compared to another layout in the repository. By default, all the weights while matching are set to 1 to favor an equal sum of all the high-level semantic features. This score then serves as the parameter for fine-grained retrieval of similar floor plans. Additionally, any feature can be given a higher weight as compared to the other features during the matching process which further, aids in preferring a feature more during retrieval. Among the publicly available floor plan datasets, the *ROBIN* [Sharma *et al.*, 2017] dataset is chosen to validate the technique shown in this Chapter.

深度框架的另一项不足在于，其未提供特征的加权组合机制，因此用户无法根据具体需求对各特征赋予不同的权重。此外，由于各层在提取和组合特征时均以隐式方式进行，且不支持用户对权重进行手动调整，因此无法通过所提出的深度框架设置选择策略。

第3章中提出的基于手工特征的非深度学习方法作为基准，是首个用于平面图检索的框架。然而，该方法存在一些局限性：（i）匹配过程分为两个阶段，即先进行布局匹配，再进行装饰匹配，因此并非统一或联合的处理方式；（ii）遗漏了一些能够表征用户需求的关键特征，例如未考虑房间尺寸。在本章中，我们提出了一种从数据库中实现平面图图像“细粒度”检索的框架。

本工作的主要贡献如下：

1. 一种用于细粒度检索任务的新型端到端框架，能够提取诸如区域及房间级装饰布局等高层次语义特征。

2. 一种用于执行特征融合的技术，以聚合高层语义特征来检索建筑平面图。

3. 在大规模数据集上开展了大量实验，以验证其鲁棒性和可扩展性。本章所提出的技术可应用于在线房产买卖或租赁场景，其中买方对房间语义相关特征具有偏好。

本章的组织结构如下：第5.1节对所提出的框架进行简要概述；第5.2节介绍细粒度特征提取过程；第5.3节阐述匹配算法；第5.4节详细讨论实验结果；随后，第5.5节给出若干关键性观察；第5.6节对本章进行总结，并将论文引向最终方案。

5.1 简要概述

图5.2展示了所提出方法的完整框架。整个框架分为两个阶段：（i）特征提取（实线）和（ii）检索（虚线）。首先对平面布局进行结构分析，为此通过识别分隔各房间的墙体，将布局划分为若干房间。随后，在平面布局中提取高层语义特征，包括房间之间的邻接关系、房间的地毯面积[Bannister, 2010]，以及每个房间内的家具数量与类型，并将其统称为“家具构成记录”。在检索阶段，从查询平面布局中提取上述特征，并对所有提取的特征按权重求和，以计算某一布局相对于知识库中另一布局的匹配得分。默认情况下，所有匹配权重均设为1，以确保各高层语义特征的贡献等同。该得分 subsequently 用作细粒度相似平面布局检索的判据。此外，在匹配过程中，可为任一特征赋予高于其他特征的权重，从而在检索时进一步突出该特征的重要性。在公开可用的平面布局数据集中，本章所展示的方法选用$ROBIN$ [Sharma $et\ al.$, 2017] 数据集进行验证。

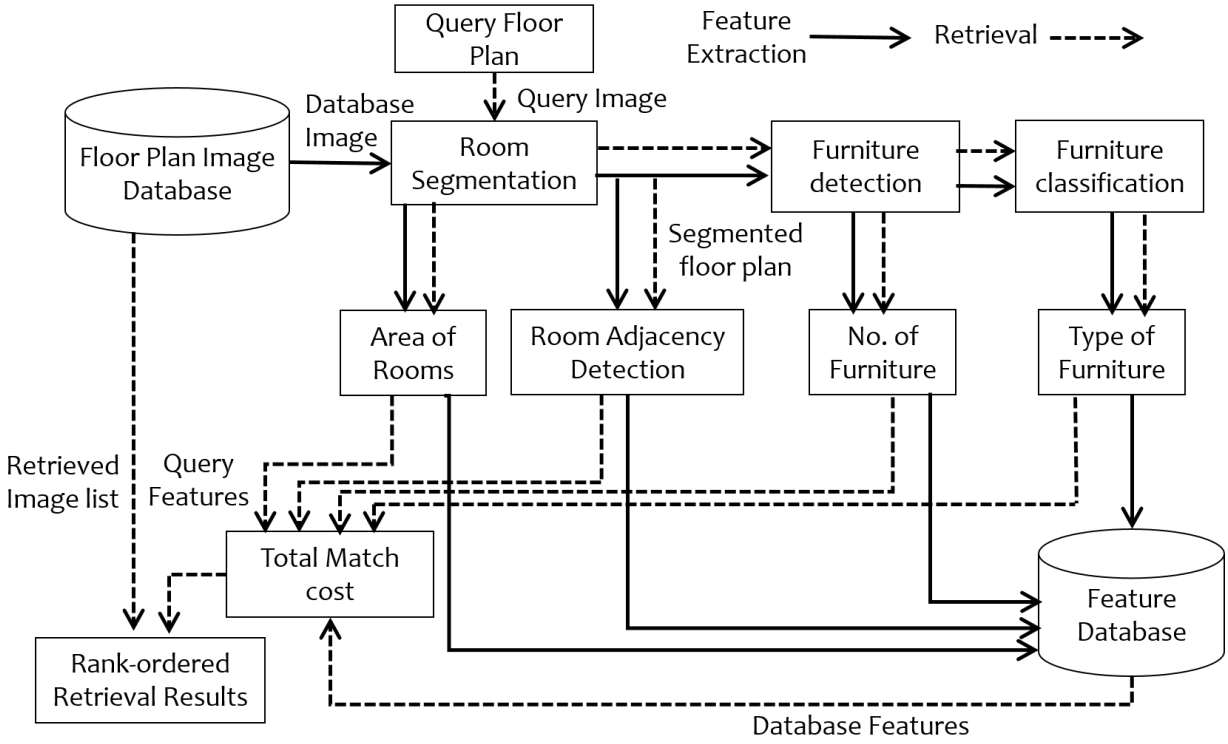


Figure 5.2. : Basic framework depicting extraction of high-level semantic features and weighted fusion for retrieval of floor plans.

5.2 HIGH LEVEL SEMANTIC FEATURES

As a preprocessing step, a morphological closing operation is carried out to segregate the layout boundary from furniture present inside the layout. The doors inside the plan are present in various orientations. They are detected and the gaps at their locations are closed to construct definite boundaries for the rooms. This recognizes the walls in the layout to represent the room structures and thus, delineates the rooms within the layout from each other. Next, connected component analysis using 8-neighborhood criteria is performed to label each room in the layout separately. The rooms in the layout are numbered to get a consistent ordering by their distance from a reference point situated at the top-left-most end of the floor plan image. A diagram showing the same is presented in Fig. 5.4.

Euclidean distance between the reference point taken to be as the top-leftmost point of the floor plan canvas and the centroid of each room is calculated. The closest room to the reference point, i.e., with the minimum distance to the reference point is labeled 1, and other rooms are labeled accordingly. Such an ordering makes the system rotation sensitive. Figure 5.3 (b) and (d) depicts the resultant room layout segmentation and labeling given a floor plan. Such an ordering helps in establishing a better correspondence between rooms while feature extraction and matching. The three features that are being extracted namely, Room Adjacency String, Carpet Area Ratio and Furniture Composition Record are described below.

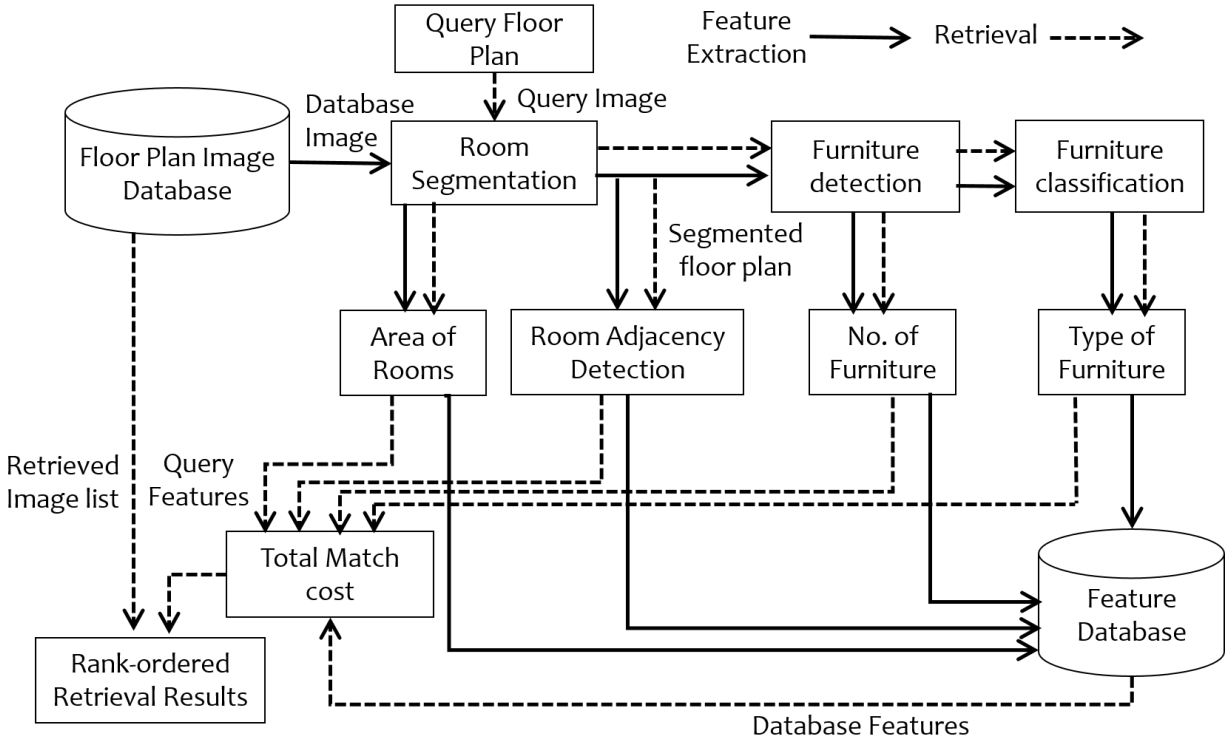


图5.2：用于提取高层语义特征并进行加权融合以实现平面图检索的基本框架。

5.2 高层次语义特征

作为预处理步骤，首先执行形态学闭运算，以将布局边界与布局内部的家具分离开来。平面图中的门具有多种朝向，通过检测这些门并封闭其所在位置的缝隙，从而为各房间构建明确的边界。这一过程能够识别布局中的墙体，以表征房间的结构，并由此在布局中区分各个房间。接下来，采用基于8邻域准则的连通组件分析，对布局中的每个房间分别进行标记。根据各房间与位于平面图左上角的参考点之间的距离，对布局中的房间进行编号，以获得一致的排序。相应的示意图如图5.4所示。

计算以平面图画布的左上角点为参考点与各房间质心之间的欧氏距离。将距离参考点最近的房间（即距离最小的房间）标记为1，其余房间依序编号。这种排序方式使系统对旋转较为敏感。图5.3（b）和（d）展示了给定平面图时所得的房间布局分割与标注结果。该排序有助于在特征提取与匹配过程中建立更为准确的房间对应关系。下文将分别介绍所提取的三项特征：房间邻接字符串、地毯面积比以及家具构成记录。

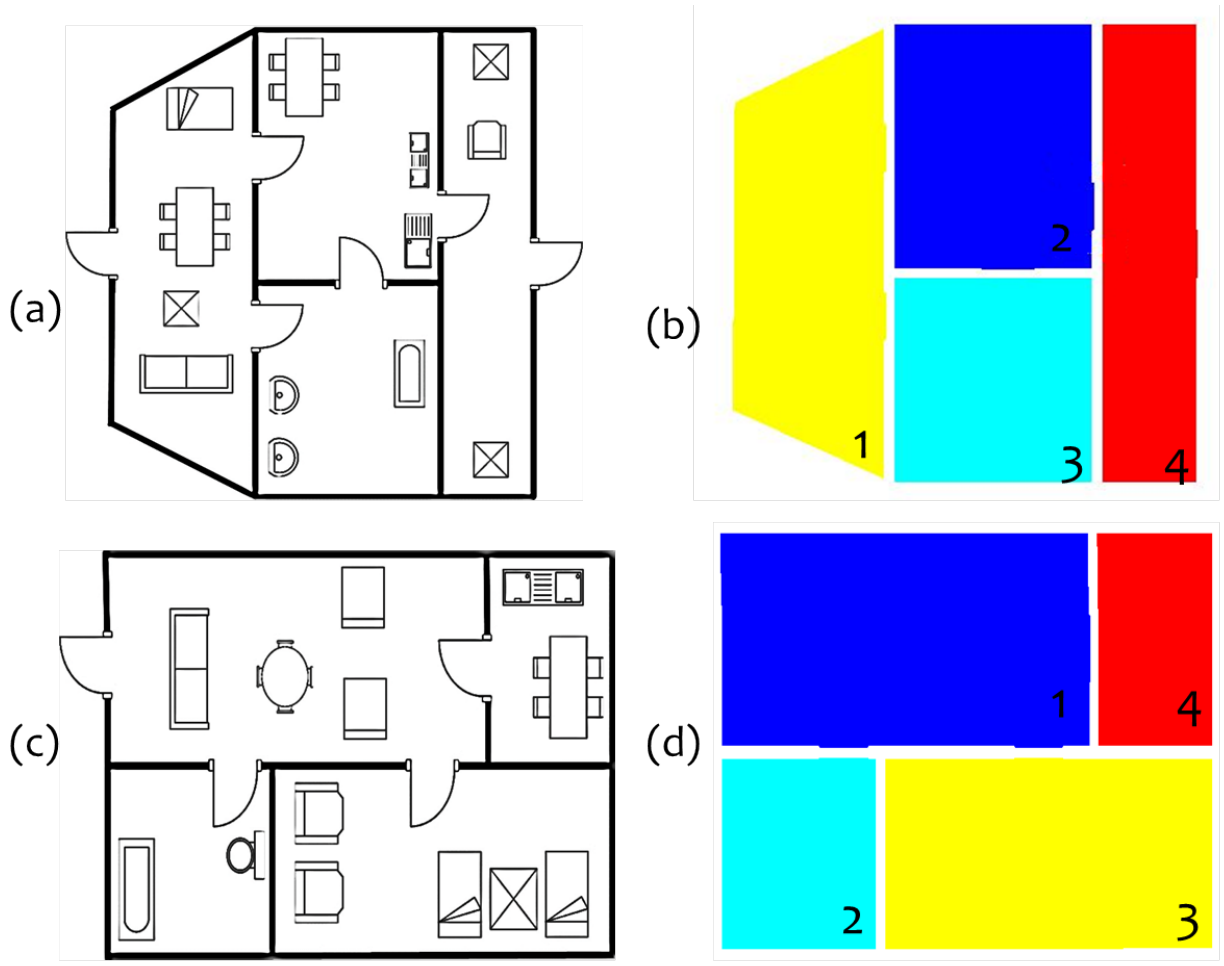


Figure 5.3. : Room level segmentation results on a pair of floor plan images from ROBIN dataset.

5.2.1 Feature 1: Room Adjacency String (RAS)

Room segmenting and labeling is followed by determining the adjacencies between the rooms to mark which rooms share boundaries in a layout using the Room Adjacency String (RAS) feature. This feature can be a preference if buyers are keen to choose the arrangement of rooms in a layout, e.g., *the master bedroom should be near the kids' bedroom.* For this purpose, an adjacency matrix is constructed from the segmented and order-wise labeled layout. For k^{th} floor plan image in the database its topological graph is denoted as $G_k = (V_k, E_k)$, where V_k is the set of vertices also denoting the number of rooms in the layout and E_k is the edge-set as depicted in Fig.5.5 (b). The adjacency matrix, for the graph G_k is denoted as Adj_k which is a $|V_k| \times |V_k|$ matrix, where, $|\cdot|$ denotes cardinality of the set V_k. For each row of the adjacency matrix corresponding to each room, a unique binary string is computed, which represents the adjacencies for that particular room. E.g. the room adjacency of room 2 in the segmented layout in Fig. 5.5 (Example 1) (a) is highlighted in Fig. 5.5 (Example 1) (c). RAS of 1011 depicts Room 2 is adjacent with Rooms 1, 3 and 4.

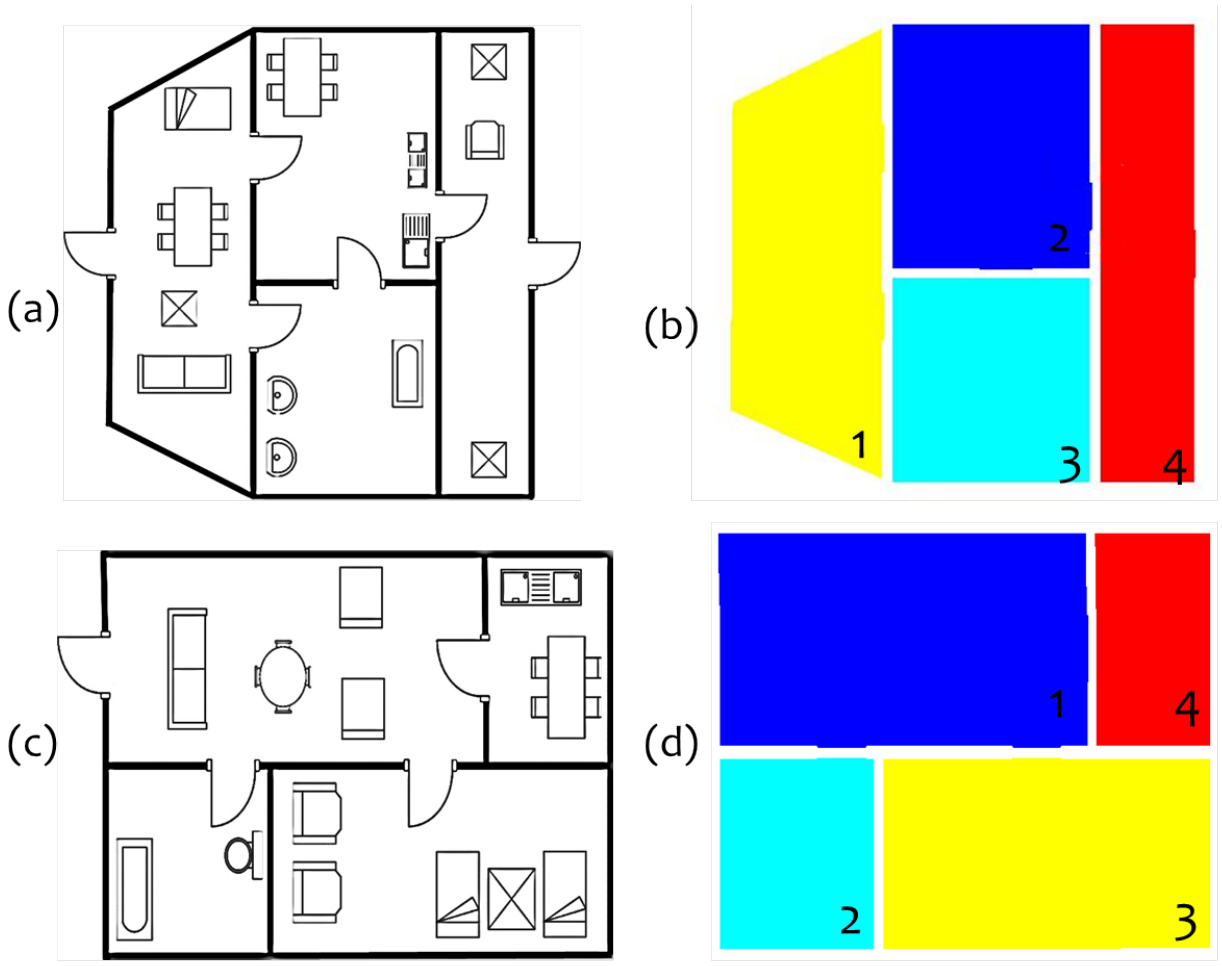


图5.3：房间　　勒在一对平面布局图上的分割结果　　来自ROBIN数据集。

5.2.1 特征1：房间邻接字符串（RAS）

在进行房间分割与标注之后，通过“房间邻接字符串”（RAS）特征确定各房间之间的邻接关系，从而标识出布局中哪些房间共用边界。这一特征可作为一项偏好选项，供购房者根据需求选择布局中的房间排列方式，例如：*the master bedroom should be near the kids' bedroom.*。为此，基于已分割并按顺序标注的布局构建邻接矩阵。对于数据库中的每一张平面图图像，其拓扑图记为k^{th}，其中$G_k = (V_k, E_k)$为顶点集，同时也表示布局中的房间数量；V_k为边集，如图5.5(b)所示。该图的邻接矩阵记为E_k，它是一个G_k阶矩阵，其中Adj_k表示集合$|V_k| \times |V_k|$的基数。针对邻接矩阵中每一行所对应的每个房间，计算出一个唯一的二进制字符串，用于表征该房间的邻接关系。例如，在图5.5（示例1）(a)所示的分割布局中，房间2的邻接关系已在图5.5（示例1）(c)中予以突出显示。其RAS值为1011，表明房间2分别与房间1、3和4相邻。

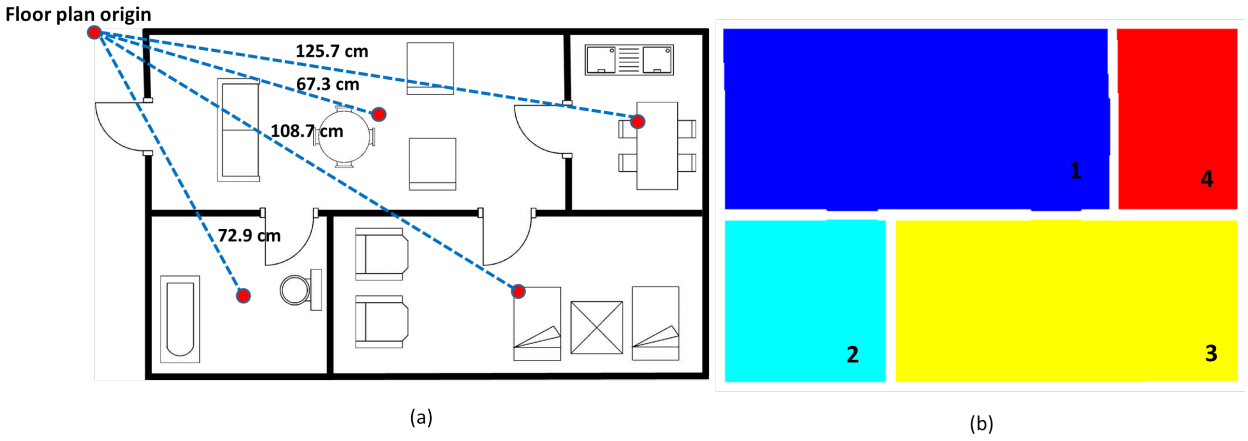


Figure 5.4. : Figure depicting the ordered labeling of rooms after segmentation. (a) Distance is calculated from a leftmost origin point up to all room centroids. Room with the lowest distance from the origin is ordered number 1 and the other rooms are labeled in similar fashion. (b) Subsequent room ordering results.

5.2.2 Feature 2: Carpet Area Ratio (CAR)

CAR is a unique feature that represents the ratios of carpet area enclosed within the walls i.e. the actual area to lay the carpet. This area does not include the thickness of the inner walls. The carpet area (Ar) of a room (L^k) is calculated using the formula for calculation of area of a polygon as specified in [Braden, 1986]:

$$Ar(L^k) = \left| \frac{(x_1y_2 - y_1x_2) + (x_2y_3 - y_2x_3) + \cdots + (x_ny_1 - y_nx_1)}{2} \right| \tag{5.1}$$

where, the pair $(x_i, y_i); \forall i = 1, \cdots, n$ denotes the n vertices of a room, taken in clockwise order, and the operator $|.|$ denotes the absolute value of the function. The motivation behind extracting this feature is the fact that usually users buying layouts have preferences in terms of the size of each room in the floor plan.

CAR ($A(\cdot)$) serves as a unique feature while a user is looking for layouts. The ordered rooms are labeled and the area between their boundaries and their respective ratios is stored which serves as the CAR. The area of a particular room (L^k) is then divided by the total area of the layout (L) to give an idea of the size of the particular room corresponding to the whole layout (refer Eq. 5.2). $Ar(L)$ is also calculated using Eq. 5.1, where (x_i, y_i) are the vertices of the polygon obtained by tracing the global layout of L.

$$A(L^k) = \frac{Ar(L^k)}{Ar(L)} \tag{5.2}$$

Figure 5.6 (b) and (d) show the areas corresponding to the different rooms in the layout shown in Fig. 5.6 (a) and (c) in the form of a pie-chart. Here, the inner entries are the actual areas of the rooms and the outer entries, represent the CAR feature of a room.

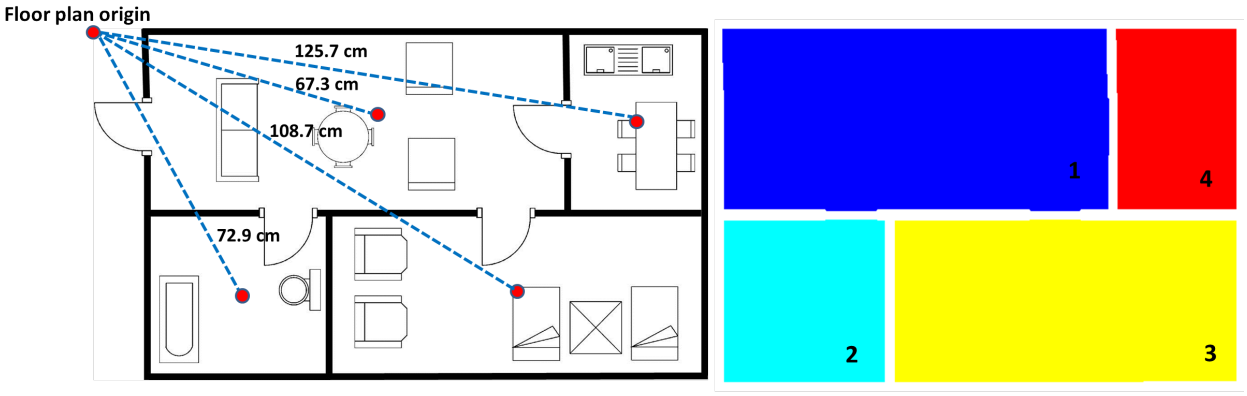


图5.4：展示分割后房间有序编号的示意图。（a）从最左侧的原点出发，计算至所有房间质心的距离。距原点距离最小的房间被编号为1，其余房间依此类推进行编号。（b）后续的房间编号结果。

5.2.2 特征2：地毯面积比（CAR）

CAR 是一项独特指标，用于表征由墙体围合的地毯铺设面积，即实际可铺设地毯的面积。该面积不包含内墙的厚度。房间（L^k）的地毯面积（Ar）按照 [Braden, 1986] 中所规定的多边形面积计算公式进行计算：

$$Ar(L^k) = \left| \frac{(x_1y_2 - y_1x_2) + (x_2y_3 - y_2x_3) + \cdots + (x_ny_1 - y_nx_1)}{2} \right| \tag{5.1}$$

其中，有序对 $x_i, y_i; \forall i = 1, \cdots, n$ 表示按顺时针方向排列的房间顶点，算子 $|.|$ 表示函数的绝对值。提取该特征的动机在于：通常，购买户型图的用户对平面图中各房间的尺寸存在偏好。

在用户寻找布局时，CAR（$A(\cdot)$）作为一种独特特征发挥作用。有序排列的房间被赋予标签，并存储其边界范围及其与整体布局的面积比，该比值即为CAR。随后，将某一特定房间的面积（L^k）除以整个布局的总面积（L），以反映该房间相对于整个布局的相对大小（参见式5.2）。$Ar(L)$ 亦通过式5.1计算得到，其中（x_i, y_i）为沿L全局布局轮廓线所获取的多边形顶点。

$$A(L^k) = \frac{Ar(L^k)}{Ar(L)} \tag{5.2}$$

图5.6（b）和（d）以饼状图的形式展示了与图5.6（a）和（c）中所示布局中不同房间相对应的面积。其中，内圈数值为各房间的实际面积，外圈数值则表示该房间的CAR特征值。

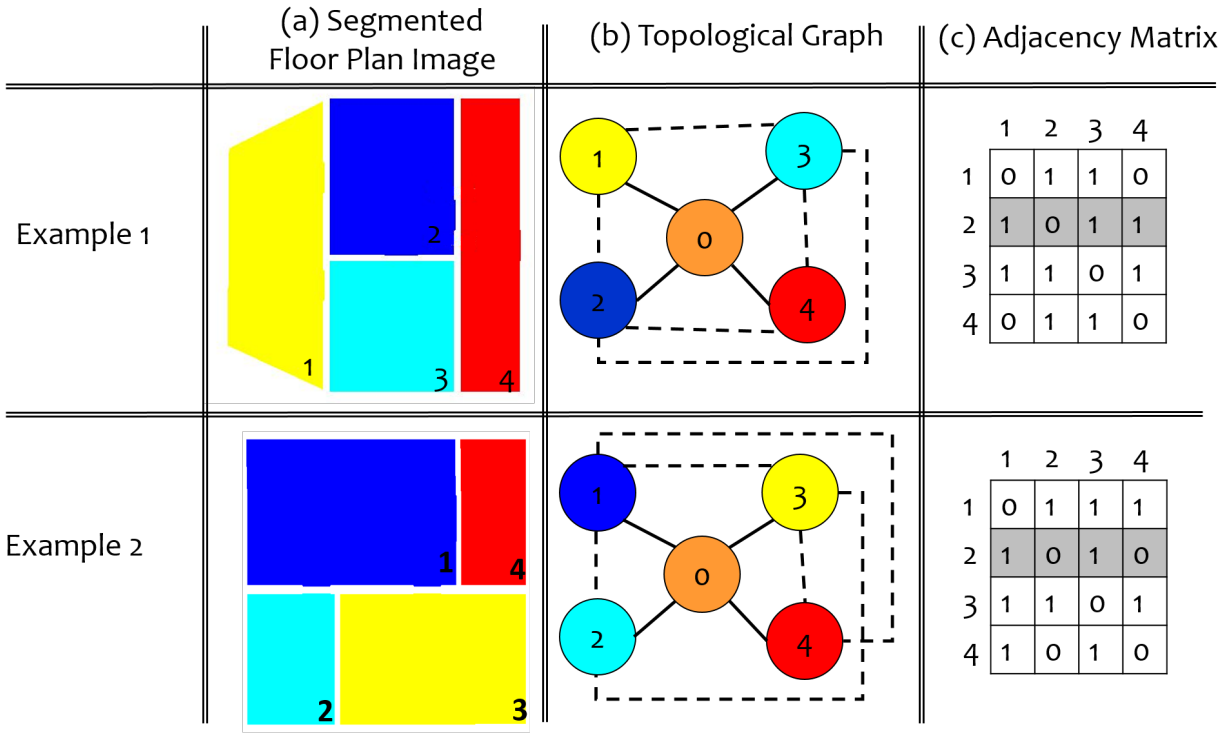


Figure 5.5. : Example of Room Adjacency String (RAS) feature: (a) Segmented rooms (b) Topological Graph (c) Adjacency matrix (Adjacency string corresponding to Room 2 is highlighted).

5.2.3 Feature 3: Furniture Composition Record (FCR)

The third feature that is computed is the Furniture Composition Record (FCR) that comprises of number and type of various furniture inside a particular room. A furniture is localized inside a room according to their position in the layout, by detecting the blobs inside the floor plan image. Further, their count (C) is stored corresponding to each room. This helps to compare the rooms according to the consumers priority of being overcrowded with decor or not. Next, the various furniture are classified by performing a connected component analysis on the detected blobs and taking the area of individual components inside the different furniture symbols. Then a ratio of the largest 3 area components is taken, and a unique signature is obtained for each furniture which can be further used for categorizing them as shown in Fig. 5.7. Algorithm 3 describes each step of the furniture categorization process in detail where the input is a room from a floor plan image without the walls. The set $\mathscr{T}$ is generated as the output of the furniture categorization step. In this step, each blob in a room is categorized and their signature is compared with the already stored furniture template signatures and assigned a type T. The union of the type of all the furniture inside a room then constitute the set $\mathscr{T}$. This set finally contains the type of furniture inside each room.

Furniture type gives a unique signature to a room. For example, a room with a bed, table, armchair etc. would primarily be a bedroom as shown in Fig. 5.8. So, while comparing a pair of layouts, it is worth to compare features of the similar type of rooms, instead of all possible rooms. Hence, categorizing a room based on type of furniture can prove helpful while matching. The feature extraction accuracies of all the 3 features is listed in Tab. 5.1. Accuracy of a particular feature is calculated by taking ground-truth information for each floor plan of the dataset in terms

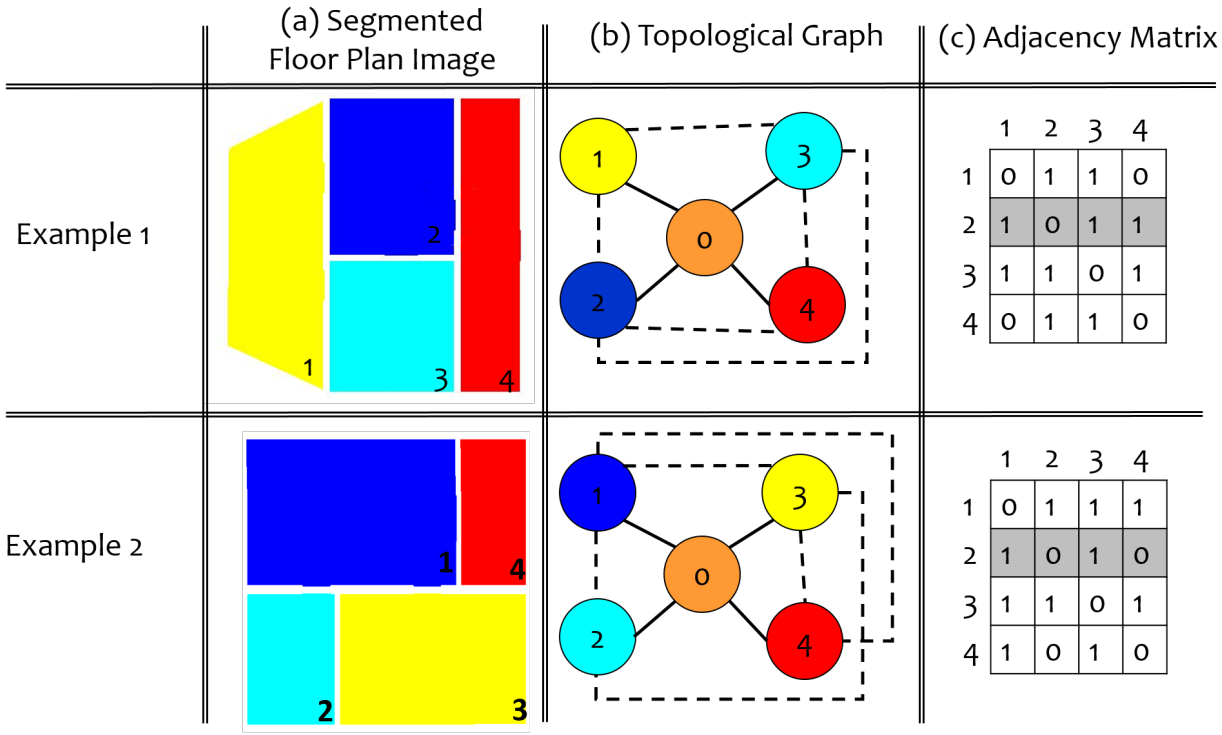


图5.5：房间邻接字符串（RAS）特征示例：（a）分割后的房间；（b）拓扑图；（c）邻接矩阵（其中与房间2对应的邻接字符串已高亮显示）。

5.2.3 特征3：家具构成记录（FCR）

第三个计算得到的特征是家具构成记录（FCR），它包含了特定房间内各类家具的数量和类型。通过检测平面图图像中的连通区域，可以确定每件家具在房间布局中的位置；随后，将各家具的数量（记为C）与相应房间关联存储。这有助于根据用户对室内装饰是否过于拥挤的偏好来比较不同房间。接下来，通过对检测到的连通区域进行连通分量分析，并测量不同家具符号内部各组件的面积，实现家具的分类。然后，计算面积最大的三个组件的面积比，为每件家具生成一个唯一标识，该标识可用于进一步对其进行归类，如图5.7所示。算法3详细描述了家具归类过程的各个步骤，其输入为不含墙体的平面图中的单个房间。家具归类步骤的输出为集合$\mathcal{T}$。在此步骤中，对房间内的每个连通区域进行分类，将其标识与已存储的家具模板标识进行比对，并为其分配一种类型（记为T）。最终，将房间内所有家具的类型取并集，得到集合$\mathcal{T}$，该集合完整地包含了每个房间内的家具类型信息。

家具类型为房间赋予了独特的特征标识。例如，如图5.8所示，配备床、桌子、扶手椅等家具的房间主要可被归类为卧室。因此，在比较两组平面布局时，应着重比较同类房间的特征，而非所有可能的房间类型。由此可见，基于家具类型的房间分类在匹配过程中具有重要参考价值。表5.1列出了三种特征的提取准确率；某一特征的准确率是通过以数据集中每张平面图的真值信息为基础进行计算得出的。

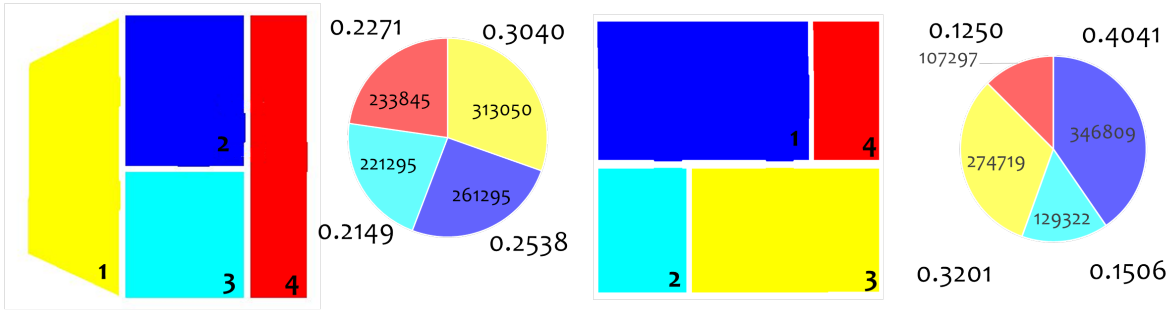


Figure 5.6. : Carpet Area Ratio (CAR) feature: (a), (c) Segmented, ordered layout, while (b), (d) CAR feature corresponding to Example 1 and 2 respectively.

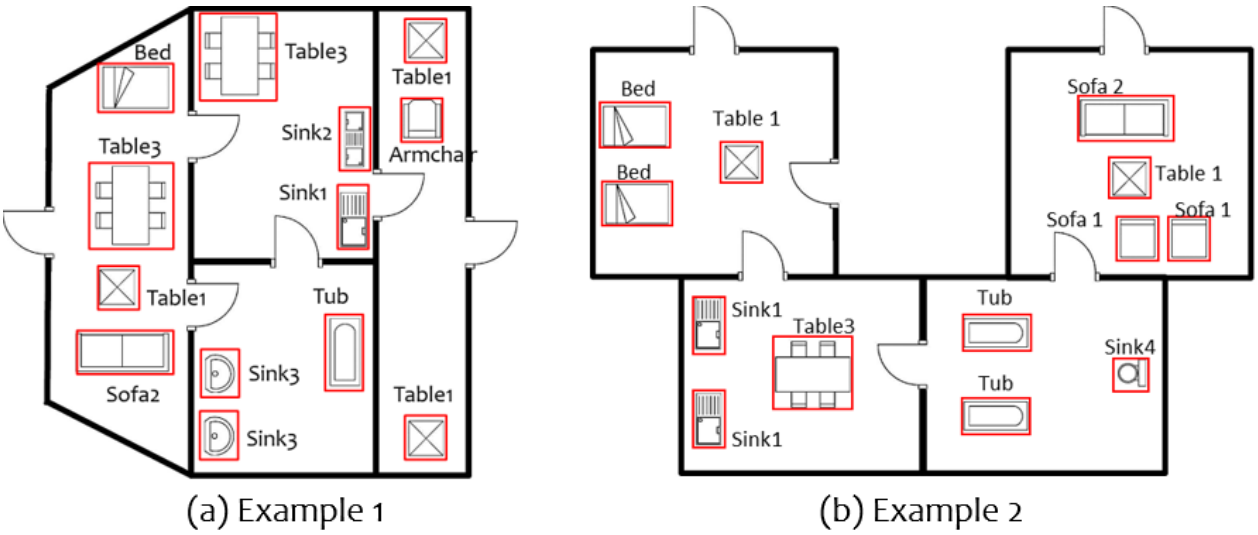


Figure 5.7. : Furniture detection and categorization in each room.

of the adjacencies, areas and furniture composition that each room in a floor plan has and further, taking a ratio of the features extracted from floor plans while running the algorithm and the ground-truth results. It was observed that each feature is extracted to a perfection using the steps mentioned above, except the furniture categorization feature, which performs to an accuracy of 89%. This accuracy while feature extraction helps in identifying the user's requirement efficiently while feature fusion and retrieval.

5.3 FINE-GRAINED MATCHING AND RETRIEVAL

Matching of a pair of floor plans using the extracted features is essential for retrieving similar layouts to the query layout. In the following sections the procedure to calculate matching score corresponding to various high level semantic features is described. Let, L_i^k and L_j^k represent

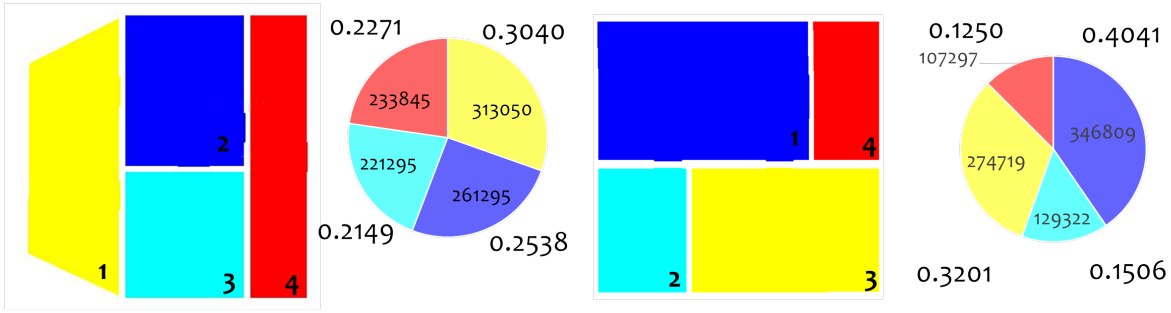


图5.6：地毯面积比（CAR）特征：（a）、（c）为分割后的有序布局，而（b）、（d）分别为对应于示例1和示例2的CAR特征。

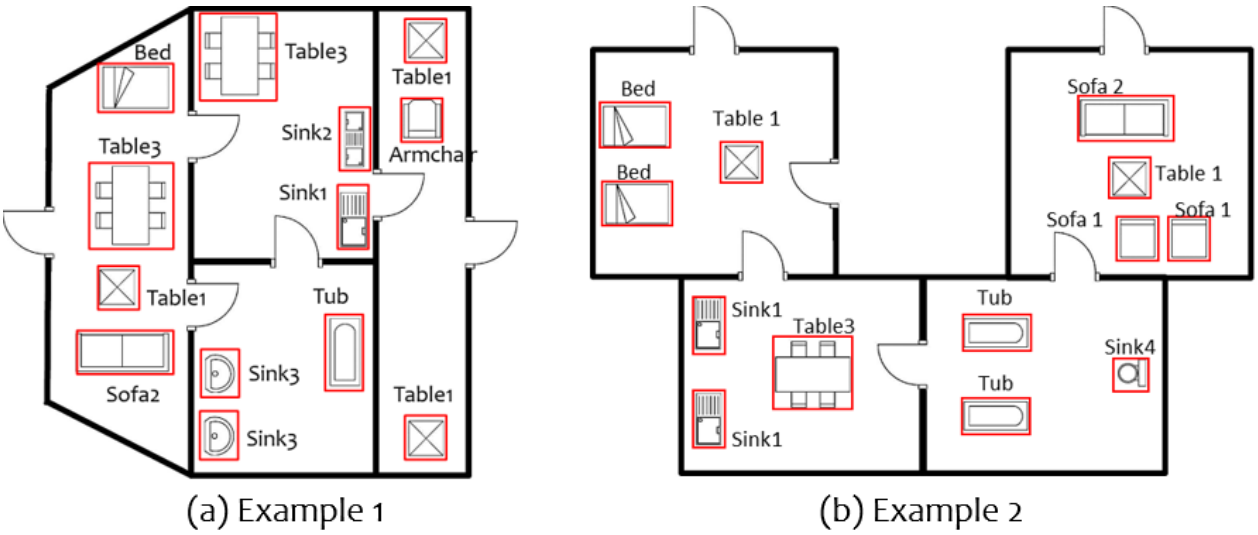


图5.7：各房间内的家具检测与分类。

针对平面图中各房间的邻接关系、面积及家具构成，并进一步计算算法运行过程中从平面图中提取的特征与真值结果之间的比值。研究发现，除家具分类特征外，其余各项特征均能通过上述步骤实现精确提取，其中家具分类特征的准确率为89%。在特征融合与检索过程中，这一特征提取精度有助于高效识别用户需求。

5.3 精细粒度匹配与检索

利用所提取的特征对一对平面布置图进行匹配，对于检索与查询布局相似的布局而言至关重要。在接下来的各节中，将详细描述用于计算与各类高层语义特征相对应的匹配得分的步骤。设L_i^k和L_j^k分别表示

Table 5.1. : Percentage Accuracy of the extracted high level semantic features

Feature	Extraction Accuracy
Room Segmentation/ Room Adjacency	100%
Carpet Area	100%
Number of Furniture	100%
Type of Furniture	89%

k^{th} room in two different layouts (L), where i and j range from 1 to n, i.e. the total number of floor plans in the dataset, while k ranges from 1 to total number of rooms (V) in a layout. If there is a difference in the number of rooms in the layout then the minimum number of rooms between the layouts is taken for comparison. Algorithm 4 describes the matching procedure in a step-by-step fashion.

5.3.1 RAS Matching Score

Edit distance, represented as $e(\cdot,\cdot)$ [Levenshtein, 1966] is used, as the measure to differentiate between a pair of adjacency strings corresponding to same numbered rooms in two different layouts as described in Sec.5.2.1. As an example room 2 for layout 1 in Fig. 5.5 (Example 1) (b) has RAS 1011 and room 2 for layout 2 in Fig. 5.5 (Example 2) (b) has RAS 1010. In this case, the two adjacency strings differ only at the fourth bit whereas, rest bits are similar. Therefore, edit distance, i.e. the operations required to convert one string to another results to 1. Similarly, in case of room 4 for both the layouts the RAS are 0110 and 1010 respectively, differing in the first

Algorithm 3 Furniture detection and categorization

Input: Room image without outer walls (I), Furniture Template Signatures (F)
Output: Furniture Count (C), Furniture Type Set ($\mathscr{T}$)

C=0, $\mathscr{T}$=$\{\phi\}$
$\mathscr{B}$=Morphological fill operation(I)
$C = |\mathscr{B}|$ ▷ $|\cdot|$: *Cardinality*
for $j = 1$ to C **do**
  $\mathscr{C} = CC(\mathscr{B}_j)$ ▷ CC()=Connected Component
  for $k = 1$ to $|\mathscr{C}|$ **do**
    $\mathscr{A}_k = Area(c_k)$, where $c_k \in \mathscr{C}$
  end for
  $A = Sort_{desc}(\mathscr{A})$
  $S(\mathscr{B}_j) = \{(A_1/A_3), (A_2/A_3), 1)\}$ ▷ $S(\cdot)$: Signature
  for $k = 1$ to $|F|$ **do**
    if $S(\mathscr{B}_j) == F_k$ **then**
      $T(\mathscr{B}_j) = T(F_k)$ ▷ $T(\cdot)$: Type of Furniture
    end if
  end for
  $\mathscr{T} = \{\mathscr{T} \cup (T(\mathscr{B}_j)\}$
end for

表5.1：所提取的高级语义特征的准确率百分比

Feature	**Extraction Accuracy**
Room Segmentation/ Room Adjacency	100%
Carpet Area	100%
Number of Furniture	100%
Type of Furniture	89%

k^{th} 在两种不同的布局（L）中，其中 i 和 j 的取值范围为 1 至 n，即数据集中楼面平面图的总数；而 k 的取值范围为 1 至某一布局中的房间总数（V）。若不同布局的房间数存在差异，则比较时采用两者之间的最小房间数。算法 4 以逐步细化的方式描述了该匹配过程。

5.3.1 RAS匹配得分

编辑距离，记为$e(\cdot,\cdot)$ [Levenshtein, 1966]，被用作度量指标，以区分两个不同布局中编号相同的房间所对应的邻接字符串，如第5.2.1节所述。例如，图5.5（示例1）（b）中布局1的房间2的邻接字符串为1011，而图5.5（示例2）（b）中布局2的房间2的邻接字符串为1010。在这种情况下，两者的邻接字符串仅在第4位上存在差异，其余位均相同，因此编辑距离，即把一个字符串转换为另一个字符串所需的编辑操作次数，为1。类似地，对于两个布局中的房间4，其邻接字符串分别为0110和1010，两者在第一位上存在差异。

Algorithm 3 Furniture detection and categorization

Input: Room image without outer walls (I), Furniture Template Signatures (F)
Output: Furniture Count (C), Furniture Type Set ($\mathscr{T}$)

C=0, $\mathscr{T}=\{\phi\}$
$\mathscr{B}$=Morphological fill operation(I)
$C = |\mathscr{B}|$ ▷ $|\cdot|$: *Cardinality*
for $j = 1$ to C **do**
  $\mathscr{C} = CC(\mathscr{B}_j)$ ▷ CC()=Connected Component
  for $k = 1$ to $|\mathscr{C}|$ **do**
    $\mathscr{A}_k = Area(c_k)$, where $c_k \in \mathscr{C}$
  end for
  $A = Sort_{desc}(\mathscr{A})$
  $S(\mathscr{B}_j) = \{(A_1/A_3), (A_2/A_3), 1)\}$ ▷ $S(\cdot)$: Signature
  for $k = 1$ to $|F|$ **do**
    if $S(\mathscr{B}_j) == F_k$ **then**
      $T(\mathscr{B}_j) = T(F_k)$ ▷ $T(\cdot)$: Type of Furniture
    end if
  end for
  $\mathscr{T} = \{\mathscr{T} \cup (T(\mathscr{B}_j)\}$
end for

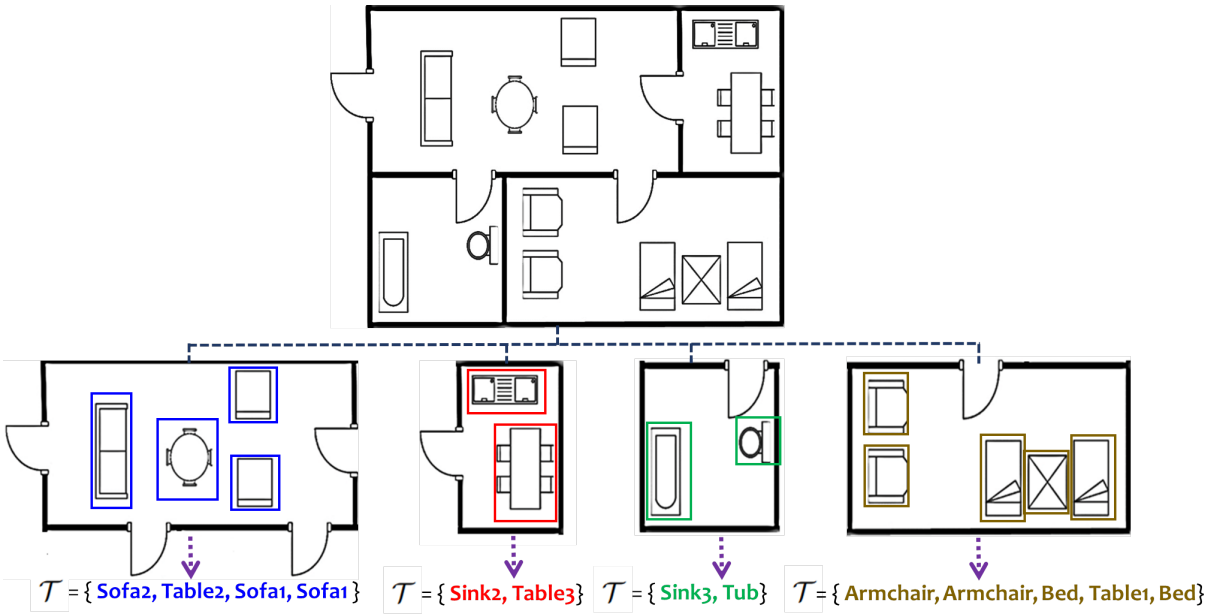


Figure 5.8. : Furniture detection and categorization in each room. Illustration underneath each room gives the array $\mathcal{T}$ containing the type of furniture in each room.

and second bit, resulting in an edit distance of 2. The RAS matching score (ρ) is computed as:

$$\rho(i,j,k) = \frac{e(L_i^k, L_j^k)}{max(V_i, V_j)} \tag{5.3}$$

Here the matrix ρ corresponds to the difference in room adjacencies between all similar numbered rooms in two layouts. Thus, higher ρ value suggests dissimilarities in adjacencies between two rooms. In a similar manner adjacency string matching of the rest of the rooms of a layout as compared to another layout is done. For normalizing this particular score in the range [0,1], it is divided by the factor equal to the maximum number of rooms between layout i and j, denoted as V_i and V_j. The idea behind normalizing in this manner is, that the maximum edit distance in the adjacency string can be obtained only if one of the strings is completely dissimilar from another, and also has a different length as compared to another. In that case the maximum operations taken to convert one string to the other would be equal to the maximum length between both the strings.

5.3.2 CAR Matching Score

The next feature for matching between two layouts is Carpet Area Ratio, $A(\cdot)$ of a particular room in both the layouts as described in Sec. 5.2.2. The CAR matching score (ψ) is computed as:

$$\psi(i,j,k) = A(L_i^k) - A(L_j^k) \tag{5.4}$$

Here the matrix ψ corresponds to the difference in Carpet Area Ratios between all similar numbered rooms in two layouts. The feature area ratio is already normalized in the range [0,1].

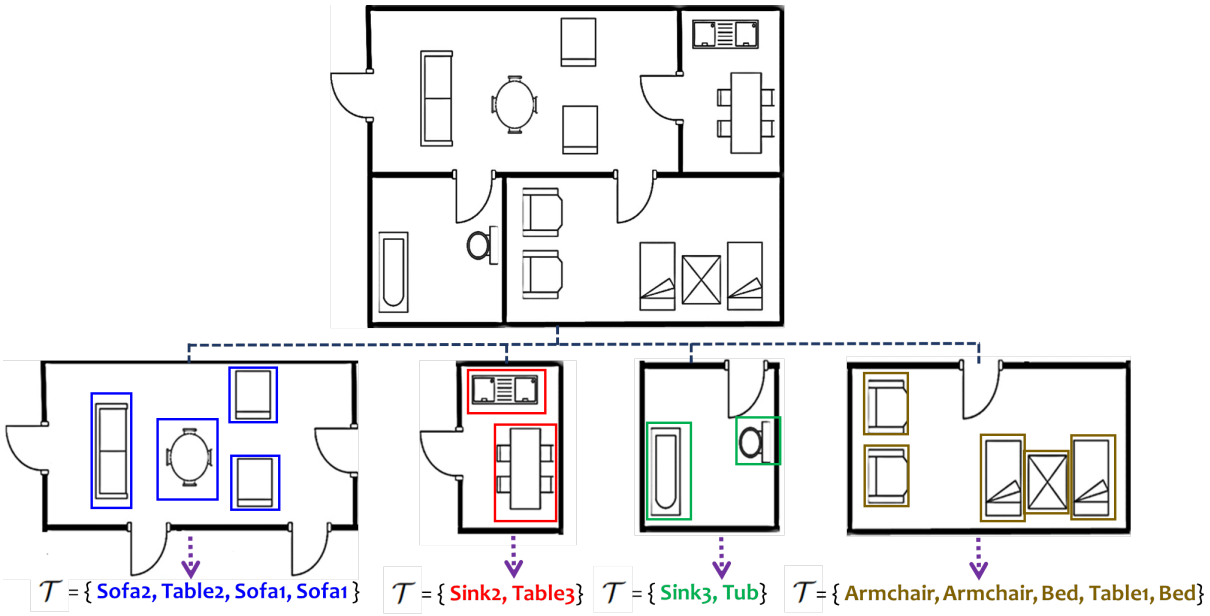


图5.8：各房间内的家具检测与分类。每个房间下方的插图给出了包含该房间内家具类型的数组$\mathcal{T}$。

以及第二个比特，从而导致编辑距离为2。RAS匹配得分（ρ）的计算公式为：

$$\rho(i,j,k) = \frac{e(L_i^k, L_j^k)}{max(V_i, V_j)} \tag{5.3}$$

此处，矩阵ρ对应于两种布局中所有同编号房间之间房间邻接关系的差异。因此，较高的ρ值表明两个房间之间的邻接关系存在差异。以类似的方式，对一个布局中的其余房间与另一个布局中的相应房间进行邻接关系字符串匹配。为了将该特定得分归一化至[0,1]区间，将其除以一个因子，该因子等于布局i与布局j之间房间数的最大值，分别记为V_i和V_j。采用这种归一化方式的原理在于：只有当其中一个字符串与另一个完全不同时，且长度也不同于另一个时，邻接关系字符串的编辑距离才能达到最大值；此时，将一个字符串转换为另一个字符串所需执行的最大操作次数即等于两字符串长度之中的较大值。

5.3.2 CAR匹配得分

两个布局之间的下一个匹配特征是地毯面积比，即如第5.2.2节所述，在两个布局中某一特定房间的$A(\cdot)$。地毯面积比匹配得分（ψ）的计算公式为：

$$\psi(i,j,k) = A(L_i^k) - A(L_j^k) \tag{5.4}$$

此处，矩阵ψ对应于两种布局中所有同编号房间的地毯面积比之差。特征面积比已在区间[0,1]内进行了归一化处理。

Algorithm 4 Fine grained retrieval using high level semantics

Input: Query Image (q), Database Image (d)
Output: Matching Score ($M(q,d)$)

for all d=1 to n **do**
  for all k=1 to min(V_q, V_d) **do**
    $\rho(q,d,k) = e(L_q^k, L_d^k)$ ▷ $e(\cdot)$: edit distance
    $\psi(q,d,k) = A(L_q^k) - A(L_d^k)$ ▷ $A(\cdot)$: Area
    $\phi(q,d,k) = C(L_q^k) - C(L_d^k)$ ▷ $C(\cdot)$:furn-count
    $\theta(q,d,k) = S(\mathscr{T}(L_q^k), \mathscr{T}(L_q^k))$ ▷ $\mathscr{T}(\cdot)$: furn-type
    $Normalize(\rho, \psi, \phi, \theta)$
  end for
  $M(q,d) = \sum_k(\rho(q,d,k)) + \sum_k(\psi(q,d,k)) + \sum_k(\phi(q,d,k)) + \sum_k(\theta(q,d,k))$
  $M(q,d) = M(q,d)/4$
Return $M(q,d)$
end for

Therefore, no post normalization required. A low value of ψ suggests high similarity in area feature of two rooms.

5.3.3 Furniture Composition Record Matching Score

The room decor component has two sub-components, namely the number of furnitures inside two same numbered rooms and type of furnitures inside the rooms.

Number of furnitures

Number of furnitures, $C(\cdot)$ inside a particular room is computed using the technique discussed in Sec. 5.2.3 and corresponding number of furnitures score, ϕ, is calculated as:

$$\phi(i,j,k) = \frac{C(L_i^k) - C(L_j^k)}{max(C(L_i^k), C(L_i^k))} \tag{5.5}$$

Here the matrix ϕ corresponds to the difference in number of furnitures between all similar numbered rooms in two layouts. This helps us compare the rooms according to the consumers priority of being overcrowded with decor or not. The difference of the number of furnitures corresponding to ϕ should be low in both the layouts if they are similar.

Type of furnitures

This particular feature categorizes type of room, for example a bedroom, bathroom or a kitchen based on its furniture arrangement. It helps us distinguish between two same numbered rooms, with different furniture types. Firstly, the classification of the furnitures in each room is done and their classified labels are stored in an array with rows corresponding to each room number and the subsequent column corresponding to the labels of the furniture in that particular room. Motivation behind matching this feature is that a buyer might specify that the third room in the

Algorithm 4 Fine grained retrieval using high level semantics

Input: Query Image (q), Database Image (d)
Output: Matching Score ($M(q,d)$)

for all d=1 to n **do**
  for all k=1 to min(V_q, V_d) **do**
    $\rho(q,d,k) = e(L_q^k, L_d^k)$ ▷ $e(\cdot)$: edit distance
    $\psi(q,d,k) = A(L_q^k) - A(L_d^k)$ ▷ $A(\cdot)$: Area
    $\phi(q,d,k) = C(L_q^k) - C(L_d^k)$ ▷ $C(\cdot)$:furn-count
    $\theta(q,d,k) = S(\mathscr{T}(L_q^k), \mathscr{T}(L_q^k))$ ▷ $\mathscr{T}(\cdot)$: furn-type
    $Normalize(\rho, \psi, \phi, \theta)$
  end for
  $M(q,d) = \sum_k(\rho(q,d,k)) + \sum_k(\psi(q,d,k)) + \sum_k(\phi(q,d,k)) + \sum_k(\theta(q,d,k))$
  $M(q,d) = M(q,d)/4$
Return $M(q,d)$
end for

因此，无需进行后处理归一化。ψ 的低值表明两个房间的区域特征具有较高的相似性。

5.3.3 家具成分记录匹配得分

房间装饰组件包含两个子组件，分别为：编号相同的两个房间内的家具数量以及房间内家具的类型。

家具数量

特定房间内的家具数量$C(\cdot)$采用第5.2.3节所述方法进行计算，相应的家具数量评分ϕ则按如下公式计算：

$$\phi(i,j,k) = \frac{C(L_i^k) - C(L_j^k)}{max(C(L_i^k), C(L_i^k))} \tag{5.5}$$

此处，矩阵ϕ对应于两种布局中所有编号相同的房间之间家具数量的差异。这有助于我们根据消费者对室内装饰是否过于拥挤的优先偏好来比较各房间。如果两种布局相似，则与ϕ对应的家具数量差异应在两种布局中均保持较低水平。

家具类型

该特定功能可根据房间内的家具布置对房间类型进行分类，例如卧室、浴室或厨房。这有助于我们在两个编号相同但家具类型不同的房间之间进行区分。首先，对每个房间中的家具进行分类，并将其分类标签存储在一个数组中：数组的每一行对应一个房间编号，后续各列则对应该房间内家具的标签。采用这一特征进行匹配的动机在于，买家可能会指定第三间房间在

layout should be a bedroom and hence, while retrieval one should retrieve floor plans with buyer's preferred placement of room in the complete floor plan. Thus, floor plan having a kitchen as the third room should be retrieved later as compared to a floor plan having the third room as the bedroom. In the below mentioned equation, the matrix θ corresponds to the difference in type of furnitures between all similar numbered rooms in two layouts. Using this, the furniture label strings are compared for similar numbered rooms in two layouts and the penalty is incremented if label strings do not match, i.e. if dissimilar furniture is found between both rooms. The more similar the two rooms are in their furniture types the lower would be the value of θ. The type of furniture in a particular layout is represented as $T(\cdot)$, string comparison is represented as $S(\cdot,\cdot)$ and the type of furniture score is represented as θ, and is computed as follows.

$$\theta(i,j,k) = \frac{S(\mathscr{T}(L_i^k), \mathscr{T}(L_j^k)}{max(C(L_i^k), C(L_i^k))} \tag{5.6}$$

5.3.4 Cumulative Match Score (CMS)

After computing the various feature matching scores for individual rooms in a query layout, CMS between a pair of layouts is obtained by adding the feature scores corresponding to all the common rooms present in both the layouts. CMS for a feature (ρ, ψ, ϕ, θ in our case) F is denoted as F^+ and is computed as:

$$F^+(i,j) = \frac{\sum_{k=1}^{min(V_i,V_j)} F(i,j,k)}{min(V_i,V_j)} \tag{5.7}$$

Using this particular equation the corresponding cumulative feature scores ρ^+, ψ^+, ϕ^+ and θ^+ are obtained. Thus, finally, a matching score for a complete layout is obtained by taking a sum of individual features for all the rooms.

5.3.5 Total Match Score

Total Match Score, M for a pair (query and database) of layouts is calculated using the weighted sum of four extracted feature scores namely RAS, CAR score, number and type of furniture score (FCR). Weights are assigned to each feature component to investigate the effect of individual feature during retrieval. This allows user to give more preference to a certain feature while retrieval. In such a case, the matching score has additional individual weight coefficients η_1, η_2, η_3 and η_4 all in the range $[0,1]$, where the default value of the coefficients is 1. In this case, the Match Score M is obtained, as shown in the equation below:

$$M(i,j) = 1 - \frac{\rho^+(i,j) + \psi^+(i,j) + \phi^+(i,j) + \theta^+(i,j)}{4} \tag{5.8}$$

It is to be noted that this score has individual components normalized in the range [0,1]. Also, the Total match score is divided by a factor 4 to be finally normalized in the range [0,1].

As discussed before, the weight coefficients can be incremented and decremented to set preferences to some features while diluting the effect of others, by tuning the values of η_1, η_2, η_3

布局应为卧室，因此在检索时，应当检索包含买家所偏好房间位置的完整平面图。由此，厨房作为第三间房的平面图应在检索排序中位于厨房而非卧室作为第三间房的平面图之后。在下述公式中，矩阵θ对应于两种布局中所有同序号房间之间家具类型的差异。基于此，对两种布局中同序号房间的家具标签字符串进行比较，若标签字符串不匹配，即两房间内存在不同类型的家具，则罚分予以增加。两房间的家具类型越相似，θ的值就越低。某一布局中的家具类型用$T(\cdot)$表示，字符串比较用$S(\cdot,\cdot)$表示，家具类型得分用θ表示，其计算方式如下。

$$\theta(i,j,k)=\frac{S(\mathscr{T}(L_i^k),\mathscr{T}(L_j^k)}{max(C(L_i^k),C(L_i^k))} \tag{5.6}$$

5.3.4 累积匹配得分（CMS）

在计算查询布局中各房间的特征匹配得分后，通过将两个布局中所有共同房间对应的特征得分相加，即可得到这两者之间的CMS。对于某一特征（在本例中为ρ,ψ,ϕ,θ），其CMS记为F^+，计算公式如下：

$$F^+(i,j)=\frac{\sum_{k=1}^{min(V_i,V_j)}F(i,j,k)}{min(V_i,V_j)} \tag{5.7}$$

利用该特定方程，可分别获得相应的累积特征得分ρ^+、ψ^+、ϕ^+和θ^+。因此，最终通过将所有房间的各特征得分相加，即可得到整个布局的匹配得分。

5.3.5 总匹配得分

对于一对布局（查询布局与数据库布局），总匹配得分M通过加权求和四个提取的特征得分来计算，即RAS得分、CAR得分以及家具数量与类型得分（FCR）。为每个特征分量分配权重，以考察各特征在检索过程中的影响。这使得用户能够在检索时对某一特定特征赋予更高的优先级。在这种情况下，匹配得分还引入了额外的个体权重系数η_1、η_2、η_3和η_4，这些系数均位于区间[0, 1]内，且默认值均为1。此时，匹配得分M可按如下公式计算：

$$M(i,j)=1-\frac{\rho^+(i,j)+\psi^+(i,j)+\phi^+(i,j)+\theta^+(i,j)}{4} \tag{5.8}$$

需要注意的是，该评分的各组成分均已归一化至[0,1]区间。此外，总匹配得分除以因子4后，最终也被归一化至[0,1]区间。

如前所述，通过调整η_1、η_2、η_3的取值，可以对某些特征赋予更高的权重，同时削弱其他特征的影响。

and η_4 in the equation mentioned below :

$$M(i,j) = 1 - \frac{\eta_1 \rho^+(i,j) + \eta_2 \psi^+(i,j) + \eta_3 \phi^+(i,j) + \eta_4 \theta^+(i,j)}{4} \tag{5.9}$$

The next section describes the qualitative and quantitative assessment of the proposed framework.

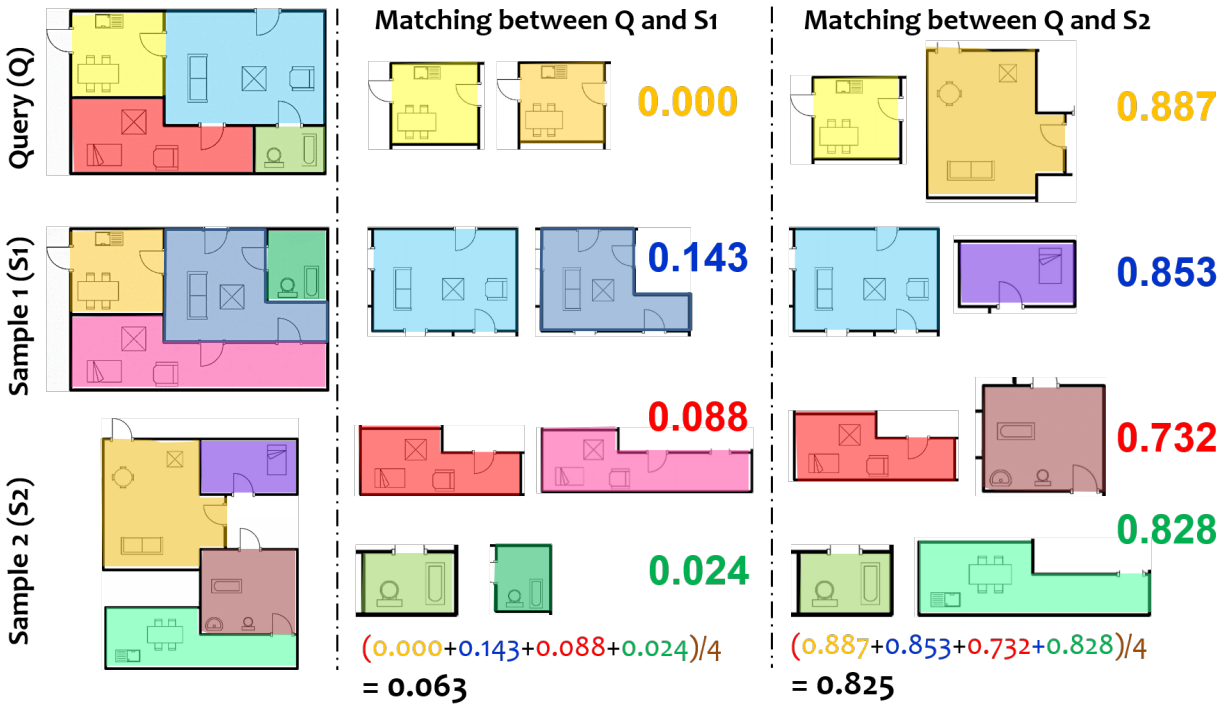


Figure 5.9. : Distance score corresponding to each room of the query layout (Q) and the database layouts (S_1 and S_2). Attributed to the difference in features, lower the score, more similar the floor plans are. Hance, finally the score is subtracted from 1 to make the final greater Matching score to represent the floor plans that match more. Color codes are used to highlight the room correspondence.

5.4 EXPERIMENTS AND RESULTS

The experiments are performed on the *ROBIN* dataset that contains three categories of layouts each with 17 subcategories containing a total of 510 floor plans. The dataset has 11 different categories of furniture inside each layout. The experiments were performed on a machine with Intel Xeon I7 processor with 16 cores.

Details of the matching process are shown in Fig. 5.9. There is a high degree of similarity (in terms of FCR) between the individual rooms of the query floor plan (Q) and that of sample floor plan S1. However, minor differences lie in the area feature (CAR) that can be observed by looking at the second room's ("blue" color) area in both the layouts. The two layouts also differ in their adjacency feature, which can be observed by looking at the fourth room (highlighted in "green"). In the case of Q, the fourth room is adjacent to the third and the second room. On the other hand,

and η_4 in the equation mentioned below :

$$M(i,j) = 1 - \frac{\eta_1 \rho^+(i,j) + \eta_2 \psi^+(i,j) + \eta_3 \phi^+(i,j) + \eta_4 \theta^+(i,j)}{4} \tag{5.9}$$

下一节将对所提出的框架进行定性和定量评估。

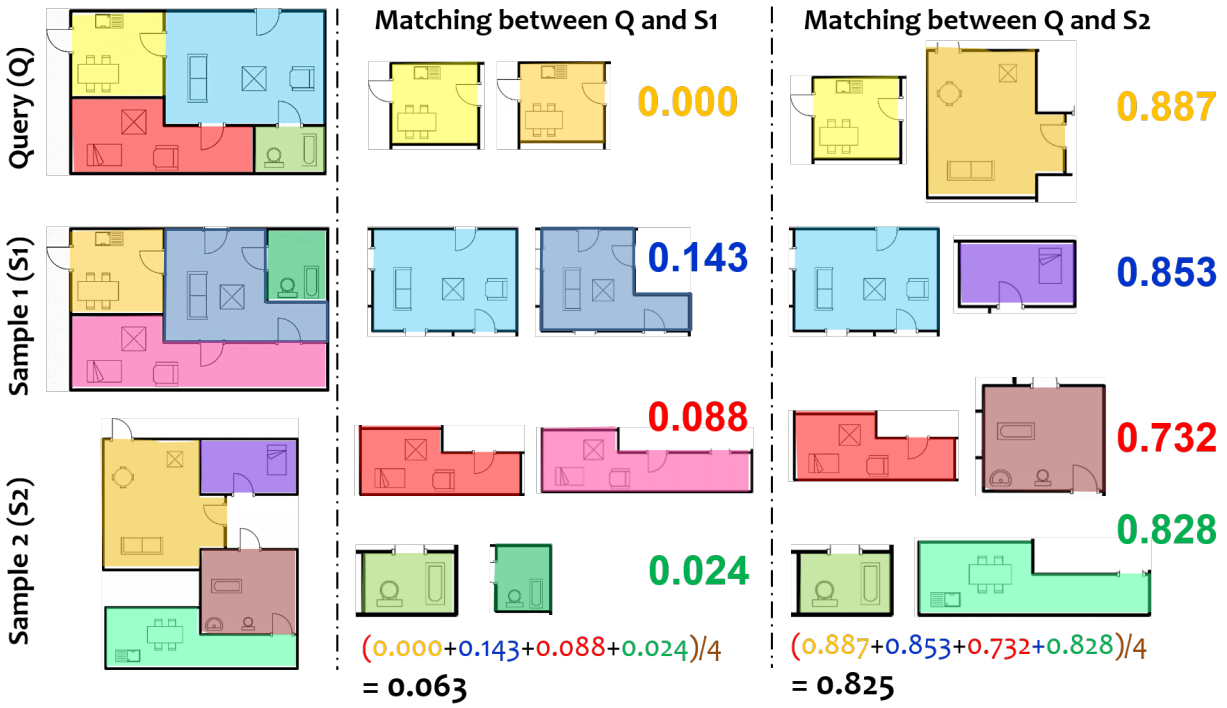


图5.9：查询布局（Q）与数据库布局（S_1和S_2）中各房间对应的距离得分。由于特征差异，得分越低，表示两者的平面布局越相似。因此，最终将得分从1中减去，以获得更高的匹配度分数，从而更好地反映布局的匹配程度。颜色编码用于突出显示房间之间的对应关系。

5.4 实验与结果

实验在包含三类布局、每类布局下设17个子类、共计510张平面图的*ROBIN*数据集上进行。该数据集中，每张布局内包含11种不同类别的家具。实验在一台配备16核Intel Xeon I7处理器的机器上完成。

匹配过程的详细情况如图5.9所示。查询平面图（Q）与样本平面图S1中的各房间在特征一致性比率（FCR）方面具有高度相似性。然而，通过比较两种布局中第二间房间（“蓝色”区域）的面积特征（CAR），可以观察到两者存在细微差异。此外，两种布局在邻接关系特征上亦不相同，这一点可通过第四间房间（“绿色”高亮部分）加以识别。就查询平面图Q而言，第四间房间分别与第三间和第二间房间相邻。另一方面，

Figure 5.10. : Retrieval results for four different query architectural floor plans taken from the *ROBIN* dataset. Note that the floor plans highlighted in red are incorrectly retrieved as global layout shape differs from query.

in the case of S1, it is adjacent only to the third room. However, database sample floor plan S2, as can be visually seen, differs in many respects as compared to Q. The total scores, as per Eq. 5.9, also justify these differences. For example in the fourth room (color coded in "green") in both Q and S2, all the features, namely the FCR, CAR as well adjacencies differ resulting in a large cost of 0.828. Such observations between Q and database floor plan S2, justify high dissimilarity score attributed to the different global shapes and thus, different CAR, RAS and FCR features of both the layouts. Hence, S2 is ranked later than S1 during retrieval.

5.4.1 Qualitative results

First of all the architectural layouts are segmented into rooms followed by detecting the various types of furniture present in each room. Thus, a semantic analysis of a floor plan is

图5.10：基于ROBIN数据集的四张不同查询建筑平面图的检索结果。请注意，以红色高亮显示的平面图因全局布局形状与查询不一致而被错误地检索出来。

就S1而言，其仅与第三个房间相邻。然而，从数据库样本平面图S2中可以直观地看出，其在诸多方面均与Q存在差异。根据式5.9计算的总得分也印证了这些差异。例如，在Q和S2的第四个房间（以“绿色”标识）中，所有特征——包括FCR、CAR以及邻接关系——均不相同，导致该部分的代价高达0.828。上述对Q与数据库平面图S2的比较进一步说明，两者在整体形态上的显著差异是造成高相异度评分的主要原因，进而导致二者在CAR、RAS和FCR等特征上的不同。因此，在检索过程中，S2的排序位于S1之后。

5.4.1 定性结果

首先，建筑布局被划分为各个房间，随后对每个房间中存在的各类家具进行识别。由此，对平面图进行语义分析便

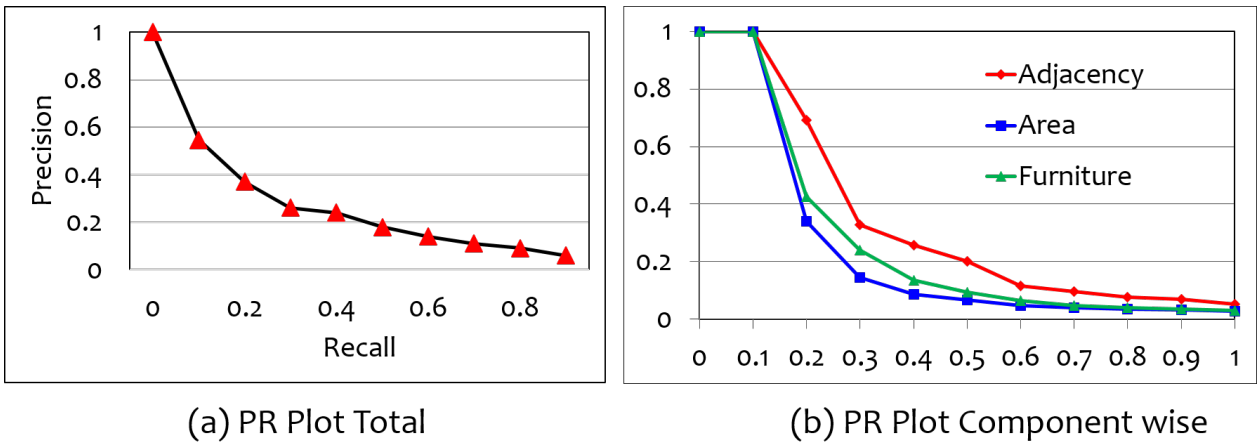
obtained which can be further used for processing. The fine-grained matching techniques proposed in this Chapter helped in the rank ordering of the matched layouts based on the differences in the adjacencies between corresponding rooms, the area of the rooms, number, and type of furniture inside the rooms. The rank-ordered retrieval results for six such queries are shown in Fig. 5.10. The first rank ordered result retrieved from the layout database is the query image itself. This is due to the fact that the match score between the retrieved layout and the query layout is maximum. The subsequent results differ in the scores in the area, adjacencies and decor features inside each room.

As is evident from Fig. 5.10(d) the rank one result is the query layout as mentioned above. Justification of the rank 2 retrieved result can be seen by observing its differences/ similarities with the query floor plan. For example, note that the adjacencies of the first room in the query layout are with room 2 and 4, that is exactly similar to the adjacencies shared by room 1 in rank 2 retrieved result. Thus, on the adjacency parameter, both layouts match completely. Although the first room is slightly bigger area-wise in the query, the furniture type and number are approximately similar in both the layouts. Hence, the match score between the query layout and the retrieved rank 2 layout is high which attributes to the result. An interesting observation is that the rank 3 result retrieves layout from a different sub-category of the dataset as it differs in the global layout shape. It is to be noted that such a retrieval is due to the fact of various common features between the query and rank 3 result which can be analyzed similarly as shown in Fig. 5.9.

5.4.2 Quantitative results

The performance of the system is quantified using a Precision and Recall (PR) metric (see Fig. 5.11). The precision values are averaged over all the query images for a particular recall value. Given a query, retrieved layouts should belong to the same sub-category of layouts as the query, i.e., it should maintain the global shape and also keeping in mind the preference set by the property seeker during requirement driven querying. Thus, if the global shape of the layouts differ then that is considered as an incorrectly retrieved result.

Figure 5.11 (a) refers to the overall PR plot according to the match score (see Eq. 5.8), i.e. giving equal weights for FCR, CAR and RAS scores and retrieving the results. As shown in



(a) PR Plot Total (b) PR Plot Component wise

Figure 5.11. : Precision and Recall (PR) plot corresponding to a) Cumulative sum of all the features with equal weights b) Preferring one feature more than the others while retrieval.

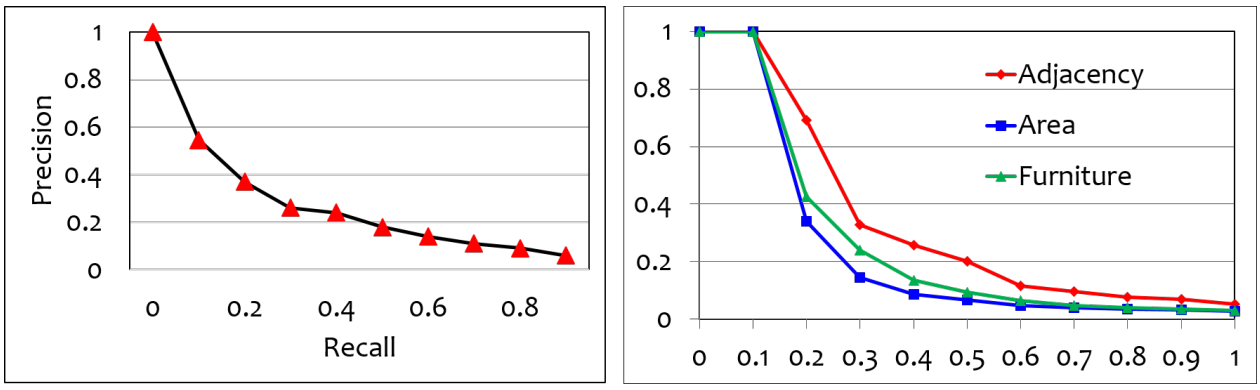
所获得的结果可进一步用于处理。本章提出的细粒度匹配技术，能够根据对应房间之间的邻接关系差异、房间面积以及房间内家具的数量和类型，对匹配的布局进行排序。图5.10展示了针对六种此类查询的排序检索结果。从布局数据库中检索出的排名第一的结果即为查询图像本身，这是因为检索到的布局与查询布局之间的匹配得分最高。后续结果则在各房间的面积、邻接关系及装饰特征等方面的得分上存在差异。

如图5.10(d)所示，排名第一的结果即为前述查询布局。排名第二的检索结果之所以获得较高置信度，可通过其与查询平面布置图之间的异同加以说明。例如，查询布局中第一间房间分别与2号和4号房间相邻，这一邻接关系与排名第二结果中1号房间的邻接关系完全一致。因此，在邻接关系这一参数上，两者的匹配程度为满分。尽管查询布局中第一间房间的面积略大，但两种布局中的家具类型及数量大致相当。由此可见，查询布局与排名第二的检索布局之间的匹配得分较高，从而导致该结果被优先返回。值得注意的是，排名第三的结果来自数据集中一个不同的子类别，其全局布局形态与查询布局存在差异。需要指出的是，这种检索结果的出现源于查询布局与排名第三结果之间存在若干共性特征，这些特征可参照图5.9所示的方式进行类似分析。

5.4.2 定量结果

系统的性能采用精确率与召回率（PR）指标进行量化（见图5.11）。对于某一给定的召回率值，精确率数值是在所有查询图像上取平均得到的。给定一个查询，检索出的布局应属于与查询相同的子类别，即在保持全局形状的同时，还需兼顾房产需求者在需求驱动式查询过程中设定的偏好。因此，若检索出的布局与查询布局的全局形状存在差异，则该结果被视为检索错误。

图5.11（a）展示了基于匹配得分的总体PR曲线（参见式5.8），即对FCR、CAR和RAS得分赋予相等权重并进行结果检索。如图所示，



(a) PR Plot Total　　(b) PR Plot Component wise

图5.11：对应于a）所有特征等权重累加，b）检索时偏重某一特征的精确率-召回率（PR）曲线。

Fig. 5.11 (a), given a query sample the rank 1 result is the query floor plan itself. This leads to the highest precision value of 1 during the initial recall. With further retrieval, the average precision value decreases due to some incorrectly retrieved results as compared to the query layout. As discussed in Sec. 5.3.5, while retrieval a user can have a preferred feature. For example, a user might be fixated over preferring area feature over the others while retrieving. The performance of the approach mentioned in this Chapter is analyzed by simultaneously increasing the weight of a feature, while decreasing weights of other features to study the impact of individual high level semantic features during retrieval, thus obtaining PR plot as in Fig. 5.11 (b), corresponding to preferring different features during retrieval. In Fig. 5.11 (b), "red" line corresponds to the PR plot given high weight to adjacency feature during retrieval. Similarly, if the user wants the furniture type and number to be very similar to the query layout then weight of the cost of types of furniture during matching can be increased. The PR plot in Fig. 5.11 (b) with the "green" line depicts the quantitative retrieval results using such an approach. Similarly, the "blue" line in the same plot depicts the results obtained when the area parameter is given more weight. The area under the PR plot is the highest when the adjacency feature ("red" line) is given the highest weight as compared to other features. Because layouts, which are falling in the same category of a global shape generally have similar room adjacencies. Even though their areas and furniture arrangement might differ a little. Thus, the retrieval accuracy is higher when adjacency parameter is favored as is evident from its PR plot.

5.5 DISCUSSION

The proposed approach, efficiently performs all the processing steps given in Fig. 5.2. As observed, the system generally succeeds in retrieving the samples from the same sub-category as the query sample, only faltering at some occasions, where, our proposed FCR and CAR features overpower the RAS feature. For run time analysis of this approach, the time taken (averaged over 510 samples) in seconds at every step was computed. Segmentation into rooms was a computationally expensive step owing to a number of intermediate processing steps. The average time taken by the segmentation step was 22.543 secs. Furniture detection and Furniture classification took 3.968 secs and 5.815 secs respectively. The four feature scores proposed by us namely CAR, RAS, No. of furnitures and Type of Furnitures take 4.253 secs, 3.453 secs, 3.234 secs and 5.736 secs respectively. It was noted that these feature scores took more time in layouts with 5-rooms, due to higher number of doors and inner components owing to the increased rooms. Calculating the cumulative matching score between two layouts and thereby performing the retrieval costed 12.257 secs at an average.

Due to a lack of availability of the implementation of such a composite framework of fine grained retrieval of floor plans in the literature, the results couldn't be compared with other retrieval based approaches. The current approach is motivated by the idea proposed in Chapter 3 in the area of floor plan retrieval [Sharma *et al.*, 2016a]. The key difference between the work discussed in this Chapter and that proposed earlier in this thesis are: (i) previous work does not establish room-wise uniform ordering and correspondence, (ii) the earlier approach took only adjacency and decor arrangement in the whole layout as the parameters during retrieval. In this Chapter, there is an inclusion of high level semantic features like ordering of rooms, area and furniture arrangement in a particular room to aid better fine grained retrieval of floor plans in the *ROBIN* dataset. For making a fair comparison, the results of the baseline method [Sharma *et al.*, 2016a] on *ROBIN* dataset are reproduced and the mean average precision values (MAP) are compared with the newly proposed method in this Chapter. As shown in Tab. 5.2, the approach proposed in this chapter has led to an increase in the mean average precision value as compared to the previous approaches.

The proposed method also has the edge over the approach proposed in Chapter 4 [Sharma

图5.11（a）显示，对于给定的查询样本，排名为1的结果即为查询的平面布局本身。这导致在初始召回阶段的精确度达到最高值1。随着检索过程的推进，由于部分检索结果与查询布局不匹配，平均精确度随之下降。如第5.3.5节所述，在检索过程中，用户可能对某一特征具有偏好。例如，用户在检索时可能更倾向于优先考虑面积特征。本章所提方法的性能通过同时提高某一特征的权重、降低其他特征的权重来进行分析，以研究各高层次语义特征在检索过程中的影响，从而得到如图5.11（b）所示的PR曲线，该曲线对应于在检索过程中对不同特征的偏好。在图5.11（b）中，“红色”曲线表示在检索过程中赋予邻接特征较高权重时的PR曲线。类似地，若用户希望检索结果中的家具类型及数量与查询布局高度相似，则可在匹配过程中提高家具类型的权重。图5.11（b）中“绿色”曲线展示了采用此类策略时的定量检索结果。同样，同一曲线中的“蓝色”曲线则反映了在面积参数权重较高的情况下的检索结果。与其它特征相比，当邻接特征（“红色”曲线）被赋予最高权重时，PR曲线下的面积最大。这是因为属于同一全局形状类别的一组布局通常具有相似的房间邻接关系，尽管其面积和家具布置可能存在细微差异。因此，从其PR曲线可以看出，当优先考虑邻接参数时，检索精度更高。

5.5 讨论

所提出的方法能够高效地完成图5.2中给出的所有处理步骤。如观察所示，该系统通常能够从与查询样本同属一个子类的样本集中检索出目标样本，仅在少数情况下出现失误，此时我们提出的FCR和CAR特征会压倒RAS特征。为分析该方法的运行时间，在每个步骤上对510个样本的耗时（取平均值）进行了统计。由于包含多个中间处理步骤，房间分割是一个计算开销较大的环节，其平均耗时为22.543秒。家具检测与家具分类分别耗时3.968秒和5.815秒。我们提出的四项特征得分——CAR、RAS、家具数量以及家具类型——分别耗时4.253秒、3.453秒、3.234秒和5.736秒。值得注意的是，在具有5个房间的布局中，这些特征得分的计算耗时有所增加，这是由于房间数量增多导致门及内部构件数量相应增加所致。最后，计算两个布局之间的累积匹配得分并据此执行检索操作，平均耗时为12.257秒。

由于文献中缺乏对这种细粒度平面图检索复合框架的具体实现，无法将其与其它基于检索的方法进行对比。当前方法的提出受到第3章关于平面图检索相关研究思路的启发[Sharma *et al.*, 2016a]。本章所讨论的工作与本论文前期提出的方案之间存在以下关键差异：(i) 前期工作未建立房间级别的统一排序与对应关系；(ii) 先前方法在检索过程中仅以整体布局中的邻接关系及装饰布置作为参数。而在本章中，引入了房间排序、面积以及特定房间内家具布置等高层语义特征，以辅助在*ROBIN*数据集上实现更精确的细粒度平面图检索。为进行公平比较，本文在*ROBIN*数据集上复现了基准方法[Sharma *et al.*, 2016a]的实验结果，并将平均精度均值（MAP）与本章新提出的方法进行了对比。如表5.2所示，本章所提出的方法较之先前方法显著提升了平均精度均值。

所提出的方法在性能上也优于第4章中提出的方法[Sharma]。

Table 5.2. : Comparison of techniques on ROBIN and SESYD Dataset

Approach	MAP (ROBIN)	MAP (SESYD)
Baseline [Sharma *et al.*, 2016a]	0.2479	1.0
OASIS with RLH [Chechik *et al.*, 2009]	0.0956	1.0
OASIS with BOW [Chechik *et al.*, 2009]	0.1892	0.99
Proposed	**0.3047**	1.0

et al., 2017], where a Convolutional Neural Network based deep framework for retrieval in floor plans is proposed. This deep framework although quantitatively performs better than the method proposed in this Chapter. However, it has a few drawbacks. For example, there is no provision of giving preferences to individual features during retrieval as the layers extract and combine the features implicitly without the provision of manipulating the weight by the user. On the other hand, the provision of a weighted combination of the features proposed in this Chapter lets the user give preferences to individual features, according to his requirement (justified by Fig. 5.11 (b)), thus making it more relevant from a user's point of view. Moreover, the CNN framework is a simple network that mainly distinguishes by the shape of an object or a layout. It would not perform well while doing fine-grain categorization of furniture components, as well as if two layouts from different classes have almost similar shapes.

As the proposed technique deals with similarity in floor plan images thus, the results are also compared with the Online Algorithm for Large Scale Image Similarity Learning (OASIS) [Chechik *et al.*, 2009] coupled with the Run Length Histogram technique (RLH) [de las Heras *et al.*, 2013] and Bag of Words technique (BOW) [Lazebnik *et al.*, 2006b]. These algorithms were chosen for comparison as they are widely popular in object retrieval in document images. It was observed that the proposed feature matching yielded a much better average precision value as compared to OASIS approach (See Tab. 5.2) because OASIS technique focuses on the difference of similarity values between relevant and irrelevant image pairs while ignoring the similarity value between highly similar images. As already discussed, reproducing the approaches mentioned in Tab. 5.2 on the SESYD public dataset yielded a perfect MAP value of 1 due to the low interclass similarity in the samples.

In [Ahmed *et al.*, 2014], the authors proposed a sketch based floor plan retrieval framework, where query layouts are represented as sketches using only primitive geometric shapes such as rectangles. In contrast, in this work, the room decor represented by complex geometric primitives is considered. Furthermore, room decor labeled fine-grained analysis was missing in the work by [Ahmed *et al.*, 2014]. This framework is rotation sensitive to allow relative position of rooms as a feature. Customers may follow Vastu tips and prefer "*master bedroom should be towards the south facing the wall, so that sleeping person's head is facing south*". Making the retrieval rotation invariant will fail to serve this purpose.

5.6 SUMMARY

The proposed framework for fine-grained retrieval of floor plans uses high-level semantic features like area, number, type of furniture and also adjacencies between the rooms. The inclusion of the high-level features during retrieval is a novel idea that can find application in scenarios where the user has specific choices in terms of the fully furnished floor plans. Also, weighted feature fusion

表5.2：ROBIN与SESYD数据集上各技术的比较

Approach	**MAP (ROBIN)**	**MAP (SESYD)**
Baseline [Sharma *et al.*, 2016a]	0.2479	1.0
OASIS with RLH [Chechik *et al.*, 2009]	0.0956	1.0
OASIS with BOW [Chechik *et al.*, 2009]	0.1892	0.99
Proposed	**0.3047**	1.0

et al.[2017]提出了一种基于卷积神经网络的深度框架，用于在建筑平面图中进行检索。尽管该深度框架在定量评估上优于本章所提出的方法，但仍存在若干不足之处。例如，在检索过程中无法对各特征赋予偏好权重，因为各层以隐式方式提取并组合特征，用户无法对其权重进行调控。相比之下，本章所提出的加权特征组合机制允许用户根据自身需求对各特征赋予相应权重（如图5.11(b)所示），从而在用户视角下更具实用性。此外，该卷积神经网络框架属于简单网络，主要依据物体或布局的形状进行区分；在进行家具部件的细粒度分类时，以及当不同类别中的两个布局具有几乎相同的形状时，其性能将大打折扣。

由于所提出的算法处理的是平面布局图像的相似性，因此还将结果与在线大规模图像相似性学习算法（OASIS）[Chechik 等，2009]、游程长度直方图技术（RLH）[de las Heras 等，2013]以及词袋模型技术（BOW）[Lazebnik 等，2006b]进行了对比。之所以选择这些算法进行比较，是因为它们在文档图像中的目标检索任务中应用广泛。实验发现，与 OASIS 方法相比，所提出的特征匹配方案获得了更高的平均精度值（参见表 5.2），这是因为 OASIS 算法主要关注相关与不相关图像对之间的相似性差异，而忽略了高度相似图像之间的相似性。如前所述，在 SESYD 公开数据集上复现表 5.2 中所述的方法时，由于样本的类间相似性较低，获得了完美的平均精度均值（MAP）1。

在[Ahmed *et al.*, 2014]中，作者提出了一种基于草图的平面布局检索框架，其中查询布局仅使用矩形等基本几何形状表示为草图。相比之下，在本研究中则考虑了由复杂几何基元表示的室内装饰。此外，[Ahmed *et al.*, 2014]的工作中缺乏对室内装饰的细粒度标注分析。该框架对旋转具有敏感性，以将房间的相对位置作为特征加以利用。客户可能会遵循吠陀建筑学（Vastu）的建议，偏好“*master bedroom should be towards the south facing the wall, so that sleeping person's head is facing south*”。若使检索过程对旋转不变，则将无法满足这一需求。

5.6 摘要

所提出的用于细粒度平面图检索的框架，同时利用了诸如面积、家具数量与类型以及房间之间的邻接关系等高层语义特征。在检索过程中引入这些高层特征是一种新颖的思路，可在用户对全配饰平面图有特定需求的场景中得到应用。此外，加权特征融合

helps in weighing features when a user is querying, and the user is fixated over certain features to be present during retrieval mandatorily.

Till this point in the thesis, only query by example in the form of image has been used. However, the primary concern in this case is that the user must have an image of a floor plan available for querying. In a real scenario, a user may not have a digitized floor plan image to search for similar floor plans. A more intuitive mode of querying is to scribble the requirement as a sketch and query accordingly. Therefore, in the next Chapter, a sketch based input query is discussed, where, the user can draw the layout and the internal furniture components as well. In the current world of smart, hand-held devices this provides an easy mode of search for the end users. Thus, the next Chapter deals with changing the mode of query from images to sketch, to ease querying and make the application more viable.

在用户进行查询时，该机制有助于对特征进行加权，并且用户会明确指定某些特征在检索过程中必须存在。

在本论文的现有阶段，仅采用了基于图像的示例查询方式。然而，这种方法的主要局限在于用户必须事先拥有可用于查询的平面图图像。在实际应用场景中，用户可能并未持有可供检索的数字化平面图图像。更为直观的查询方式是通过手绘草图来表达需求并进行相应查询。因此，在下一章中，我们将探讨基于草图的输入查询方法，用户不仅能够绘制建筑布局，还可以标注内部家具等构件。在当前智能手持设备普及的时代，这种方式为终端用户提供了便捷的搜索途径。由此可见，下一章将重点研究从基于图像的查询模式向基于草图的查询模式的转变，以简化查询流程并提升应用的实用性。

6

Domain invariant feature transform for interactive floor plan retrieval

Content-based image retrieval (CBIR) emerged as an attempt to deal not only with the absence or insufficiency of annotations for most of the images, but also to support alternative retrieval approaches, relying on visual perception, which is more appropriate in many scenarios. Within CBIR, sketch-based image retrieval (SBIR) aims at retrieving images that are similar to a sketch made by the user (typically a simple set of drawing lines). Thus, SBIR is particularly adapted in situations where a user has a perceptual image of what he is searching. In this scenario, a sketch image is useful specially when the image dataset is not annotated and the user has no similar example image to use as a query input. The challenges associated with SBIR are:

1. Finding a relevant visual content representation associated with a similarity measure that allows effective comparison with a query that is not a picture, but rather a drawing made by a user who sometimes is not very skillful.

2. Making retrieval scalable to large image datasets by building an appropriate index structure which is able to exploit the content representation and similarity measure in a better way.

3. Sketches are highly abstract and carry a huge level of variation from person-to-person.

4. Sketches don't provide much of visual cues like colour as present in images.

5. The inherent difference between two domains (sketch and image) also makes the problem more challenging.

In Chapter 3, 4 and 5, the proposed methodologies revolve around extracting semantic and structural features from floor plan images. Exploring both hand-crafted and deep learning features has been done in the previous chapters to extract and represent meaningful content in images and finally retrieve similar floor plan images from the database. The important thing to be noted here is that the mode of the query until this chapter is through an image. In this Chapter, the query mode is interactive, i.e., it can either be an image based query or a sketch based query. This Chapter is organized as follows: Sec. 6.1 gives a brief overview of the approaches proposed in this Chapter, Sec. 6.2 gives an insight into the first approach using Cyclic GAN that has been proposed in this Chapter and Sec. 6.3 details out the second approach using autoencoders and Cyclic GANs for the task of retrieval. Section 6.4 gives a qualitative and quantitative analysis of both approaches. Finally, Sec. 6.5 concludes the Chapter.

6.1 BRIEF OVERVIEW

Sketching using pen and paper is more intuitive and easy to represent ideas as compared to other modalities. With the popularity of digital hand-held devices, users are keen on using sketches to exchange ideas. Users can sketch the floor plans on a tablet in the same fashion as they would on

6

用于交互式平面图检索的领域不变特征变换

基于内容的图像检索（CBIR）应运而生，旨在应对大多数图像缺乏标注或标注不足的问题，并支持基于视觉感知的替代检索方法，这在许多场景下更为适用。在CBIR框架下，基于草图的图像检索（SBIR）旨在检索与用户绘制的草图（通常为一组简单的线条）相似的图像。因此，当用户对其所搜索的内容具有明确的感知形象时，SBIR尤为适用。在这种情况下，尤其是在图像数据集未加标注且用户无法提供类似示例图像作为查询输入时，草图图像显得尤为有用。与SBIR相关的挑战包括：

1. 寻找一种与相似性度量相关的视觉内容表示方法，该方法能够实现与查询的有效比对；所述查询并非图片，而是一位用户绘制的草图，且该用户的绘图技巧有时并不娴熟。

2. 通过构建一种能够更有效地利用内容表示和相似性度量的适当索引结构，使检索技术可扩展至大规模图像数据集。

3. 草图具有高度的抽象性，且不同个体之间的差异极大。

4. 素描不像图像那样提供丰富的视觉线索，如颜色信息。5. 素描与图像这两个领域之间的固有差异也使该问题更具挑战性。

在第3、4和5章中，所提出的多种方法均围绕从平面图图像中提取语义特征与结构特征展开。前几章已分别探讨了手工设计特征与深度学习特征的提取与表示，旨在从图像中挖掘并表征有意义的内容，最终实现从数据库中检索相似的平面图图像。值得注意的是，截至本章为止，查询方式均为基于图像的查询。而在本章中，查询模式则为交互式，即既可以是基于图像的查询，也可以是基于草图的查询。本章的组织结构如下：第6.1节简要概述本章所提出的方法；第6.2节深入介绍本章提出的基于循环生成对抗网络（Cyclic GAN）的第一种方法；第6.3节详细阐述用于检索任务的第二种方法，该方法结合了自编码器与循环生成对抗网络；第6.4节对两种方法分别进行定性和定量分析；最后，第6.5节对本章予以总结。

6.1 简要概述

与其它表达方式相比，使用纸笔进行草图绘制更为直观，且更易于呈现创意。随着便携式数字设备的普及，用户愈发倾向于通过草图来交流想法。用户可以在平板上以与在纸上相同的方式绘制平面图。

paper. With the increase in the usage of touchscreens and hand-held devices, sketch-based querying for finding a dream home can be a convenient proposition. Sketch-based floor plan retrieval can benefit the architects and the users while searching for floor plans in the early design phases of a building. However, sketch-based processing and retrieval come with their challenges. In this Chapter, techniques for retrieval of floor plans given an interactive query mode are proposed. Two approaches to facilitate sketch based retrieval of floor plans are presented in this Chapter. Figure 6.1 shows the flow charts of the two approaches.

1. **Approach 1:** An initial attempt aims at using a Cyclic Generative Adversarial Network (Cyclic GAN) for domain mapping between floor plan sketches and floor plan images along with Convolutional Neural Network (CNN) to extract features from the mapped query and retrieval set.

2. **Approach 2:** Although approach 1 performs well for sketch-based retrieval of floor plans, there is still a scope of improvement in terms of the precision values during retrieval. Also, there is no provision of image-based retrieval in the proposed approach 1. This serves as the motivation for an efficient unified framework to meet the requirements of both image to image as well as the sketch to image retrieval. Therefore, to improve the results of the retrieval mode and to propose a composite solution to multimodal retrieval, a unified framework for retrieval of similar floor plan images given query mode as floor plan sketches or images is proposed. This approach uses the conjunction of CNN for query classification, Cyclic GAN for sample generation and Autoencoder for domain mapping and proves to be efficient than approach 1 in terms of retrieval.

In the next section the methodology associated with Approach 1, taking into account Cyclic GANs and CNN for feature extraction and domain mapping is discussed in detail.

6.2 APPROACH 1: CYCLIC GAN FOR DOMAIN ADAPTATION AND SKETCH BASED FLOOR PLAN RETRIEVAL

In this section, a brief overview of the first approach for similar floor plan retrieval using sketch modality is discussed. Figure 6.2 depicts the complete framework of the proposed technique. If the query is a sketch, the steps carried out are as follows. The entire framework is divided into three main phases. They are: (1) domain mapping through Cycle Generative Adversarial Networks (Cyclic GANs); (2) feature representation through Convolutional Neural Networks and (3) matching and retrieval. Given an input sketch S, the goal is to retrieve similar looking floor plan images (I) from the ROBIN dataset [Sharma *et al.*, 2017] corresponding to the query sketch. The Cyclic GAN is trained using images from ROBIN and sketches from S-ROBIN to learn domain representation. As direct retrieval of floor plan images given a sketch is not straightforward (refer the low performance of CNN1 in Fig. 6.14. (b)). Therefore, first the ROBIN dataset images are converted into their sketch counterparts (called GAN generated sketch dataset (GS-ROBIN)) using a Cyclic GAN trained to map the sketch and image domain. Thereafter, the post-processing of the generated sketches (GS) is done to get their enhanced versions. Then a CNN is trained over the generated sketches to learn the deep feature representation for the floor plan retrieval task. The learned deep representation helps to extract deep features from the GS-ROBIN database. The domain mapping is necessary because of large variation between the quality of images/sketches generated using Cyclic GAN and original images/sketches from the ROBIN/S-ROBIN dataset, as shown in Fig. 6.3. During the retrieval stage, the framework extracts the deep features from the query sketch also using the same deep representation. The extracted deep features are then matched with the database features. A similarity score for a particular layout is thus calculated

随着触摸屏和手持设备的使用日益普及，基于草图的查询方式在寻找理想住宅时显得尤为便捷。在建筑的早期设计阶段，基于草图的平面图检索能够同时惠及建筑师与用户。然而，基于草图的处理与检索也面临诸多挑战。本章提出了一种基于交互式查询模式的平面图检索技术，并介绍了两种用于促进基于草图的平面图检索的方法。图6.1展示了这两种方法的流程图。

1. 方法一：初步尝试采用循环生成对抗网络（Cyclic GAN）实现平面草图与平面图像之间的域映射，并结合卷积神经网络（CNN）从映射后的查询集和检索集中提取特征。

2. 方法二：尽管方法一在基于草图的平面图检索任务中表现良好，但在检索过程中的精度指标仍有提升空间。此外，所提出的方法一并未提供基于图像的检索功能。这一局限性促使我们设计一种高效的统一框架，以同时满足基于图像的图像检索以及基于草图的图像检索需求。因此，为改进检索效果并提出一种面向多模态检索的综合解决方案，本文提出了一种统一框架，该框架可根据查询模式（平面图草图或平面图图像）检索相似的平面图图像。该方法结合了用于查询分类的卷积神经网络（CNN）、用于样本生成的循环生成对抗网络（CycleGAN）以及用于领域映射的自编码器，在检索性能方面优于方法一。

在下一节中，将详细讨论与方法一相关的技术路线，该路线综合考虑了循环生成对抗网络（Cyclic GAN）以及卷积神经网络（CNN）在特征提取与领域映射中的应用。

6.2 方法一：基于循环生成对抗网络的域适应与基于草图的平面布局检索

在本节中，将简要概述基于草图模态进行相似平面布局检索的第一种方法。图6.2展示了所提出方法的完整框架。若查询为草图，则具体步骤如下：整个框架可分为三个主要阶段，分别为：（1）通过循环生成对抗网络（Cyclic GAN）进行域映射；（2）通过卷积神经网络进行特征表示；以及（3）匹配与检索。给定输入草图S，目标是从ROBIN数据集[Sharma *et al.*, 2017]中检索与该草图外观相似的平面布局图像I。循环生成对抗网络利用ROBIN数据集中的图像及S-ROBIN数据集中的草图进行训练，以学习域间的表征。由于直接根据草图检索平面布局图像并不容易（参见图6.14(b)中CNN1的较低性能），因此首先通过一个经训练用于实现草图与图像域之间映射的循环生成对抗网络，将ROBIN数据集中的图像转换为其对应的草图版本（称为GAN生成草图数据集，GS-ROBIN）。随后，对生成的草图（GS）进行后处理，得到其增强版本。接着，在这些生成的草图上训练卷积神经网络，以学习用于平面布局检索任务的深度特征表示。所学得的深度表示有助于从GS-ROBIN数据库中提取深度特征。由于如图6.3所示，使用循环生成对抗网络生成的图像/草图与ROBIN/S-ROBIN数据集中原始图像/草图在质量上存在较大差异，因此域映射是必要的。在检索阶段，该框架同样采用上述深度表示从查询草图中提取深度特征，并将其与数据库中的特征进行匹配，从而计算出特定布局的相似度得分。

for all the query samples. To know which individual layer of the deep framework performs best, similarity using all the individual deep features/layers is calculated, and at the same time, the effect of CNN trained on the ROBIN dataset, the S-ROBIN dataset, and the GS-ROBIN dataset is studied. Details of the same are discussed in Sec. 6.4.5.

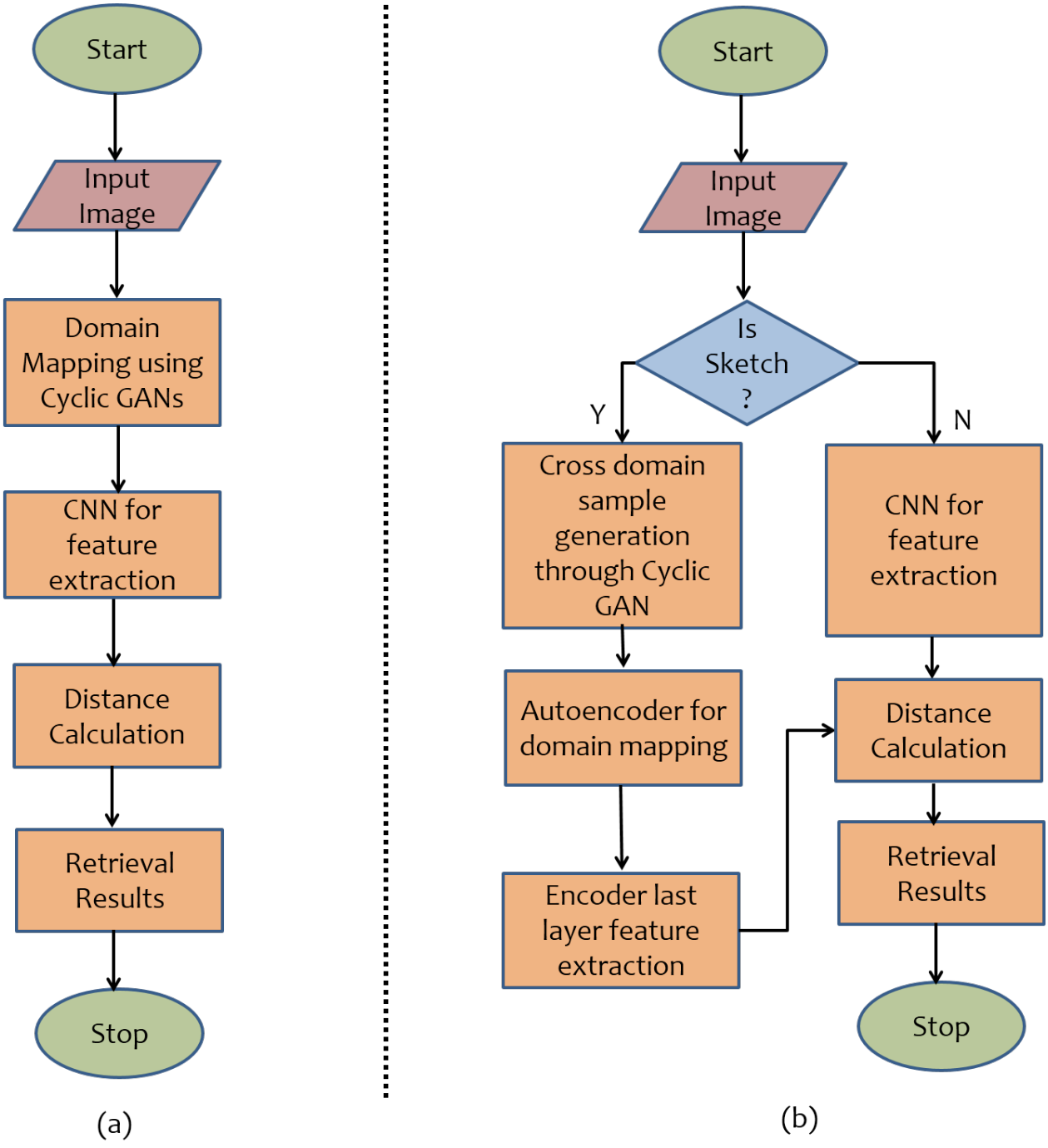


Figure 6.1. : Flowchart comparing the different sub-components in two approaches for sketch based retrieval. (a) Approach 1: Basic framework using cyclic GAN for domain mapping and sketch based retrieval (b) Approach 2: An improved unified framework using cyclic GAN and autoencoders in conjunction for floor plan retrieval.

针对所有查询样本。为确定深度网络中哪一层的性能最佳，计算了基于各单独深度特征层的相似度；同时，还研究了分别在ROBIN数据集、S-ROBIN数据集和GS-ROBIN数据集上训练的CNN模型的效果。相关细节详见第6.4.5节。

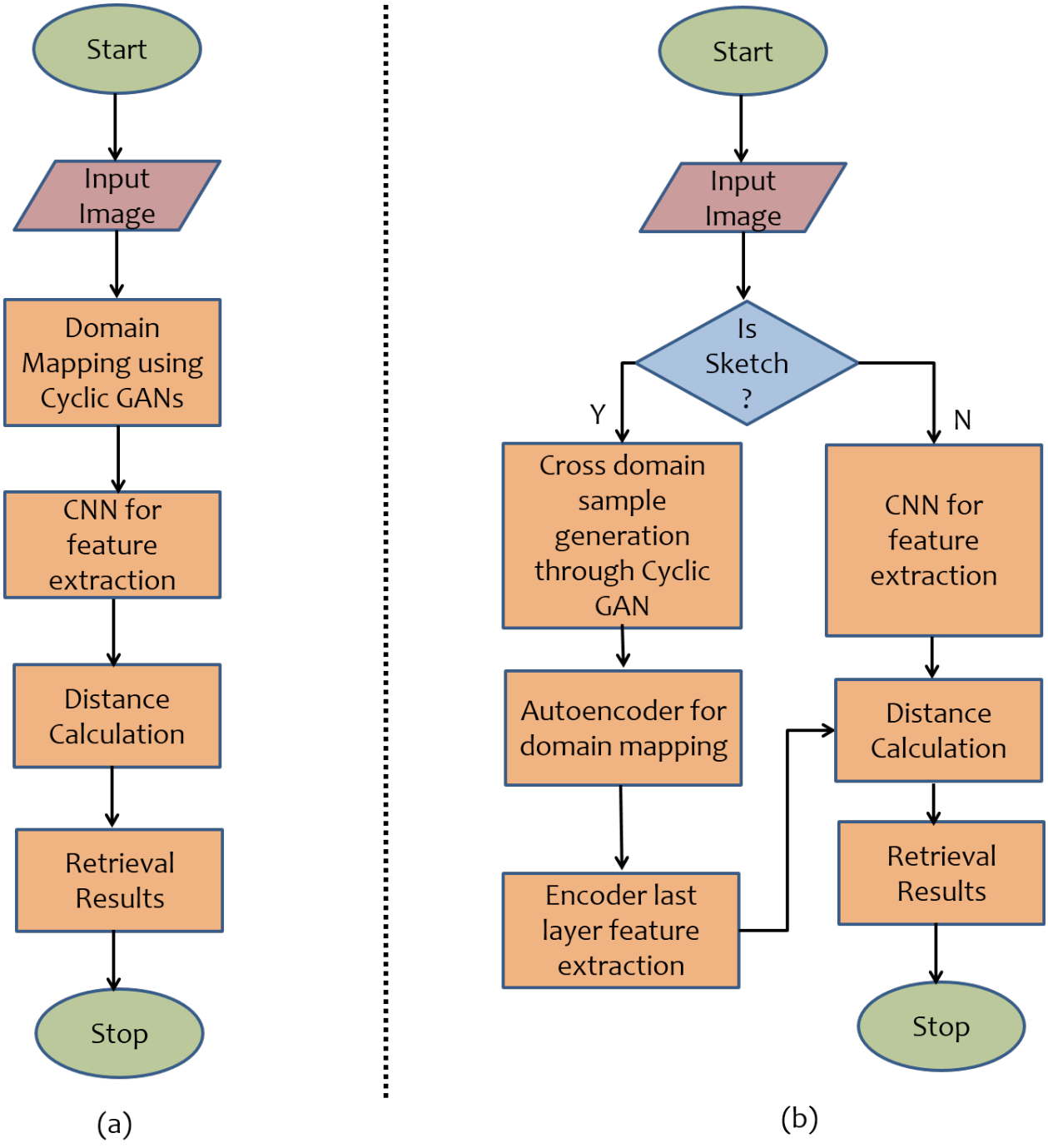


图6.1：比较两种基于草图检索方法中不同子组件的流程图。（a）方法一：采用循环生成对抗网络进行域映射并实现基于草图检索的基本框架；（b）方法二：一种改进的统一框架，结合使用循环生成对抗网络与自编码器以实现平面图检索。

6.2.1 Domain Mapping through Cyclic GAN

In this section, the use of Cyclic GAN for facilitating the "domain shift" between sketch and image floor plan domain is illustrated. In this case, the purpose is to choose a floor plan image domain (I) and then transform it into an image of sketched floor plan domain (S), and vice versa. Since one-to-one mapping does not exist between the sketch and the image floor plans in the training set, the problem becomes more challenging Zhu *et al.* [2017]. Having this relaxation of the non-existence of one-to-one mapping makes this formulation quite powerful. The need for a paired image in the target-domain is eliminated by the two-step transformation that involves mapping of source domain image to the target and back to the original image. Mapping the image to the target domain is improved by competing for the generator with the discriminator in the Cyclic GAN model. For training the Cyclic GAN model, a training set consisting of 70% samples from the ROBIN and S-ROBIN datasets is used.

Figure 6.4 depicts the overall objective of the proposed model for domain adaptation. Let, the image domain be denoted as I, and the sketch domain be denoted as S. An i^{th} floor plan image sample is denoted as f_I^i, where $i = 1,2,\ldots,m$. Here m is the number of floor plan image samples taken from ROBIN. On the other hand, the j^{th} floor plan sketch sample from S-ROBIN is denoted as f_S^j, where $j = 1,2,\ldots,n$. In this case, n is the number of sketches in S-ROBIN. To convert a floor plan sketch to its corresponding image version, a mapping function $G(.)$ is introduced. The task of $G(.)$ is to take a floor plan sketch from S-ROBIN and convert it to an image, which closely resembles a floor plan image ($\tilde{f}_I^j$) from ROBIN. This process is denoted in Fig. 6.4. Similarly, to convert a floor plan image from ROBIN to that of a sketch (belonging to S-ROBIN), another mapping function $F(.)$ is introduced. This function performs the same task as that of $G(.)$, but in the reverse order. The hypothesis is that the generated images, i.e., outputs of $G(.)$ and $F(.)$ will eventually match the target distributions. To compare between the actual and the desired distributions, two adversarial discriminators D_S and D_I have been introduced. The task of D_I is to check how good the mapping is from a given sketch to an image, whereas D_S aims to achieve the reverse. During the training phase, examples from both S-ROBIN and ROBIN dataset are given to the network to learn the mapping functions $G(.)$ and $F(.)$.

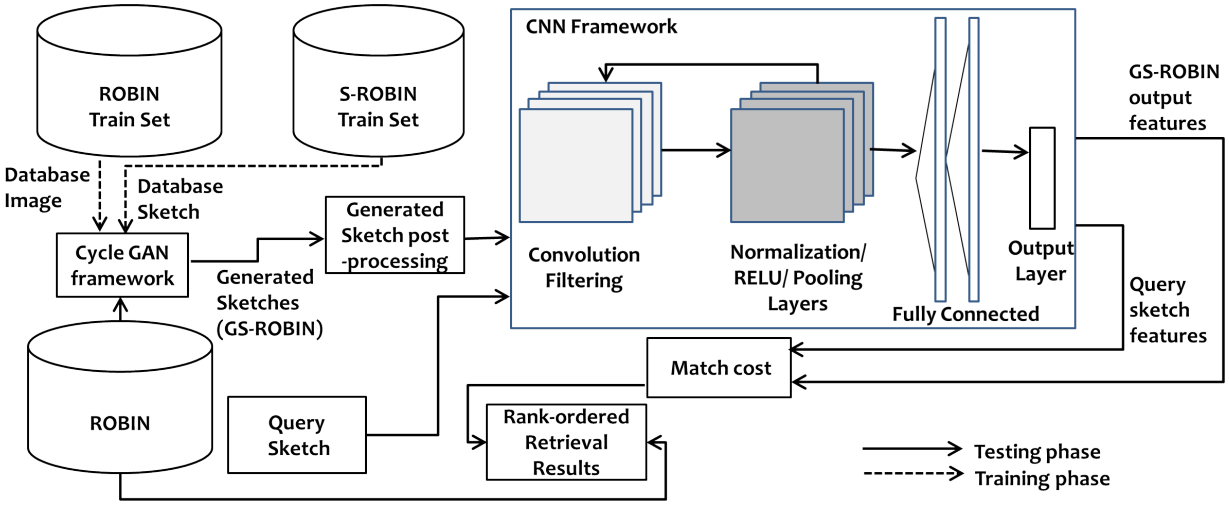


Figure 6.2. : Framework for sketch based retrieval of floor plans. Domain adaptation is facilitated by a Cyclic GAN.

6.2.1 通过循环生成对抗网络进行领域映射

在本节中，阐述了利用循环生成对抗网络（Cyclic GAN）来缓解草图与图像平面图域之间的“领域偏移”问题。具体而言，其目的在于选定一个平面图图像域（I），并将其转换为草图平面图图像域（S），反之亦然。由于训练集中草图与图像平面图之间不存在一一映射关系，这一问题变得更加复杂，如Zhu *et al.* [2017]所指出的。通过放宽对一一映射关系的严格要求，该方法具有较强的适用性。借助包含源域到目标域再到原始图像的两步变换，无需提供目标域中的配对图像。在循环生成对抗网络模型中，通过生成器与判别器之间的对抗机制，可进一步提升图像到目标域的映射效果。用于训练循环生成对抗网络模型的训练集由ROBIN和S-ROBIN数据集中的70%样本构成。

图6.4展示了所提出领域适应模型的总体目标。设图像域记为I，草图域记为S。一个楼层平面图图像样本记为f_I^i，其中$i = 1, 2, \ldots, m$。此处m表示从ROBIN中抽取的楼层平面图图像样本数量。另一方面，来自S-ROBIN的楼层平面图草图样本记为f_S^j，其中$j = 1, 2, \ldots, n$。在这种情况下，n表示S-ROBIN中的草图数量。为了将楼层平面图草图转换为其对应的图像版本，引入了一个映射函数$G(.)$。$G(.)$的任务是从S-ROBIN中获取一幅楼层平面图草图，并将其转换为一幅与ROBIN中的楼层平面图图像（$\tilde{f}_I^j$）高度相似的图像。这一过程在图6.4中予以说明。类似地，为了将ROBIN中的楼层平面图图像转换为属于S-ROBIN的草图，又引入了另一个映射函数$F(.)$。该函数执行与$G(.)$相同的功能，但方向相反。其假设是：由$G(.)$和$F(.)$生成的图像最终将与目标分布相匹配。为了比较实际分布与期望分布，引入了两个对抗判别器D_S和D_I。D_I的任务是评估给定草图到图像的映射效果，而D_S则致力于实现反向映射。在训练阶段，网络同时接收来自S-ROBIN和ROBIN数据集的样本，以学习映射函数$G(.)$和$F(.)$。

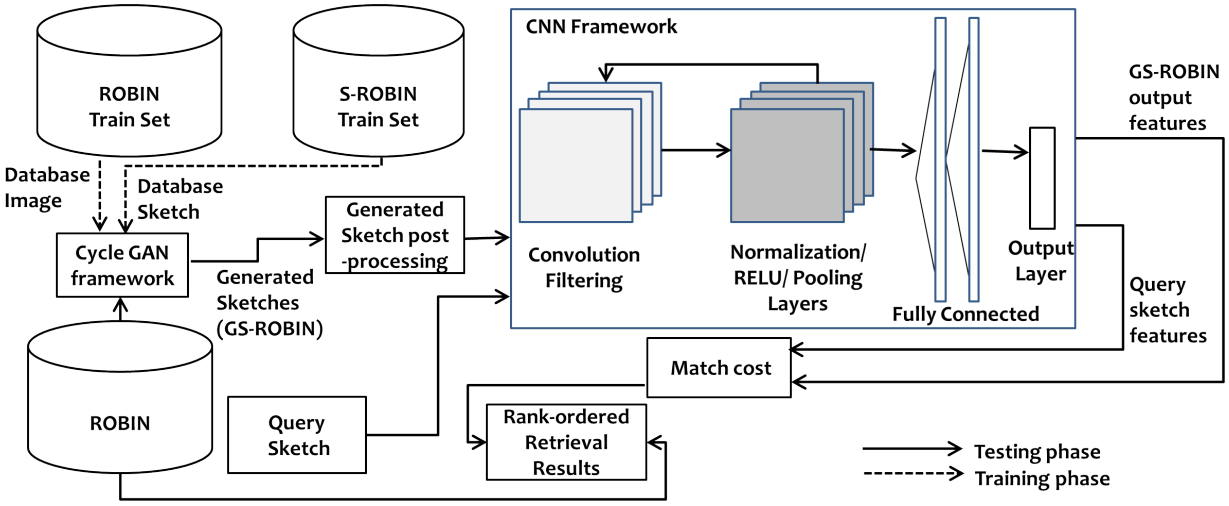


图6.2：基于草图的平面图检索框架。领域自适应由循环生成对抗网络（CycleGAN）实现。

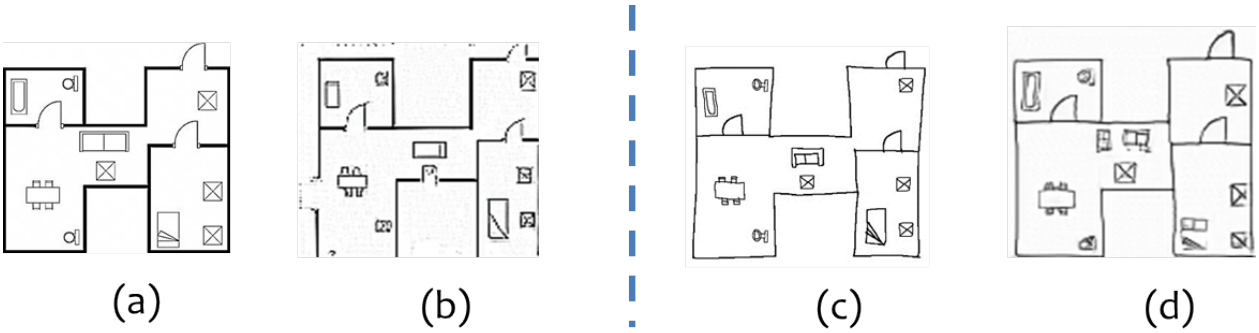


Figure 6.3. : Comparing the original sketches/images with outputs generated from Cyclic GAN. (a) Original floor image from ROBIN dataset. (b) Generated floor plan image using Cyclic GAN. (c) Original floor plan sketch from S-ROBIN dataset. (d) Generated floor plan sketch using Cyclic GAN. Kindly note the quality of generated sketches is better than that of generated images.

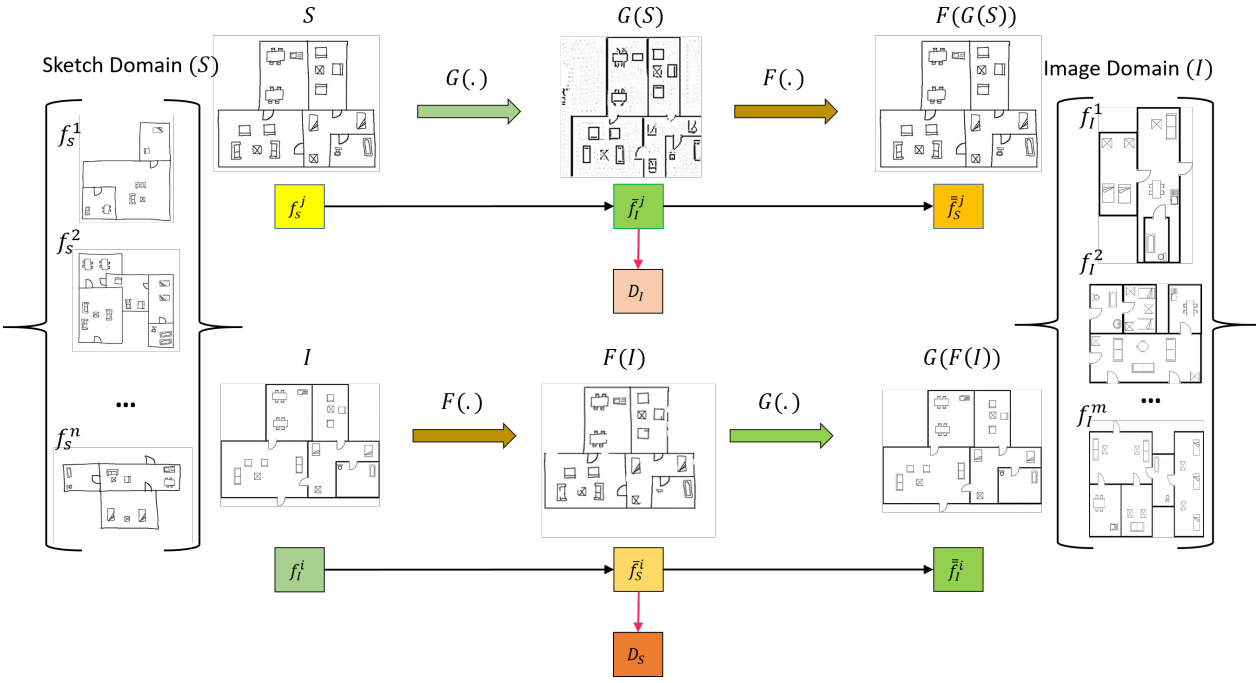


Figure 6.4. : Cyclic GAN framework, corresponding to the proposed approach with converting the sketches from the S-ROBIN dataset into GAN generated images (constituting GI-ROBIN dataset) closely similar in features to the to-be retrieved images and images from the ROBIN dataset to sketches (constituting GS-ROBIN dataset).

Let the probability distribution of a sample $f_S \in S$ be denoted as $f_S \rightarrow p(f_S)$, and for a sample $f_I \in I$ be denoted as $f_I \rightarrow p(f_I)$. Then the adversarial loss function for mapping a floor plan sketch to an image ($S2I_A(.)$) can be written as:

$$S2I_A(G, D_I, S, I) = E_{f_I \rightarrow p(f_I)}[logD_I(f_I)] + E_{f_S \rightarrow p(f_S)}[log(1 - D_I(F(f_S)))] \tag{6.1}$$

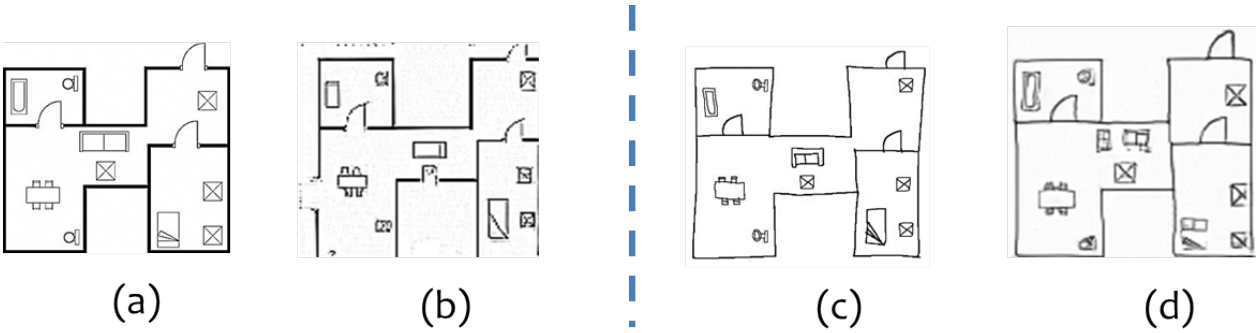


图6.3：对比原始草图/图像与由循环生成对抗网络（Cyclic GAN）生成的输出。（a）ROBIN数据集中的原始楼层图像。（b）利用循环生成对抗网络生成的楼层平面图图像。（c）S-ROBIN数据集中原始的楼层平面图草图。（d）利用循环生成对抗网络生成的楼层平面图草图。请注意，生成的草图质量优于生成的图像质量。

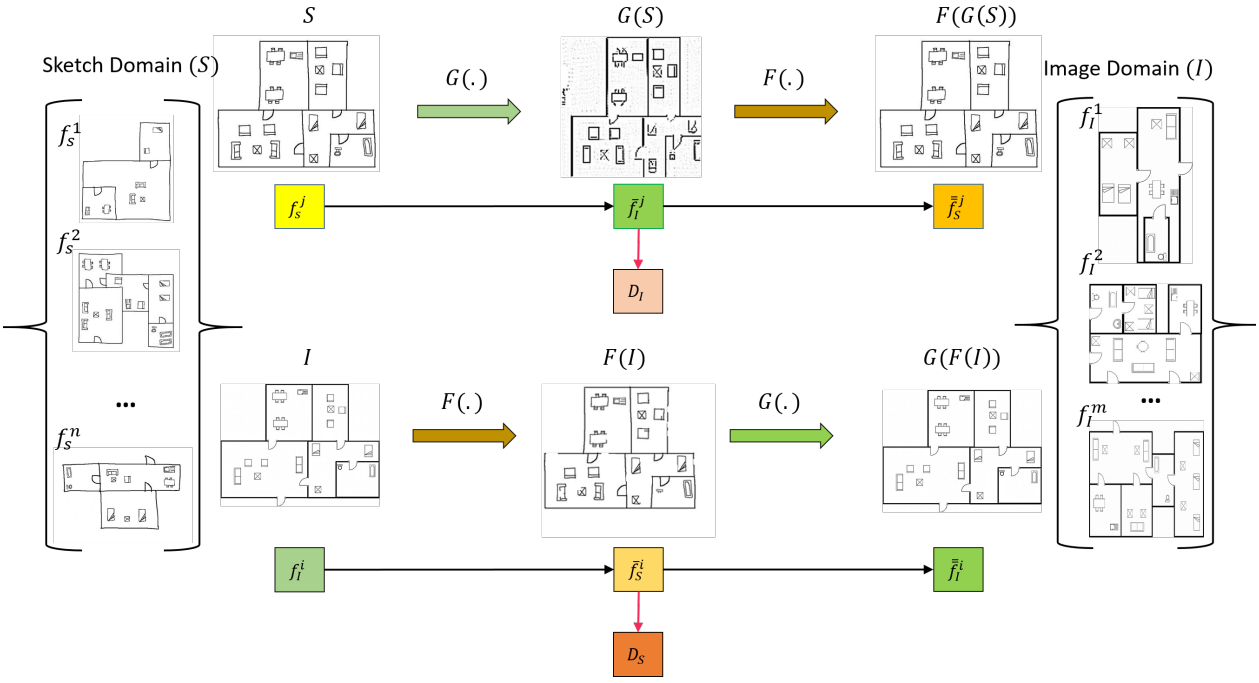


图6.4：循环生成对抗网络框架，对应于所提出的方案，该方案将S-ROBIN数据集中的草图转换为由生成对抗网络生成的图像（构成GI-ROBIN数据集），这些图像在特征上与待检索图像及ROBIN数据集中图像高度相似；同时还将ROBIN数据集中的图像转换为草图（构成GS-ROBIN数据集）。

设样本$f_S \in S$的概率分布记为$f_S \rightarrow p(f_S)$，样本$f_I \in I$的概率分布记为$f_I \rightarrow p(f_I)$。则将平面布局草图映射为图像的对抗损失函数（$S2I_A(.)$）可表示为：

$$S2I_A(G, D_I, S, I) = E_{f_I \rightarrow p(f_I)}[logD_I(f_I)] + E_{f_S \rightarrow p(f_S)}[log(1 - D_I(F(f_S)))] \tag{6.1}$$

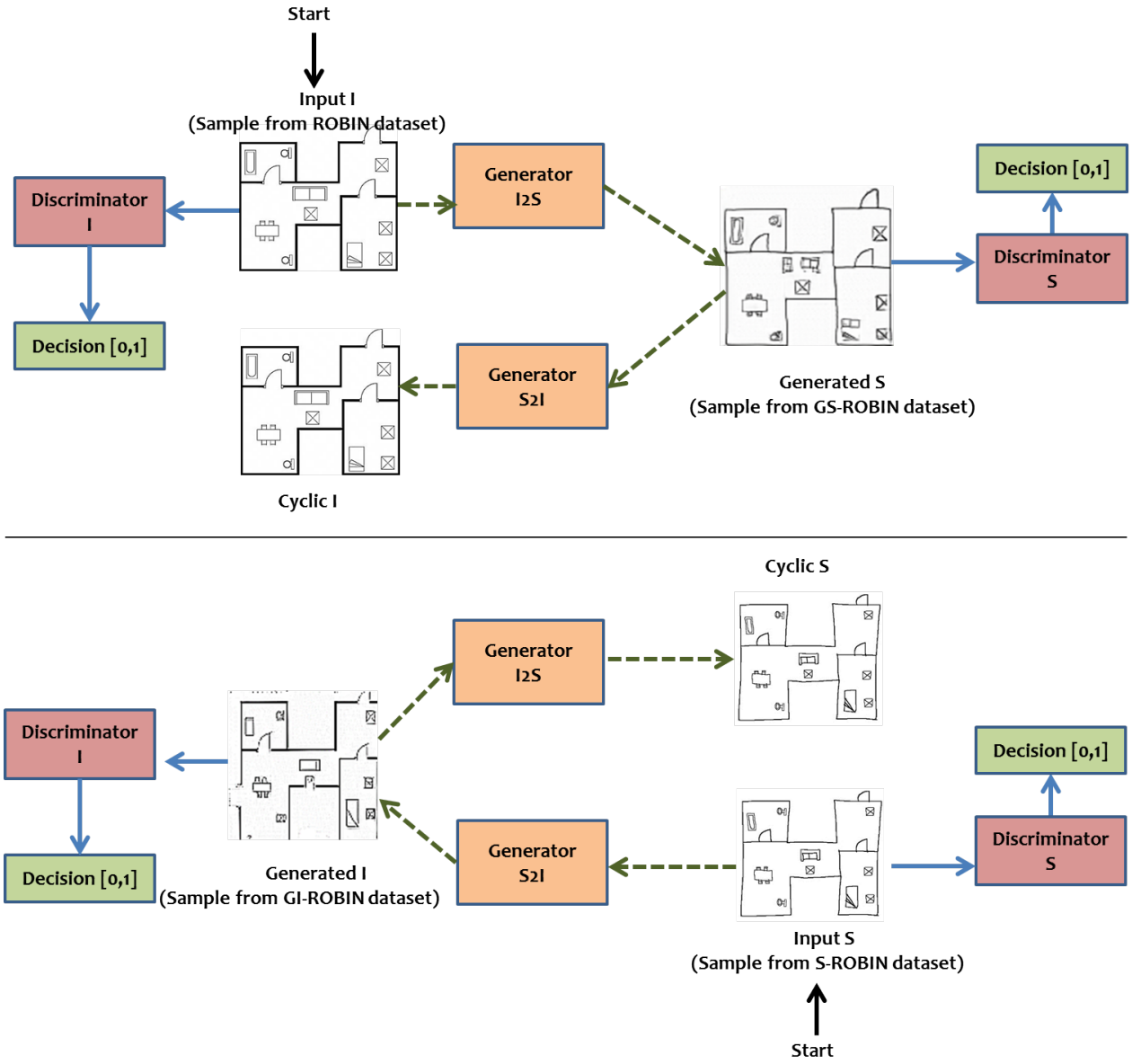


Figure 6.5. : Network architecture for the Cyclic GAN framework, representing the inputs from ROBIN and S-ROBIN dataset. The outputs being generated constitute the GS-ROBIN dataset.

In Eq. 6.1, two operations are taking place. Here the discriminator term D_I, tries to maximize the objective function by distinguishing between the generated floor plan images and the given sketches. Whereas, the generator function $G(.)$ tries to minimize the objective and synthesize image samples as close as possible to the sketch. Similarly, the adversarial loss function for the reverse mapping process, i.e. image to sketch $I2S_A(.)$ is formulated as:

$$I2S_A(F, D_S, I, S) = E_{f_S \to p_{(f_S)}}[logD_S(f_S)] + E_{f_I \to p_{(f_I)}}[log(1 - D_S(F(f_I)))] \tag{6.2}$$

As shown in Fig. 6.4, the generated target domain image using $G(.)$, and $F(.)$, is close to that of the source domain. The learned mapping functions are able to map to any random permutation

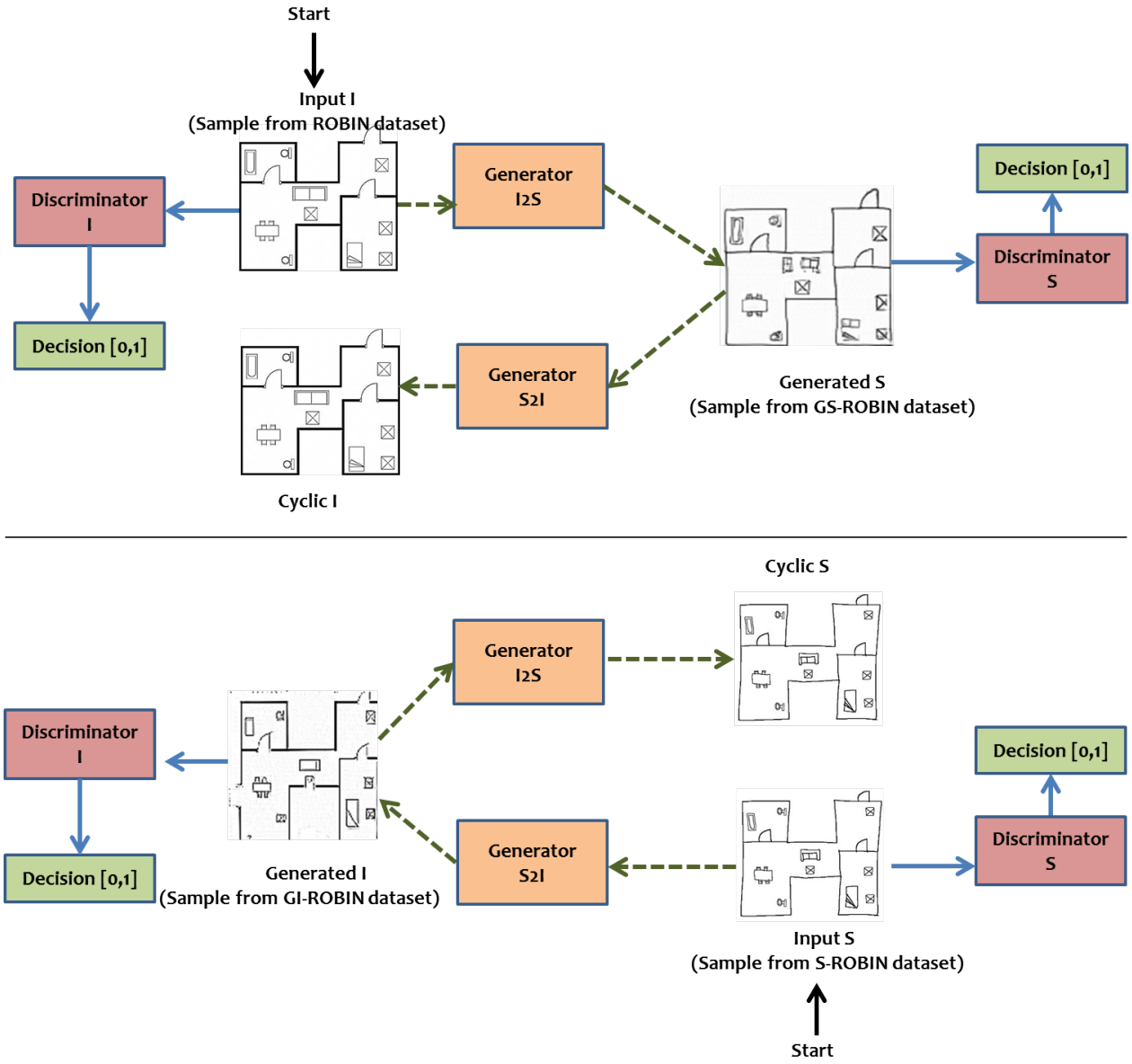


图6.5：循环生成对抗网络框架的网络架构，展示了来自ROBIN和S-ROBIN数据集的输入。所生成的输出构成了GS-ROBIN数据集。

在式6.1中，同时进行了两项操作。其中，判别器项D_I通过区分生成的平面布局图像与给定的草图来试图最大化目标函数；而生成器项$G(.)$则试图最小化该目标函数，并生成尽可能接近草图的图像样本。类似地，反向映射过程，即从图像到草图的对抗损失函数$I2S_A(.)$可表述为：

$$I2S_A(F, D_S, I, S) = E_{f_S \to p(f_S)}[logD_S(f_S)] + E_{f_I \to p(f_I)}[log(1 - D_S(F(f_I)))] \tag{6.2}$$

如图6.4所示，利用$G(.)$和$F(.)$生成的目标域图像与源域图像较为接近。所学习的映射函数能够映射至任意随机排列。

of images. However, it doesn't ensure that for a given floor plan image f_I^i, a corresponding f_S^i is obtained. In other words, given an image sample $f_s \in S$, if the generator function $G(.)$ is able to generate f_I^G (here the sub-script I stands for the image domain, and the super-script G denotes that it is a generated image), then the mapping function $F(.)$ must be able to take f_I^G and bring back f_S. So the sketch to image generation cycle completes when the proposed network is able to re-generate the input samples, from the synthesized (or GAN generated) samples, and thus termed as "cycle-consistent". Similar consistency is also desired for image to sketch domain translation. The cycle consistency loss ($CCL(.)$) function between the two mapping functions is formulated as:

$$CCL(G,F) = E_{f_S \to p(f_S)}[||F(G(f_S)) - f_S||_1] + E_{f_I \to p(f_I)}[||G(F(f_I)) - f_I||_1] \tag{6.3}$$

The final objective function ($X(.)$), that combines the two adversarial loss functions and the cycle consistency loss function is given by :

$$X(G,F,D_I,D_S) = S2I_A(G,D_I,S,I) + I2S_A(F,D_S,I,S) + \alpha CCL(G,F) \tag{6.4}$$

Here the hyper-parameter α controls the relative significance of the pair of loss functions, i.e. the adversarial loss and cyclic loss. Using this objective function $X(.)$, the optimum value for the two mapping functions $G(.)$ and $F(.)$ (by minimizing over G and F, while maximizing D_I and D_S) is obtained. The value of α was empirically determined to be 10, which yields the best retrieval accuracy for all the experiments performed for this problem. Another diagram to depict the Cyclic GAN framework is shown in Fig. 6.5.

Figure 6.6 depicts the schematic structure of the generator and discriminator network. There are two parts to the network structure, the generator network and the discriminator network. There are three components to the generator network, namely, an encoder, a transformer, and a decoder. The encoder is a convolutional neural network that extracts the features from a given floor plan dataset. The second building block is a transformer network. This layer has been constructed using 6 ResNet blocks to transform from I domain to S domain (refer Fig. 6.4). In Fig. 6.6, the internal structure of an individual ResNet block is shown. Each block is a neural network layer

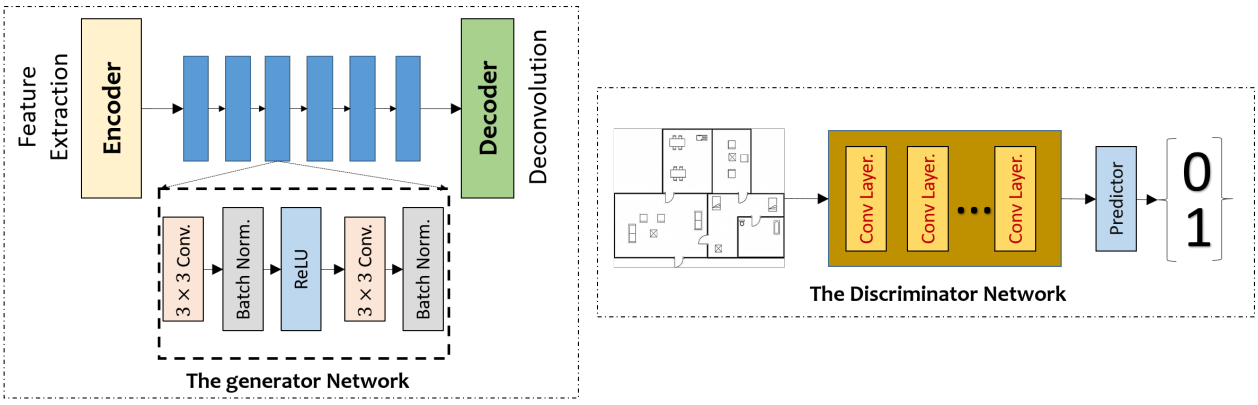


Figure 6.6. : Schematic structure of the Cyclic GAN network used in the proposed framework for image floor plan to synthesized sketched floor plan mapping, and vice-versa.

图像。然而，这并不能保证对于给定的平面布局图像f_I^i，能够得到相应的f_S^i。换言之，给定一个图像样本$f_s \in S$，如果生成器函数$G(.)$能够生成f_I^G（（其中下标I表示图像域，上标G表示该图像是由生成器生成的）），那么映射函数$F(.)$必须能够接收f_I^G并将其还原为f_S。因此，当所提出的网络能够从合成图像（或GAN生成的图像）中重新生成输入样本时，草图到图像的生成循环即告完成，从而被称为“循环一致性”。同样地，在图像到草图的域间转换中也期望实现这种一致性。两个映射函数之间的循环一致性损失函数$CCL(.)$可表述为：

$$CCL(G,F) = E_{f_S \to p(f_S)}[||F(G(f_S)) - f_S||_1] + E_{f_I \to p(f_I)}[||G(F(f_I)) - f_I||_1] \tag{6.3}$$

最终的目标函数（式$X(.)$）由两个对抗损失函数与循环一致性损失函数共同构成，其表达式为：

$$X(G,F,D_I,D_S) = S2I_A(G,D_I,S,I) + I2S_A(F,D_S,I,S) + \alpha CCL(G,F) \tag{6.4}$$

在此，超参数α用于控制两个损失函数——对抗损失与循环损失——的相对重要性。借助该目标函数$X(.)$，通过分别对G和F进行最小化，并同时对D_I和D_S)进行最大化，即可获得两个映射函数$G(.)$和$F(.)$ (的最优解。经实验确定，α的取值为10，此值在针对该问题所开展的所有实验中均能获得最佳的检索精度。循环GAN框架的另一示意图如图6.5所示。

图6.6展示了生成器网络与判别器网络的结构示意图。该网络由两部分组成，即生成器网络和判别器网络。生成器网络包含三个组成部分，分别为编码器、Transformer模块和解码器。编码器是一个卷积神经网络，用于从给定的平面布局数据集中提取特征。第二个构建模块是Transformer网络，该层由6个ResNet块构成，用于实现从I域到S域的转换（参见图6.4）。在图6.6中，展示了单个ResNet块的内部结构，每个块均为一个神经网络层。

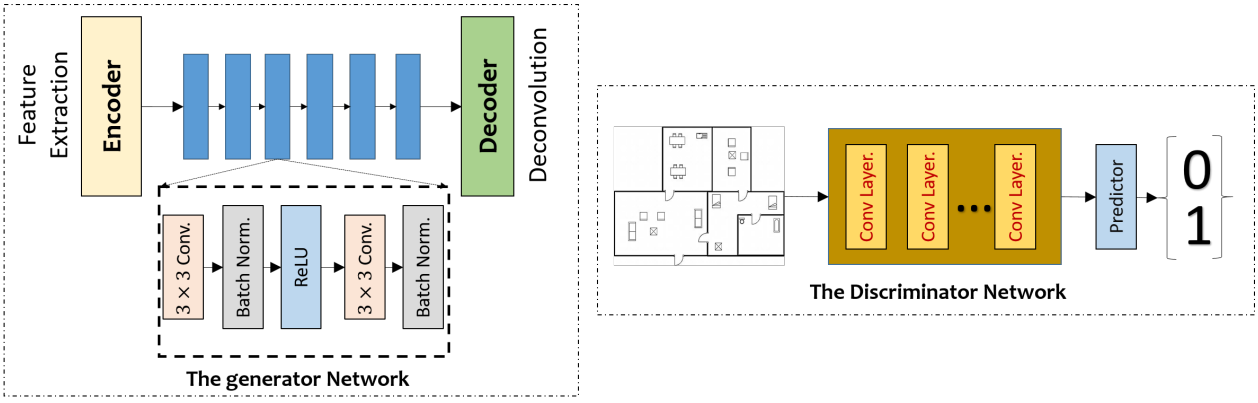


图6.6：所提出的用于从建筑平面图到合成草图式平面图映射及其逆向映射框架中循环生成对抗网络的示意图结构。

which consists of two convolution layers where a residue of input is added to the output. This is done to ensure properties of previous layer inputs are available for later layers as well, so that their output does not deviate much from original input resulting into abrupt results. The third component is a decoder network. The decoder block recreates the low level features from the feature vector by applying a deconvolution layer. The next part of the network is a discriminator network. The input to the discriminator network is an image. The output of the discriminator network is a prediction value 0 or 1, that denotes whether the generated image is original or the output from the generator. The discriminator is implemented as a series of convolutional neural networks. After this training and testing phase, a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset are obtained. Thus, GI-ROBIN dataset generated from the S-ROBIN dataset, and GS-ROBIN dataset generated from the ROBIN dataset, are obtained which are further used for retrieval purposes. To improve the quality of the generated images and sketches, they are sharpened as a post-processing step.

After this training and testing phase, a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset are obtained. It was observed that if the obvious path, of converting the query sketches into corresponding image counterparts and then retrieving from the ROBIN dataset was taken, the retrieval performance was quite weak giving an average precision value as low as 0.1631 and 0.1219, upon being trained on ROBIN as well as S-ROBIN dataset respectively (refer Fig. 6.14. (a)). Therefore, the less intuitive path of converting the ROBIN dataset into sketches and then performing retrieval was taken to yield better results.

6.2.2 Feature Representation through CNN

CNNs have proved to be very effective for feature extraction and classification purposes. In the recent past, research gaining insight into CNN models includes exploring new non-linear activation functions, new training techniques, optimal network configurations and making them system time efficient and reducing overfitting.

In this work as proposed in Chapter 4, a combination of convolutional, pooling, normalization and fully connected layers, similar to described in the work AlexNet [Krizhevsky *et al.*, 2012a], is taken to obtain an effective feature representation from floor plans. Changes are made in the model as compared to AlexNet by not using relighting data augmentation during training; and by changing the order of pooling and normalization layers [Jia *et al.*, 2014]. For learning a representation of CNN (deep) layers, training of the deep model was done on ROBIN, S-ROBIN and GS-ROBIN dataset. It was observed that the deep learning framework trained on GS-ROBIN dataset performs the best during retrieval (see Fig. 6.14. (b)). The effectiveness of each layer was also investigated by taking the activations of all the layers (convolution, pooling, norm, and fully connected). The major motivation behind using lower convolution neural network layers is to understand the effectiveness of low-level features as compared to higher-level feature representation for floor plan retrieval tasks. Since one only needs to compute the feed-forward network based on the matrix multiplication for one time, the whole scheme is highly time efficient. Therefore, this is how the features from both the sketched query samples as well as the ROBIN generated sketches (GS-ROBIN dataset) are generated. In matching and retrieval stage, query samples are matched with database samples to effectively retrieve the rank-ordered samples from the database.

该网络由两层卷积组成，其中将输入的残差加到输出上。这样做是为了确保前一层的特征信息也能被后续层所利用，从而避免其输出与原始输入产生较大偏差，导致结果出现突变。第三部分是解码器网络。解码器模块通过应用反卷积层，从特征向量中重建低层次特征。网络的下一部分是判别器网络。判别器的输入是一幅图像，其输出为一个预测值0或1，用于判断生成的图像是真实的还是由生成器产生的。判别器采用一系列卷积神经网络实现。经过训练与测试阶段后，分别得到一组与草图对应的生成图像，以及一组与ROBIN数据集对应的草图。由此，基于S-ROBIN数据集生成了GI-ROBIN数据集，基于ROBIN数据集生成了GS-ROBIN数据集，并将其进一步用于检索任务。为提升生成图像和草图的质量，在后处理阶段对其进行了锐化处理。

经过这一训练与测试阶段，得到了一组与草图对应的生成图像，以及一组与ROBIN数据集对应的草图。观察发现，若采用将查询草图转换为相应图像并直接在ROBIN数据集中进行检索的常规路径，则检索性能较为薄弱，在分别以ROBIN和S-ROBIN数据集进行训练时，平均精度值分别低至0.1631和0.1219（参见图6.14(a)）。因此，转而采用将ROBIN数据集先转换为草图、再进行检索这一相对不直观的路径，从而获得了更好的效果。

6.2.2 通过卷积神经网络进行特征表示

卷积神经网络已被证明在特征提取与分类任务中具有极高的有效性。近年来，针对卷积神经网络模型的研究工作主要包括：探索新型非线性激活函数、开发新的训练方法、确定最优网络结构，并在此基础上提升系统运行效率、抑制过拟合现象。

在本工作中，如第4章所述，采用了一种由卷积层、池化层、归一化层和全连接层组成的网络结构，该结构与AlexNet [Krizhevsky $et\ al.$, 2012a] 中所描述的类似，用于从平面图中提取有效的特征表示。与AlexNet相比，本模型在训练过程中未使用光照增强的数据增广技术，并调整了池化层与归一化层的先后顺序[Jia $et\ al.$, 2014]。为学习CNN（深度）层的特征表示，深度模型分别在ROBIN、S-ROBIN及GS-ROBIN数据集上进行了训练。实验结果表明，基于GS-ROBIN数据集训练的深度学习框架在检索任务中表现最佳（参见图6.14(b)）。此外，通过分析各层（卷积层、池化层、归一化层及全连接层）的激活值，进一步考察了各层的有效性。采用较低层次的卷积神经网络层的主要动机在于，比较低级特征与高级特征表示在平面图检索任务中的效能差异。由于只需进行一次基于矩阵乘法的前向传播计算，整个方案具有极高的时间效率。因此，无论是草图查询样本，还是由ROBIN生成的草图（GS-ROBIN数据集），其特征均以此方式提取。在匹配与检索阶段，将查询样本与数据库中的样本进行匹配，从而高效地从数据库中检索出按相关性排序的样本。

6.2.3 Matching and Retrieval

As the retrieval is performed on the generated sketches from the ROBIN dataset, therefore, let $(x_\phi^{GS}, y_\phi^{GS})$, $\phi = 1,2...,m$, represent the generated sketches, where, m is the total number of samples in the GS-ROBIN dataset, x_ϕ^{GS} is the observed variable and y_ϕ^{GS} is the corresponding class label in the retrieval database. Let $f(x_\phi^{GS}) : \{f_1(x_\phi^{GS}), f_2(x_\phi^{GS}), .., f_H(x_\phi^{GS})\}$ be the set of hidden layer features extracted from x_ϕ^{GS} GS-ROBIN dataset sample, where $f_p(x_\phi^{GS})$ represents p^{th} hidden layer feature extracted from x_ϕ^{GS} sample in database and H represents the total number of hidden layers in deep feature hierarchy. For experimentation purpose, the features have been extracted from 12 hidden layers of CNN model. For each sample in the database, features from each hidden layer $(f_p(x_\phi^{GS}))$ are extracted and stored in the feature database as discussed in Sec. 6.2.2. In a similar manner, features are extracted from the query sketches as well. Firstly, the performance of individual hidden layer for proposed floor plan retrieval task is observed. Let $f_s(q) : \{f_{s1}(q), f_{s2}(q), .., f_{sH}(q)\}$ be the the set of hidden layer features extracted from a query sketch (q). Then, the Matching Score (M) between query sketch (q) and a sample from the database (x_ϕ^{GS}) using p^{th} hidden layer feature is calculated (and used for ranking), as:

$$M_{approach_1} = \sum_{i=1}^{|f_p(x_\phi^{GS})|} |f_p^i(x_\phi^{GS}) - f_{sp}^i(q)|_2 \tag{6.5}$$

Secondly, the best hidden layer representation for each retrieval sample is obtained by combining the matching score outputs of each hidden layer using a *min* operation. The Matching Score (M) between a query sketch (q) and a generated sketch from the ROBIN dataset (x_ϕ^{GS}) using all the hidden layer features is calculated (and used for ranking), as:

$$M_{approach_1} = argmin_p \sum_{i=1}^{|f_p(x_\phi^{GS})|} |f_p^i(x_\phi^{GS}) - f_{sp}^i(q)|_2, \quad p \in (1,...,H) \tag{6.6}$$

Using Eq. 6.5 and 6.6, given a query sketch the similar generated sketches from the GS-ROBIN dataset are obtained. It is to be noted, that as these sketches have been generated from ROBIN dataset, therefore, transitively they correspond to the images they are generated from. Thus, through this indirect relation, it is shown that the images belonging to the ROBIN dataset [Sharma *et al.*, 2017] are obtained as the retrieved result given the query sketch.

In the next section, Approach 2, catering to sketch based retrieval is discussed in detail. This approach additionally proposes a unified framework for dealing with both image and sketch based queries and finally performaing efficient matching and retrieval of similar floor plans given the dual query mode.

6.3 APPROACH 2: UNIFIED FRAMEWORK FOR RETRIEVAL USING AUTOENCODER AND CYCLIC GAN

Figure 6.7 depicts the complete framework of the proposed technique. Firstly, a query is classified as a floor plan sketch or image using a Convolutional Neural Network (CNN). If

6.2.3 匹配与检索

由于检索是在ROBIN数据集生成的草图上进行的，因此令$x_{\phi}^{GS}, y_{\phi}^{GS}$、$\phi$ =、,、 ...,m分别表示这些生成的草图，其中m为GS-ROBIN数据集中样本的总数，x_{ϕ}^{GS}为观测变量，y_{ϕ}^{GS}为检索数据库中对应的类别标签。令$f(x_{\phi}^{GS})$: $\{f_1(x_{\phi}^{GS}), f_2(x_{\phi}^{GS}), .., f_H(x_{\phi}^{GS})\}$为从GS-ROBIN数据集中的某个样本提取的隐藏层特征集合，其中$f_p(x_{\phi}^{GS})$表示从数据库中的第{i}个样本提取的第{j}个隐藏层特征，H则表示深度特征层级结构中的隐藏层总数。在实验中，特征是从CNN模型的12个隐藏层中提取的。对于数据库中的每个样本，均按第6.2.2节所述，提取各隐藏层的特征并存储于特征数据库中。同样地，查询草图的特征也被提取。首先，考察单个隐藏层在所提出的平面布局检索任务中的性能。令$f_s(q)$: $\{f_{s1}(q), f_{s2}(q), .., f_{sH}(q)\}$为从某一查询草图（$q$）中提取的隐藏层特征集合。随后，利用第{k}个隐藏层的特征计算查询草图（q）与数据库中某样本（x_{ϕ}^{GS}）之间的匹配得分（M），并以此作为排序依据，计算公式如下:

$$M_{approach_1} = \sum_{i=1}^{|f_p(x_{\phi}^{GS})|} |f_p^i(x_{\phi}^{GS}) - f_{sp}^i(q)|_2 \tag{6.5}$$

其次，通过将各隐藏层的匹配分数输出以min运算进行组合，得到每个检索样本的最佳隐藏层表示。利用所有隐藏层特征，计算查询草图(q)与ROBIN数据集中生成草图(x_{ϕ}^{GS})之间的匹配分数(M)（并用于排序），其计算公式为:

$$M_{approach_1} = argmin_p \sum_{i=1}^{|f_p(x_{\phi}^{GS})|} |f_p^i(x_{\phi}^{GS}) - f_{sp}^i(q)|_2, \quad p \in (1, ..., H) \tag{6.6}$$

利用式6.5和6.6，给定一个查询草图，即可从GS-ROBIN数据集中获取与其相似的生成草图。需要指出的是，由于这些草图均源自ROBIN数据集，因此它们在传递意义上与所对应的图像存在关联。由此，通过这一间接关联关系可以证明：当以某个查询草图为输入时，检索结果即为属于ROBIN数据集[Sharma *et al.*, 2017]的图像。

在下一节中，将详细讨论适用于基于草图检索的方案2。该方案进一步提出了一种统一框架，用于同时处理基于图像和基于草图的查询，并在双模式查询下实现对相似平面图的高效匹配与检索。

6.3 方法二：基于自编码器与循环生成对抗网络的统一检索框架

图6.7展示了所提出方法的完整框架。首先，利用卷积神经网络（CNN）将查询分类为平面布局草图或图像。如果

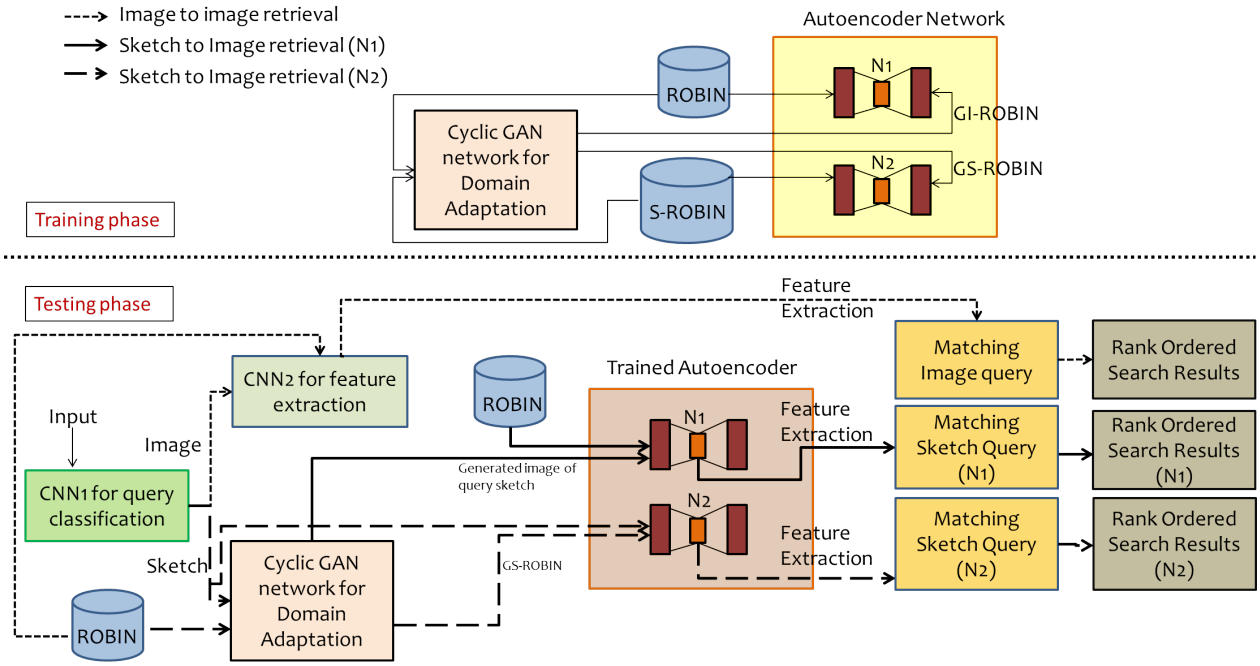


Figure 6.7. : Unified framework for multimodal retrieval.

the query is a floor plan image, the query and the database images are passed through a CNN for feature extraction, and retrieval is carried out using a distance metric to define similarity as proposed in Chapter 4. However, if the query is a sketch, the steps are taken as follows. The entire framework is divided into three main phases. They are: (1) cross-domain sample generation through Cyclic Generative Adversarial Networks (Cyclic GANs); (2) domain mapping through autoencoder and Cyclic GANs, and (3) matching and retrieval. Given an input sketch S, the goal is to retrieve similar looking floor plan images (I) from the ROBIN dataset [Sharma *et al.*, 2017] corresponding to the query sketch. The Cyclic GAN is trained using images from ROBIN and sketches from S-ROBIN to learn domain representation. Direct retrieval of floor plan images, given a sketch, is not straightforward (refer to the low performance of DANIEL in Fig. 6.18) proposition. Therefore, firstly the ROBIN dataset images are converted into their sketch counterparts (called GAN generated sketch dataset (GS-ROBIN)) and the S-ROBIN dataset into their image counterparts (called GAN generated image dataset (GI-ROBIN)) using a Cyclic GAN trained to map the sketch and image domain (as shown in Fig. 6.4).

Further, to map both the floor plan sketches/ retrieval set floor plan images and the cyclic GAN generated outputs, to a common space, experiments are performed by training two autoencoders, taking input as images from ROBIN (network N1) and sketches from S-ROBIN (network N2). The basic autoencoder network is tweaked a little, by keeping the target as GAN generated images (GI-ROBIN) in case of N1 and GAN generated sketches (GS-ROBIN) in the case of N2. This autoencoder learning helps in domain mapping of (a) Images from ROBIN and Generated Images from S-ROBIN (i.e., GI-ROBIN) using autoencoder N1, (b) Sketches from S-ROBIN and Generated Sketches from ROBIN (i.e., GS-ROBIN) using autoencoder N2. Further, the compressed features of both query sketch and floor plan images in the database are extracted from the last layer of the encoder of the trained autoencoder network and used for matching and retrieval purposes. In the next section, details about the various modules of the unified framework are discussed.

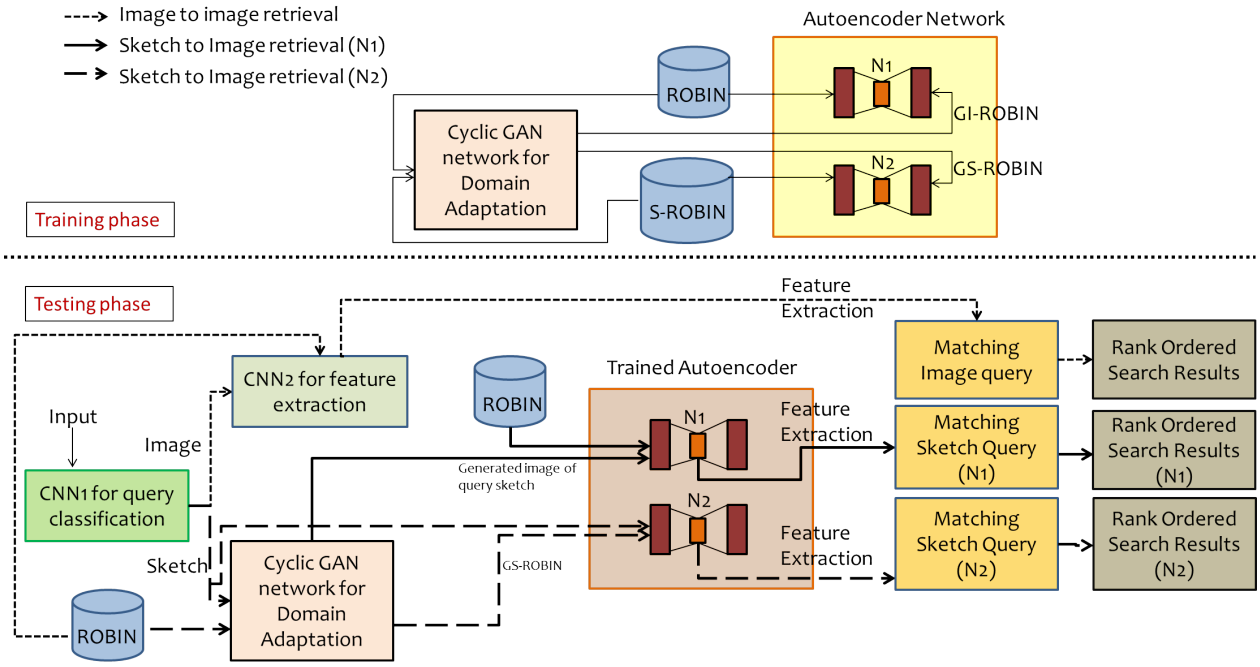


图6.7：多模态检索的统一框架。

查询为平面布局图像时，查询图像与数据库中的图像均通过卷积神经网络进行特征提取，并按照第4章所提出的距离度量方法定义相似性以实现检索。然而，若查询为草图，则处理步骤如下：整个框架可分为三个主要阶段，分别为：（1）基于循环生成对抗网络（Cyclic GAN）的跨域样本生成；（2）基于自编码器与循环生成对抗网络的域映射；以及（3）匹配与检索。给定输入草图S，目标是从ROBIN数据集[Sharma *et al.*, 2017]中检索出与该草图外观相似的平面布局图像I。循环生成对抗网络利用ROBIN数据集中的图像及S-ROBIN数据集中的草图进行训练，以学习各域的表征。直接根据草图检索平面布局图像并非易事（参见图6.18中DANIEL模型的低性能表现）。因此，首先通过一个经训练用于映射草图域与图像域的循环生成对抗网络，将ROBIN数据集中的图像转换为其对应的草图形式（称为GAN生成草图数据集，GS-ROBIN），并将S-ROBIN数据集中的草图转换为其对应的图像形式（称为GAN生成图像数据集，GI-ROBIN）（如图6.4所示）。

此外，为了将平面草图/检索集合中的平面图像以及循环生成对抗网络（cGAN）的生成结果映射到同一公共空间，我们通过训练两个自编码器来进行实验：以ROBIN中的图像作为输入（网络N1），以S-ROBIN中的草图作为输入（网络N2）。对基础自编码器网络进行了适当调整：对于网络N1，其目标为cGAN生成的图像（GI-ROBIN）；对于网络N2，其目标为cGAN生成的草图（GS-ROBIN）。这种自编码器学习有助于实现以下领域的映射：（a）利用网络N1将ROBIN中的图像与S-ROBIN生成的图像（即GI-ROBIN）进行域间映射；（b）利用网络N2将S-ROBIN中的草图与ROBIN生成的草图（即GS-ROBIN）进行域间映射。进一步地，从已训练自编码器网络的编码器最后一层提取查询草图和数据库中平面图像的压缩特征，并将其用于匹配与检索。在下一节中，将详细讨论该统一框架的各个模块。

6.3.1 CNN for query classification

In this module of the framework a Convolutional Neural Network (CNN), trained on both the floor plan sketch and image datasets, ROBIN and S-ROBIN, is used to classify whether a query is a sketch or an image. A combination of convolutional, pooling, normalization, ReLU and fully connected layers (see Fig. 6.8) along with dropout regularization technique, similar to described in the work AlexNet [Krizhevsky *et al.*, 2012a], is used to obtain an effective feature representation from both floor plan images and sketches.

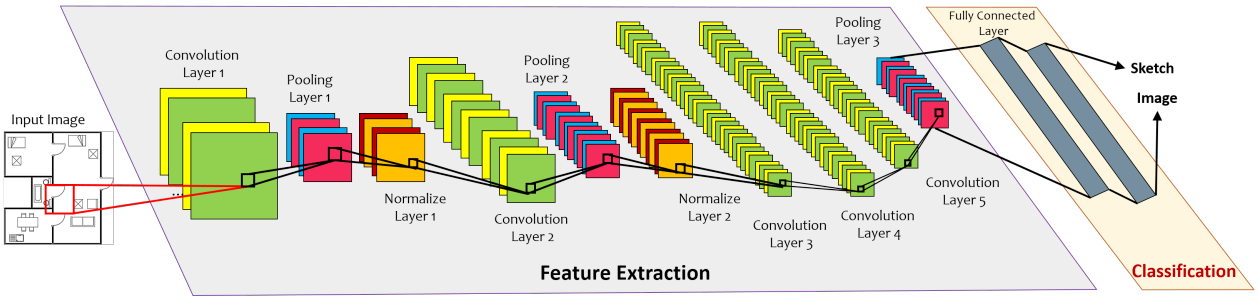


Figure 6.8. : Layerwise depiction of the Convolutional Neural Network (CNN) for query classification

As proposed in Chapter 4, to create a distinction from the original AlexNet framework, the model is not trained with the relighting data-augmentation; and pooling is done before normalization [Jia *et al.*, 2014]. To obtain the feature representation from both the floor plan sketches and images, they are fed into the input layer of the learned CNN model. The sketched floor plans are labeled as 1 and the floor plan images are labeled as 0. The model takes the activations of all the layers viz. a viz. convolution, pooling, normalization, and fully connected to extract the low-level features from the sketch and image floor plan datasets. Since the feed-forward network performs matrix multiplication only once, it makes the whole scheme highly time efficient. The datasets are divided into training and testing sets, which are comprised of 90% and 10% samples respectively and the network is appropriately trained. Finally, on giving the test set to the trained network, the probability of each sample from the mixed bag of S-ROBIN and ROBIN is evaluated to determine which class (0 or 1) they lie into and finally, each test sample is classified into a floor plan image or a sketch. If it is a floor plan image, retrieval proceeds as in [Sharma *et al.*, 2017]. However, if it is a floor plan sketch then domain adaptation as proposed in the subsequent subsections is done.

The next task is cross-domain sample generation using Cyclic GANs, which is done in a similar manner as discussed in Sec. 6.2.1. After the training and testing phase of the Cyclic GAN, a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset is obtained. Thus, after this step, two sets, GI-ROBIN dataset generated from the S-ROBIN-dataset, and GS-ROBIN dataset generated from the ROBIN dataset, are obtained which are used for retrieval purposes.

6.3.2 Autoencoder for domain mapping

If the query is classified as a sketch, it is first converted into its image representation using Cyclic GAN as discussed in Sec. 6.2.1. As observed in Fig. 6.3, function $G(S)$ generates an image corresponding to the floor plan sketch S that looks approximately similar to how a corresponding image would be in the ROBIN dataset. However, the approximation is still a little far from the exact actual image representation (Fig. 6.3 (b)). Therefore, if direct matching between a generated image from the query sketch and samples of the ROBIN dataset (retrieval set) is performed, the average

6.3.1 用于查询分类的卷积神经网络

在该框架模块中，采用了一种卷积神经网络（CNN），该网络同时基于平面草图数据集和图像数据集进行训练，用于对查询样本进行分类，以判断其为草图还是图像。该网络由卷积层、池化层、归一化层、ReLU激活层以及全连接层组合而成（参见图6.8），并辅以类似于AlexNet [Krizhevsky *et al.*, 2012a] 中所描述的Dropout正则化技术，从而从平面图图像和草图中提取有效的特征表示。

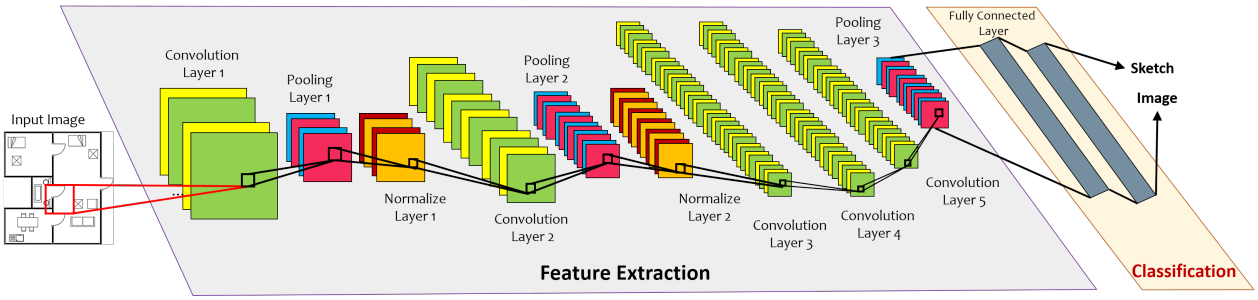


图6.8：用于查询分类的卷积神经网络（CNN）的逐层示意图

如第4章所述，为与原始的AlexNet框架有所区分，该模型未采用光照重调的数据增强策略；并且在归一化之前进行池化操作[Jia *et al.*, 2014]。为了从平面草图和图像中分别获取特征表示，将二者分别输入已训练好的CNN模型的输入层。其中，草图被标记为1，平面图被标记为0。该模型利用各层的激活值——包括卷积、池化、归一化以及全连接层——从草图和平面图数据集中提取低级特征。由于前向传播网络仅执行一次矩阵乘法运算，因此整个方案具有极高的时间效率。数据集被划分为训练集和测试集，分别包含90%和10%的样本，并对网络进行相应的训练。最后，将测试集输入已训练的网络后，针对来自S-ROBIN和ROBIN混合数据集中的每个样本，计算其属于某一类（0或1）的概率，从而最终将每个测试样本分类为平面图或草图。若为平面图，则按照[Sharma *et al.*, 2017]中的方法继续进行检索；若为平面草图，则按照后续小节中提出的领域自适应方法进行处理。

下一步任务是利用循环生成对抗网络（Cyclic GAN）进行跨域样本生成，其具体实现方式与第6.2.1节所述类似。在循环GAN的训练与测试阶段结束后，分别得到一组由草图生成的图像，以及一组对应于ROBIN数据集的草图。因此，在此步骤之后，将获得两组数据：一组是由S-ROBIN数据集生成的GI-ROBIN数据集，另一组是由ROBIN数据集生成的GS-ROBIN数据集，这两组数据均用于检索任务。

6.3.2 用于领域映射的自编码器

如果查询被归类为草图，则首先按照第6.2.1节所述，利用循环生成对抗网络将其转换为其图像表示。如图6.3所示，函数$G(S)$生成的与平面布局草图S对应的图像，在外观上与ROBIN数据集中相应图像大致相似。然而，该近似结果与实际图像仍存在一定差距（图6.3(b)）。因此，若直接在查询草图生成的图像与ROBIN数据集样本（检索集合）之间进行匹配，则平均

precision values obtained were as low as 0.2, which although, is still better than matching sketches and images directly (average precision in this case was obtained to be 0.04) but requires significant improvement. Therefore, to overcome these shortcomings, the introduction of autoencoders is proposed. This helps in mapping the sketch query and sketches generated from the floor plan images, or in another case the image generated from query sketch and the retrieval set floor plan images onto a common space for matching purposes. An autoencoder neural network is an unsupervised learning algorithm that applies backpropagation and sets the target values to be equal to the inputs, i.e., it uses $y^i = x^i$. The autoencoders here are not being used to generate output image similar to the input image at the output layer, rather the use of autoencoders is proposed to learn a common subspace representation of two similar looking domains. Therefore, in the proposed case, additional output image (generated using Cyclic GAN) corresponding to the input image is used for training the autoencoders (Fig. 6.9). This additional output image helps to learn intermediate representation of both input and output domains. This intermediate representation acts as a common subspace representation in our case. The novelty in this case lies in the fact that, unlike traditional autoencoders, which try to generate an output similar to the input fed to them by computing the loss between the input and the target output and updating the weights, the training process of the proposed autoencoder is slightly modified. In the proposed case, a new way of learning representation is performed, where, predefined outputs are used along with the decoder generated output for autoencoder training. The aim is to minimize the difference between the decoder generated output and provided output (generated through Cyclic GAN). The autoencoder is trained in two ways, firstly, the autoencoder labeled as N1 is trained keeping the input as images from ROBIN dataset (i $\in$ ROBIN), whereas, the additional target output as images generated from sketches through the Cyclic GAN network (i' $\in$ GI-ROBIN). This helps in learning the intermediate representation till the loss between the decoder generated output and the sketch generated image domain converges to a small value. For this task, an autoencoder is used as shown in Fig. 6.9.

An autoencoder consists of two parts, the encoder and the decoder, which can be represented as $\mathscr{E}$ and $\mathscr{D}$ respectively. Where, $\mathscr{E} : i \rightarrow z$ and $\mathscr{D} : z \rightarrow \hat{i}$, and z is the compressed representation of the input. Note that during training the target has been set as the images generated form sketches of the S-ROBIN dataset, so that the autoencoder learns to map both domains.

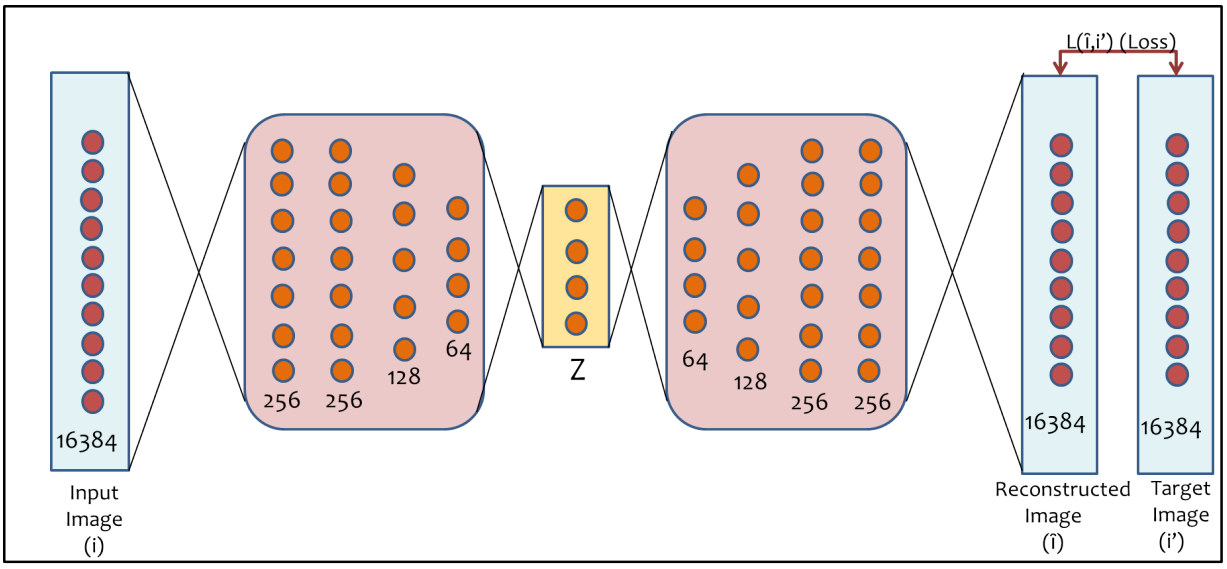


Figure 6.9. : Layerwise depiction of the autoencoder network.

所获得的精度值低至0.02，尽管这一结果仍优于直接匹配草图与图像（后者的平均精度为0.04），但仍需显著改进。为此，本文提出引入自编码器，以实现将草图查询与由平面图生成的草图，或由查询草图生成的图像与检索集合中的平面图映射到同一公共空间，从而进行匹配。自编码器是一种无监督学习算法，采用反向传播机制，并将目标值设为与输入相同，即使用$y^i = x^i$。在本研究中，自编码器并非用于在输出层生成与输入图像相似的输出图像，而是旨在学习两个外观相似领域的共同子空间表示。因此，在所提出的方案中，利用与输入图像对应的额外输出图像（通过循环生成对抗网络生成）对自编码器进行训练（见图6.9）。该额外输出图像有助于学习输入域与输出域的中间表示，而这一中间表示在本研究中充当了共同子空间的表示。本方法的新颖之处在于：不同于传统自编码器通过计算输入与目标输出之间的损失并更新权重来尝试生成与输入相似的输出，本研究对自编码器的训练过程进行了适当调整。在本方案中，采用了一种新的表征学习方式，即在自编码器训练过程中同时使用预定义的输出以及解码器生成的输出。其目标是使解码器生成的输出与所提供的、通过循环生成对抗网络生成的输出之间的差异最小化。自编码器的训练分为两种方式：首先，将标记为N1的自编码器以ROBIN数据集中的图像作为输入（i ∈ ROBIN），并将通过循环生成对抗网络从草图生成的图像作为附加的目标输出（i’ ∈ GI-ROBIN）。这有助于学习中间表示，直至解码器生成的输出与草图生成的图像域之间的损失收敛至一个较小的值。针对该项任务，采用了如图6.9所示的自编码器架构。

自编码器由编码器和解码器两部分组成，分别记为$\mathscr{E}$和$\mathscr{D}$。其中，$\mathscr{E}: i \to z$，$\mathscr{D}: z \to \hat{i}$，而z为输入的压缩表示。需要注意的是，在训练过程中，目标被设定为由S-ROBIN数据集中的草图生成的图像，从而使自编码器能够学习对两个域进行映射。

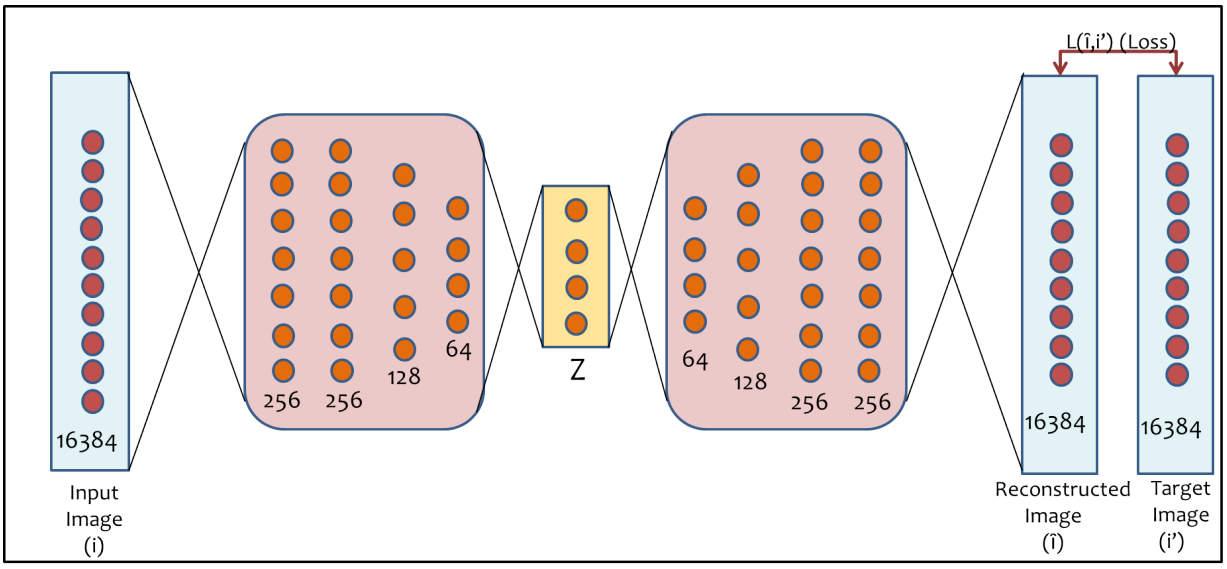


图 图6.9：自编码器的逐层可视化　　网络。

Further, the squared loss function to minimize the reconstruction error to train the network is defined as follows:

$$\mathscr{L}(\hat{i}, i') = ||\hat{i} - i'||^2 \tag{6.7}$$

Note that, $\hat{i}$ is the reconstructed image from the trained autoencoder corresponding to the input. The distinction of our autoencoder network from a simple autoencoder is that the loss is calculated between the GAN generated floor plan image and the reconstructed image. This helps in efficient mapping of both sketch and image domains by taking aid from the Cyclic GAN network, for conversion of sketch query to image or retrieval set images to sketch. The parameters for the autoencoder network are listed in Tab. 6.3.

After the network is fully trained, features from the last layer of the encoder are extracted for both the query sketch generated image that has been obtained using a Cyclic GAN and images from ROBIN dataset that serve as the retrieval set. This helps to obtain a compressed feature vector for both, which is further used for matching and retrieval. This feature extraction helps in mapping both domain of images (I) and domain of sketches (S) to a common domain called (C) where, now the matching and retrieval process takes place.

On similar lines, the autoencoder network, labeled N2, was also trained keeping the input as sketches from the S-ROBIN dataset and the target as generated sketches (GS-ROBIN dataset) obtained by the Cyclic GAN network. It was interesting to note, that due to the quality of the generated sketches from the Cyclic GAN being better, the autoencoder learned a better representation and mapping of both the domains using the N2 network. As per our knowledge, this is a novel way to learn a common embedding space between the sketch and image domain, using a modification in the autoencoder framework by not keeping the input and the expected output to be the same.

6.3.3 Matching and Retrieval

During testing, if the query is a floor plan image then matching and retrieval proceeds as proposed in Sec. 4.3. However, if the query encountered in the framework is classified as a sketch, then two paths can be taken for retrieval purposes (solid-line and dashed line) as shown in Fig. 6.7. The query sketch can be converted into its image representation using the Cyclic GAN. Compressed vector can be obtained from the final layer of the encoder, by passing query sketch generated image and the ROBIN dataset through the autoencoder network N1. The features thus obtained can be used for matching and retrieval purposes. On similar lines, the other path that can be adopted is if the query is a sketch, then the query sketch, along with the sketch counterparts of the ROBIN dataset (GS-ROBIN dataset) can be passed through N2 autoencoder network, where again, compressed vector obtained from the last layer of the trained encoder can be used as a feature vector for both query sketch and the retrieval set. To illustrate the matching process for one such path, let $f(x_i) : \{f_1(x_i), f_2(x_i), .., f_n(x_i)\}$ be the feature vector extracted for x_i image where $i \in$ ROBIN dataset, and n represents the size of this feature vector, which in our case is 64. For each sample in the database, features are extracted and stored in the feature database. In a similar manner, features are extracted from the query sketch generated image. Then, the Matching Score ($M_{Approach_2}$) between query sketch generated image(gi) and a sample from the database (x_i) is calculated (and used for ranking), using three different distance metrics:

Further, the squared loss function to minimize the reconstruction error to train the network is defined as follows:

$$\mathscr{L}(\hat{i},i') = ||\hat{i} - i'||^2 \tag{6.7}$$

需要注意的是，$\hat{i}$ 是与输入对应的、由训练好的自编码器重构得到的图像。我们的自编码器网络与简单自编码器的区别在于，其损失函数是在 GAN 生成的平面布局图像与重构图像之间计算的。这一设计借助循环 GAN 网络，实现了草图域与图像域之间的高效映射，从而支持将草图查询转换为图像，或将检索集中的图像转换为草图。自编码器网络的参数详见表 6.3。

在网络训练完成后，分别从编码器的最后一层提取特征：一是通过循环生成对抗网络（Cyclic GAN）生成的查询草图图像，二是作为检索集的ROBIN数据集中的图像。这有助于为两者分别获得一个压缩的特征向量，并进一步用于匹配与检索。该特征提取过程将图像域（I）和草图域（S）映射到一个共同的域（C），从而在该共同域中进行匹配与检索。

与此类似，自编码器网络N2也在训练时将S-ROBIN数据集中的草图作为输入，以由循环生成对抗网络得到的生成草图（GS-ROBIN数据集）作为目标。值得注意的是，由于循环生成对抗网络生成的草图质量更高，自编码器通过网络N2学到了两个域之间更优的表示与映射关系。据我们所知，这是一种新颖的方法，通过在自编码器框架中进行改进，不强制要求输入与期望输出相同，从而在草图域与图像域之间学习一个共同的嵌入空间。

6.3.3 匹配与检索

在测试过程中，若查询为平面布局图像，则按照第4.3节所述的方法进行匹配与检索。然而，若框架中遇到的查询被分类为草图，则可根据图6.7所示，沿两条路径（实线与虚线）开展检索工作。可通过循环生成对抗网络（Cyclic GAN）将查询草图转换为其图像表示；通过将生成的查询草图图像与ROBIN数据集输入至自编码器网络N1，可从编码器的最终层获得压缩向量，该特征可用于匹配与检索。与此类似，另一条可行的路径是：若查询为草图，则可将其连同ROBIN数据集中的草图对应样本（即GS-ROBIN数据集）一并输入至自编码器网络N2；同样地，可将训练好的编码器最后一层输出的压缩向量作为查询草图及检索集合的特征向量。为说明其中一条路径的匹配过程，设$f(x_i) : \{f_1(x_i), f_2(x_i), .., f_n(x_i)\}$为从$x_i$图像中提取的特征向量，其中$i \in$代表ROBIN数据集，$n$表示该特征向量的维度，在本例中为64。对于数据库中的每个样本，均提取其特征并存储于特征数据库中。以相同方式，对查询草图生成的图像亦提取特征。随后，利用三种不同的距离度量，计算查询草图生成图像（gi）与数据库中某一样本（x_i）之间的匹配得分（$M_{Approach_2}$），并据此进行排序。

Euclidean Distance Metric

$$M_{Approach_2} = \{ \sum_{a=1}^{|f(x_i)|} \{f_a(gi) - f_a(x_i)\}^2 \}^{\frac{1}{2}} \tag{6.8}$$

Manhattan Distance

$$M_{Approach_2} = \sum_{a=1}^{|f(x_i)|} |f_a(gi) - f_a(x_i)|_2 \tag{6.9}$$

Cosine Distance

$$M_{Approach_2} = \sum_{a=1}^{|f(x_i)|} \frac{\sum_{a=1}^{|f(x_i)|} \{f_a(gi).f_a(x_i)\}}{\{\sum_{a=1}^{|f(x_i)|} \{f_a(gi)\}^2\}^{\frac{1}{2}}.\{\sum_{a=1}^{|f(x_i)|} \{f_a(x_i)\}^2\}^{\frac{1}{2}}} \tag{6.10}$$

The distance is calculated using 10-fold cross-validation where, the query sketch samples consist of 70% of the sketched floor plans from S-ROBIN dataset and the retrieval set comprises of all the 510 floor plans in the ROBIN dataset. Based on this distance the database floor plan images are rank ordered. Shorter the distance, more similar a pair of floor plan is. The rank ordered samples are shown to the user. Next, the details of the experiments performed, implementation details, and the results obtained from them are discussed.

6.4 EXPERIMENTS AND RESULTS

6.4.1 Dataset Creation

The experimentation was done on 2 datasets : (1) ROBIN dataset[Sharma *et al.*, 2017] and (2) S-ROBIN dataset. For the framework all the floor plan images of ROBIN to sketches were replicated. Since, there is no publicly available sketch floor plan dataset for the DAR community, a new dataset was created and named as S-ROBIN as discussed in Annexure A. S-ROBIN dataset has 510 real world floor plan sketches divided into 3 broad categories and 17 sub-categories differing in the overall global outer shape of the layouts. It is to be noted that class-belongingness of each sketched layout is determined in the dataset by its outer shape, for example all the hand-drawn floor plans in one sub-category will have the same L-shaped outer layout. To capture these hand drawn sketched floor plans, the volunteers were asked to draw the floor plans on a digital platform using a Wacom Tablet. The resolution ranges from 591×517 pixels to 1483×884. Since working in the off-line mode is considered, the hand drawn floor plans are stored as an image and used for experimentation. All the images were saved in a gray scale format.

6.4.2 Implementation Details

For training the Cyclic GAN, the datasets ROBIN and S-ROBIN were divided into training (70%) and testing (30%) sets. Further, the Cyclic GAN with the help of the losses discussed in Sec. 6.2.1, learns the mapping between both domains. During testing, due to the cyclic nature of the network, both the generated sketches from [Sharma *et al.*, 2017] and the corresponding images generated from the S-ROBIN dataset are obtained. It was observed that visually the sketches generated from ROBIN were of better quality. To make it clear to the reader, let query sketch

欧几里得距离度量

$$M_{Approach_2} = \{ \sum_{a=1}^{|f(x_i)|} \{f_a(gi) - f_a(x_i)\}^2 \}^{\frac{1}{2}} \tag{6.8}$$

曼哈顿距离

$$M_{Approach_2} = \sum_{a=1}^{|f(x_i)|} |f_a(gi) - f_a(x_i)|_2 \tag{6.9}$$

余弦距离

$$M_{Approach_2} = \sum_{a=1}^{|f(x_i)|} \frac{\sum_{a=1}^{|f(x_i)|} \{f_a(gi).f_a(x_i)\}}{\{\sum_{a=1}^{|f(x_i)|} \{f_a(gi)\}^2\}^{\frac{1}{2}}.\{\sum_{a=1}^{|f(x_i)|} \{f_a(x_i)\}^2\}^{\frac{1}{2}}} \tag{6.10}$$

距离的计算采用10折交叉验证，其中查询草图样本由S-ROBIN数据集中的70%已生成草图的平面图构成，而检索集合则包含ROBIN数据集中全部510张平面图。根据该距离对数据库中的平面图图像进行排序，距离越短，两幅平面图的相似度越高。排序后的样本将展示给用户。接下来，将详细讨论所开展的实验、实现细节及其所得结果。

6.4 实验与结果

6.4.1 数据集构建

实验在两个数据集上进行：（1）ROBIN数据集[Sharma，2017年版]和（2）S-ROBIN数据集。针对所提出的框架，ROBIN数据集中所有平面图均被转换为草图并予以复制。鉴于面向DAR领域的公开可用草图平面图数据集尚不存在，我们创建了一个新数据集，并将其命名为S-ROBIN，具体详见附录A。S-ROBIN数据集包含510幅真实世界的平面图草图，分为三大类和17个小类，各小类在布局的整体外部轮廓上存在差异。需要指出的是，该数据集中每幅草图的类别归属均由其外部轮廓决定，例如，某一小类中的所有手绘平面图均具有相同的L形外部轮廓。为采集这些手绘草图，志愿者被要求使用Wacom数位板在数字化平台上绘制平面图，图像分辨率介于591×517像素至1483×884像素之间。由于考虑离线工作模式，手绘平面图以图像形式存储，并用于实验。所有图像均以灰度格式保存。

6.4.2 实施细节

在训练循环GAN时，数据集ROBIN和S-ROBIN被划分为训练集（70%）与测试集（30%）。此外，该循环GAN借助第6.2.1节中讨论的损失函数，学习两个域之间的映射关系。在测试阶段，由于网络的循环特性，既可得到来自[Sharma等，2017]的生成草图，也可得到由S-ROBIN数据集生成的对应图像。实验发现，从视觉效果上看，由ROBIN生成的草图质量更高。为便于读者理解，设查询草图为

be S, the ROBIN dataset images to be retrieved be $\{I\}$ and the generated sketches from ROBIN ($\{I\}$) be $\{GS\}$. Then as S and $\{GS\}$ are in the same domain, similar sketches in $\{GS\}$ are retrieved smoothly through the framework. Further, it is to be noted, these generated sketches $\{GS\}$ exactly correspond to their image counterparts from $\{I\}$. Hence, it is safe to say that through the framework relationship between S and $\{GS\}$ has been determined. Also, relationship between $\{GS\}$ and $\{I\}$ pre-exists. Therefore, transitively, relationship between S and $\{I\}$ has thus been established. It is this indirect relationship that helps to justify that ROBIN dataset images similar to the query sketch can be shown as the retrieval result obtained through the framework.

Approach 1:

For feature representation in Approach 1, three CNNs were trained using the ROBIN dataset, S-ROBIN dataset and the GS-ROBIN dataset, respectively. All the three datasets are again divided into training set with 70% samples and test set with 30% samples. As a retrieval database, the entire GS-ROBIN dataset is used which is generated from the ROBIN dataset and thus, transitively represents it. Each floor plan image has been resized to a size of 128×128 for both matching and feature representation task. For deep model configuration, the architecture described in [Krizhevsky *et al.*, 2012a] is followed.

Approach 2:

After the classification task, if the query is a floor plan image, feature extraction and retrieval proceeds as in [Sharma *et al.*, 2017]. If the query is a sketched floor plan then the Cyclic GAN network comes into picture to convert the sketched floor plans into their image counterparts, which are then used for training the autoencoder, to bridge the difference between the sketched and the image domain.

6.4.3 Parameter Setting

Convolutional Neural Network:

The parameters for the CNN network for both feature extraction in Approach 1 and query classification in Approach 2, are similar as mentioned in Chapter 4. Table 6.1 lists out these parameters again.

Cyclic GAN:

Setting appropriate values for the hyper parameter plays a pivotal role for the success of the Cyclic GAN and the deep learning implementation. Table 6.2 lists out the parameters and their corresponding values used while implementing the framework. The effect of higher number of iterations to generate good quality image is depicted in Fig. 6.10. It is to be noted that the input is given in the form of an image and number of filters in the 1^{st} layer of the generator determine the output features of the convolution layer. This is then passed to the next convolution layer, which subsequently leads to extraction of progressively higher level features aiding in domain mapping. It can be observed that given an original floor plan image our proposed network is able to generate a sketched version of it with very high accuracy. As the number of iterations increases the quality improves. It was empirically found that after the optimum value of number of iterations is achieved, the quality of the image doesn't improve significantly, and hence to have a trade-off between time and quality, an appropriate value for the number of iterations is chosen, which in our case was chosen to be 60000 iterations.

Autoencoder framework:

设S为待检索的ROBIN数据集图像，由ROBIN生成的草图记为{*I*}，对应的图像记为{*GS*}。由于S与{*GS*}属于同一领域，因此通过该框架能够顺利检索出{*GS*}中与其相似的草图。此外，值得注意的是，这些生成的草图{*GS*}与其在{*I*}中的对应图像完全一致。由此可以认为，通过该框架已确定了S与{*GS*}之间的关系。同时，{*GS*}与{*I*}之间也预先存在某种关系。因此，根据传递性，S与{*I*}之间的关系得以建立。正是这种间接关系，为以下结论提供了依据：通过该框架所获得的检索结果中，能够展示与查询草图相似的ROBIN数据集图像。

方法一：

在方法1中，针对特征表示，分别使用ROBIN数据集、S-ROBIN数据集和GS-ROBIN数据集训练了三个卷积神经网络。这三个数据集均按70%的样本划分为训练集，30%的样本划分为测试集。作为检索数据库，采用由ROBIN数据集生成的整个GS-ROBIN数据集，因此该数据库间接表征了ROBIN数据集。对于匹配与特征表示任务，每幅平面图均被缩放至128×128的尺寸。深度模型的配置遵循[Krizhevsky, 2012a]中所描述的架构。

方法二：

在分类任务完成后，若查询为平面图图像，则特征提取与检索过程按照[Sharma等，2017]中的方法进行。若查询为手绘平面图，则采用循环生成对抗网络（Cyclic GAN）将其转换为对应的图像形式，并以此训练自编码器，以弥合手绘与图像域之间的差异。

6.4.3 参数设置

卷积神经网络：

在方法一的特征提取和方法二的查询分类中，卷积神经网络的参数设置与第4章所述基本一致。表6.1再次列出了这些参数。

循环生成对抗网络：

为超参数设置合适的取值对于循环生成对抗网络及深度学习模型的成功至关重要。表6.2列出了在框架实现过程中所使用的各参数及其对应取值。图6.10展示了增加迭代次数对生成高质量图像的影响。需要指出的是，输入以图像形式提供，而生成器第1层中的滤波器数量决定了该卷积层的输出特征；随后这些特征被传递至下一层卷积层，进而逐步提取更高层次的特征，从而有助于域间映射的实现。可以观察到，给定一张原始平面图，我们提出的网络能够以极高的精度生成其草图版本；随着迭代次数的增加，图像质量不断提升。经实验发现，当迭代次数达到最优值后，图像质量的提升已不显著；因此，为在时间和质量之间取得平衡，需合理选择迭代次数，在本研究中该值被设定为60,000次。

自编码器框架：

Layers	1	2	3	4	5	6
Type	conv1+ maxpool1+ norm1	conv2+ maxpool2+ norm2	conv3	conv4	conv5+ maxpool3	full (fc6,fc7)
Channels	96	256	384	384	256	4096
Filter Size	11*11	5*5	3*3	3*3	3*3	-
Convolution Stride	4*4	1*1	1*1	1*1	1*1	-
Pooling Size	3*3	3*3	-	-	3*3	-
Pooling Stride	2*2	2*2	-	-	2*2	-
Padding Size	-	1*1	1*1	1*1	1*1	-

Table 6.1. : Network Parameters for the proposed framework

Table 6.2. : Parameter Values used while implementing Cyclic GAN

Parameter	**Value**
# filters in the 1^{st} layer of the generator	64
# filters in the 1^{st} layer of the discriminator	64
batch size	1
pool size	50
img width and img height	128

The autoencoder used in Approach 2, plays a pivotal part in mapping both the sketched and image floor plans onto a common space for efficient retrieval. The images/sketches are reshaped to a size 128×128 and vectorized into a 16384×1 vector, before being fed to the autoencoder network.

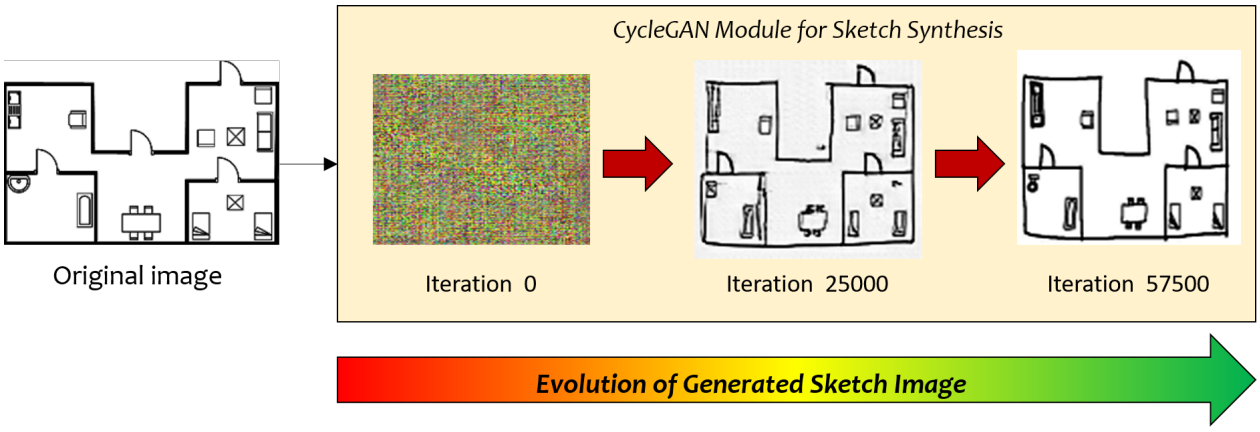


Figure 6.10. : A diagram depicting stages of sketch synthesis from an image through our Cyclic GAN framework. Note how, the discriminator and generator improve with increasing the number of training iterations.

Layers	1	2	3	4	5	6
Type	conv1+ maxpool1+ norm1	conv2+ maxpool2+ norm2	conv3	conv4	conv5+ maxpool3	full (fc6,fc7)
Channels	96	256	384	384	256	4096
Filter Size	11*11	5*5	3*3	3*3	3*3	-
Convolution Stride	4*4	1*1	1*1	1*1	1*1	-
Pooling Size	3*3	3*3	-	-	3*3	-
Pooling Stride	2*2	2*2	-	-	2*2	-
Padding Size	-	1*1	1*1	1*1	1*1	-

表6.1：所提出框架的网络参数

表6.2：循环GAN实现过程中所使用的参数值

Parameter	**Value**
# filters in the 1^{st} layer of the generator	64
# filters in the 1^{st} layer of the discriminator	64
batch size	1
pool size	50
img width and img height	128

方法2中所采用的自编码器在将手绘和图像形式的平面图映射至同一特征空间以实现高效检索方面发挥了关键作用。在将其输入自编码器网络之前，图像/草图首先被重新调整为128×128的尺寸，并被向量化为一个16384维的列向量。

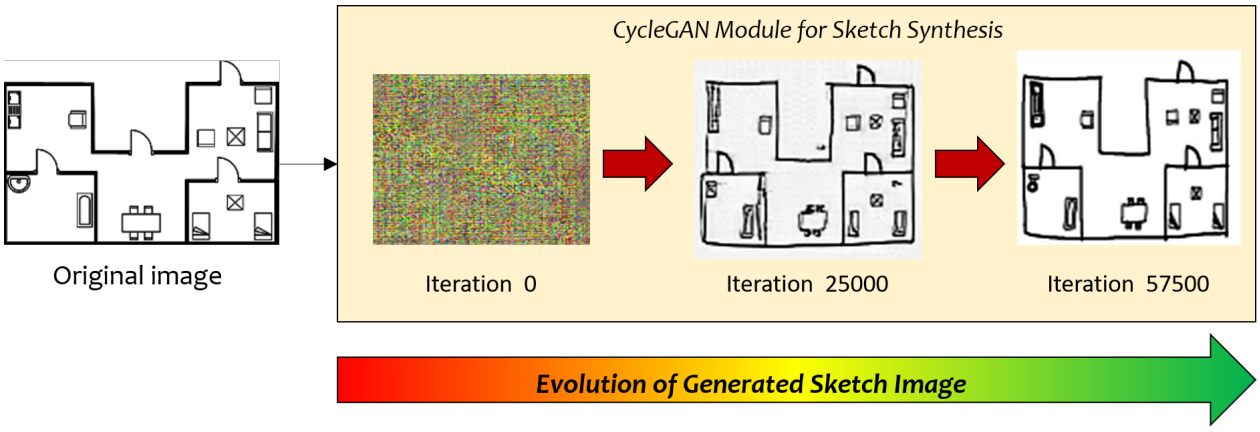


图6.10：展示了通过我们的循环生成对抗网络框架，从图像进行草图合成的各阶段示意图。请注意，随着训练迭代次数的增加，判别器和生成器的性能均有所提升。

100,000 iterations were undertaken for the autoencoder to train properly and 70% of the samples from the ROBIN and S-ROBIN datasets were used as training set for training both CNN, Cyclic GAN and the autoencoder. Further, the details of the parameters involved in the autoencoder network are listed in Tab. 6.3.

Table 6.3. : Parameter Values for Encoder Network

Component	Parameter	Value
Encoder	No. filters in the 1^{st} layer	256
	No. filters in the 2^{nd} layer	256
	No. filters in the 3^{rd} layer	128
	No. filters in the 4^{th} layer	64
Decoder	No. filters in the 1^{st} layer	64
	No. filters in the 2^{nd} layer	128
	No. filters in the 3^{rd} layer	256
	No. filters in the 4^{th} layer	256

6.4.4 Qualitative Results

Approach 1:

Approach 1 is able to successfully retrieve a rank order set of floor plan images from the ROBIN dataset given a query in the form of sketch, using the proposed domain mapping framework consisting of Cyclic GAN and CNN. The top 5 rank ordered retrieval results for three different hand-drawn queries, are shown in Fig. 6.11 (a),(b) and (c). The samples which are erroneously retrieved, i.e. not from the same class as that of the query class, are highlighted with a "red" bounding box. For all the results shown in Fig. 6.11, the features extracted from the first normalization layer of the CNN stack have been used, owing to the fact that it yields the best performance. The first rank ordered result retrieved from the ROBIN dataset, is the image exactly matching the query sketch. This is due to maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower. In case of Fig. 6.11 (a) and (b), all the top 5 retrieved samples belong to the query class. Global layouts of all the floor plans are identical and so is the number of rooms in the floor plans. It is to be noted that there is a difference in the shape and position of the individual rooms, as well as, the furniture components for the rank ordered samples. There are 4 rooms in each floor plan (refer Fig. 6.11 (a)), however this framework is able to rank order the samples correctly by ordering the most similar ones in terms of the features present inside the floor plans.

Figure 6.11(c) depicts one of the cases where this algorithm does not perform perfectly. As it can be noted, for a given query sketch, the 5^{th} ranked result is erroneously retrieved (highlighted by "red") by this technique. Justification of the Rank 5 erroneously retrieved result can be understood through qualitative analysis. Observe that the rooms in both the layouts share very similar adjacency. The bottom part of the floor plan is almost identical. Moreover, the size and type of furniture present in the individual rooms is quite similar in case of the query and the Rank 5 sample. Hence, similarity score between the query floor plan and the retrieved Rank 5 layout is high.

Approach 2:

The framework proposed in this approach can successfully retrieve a rank order set of floor plan images from the ROBIN dataset given a query in the form of floor plan sketch/image. It is

为使自编码器能够充分训练，共进行了100,000次迭代；ROBIN和S-ROBIN数据集中的70%样本被用作训练集，用于训练卷积神经网络（CNN）、循环生成对抗网络（Cyclic GAN）以及自编码器。此外，自编码器网络所涉及的参数详情见表6.3。

表6.3：编码器网络的参数值

Component	**Parameter**	**Value**
Encoder	No. filters in the 1^{st} layer	256
	No. filters in the 2^{nd} layer	256
	No. filters in the 3^{rd} layer	128
	No. filters in the 4^{th} layer	64
Decoder	No. filters in the 1^{st} layer	64
	No. filters in the 2^{nd} layer	128
	No. filters in the 3^{rd} layer	256
	No. filters in the 4^{th} layer	256

6.4.4 定性结果

方法一：

方法1能够利用由循环生成对抗网络（Cyclic GAN）和卷积神经网络（CNN）构成的所提出的领域映射框架，根据草图形式的查询，从ROBIN数据集中成功检索出按排序顺序排列的平面布局图像集合。图6.11(a)、(b)和(c)分别展示了针对三幅不同手绘查询的前5位排序检索结果。其中，被错误检索的样本——即不属于与查询类别相同的类别——以红色边界框标出。对于图6.11中所示的所有结果，均采用了从CNN堆栈的第一层归一化层提取的特征，因为该层能够获得最佳性能。从ROBIN数据集中检索出的排名第一的图像与查询草图完全匹配，这是由于检索到的布局与查询布局之间的匹配分数最高所致。后续结果的匹配分数依次递减，因此排名较低。在图6.11(a)和(b)中，前5位检索到的样本均属于查询类别。所有平面布局的整体格局相同，且房间数量也一致。需要指出的是，尽管如此，各排序样本的单个房间形状与位置以及家具配置仍存在差异。每个平面布局均包含4个房间（参见图6.11(a)），然而本框架仍能根据平面布局内部所含特征的相似程度，对样本进行正确排序。

图6.11(c)展示了该算法未能完美运行的一个案例。可以看出，对于给定的查询草图，排名第5的结果（用红色标注）被该方法错误地检索出来。对该第5名结果出现错误的原因可通过定性分析加以解释。观察发现，两个布局中的房间在邻接关系上极为相似；平面图的下部几乎完全一致。此外，查询样本与排名第5的样本中各房间内的家具尺寸和类型也十分接近，因此查询平面图与被检索出的第5名布局之间的相似度较高。

方法二：

该方法所提出的框架能够成功地从ROBIN数据集中，根据以平面草图/图像形式提供的查询，检索出一组按相关性排序的平面布局图像。它

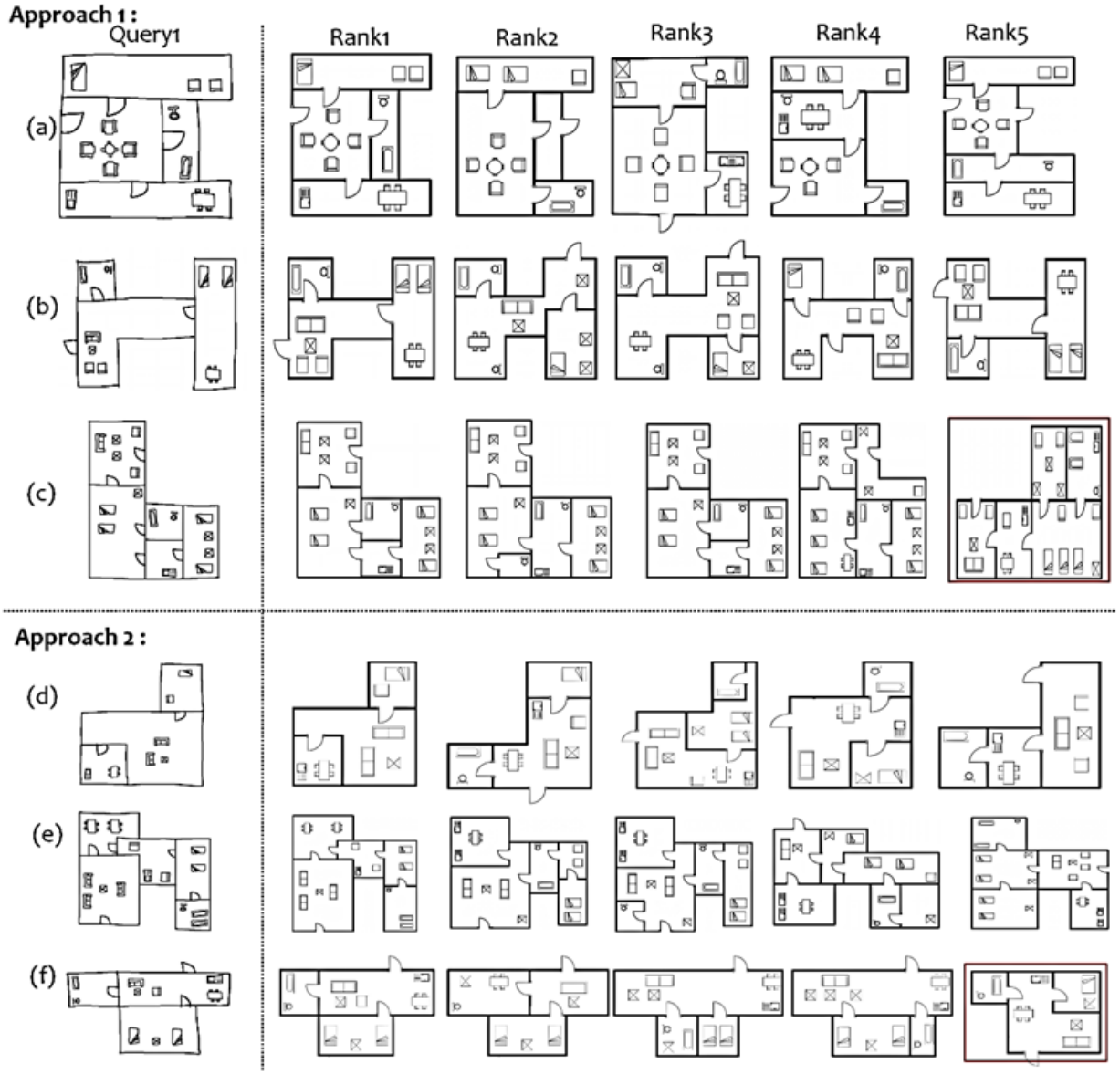


Figure 6.11. : Top five rank ordered retrieval results for the two proposed approaches for three different hand-drawn query floor plans.

to be noted that the retrieval set is always the ROBIN dataset. Experimentation was done taking random queries from both S-ROBIN and ROBIN dataset. The top 5 rank ordered retrieval results for three different hand-drawn queries are shown in Fig. 6.11 (d), (e) and (f). The samples which are erroneously retrieved, i.e., not from the same class as that of the query class, are highlighted by a bounding box. As shown in Fig. 6.11, it can be observed that the conjunction of Cyclic GAN and autoencoder for mapping as proposed in the framework, is efficiently able to capture the similarity between the query sketch and the floor plan images. In the case of Fig. 6.11 (d) and (e), all the top 5 retrieved samples belong to the query class. Global layouts of the all the floor plans are identical

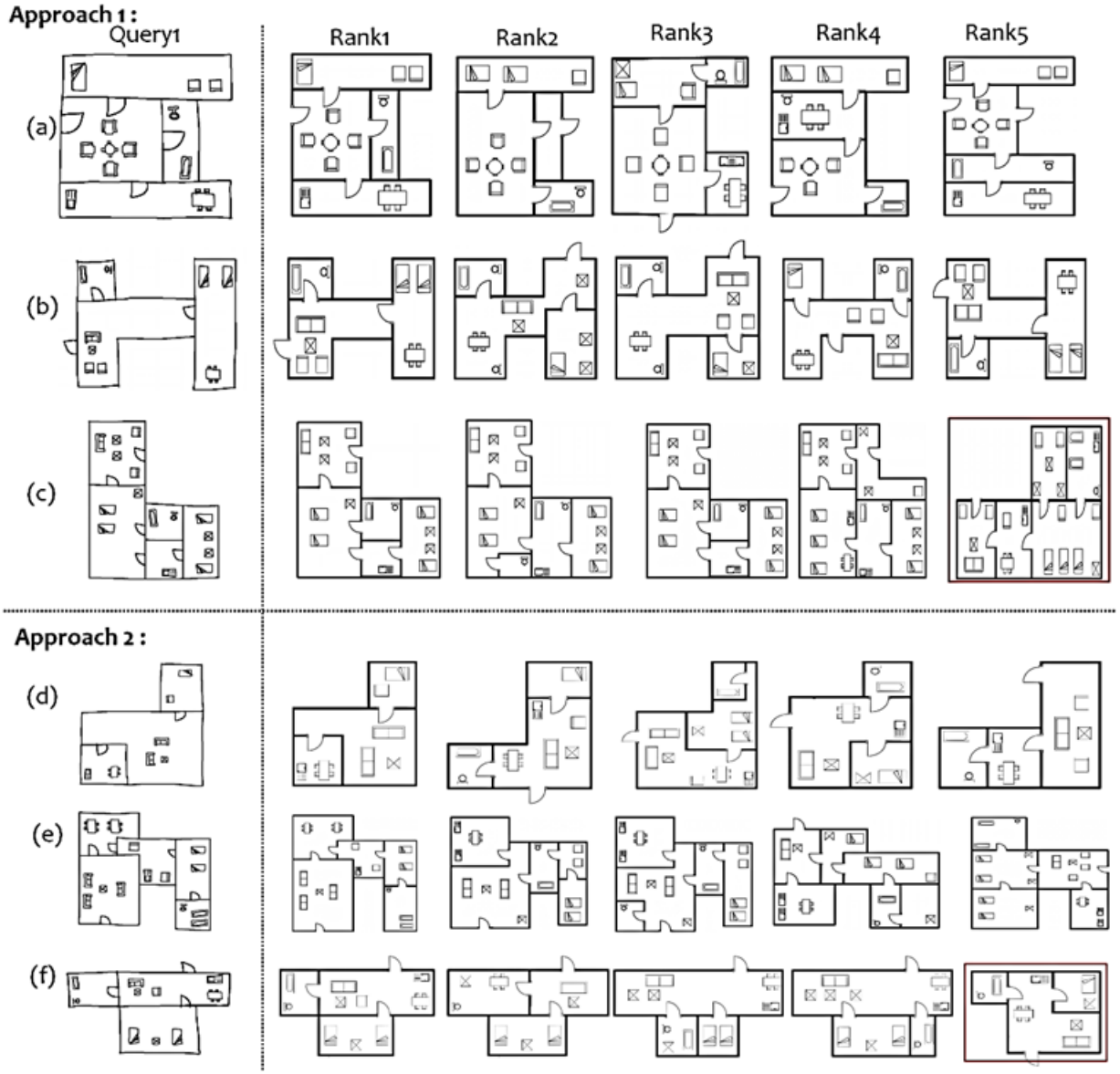


图6.11：针对三种不同的手绘查询平面图，两种所提方法的前五位排序检索结果。

需要注意的是，检索集合始终为 ROBIN 数据集。实验采用从 S-ROBIN 和 ROBIN 两个数据集中随机抽取的查询进行。图 6.11（d）、（e）和（f）展示了三幅不同手绘查询的前 5 位排序检索结果。其中，被错误检索的样本——即不属于与查询样本相同类别——以边界框加以标注。如图 6.11 所示，可以观察到，本框架中提出的基于循环生成对抗网络与自编码器的映射组合能够有效捕捉查询草图与平面布局图像之间的相似性。在图 6.11（d）和（e）的情形下，前 5 位检索到的样本均属于查询类别，且所有平面布局的整体格局完全一致。

and so is the number of rooms in the floor plans. It is to be noted that there is a difference in the shape and position of the individual rooms, as well as, the furniture components for the rank ordered samples. There are three rooms in each floor plan (refer Fig. 6.11 (d)). However, this framework can rank order the samples correctly by ordering the most similar ones in terms of the features present inside the floor plans.

Figure 6.11(f) depicts one of the cases where the algorithm does not perform perfectly. As it can be noted, for a given query sketch, the 5^{th} ranked result is erroneously retrieved (highlighted by "red") by this technique. Justification of the Rank 5 erroneously retrieved result can be understood through qualitative analysis. Observe that the rooms in both the layouts share very similar adjacency. The similarity lies in specific parts such as the 1^{st}, 2^{nd}, and 3^{rd} rooms have the same furniture components as the query floor plan, although the overall layout differs.

Similarly, qualitative analysis of the framework for the query being an image was also done as shown in Fig. 6.12. For all the results shown in Fig. 6.12, the features extracted from the first normalization layer of the CNN stack are used, as it yields the best performance. The first rank ordered result retrieved from the ROBIN dataset, is the image exactly matching the query image. The reason is the maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower.

Both the proposed frameworks are also able to perform well for queries belonging to unseen classes. Figure 6.13 (1) and (2) depict two such scenarios where the sketch query does not match any of the samples in the ROBIN dataset. Still, this framework is capable of retrieving the closest

Figure 6.12. : Top five rank ordered retrieval result of proposed framework for three different query floor plan images using Approach 2 proposed in the Chapter.

同样地，平面图中的房间数量也保持一致。需要注意的是，对于按等级排序的样本而言，各房间的形状与位置以及家具构件均存在差异。每个平面图包含三个房间（参见图6.11(d)）。然而，该框架仍能通过依据平面图中所含特征的相似性对样本进行正确排序，从而实现等级划分。

图6.11(f)展示了算法未能完美运行的一个案例。可以看出，对于给定的查询草图，该方法错误地检索出了排名第五的结果（以红色标注）。对这一排名第五的误检结果的成因可通过定性分析加以解释。观察发现，两个布局中的房间在邻接关系上极为相似；具体而言，1st、2nd和3rd号房间与查询平面图具有相同的家具配置，尽管整体布局存在差异。

同样，针对查询为图像的框架也进行了定性分析，如图6.12所示。在图6.12所示的所有结果中，均采用了从CNN堆叠的第一层归一化层提取的特征，因为该特征组合能够获得最佳性能。从ROBIN数据集中检索出的排名第一的结果与查询图像完全匹配，其原因为该布局与查询布局之间的匹配得分最高。后续结果的匹配得分依次递减，因此排名较低。

所提出的两种框架均能对属于未见类别的查询取得良好的检索效果。图6.13（1）和（2）展示了两种此类情形，其中草图查询与ROBIN数据集中的任何样本均不匹配。即便如此，该框架仍能检索出最相似的

图6.12：采用本章提出的方案2，对三张不同的查询平面布局图像，所提框架的前五位检索结果排序。

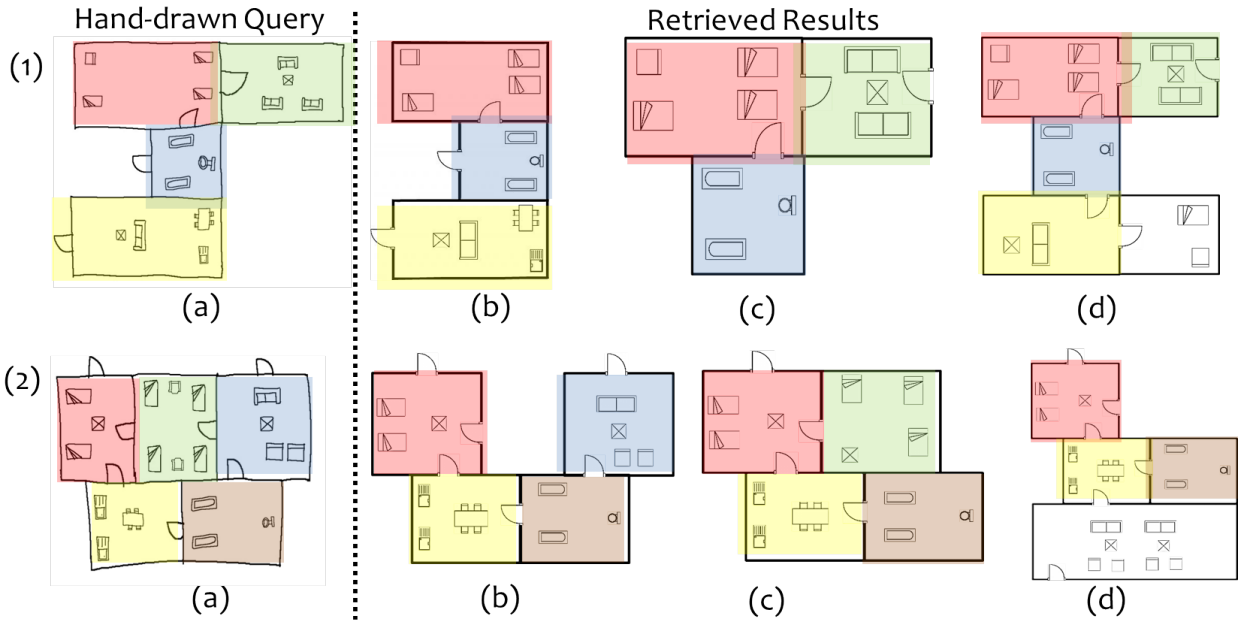


Figure 6.13. : Two examples showing results for querying the system for partial matches. (a) Query Sketch, (b),(c),(d) Retrieved results. Colour Coding establishes room correspondence between the query and the results.

similar results, in terms of room furniture and room placement. E.g., note the similarity between the rooms highlighted in "red", "blue" and "yellow" of the query sketch Fig. 6.13 (1) (a) and the retrieved result of Fig. 6.13 (1) (b). The decor components and room placement are quite similar. The "green" part of Fig. 6.13 (1) (a) is missing in Fig. 6.13 (1) (b). Similar observations can be made for the other results also. Although the shape of the layout is a subset of the shape of the query sketch and does not match completely, this framework is able to correctly retrieve and rank-order the results. Thus, this can help to suggest varied finished designs to architects/property buyers if they have some abstract ideas in mind.

6.4.5 Quantitative Results

The performance of both the approaches is quantified using the Precision (P) and Recall (R) metric. The precision values have been averaged over all the queries for the particular recall values. Given a query sketch, retrieved layouts should belong to the same sub-category of layouts as the query, keeping in mind the preference set by the property seeker during querying. Here, the global outer shape of the layout is the criteria for class belongingness of a given floor plan.

Approach 1:

As discussed, direct retrieval of floor plan images given a sketch is not straightforward (refer to the low performance of the CNN framework in Fig. 6.18 using DANIEL proposed in Chapter 4). Moreover, converting query sketches into corresponding image counterparts and then retrieving from the ROBIN dataset yielded weak performance (refer CNN A, B (trained on ROBIN and S-ROBIN respectively) in Fig.6.14 (a)). Therefore, sketch counterparts of ROBIN dataset generated through CyclicGAN are used for retrieval. The experimentation was carried out with CNNs trained on ROBIN (CNN_1), S-ROBIN (CNN_2) and the GS-ROBIN dataset (CNN_3), for feature

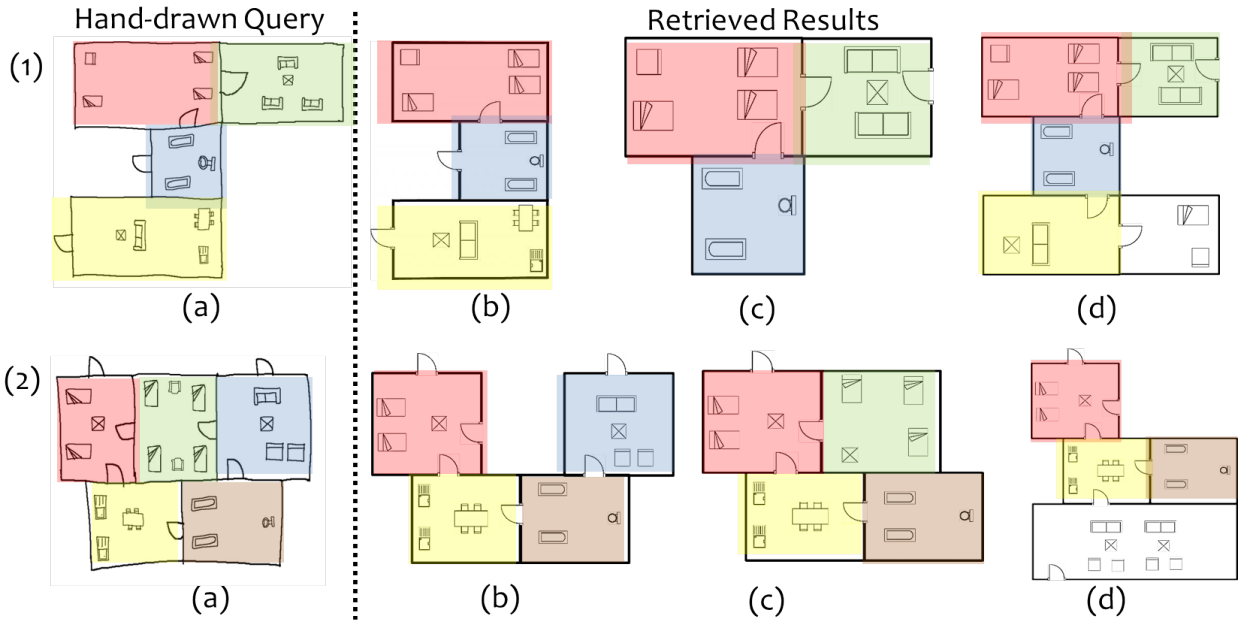


图6.13：两个示例展示了系统在进行部分匹配查询时的检索结果。（a）查询草图，（b）、（c）、（d）为检索到的结果。颜色编码用于建立查询与结果之间的空间对应关系。

在房间家具和空间布局方面，也得到了类似的结果。例如，对比查询草图图6.13(1)(a)中以“红色”、“蓝色”和“黄色”标注的区域与检索结果图6.13(1)(b)中的对应部分，可以发现其装饰元素和空间布置高度相似；而图6.13(1)(a)中的“绿色”区域在图6.13(1)(b)中则未出现。对其他检索结果亦可作出类似的观察。尽管检索结果的平面形状仅为查询草图形状的子集，并未完全匹配，但该框架仍能正确地完成检索并进行排序。因此，当建筑师或购房者仅有一些抽象的设计构想时，该方法有助于为其推荐多样化的最终设计方案。

6.4.5 定量结果

两种方法的性能均采用精确率（P）和召回率（R）指标进行量化。对于特定的召回率值，精确率数值已在所有查询上进行了平均。给定一个查询草图，检索到的平面布局应与查询属于同一子类别，同时需考虑房产需求者在查询时设定的偏好。在此，平面布局的整体外部轮廓是判定某一平面图所属类别的依据。

方法一：

如前所述，给定草图直接检索平面图图像并非易事（参见第4章提出的DANIEL框架在图6.18中表现不佳）。此外，将查询草图转换为对应的图像样本后再从ROBIN数据集中进行检索，其性能同样欠佳（参见图6.14(a)中的CNN A和CNN B，分别在ROBIN和S-ROBIN数据集上训练）。因此，本研究采用通过CyclicGAN生成的ROBIN数据集草图对应样本进行检索。实验分别使用在ROBIN数据集（CNN_1）、S-ROBIN数据集（CNN_2）以及GS-ROBIN数据集（CNN_3）上训练的卷积神经网络进行特征

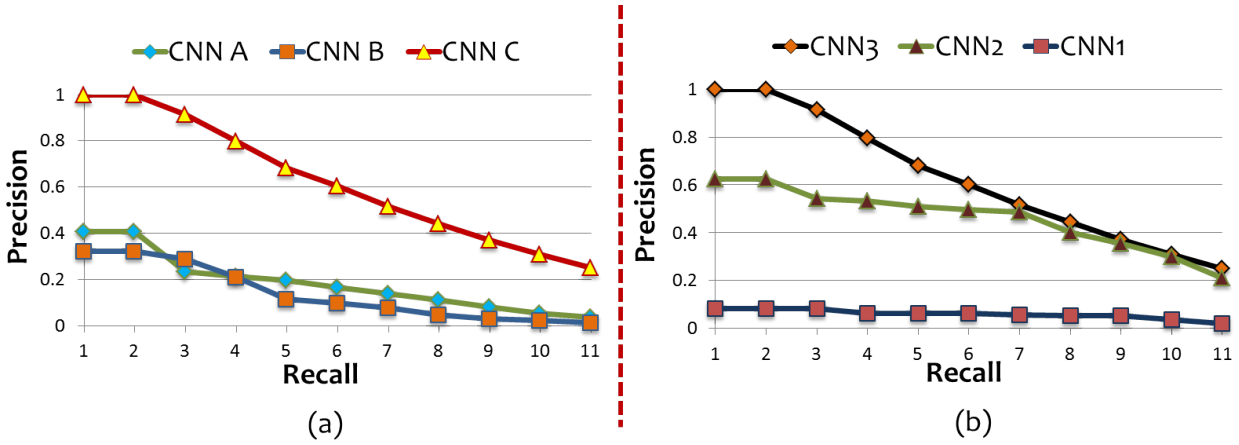


Figure 6.14. : (a) Effect of converting query sketch into image through Cyclic GAN and performing retrieval (CNN A (trained on ROBIN) and CNN B (trained on S-ROBIN)) as opposed to converting the retrieval database into sketches (CNN C). (b) Effect of training CNNs on ROBIN (CNN_1), S-ROBIN (CNN_2) and GS-ROBIN dataset (CNN_3)

extraction purpose, and interesting quantitative results were obtained showing that CNN trained on the GS-ROBIN dataset performs the best in terms of retrieval of similar floor plans (refer Fig.6.14 (b)). To understand the effectiveness of the proposed CNN layers, the PR and Mean Average Precision (MAP) of floor plan retrieval task was evaluated using each layer separately (as shown in Fig. 6.15). It was analyzed that normalized and pooling layers are quite powerful as compared to convolutional layer for floor plan retrieval task. It was observed that the normalized layer 1 yielded the best average precision value of 0.63 while retrieval. An analysis was also performed for the proposed min based approach (refer Eq. 6.6) during retrieval task and it was observed that the precision values through the min-based approach were obtained to be 0.525.

Approach 2:

Using the unified framework proposed in Approach 2, the mean average precision (MAP) value obtained if the query is an image is 0.56 as reported in Chapter 4.

However, if the query is a sketched floor plan, as discussed before, direct retrieval of floor plan images given a sketch is not straightforward (refer the low performance of CNN framework, DANIEL in Fig. 6.18). The experimentation was done with training the autoencoder network using sketches as well as images as described in Sec. 6.3.2. The PR plot obtained for retrieval using network N1 and N2 is shown in Fig. 6.16.

As shown in Fig. 6.11, given a query sketch the rank 1 result is the corresponding query floor plan image itself. This leads to the highest precision value of 1 during the initial recall. With further retrieval the average precision value decreases due to some incorrectly retrieved results belonging to other categories of layouts as compared to the query layout. It is to be noted that the area under the PR curve (refer Fig. 6.16) is greater for network N2, trained on sketch datasets due to the fact that the generated sketches from Cyclic GAN network are qualitatively better. The average precision value is obtained to be 0.642 using N2 network for domain mapping and 0.621 using the N1 network. Also, Tab. 6.4 lists out how each distance metric shown in Sec. 6.3.3 performs while matching and retrieval.

Figure 6.17, compares both the approaches and their analyzed sub-components. In Approach

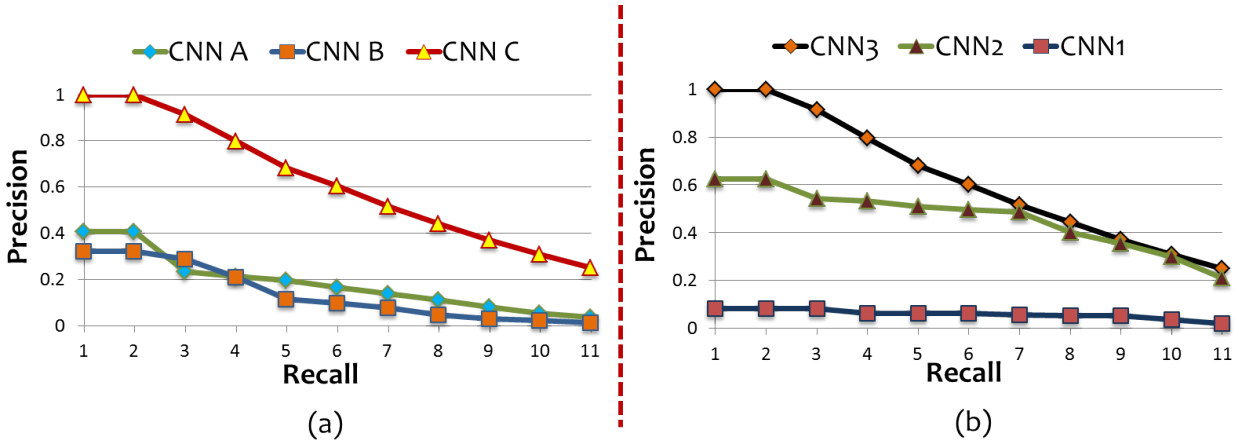


图6.14：（a）通过循环生成对抗网络（Cyclic GAN）将查询草图转换为图像并进行检索（CNN A，基于ROBIN数据集训练；CNN B，基于S-ROBIN数据集训练），与将检索数据库转换为草图后进行检索（CNN C）的效果对比。（b）分别在ROBIN（CNN_1）、S-ROBIN（CNN_2）和GS-ROBIN数据集（CNN_3）上训练卷积神经网络的效果。

出于提取目的，我们获得了有意义的定量结果，表明在相似平面图检索方面，基于GS-ROBIN数据集训练的CNN表现最佳（参见图6.14(b)）。为探究所提出的CNN各层的有效性，我们分别对每一层进行了平面图检索任务的PR曲线和平均精度均值（MAP）评估（如图6.15所示）。分析表明，在平面图检索任务中，归一化层和池化层相较于卷积层具有更强的表征能力。实验发现，归一化层1在检索过程中实现了最高的平均精度值0.63。此外，我们还针对所提出的基于最小值的方法（参见式6.6）开展了检索任务的分析，结果表明，该方法获得的精度值为0.525。

方法二：

在方法2所提出的统一框架下，若查询为图像，所获得的平均精度均值（MAP）为0.556，如第4章所述。

然而，如果查询为手绘平面图，如前所述，根据草图直接检索平面图图像并非易事（参见图6.18中CNN框架DANIEL的较低性能）。实验采用第6.3.2节所述的方法，分别以草图和图像作为训练样本对自编码器网络进行训练。利用网络N1和N2进行检索所得到的PR曲线如图6.16所示。

如图6.11所示，对于给定的查询草图，排名为1的结果即为其对应的查询平面布局图像本身。这使得在初始召回阶段的精确率值达到最高。随着检索过程的深入，由于部分检索结果属于与查询布局不同的类别，平均精确率随之下降。值得注意的是，PR曲线下的面积（参见图6.16）在基于草图数据集训练的网络N2上更大，这是因为由循环生成对抗网络生成的草图在质量上更为优越。采用网络N2进行领域映射时，平均精确率为0.642；而采用网络N1时，该值为0.621。此外，表6.4列出了第6.3.3节中所提及的各类距离度量在匹配与检索过程中的具体表现。

图6.17比较了两种方法及其所分析的子组件。在方法

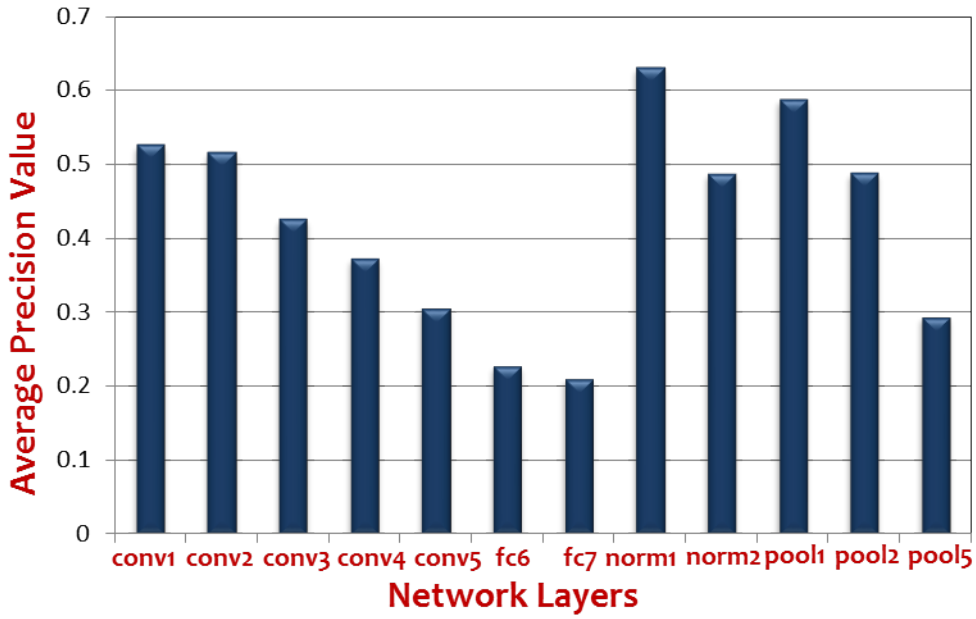


Figure 6.15. : Precision and Recall (PR) plot, comparing result of the framework, using all the hidden layers CNN features.

Table 6.4. : Mean Average Precision values for the proposed framework considering different distance metrics.

Distance Metric	Average Precision Value
Euclidean Distance	0.642
Cosine Distance	0.636
Manhattan Distance	0.625

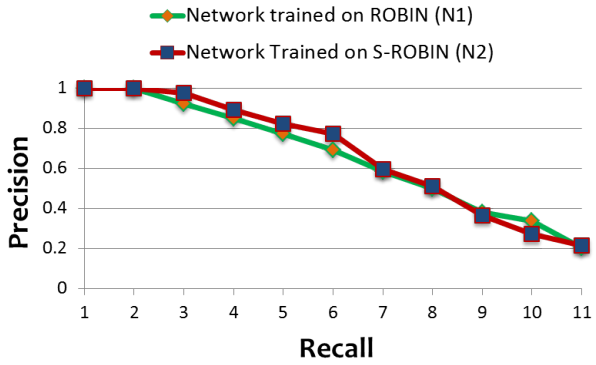


Figure 6.16. : Precision and Recall (PR) plot, comparing result of this framework on autoencoder trained with ROBIN dataset (N1) and S-ROBIN dataset (N2).

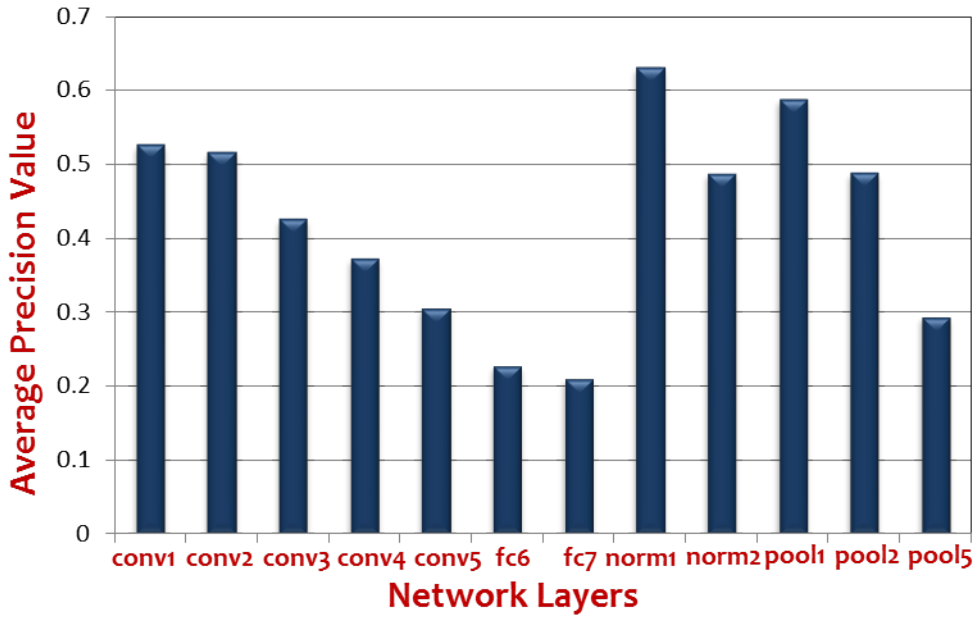


图6.15：精确率-召回率（PR）曲线，比较了采用所有隐藏层CNN特征的框架的实验结果。

表6.4：在采用不同距离度量时，所提出框架的平均精度均值。

Distance Metric	Average Precision Value
Euclidean Distance	0.642
Cosine Distance	0.636
Manhattan Distance	0.625

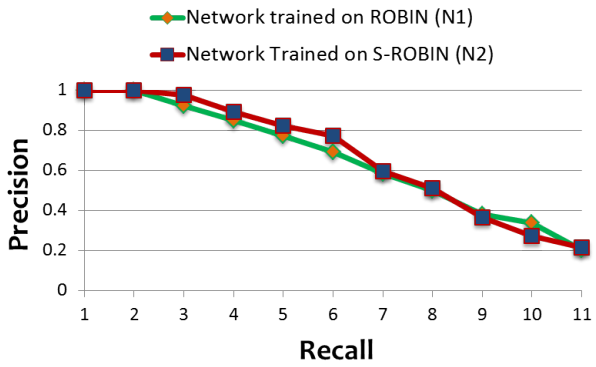


图6.16：精确率-召回率（PR）曲线，比较了本框架在分别基于ROBIN数据集（N1）和S-ROBIN数据集（N2）训练的自编码器上的实验结果。

1, retrieval results are demonstrated using (1) the min-based approach taking the minimum of distances obtained by retrieval from all layers of the CNN framework and then calculating the Precision and Recall. (2) The best precision obtained from each layer of the CNN framework. This layer was obtained to be normalization one layer, which out-performed all other layers of the CNN framework in terms of retrieval. In Approach 2, N1 autoencoder is trained using images from ROBIN dataset as input and the target images as sketch generated images, (GI-ROBIN dataset) obtained through Cyclic GAN network. N2 autoencoder network, on the other hand, is trained using sketches from S-ROBIN dataset as input and the target images as the image generated sketches, GS-ROBIN dataset obtained through Cyclic GAN network.

It can be observed looking at the area under the curve in Fig. 6.17, that Approach 2, using N2 autoencoder network outperformed all the other approaches while retrieval and gave a MAP value of 0.642 during retrieval.

Comparative studies with state-of-the-art (SOA) techniques were carried out to show the effectiveness of the proposed approaches. Figure 6.18 depicts the PR plot comparing the result of best of the two approaches (Approach 1 using norm1 layer and Approach 2 using N2 network), with the other SOA techniques. There is a significant increase in the performance while using the two proposed approaches as compared to other features or techniques keeping sketch as query mode during retrieval.

Kindly note, the under-performance of the basic deep learning framework [Sharma *et al.*, 2017] (average precision value : 0.04393) using straight-forward feature extraction from CNNs.

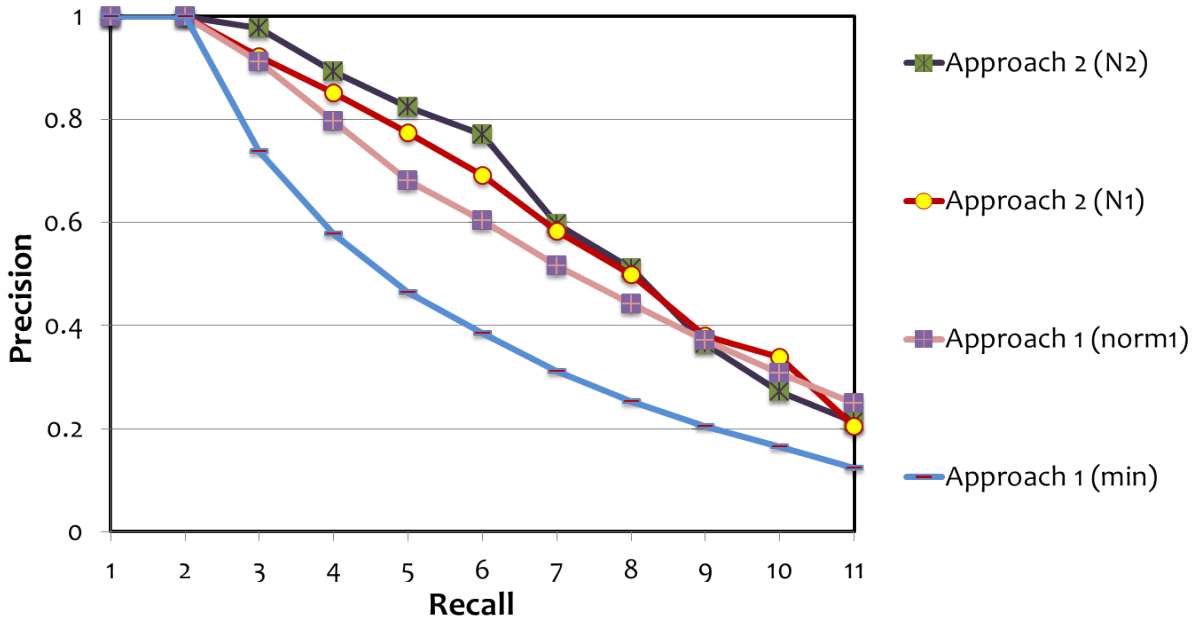


Figure 6.17. : PR plot, comparing the two approaches. Approach 1 extracts features and performs retrieval taking minimum of the precision values obtained by each layer (Approach 1 (min)) and taking the best layer (normalization 1 layer) giving the highest MAP value (Approach 1 (norm1)). Whereas, Approach 2 has two autoencoder networks, N1 trained on ROBIN and GI-ROBIN and N2 trained on S-ROBIN and GS-ROBIN.

1. 检索结果的展示采用以下两种方式：（1）基于最小值的方法，即取CNN框架各层检索得到的距离中的最小值，然后计算查准率和查全率；（2）选取CNN框架各层中获得的最佳查准率对应的层。实验发现，该层为归一化层，在检索性能上优于CNN框架的其他所有层。在方法二中，N1自编码器以ROBIN数据集中的图像作为输入，以通过循环生成对抗网络（Cyclic GAN）生成的草图图像（GI-ROBIN数据集）作为目标图像进行训练；而N2自编码器则以S-ROBIN数据集中的草图作为输入，以通过循环生成对抗网络生成的图像草图（GS-ROBIN数据集）作为目标图像进行训练。

从图6.17中曲线下面积的分布可以看出，采用N2自编码器网络的方案2在检索过程中表现最优，其平均精度均值（MAP）达到0.642。

通过与当前最先进（SOA）技术的对比研究，验证了所提出方法的有效性。图6.18展示了PR曲线，将两种方法中表现最优者（即采用norm1层的方案1和采用N2网络的方案2）与其他SOA技术进行了比较。在以草图作为查询模式进行检索时，与使用其他特征或技术相比，采用这两种所提出方法可显著提升检索性能。

请注意，基于深度学习的基本框架[Sharma v1, 2017]在直接从卷积神经网络中提取特征时表现欠佳（平均精度值为0.4393）。

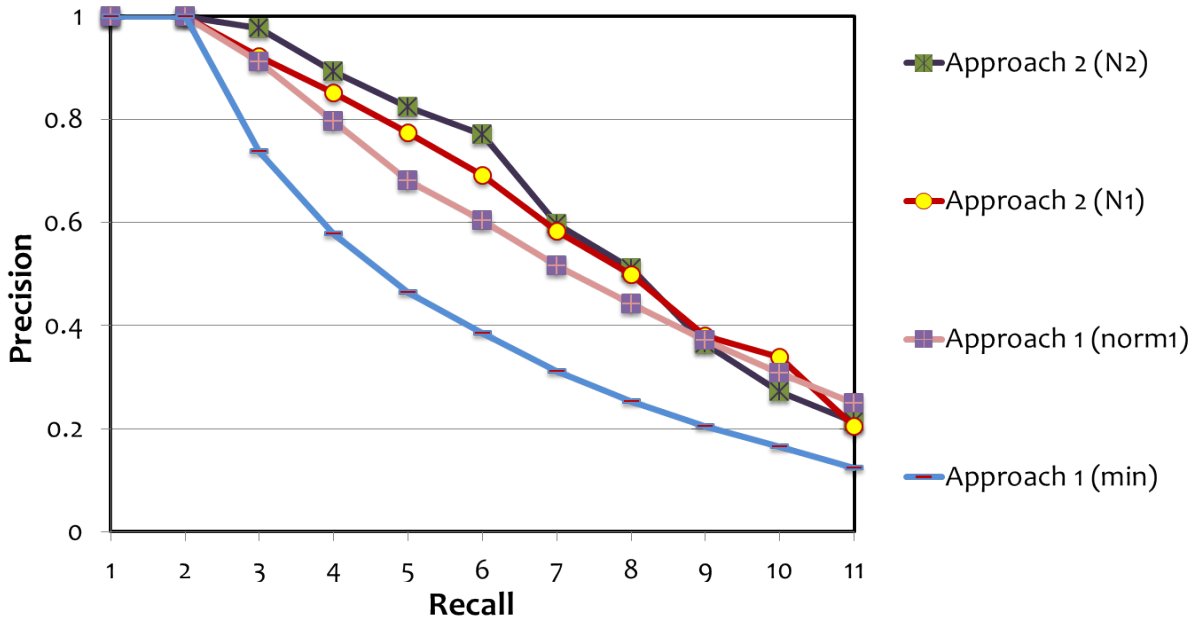


图6.17：PR曲线，比较两种方法。方法1通过提取特征并进行检索，分别采用各层所获得的精度值中的最小值（方法1（min））以及给出最高MAP值的最佳层（归一化至1层）（方法1（norm1））。而方法2则包含两个自编码器网络：N1在ROBIN和GI-ROBIN上训练，N2在S-ROBIN和GS-ROBIN上训练。

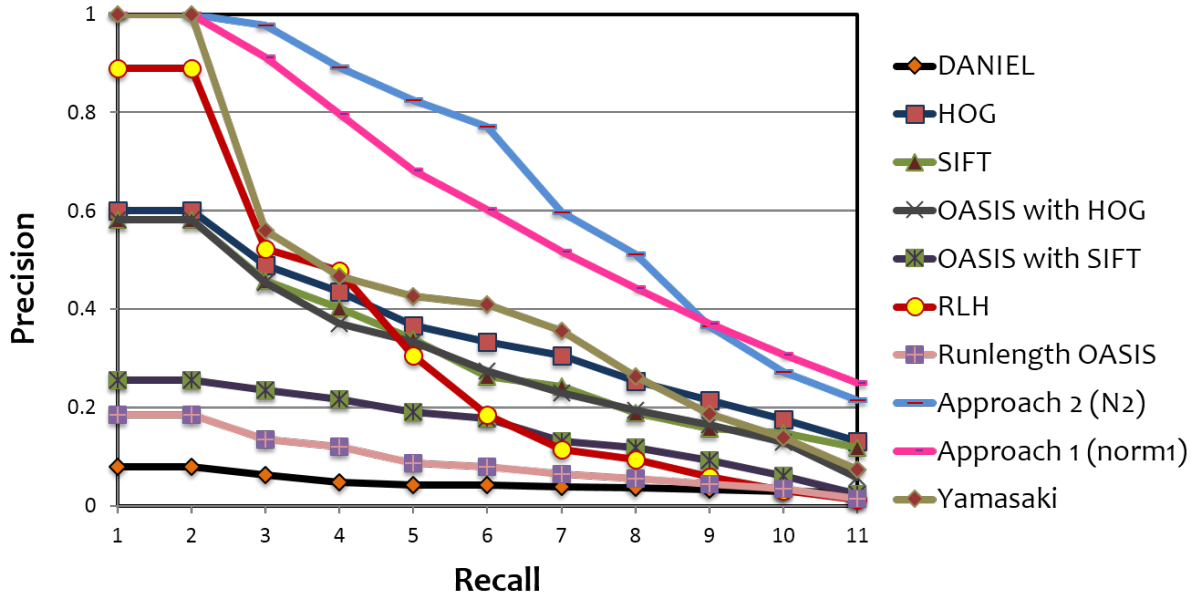


Figure 6.18. : PR plot, comparing the best of the proposed two approaches (Approach 1 (norm1)) and (Approach 2 (N2)) with SOA techniques.

The proposed approach was also compared with the very recent technique mentioned in [Yamasaki *et al.*, 2018] that segments the floor plans using CNN and matches them using Maximum Common Subgraph. It was observed that the proposed technique performs considerably well as compared to [Yamasaki *et al.*, 2018] (improvement in average precision value by 0.254), because the matching criteria in Yamasaki *et al.* [2018] only pertains to matching adjacencies and is also not well scaled for domain adaptation from sketch to image. Also, SIFT, HOG, Runlength Histogram and OASIS approaches perform weakly as they are not able to capture the abstract and sparse nature of sketches and are unable to map between the features of images and sketches. Thus, the comparative results justify the importance of the introduction of both the domain-mapping approaches given a sketch based query and image-based retrieval samples.

6.5 SUMMARY

In this chapter, two frameworks are proposed to handle multimodal retrieval in floor plans. Firstly, a deep learning framework using GAN model is proposed for sketch-based retrieval of building floor plan images. Sketch-based queries make the framework more convenient for the end user to access. Moreover, sketch based queries help in capturing the user's intent in the best possible manner. Due to the limitation on the available images in the database, it is not always necessary to find a perfect mapping of what the user desires to query in the database. Hence, drawing one's ideas on a device which can capture the basic structure of a floor plan and return similar images can prove to be beneficial as an application. The framework is capable of achieving an average precision value of 0.63 upon experimenting with 510 real-world floor plan images and their sketch replicas. Also, the framework is capable of partial matching of sketched floor plans. These abstract floor plans match in certain features, if not all, to the floor plan images in the database. Even if the user has some basic idea of what he desires his floor plans to be then the user can use this

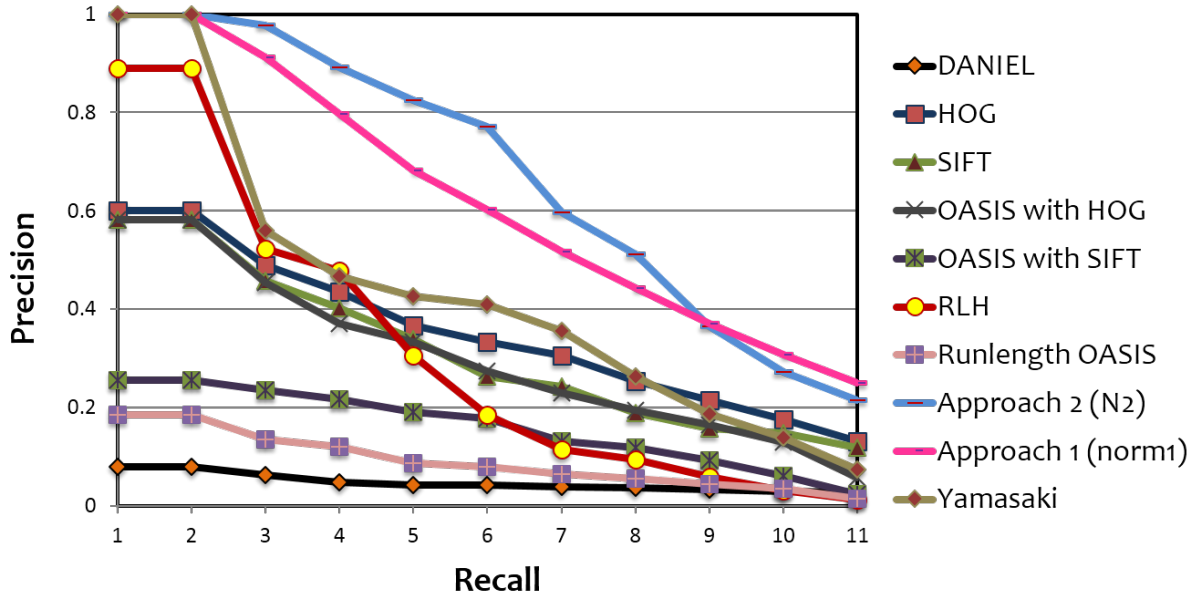


图6.18：PR曲线，比较所提出的两种方法中表现最佳的方案（方法1，即norm1）和（方法2，即N2）与现有最优算法（SOA）的性能。

所提出的方法还与文献[Yamasaki，2018]中提及的最新技术进行了对比，该技术采用卷积神经网络对平面图进行分割，并利用最大公共子图匹配算法进行匹配。实验结果表明，与文献[Yamasaki，2018]相比，所提出的方法性能显著更优（平均精度提升了0.254），这是因为Yamasaki[2018]中的匹配准则仅关注邻接关系的匹配，且在从草图到图像的领域适应任务中扩展性较差。此外，SIFT、HOG、游程直方图以及OASIS等方法的表现较为逊色，其原因在于这些方法难以捕捉草图的抽象性和稀疏性特征，也无法实现图像与草图特征之间的有效映射。因此，对比实验结果充分证明了在基于草图的查询与基于图像的检索场景下，引入两种领域映射方法的重要性。

6.5 摘要

在本章中，我们提出了两种用于处理建筑平面图多模态检索的框架。首先，我们提出了一种基于生成对抗网络（GAN）模型的深度学习框架，用于实现基于草图的建筑平面图图像检索。基于草图的查询使该框架对终端用户而言更加便捷，并且能够以最佳方式捕捉用户的意图。由于数据库中可用图像数量有限，往往难以在数据库中找到与用户查询完全匹配的图像。因此，在能够捕捉平面图基本结构的设备上绘制构思并返回相似图像的应用程序，具有显著的实际价值。经在510张真实世界建筑平面图及其草图副本上进行实验，该框架的平均精度达到0.63。此外，该框架还支持草图平面图的部分匹配；这些抽象的平面图虽然未必与数据库中的所有特征完全一致，但在某些特征上仍能与之匹配。即使用户对其期望的平面图仅有初步构想，亦可借助该框架实现检索。

framework for retrieval of similar floor plans.

The second framework applies domain adaptation techniques that help to map both sketches and images to a single framework and are capable of performing better retrieval. There was a need to develop a unified framework capable of both image to image and sketch to image retrieval to serve as a composite tool for multimodal retrieval in floor plans. It was observed that autoencoders in conjunction with Cyclic GANs performed better during retrieval of architectural floor plans giving an average precision value of 0.642 which is improvement by a value 0.1 over the network containing only Cyclic GANs for domain adaptation.

用于检索相似平面布局的框架。

第二种框架采用领域适应技术，能够将草图与图像映射到统一的表示空间，并实现更优的检索性能。为构建一种兼具图像到图像检索与草图到图像检索能力的统一框架，以作为平面图多模态检索的综合性工具，有必要进行相关研究。实验发现，自编码器与循环生成对抗网络（Cyclic GAN）相结合，在建筑平面图检索任务中表现更为优异，平均精度达到0.642，较仅采用循环生成对抗网络进行领域适应的模型提升了0.1。

Annexure A
Floor plan datasets ROBIN and S-ROBIN

Dataset plays a pivotal role in any image processing problem. Availability of a public dataset sometimes makes the life of a researcher easy. However, this is true for only generic problems, and not so for problems which are of some specific nature. Floor plan image analysis is such a problem where a large collection of samples is hardly available. For this reason two public datasets have already been proposed by researchers. They are: (i) SESYD [Delalandre *et al.*, 2010b] and (ii) CVC-FP [de las Heras *et al.*, 2015]. The former has 10 classes of floor plans, where there are 100 samples/class. The latter has 122 scanned floor plan documents divided into 4 categories based on the origin and style. The SESYD dataset is inappropriate for application to a retrieval task due to very less difference between a pair in a class of floor plan images. Also, in the CVC-FP dataset, the samples are insufficient in number for the task of floor plan retrieval.

To train and evaluate the contributions presented in this thesis, two floor plan databases, named Repository Of BuildIng floor plaNs (ROBIN) and Sketched-Repository Of BuildIng floor plaNs (S-ROBIN) are created. S-ROBIN are the sketch replicas of the ROBIN dataset containing floor plan images. These datasets have significantly aided in the content analysis and retrieval of similar floor plans as proposed in this thesis. The primary motivation while creating the ROBIN dataset was to have: i) sufficient number of sample to execute the experiments, and ii) variation (inter and intra class) among the samples. While searching for a dream house, the customer generally has the follwing sequence of requirements in his/her mind; i) the number of rooms inside the house, ii) the global shape of the house, iii) what is the pattern in which the rooms are connected to each other, and iv) if a user is looking for a well furnished home. then which furniture should be placed in the house and in which arrangement. In the following subsections, each of the points mentioned above are elaborated in detail.

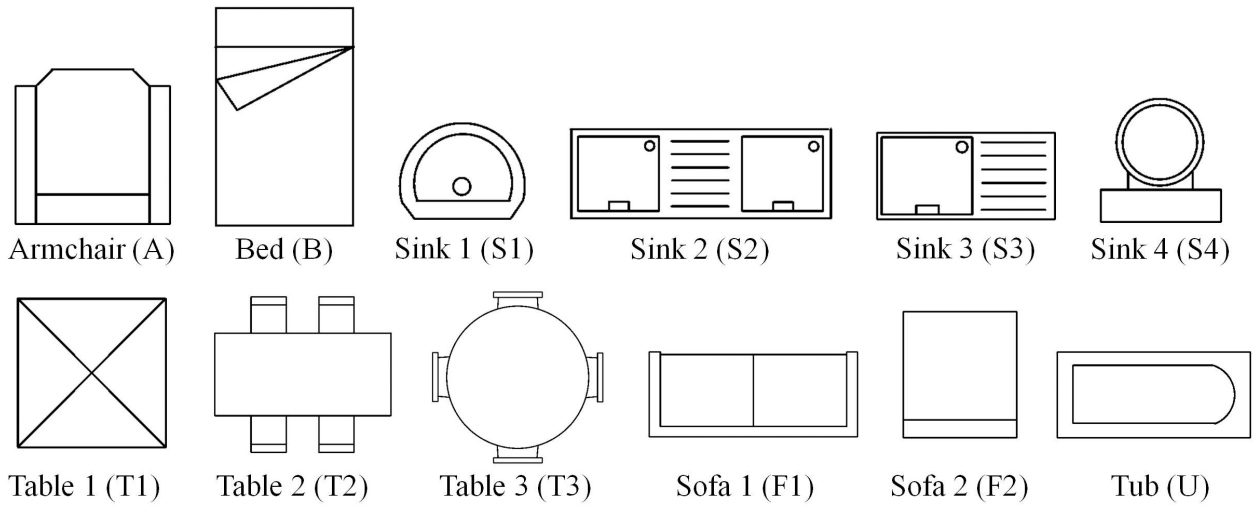


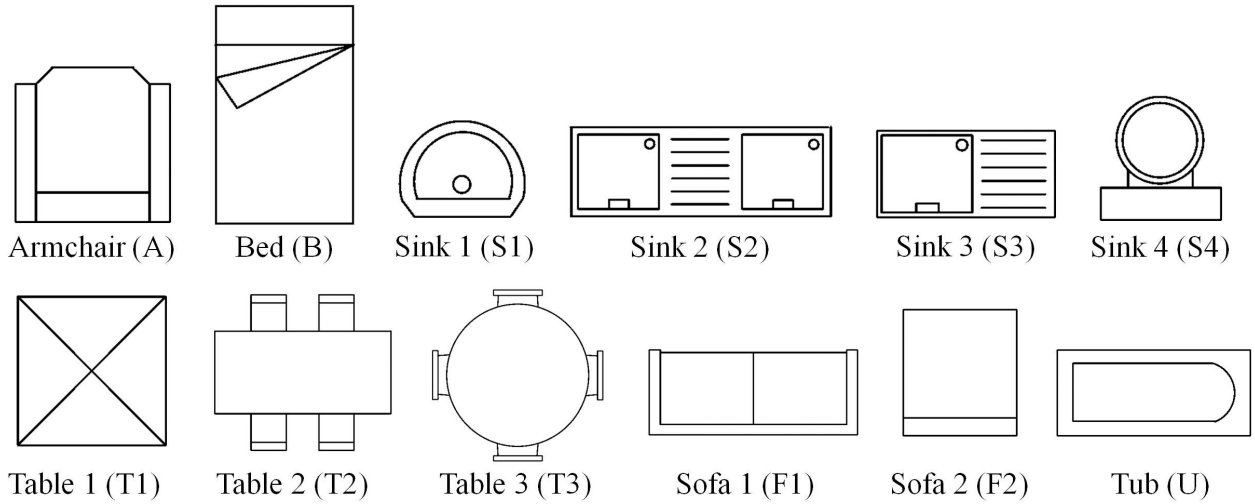
Figure A.1. : Symbol set used as furniture in ROBIN and S-ROBIN dataset.

附件A

Floor plan datasets ROBIN and S-ROBIN

数据集在任何图像处理问题中都起着至关重要的作用。公开数据集的可用性有时能够简化研究人员的工作。然而，这仅适用于通用问题，而对于某些特定类型的问题则不然。楼层平面图图像分析便属于此类问题，其中可供使用的样本数量极为有限。正因如此，研究者已提出了两个公开数据集，分别为：(i) SESYD [Delalandre *et al.*, 2010b] 和 (ii) CVC-FP [de las Heras *et al.*, 2015]。前者包含10类楼层平面图，每类有100个样本；后者则包含122份扫描得到的楼层平面图文档，按来源与风格划分为4个类别。由于同一类别中的楼层平面图样本之间差异甚微，SESYD数据集并不适于检索任务。此外，在CVC-FP数据集中，样本数量亦不足以支持楼层平面图检索任务。

为训练并评估本论文所提出的各项方法，我们构建了两个平面图数据库，分别命名为“建筑平面图资源库”（Repository Of BuildIng floor plaNs，简称ROBIN）和“草图版建筑平面图资源库”（Sketched-Repository Of BuildIng floor plaNs，简称S-ROBIN）。其中，S-ROBIN是ROBIN数据集的草图复制品，包含平面图图像。这些数据集在本论文所提出的相似平面图内容分析与检索任务中发挥了重要作用。创建ROBIN数据集的主要动机在于：i）确保实验所需的样本数量充足；ii）使样本在类间及类内均具有足够的多样性。在寻找理想住宅时，客户通常会在脑海中形成如下需求序列：i）房屋内部的房间数量；ii）房屋的整体形态；iii）各房间之间的连接模式；以及iv）若用户寻求精装修住宅，则还需明确应选用何种家具及其摆放布局。在接下来的小节中，上述各要点将予以详细阐述。



图A.1：在ROBIN和S-ROBIN数据集中用作家具的符号集。

A.1 NUMBER OF ROOMS

In ROBIN there are three categories of floor plans, depicting the three broad classes of houses. There are houses with 3, 4 and 5 rooms. The purpose and the subsequent label of each room varies depending on the various decors present in them. To be consistent with the existing floor plan datasets, the furniture symbols used in ROBIN dataset are similar to the ones used in SESYD dataset [Delalandre *et al.*, 2010a] as shown in Fig. A.1.

A.2 GLOBAL SHAPE OF THE HOUSE

The second important criteria kept in mind while designing the dataset was the global or overall shape of the layout. To design the floor plans the rooms were arranged either in the horizontal or the vertical direction with respect to each other. While designing, due consideration was given to the fact that, there is no arbitrary shaped floor plan image and also that curvilinear structures in the layout walls are avoided. Moreover, to increase the variability among the samples, difference in arrangement of rooms across 3, 4 and 5 room layout categories was also introduced. Figure A.2 (a), (b) and (c) depict the example of variation in global shapes across a layout category.

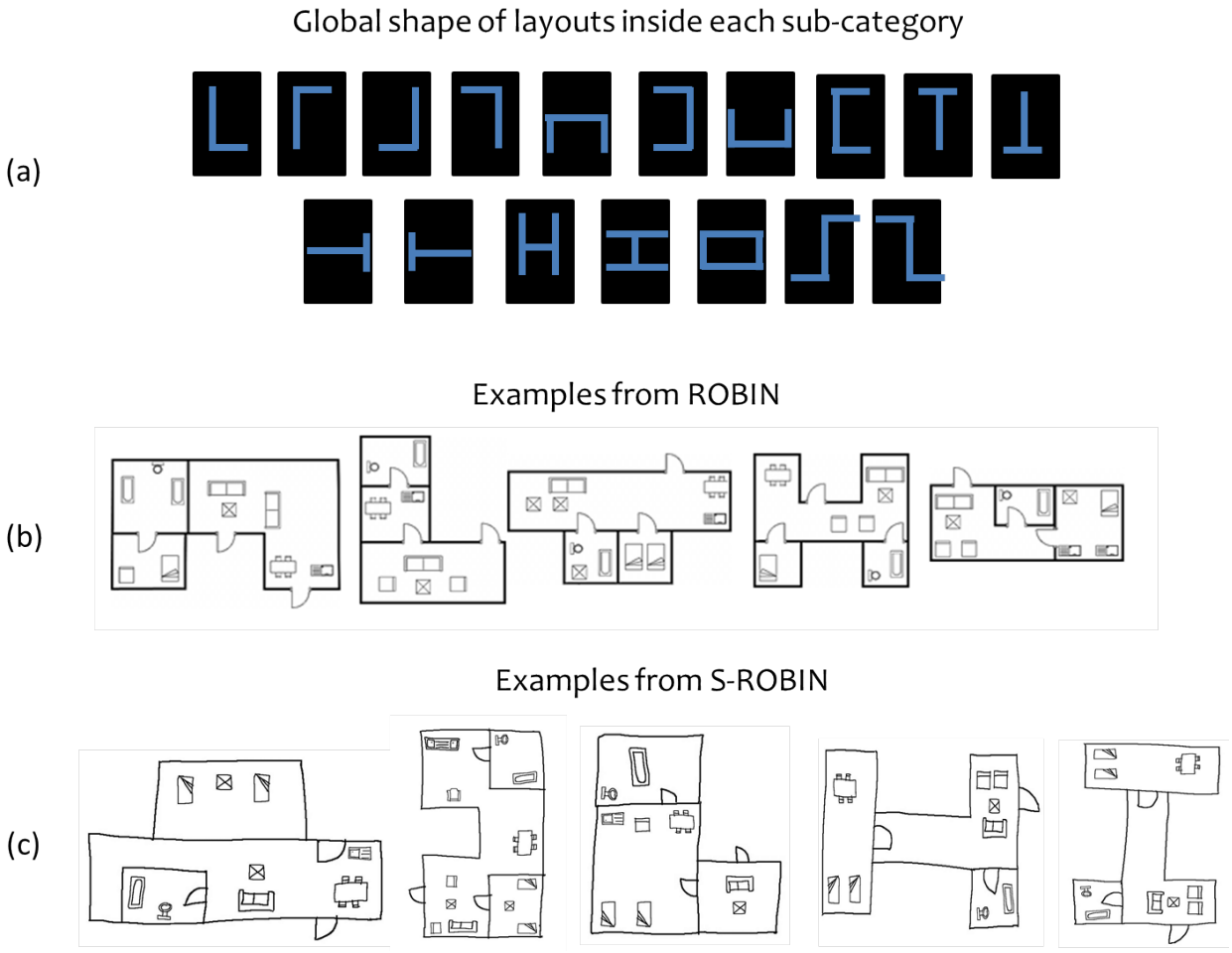


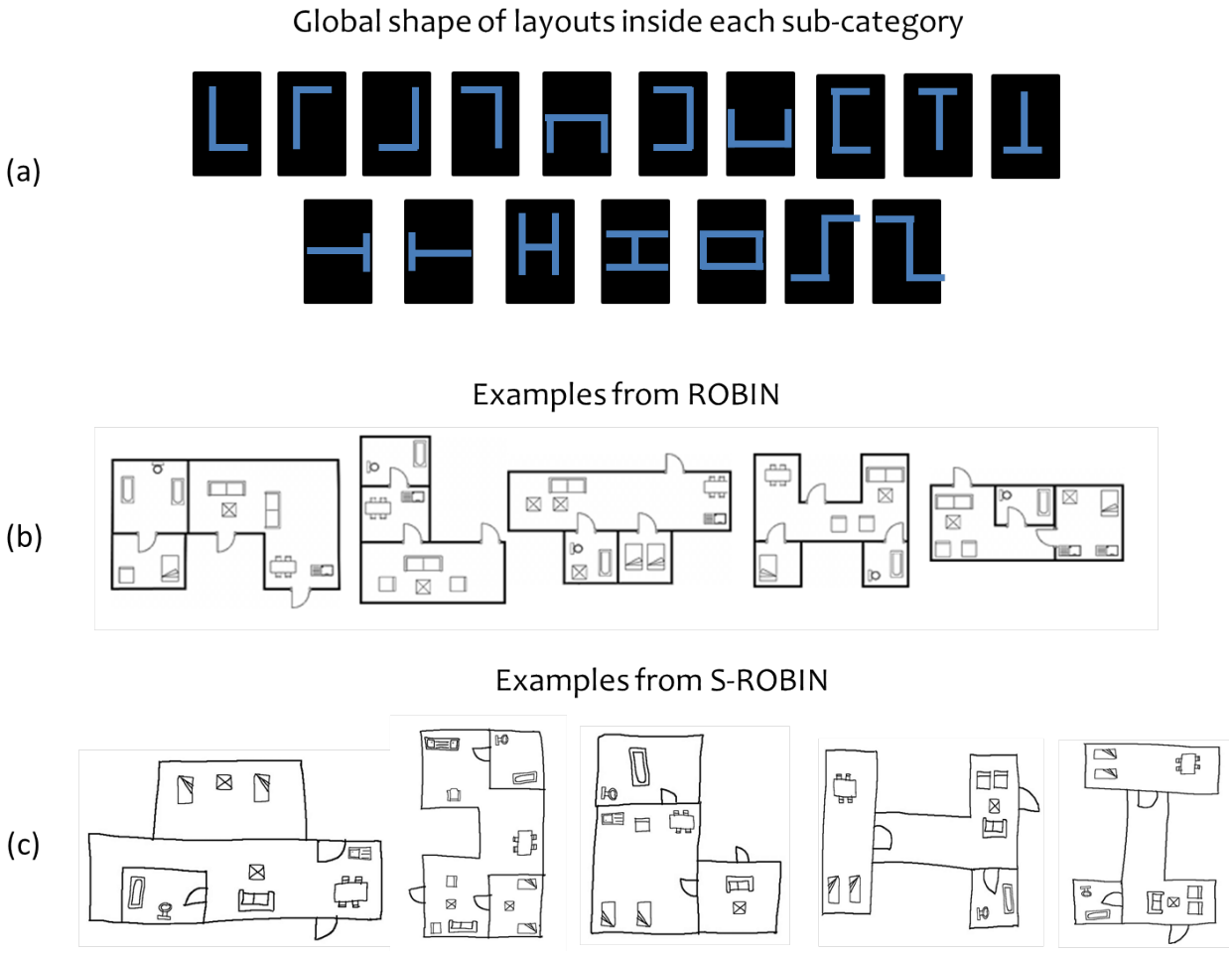
Figure A.2. : 3-Room layouts in the ROBIN and S-ROBIN dataset.

A.1 房间数量

在ROBIN数据集中，共有三类平面布局，分别对应三种主要的住宅类型。这些住宅分别包含3间、4间和5间房间。各房间的功能及其后续标注会根据其内部所配置的不同装饰而有所差异。为与现有平面布局数据集保持一致，ROBIN数据集中使用的家具符号与SESYD数据集[Delalandre, 2010a]中的符号相似，如图A.1所示。

A.2 房屋的整体形状

在设计数据集时，所遵循的第二个重要原则是布局的整体形态。在绘制平面图时，各房间之间均按水平或垂直方向进行排列。设计过程中充分考虑了以下两点：一是不存在任意形状的平面图；二是避免在墙体布局中出现曲线结构。此外，为提高样本的多样性，还在3室、4室和5室这三类布局中引入了不同的房间布置方式。图A.2（a）、（b）和（c）分别展示了同一布局类别中整体形态的差异示例。



图A.2：ROBIN和S-ROBIN数据集中3个房间的布局。

A.3 CONNECTIVITY WITHIN A HOUSE

Doors are placed at the desired places in the floor plans to ensure connectivity between the rooms. Orientation of each door is given special consideration to ensure proper direction of movement within the house. For every floor plan, there is atleast one door lying on the outer wall of the floor plan that opens inside the house and provides entry/exit for the house. Windows are appropriately placed on the walls of the floor plan to ensure proper ventilation and aesthetics. This makes the floor plan more realistic. Figure A.3 (b) can be observed for the arc-type door symbols satisfying entry-exit criteria in the floor plans.

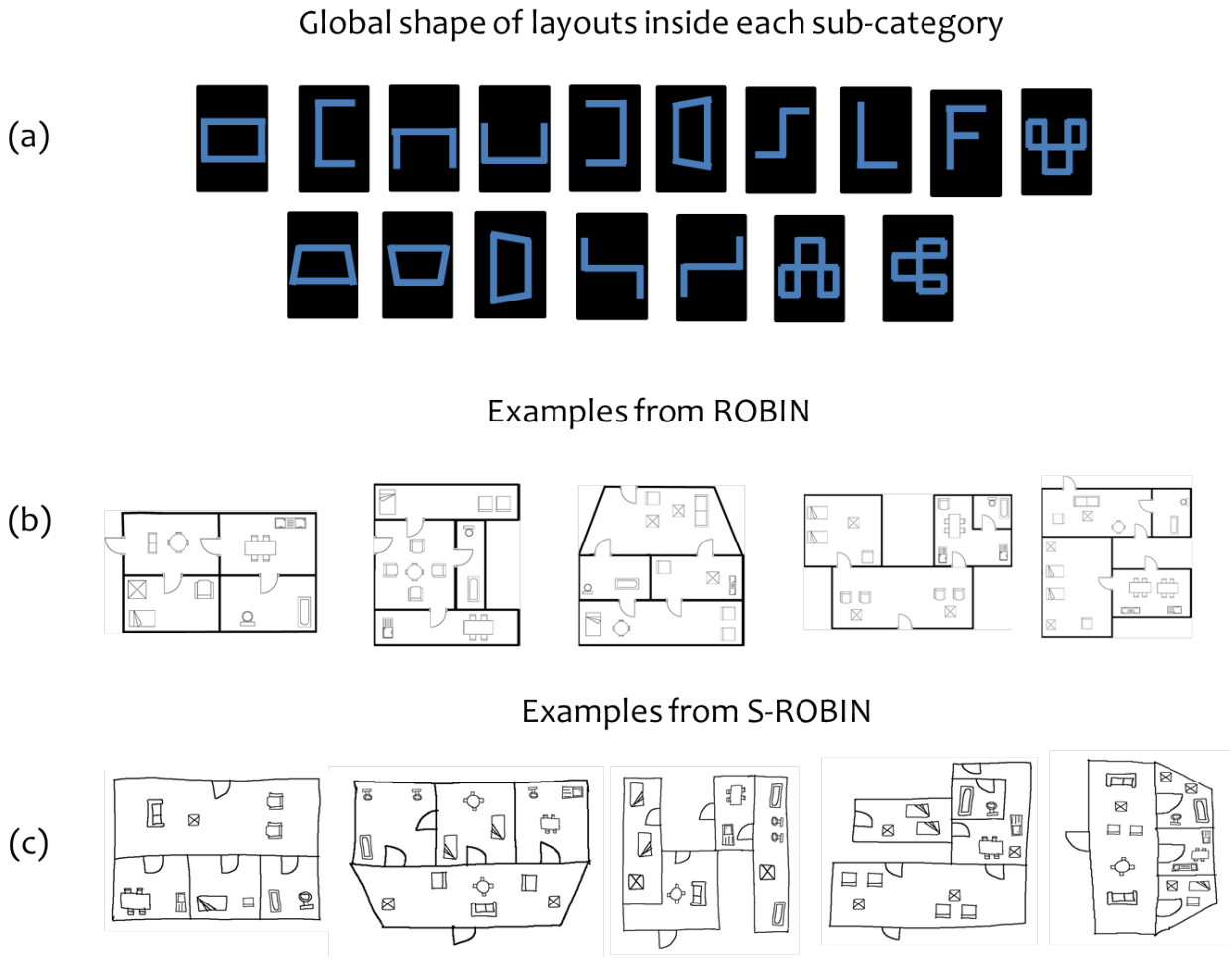


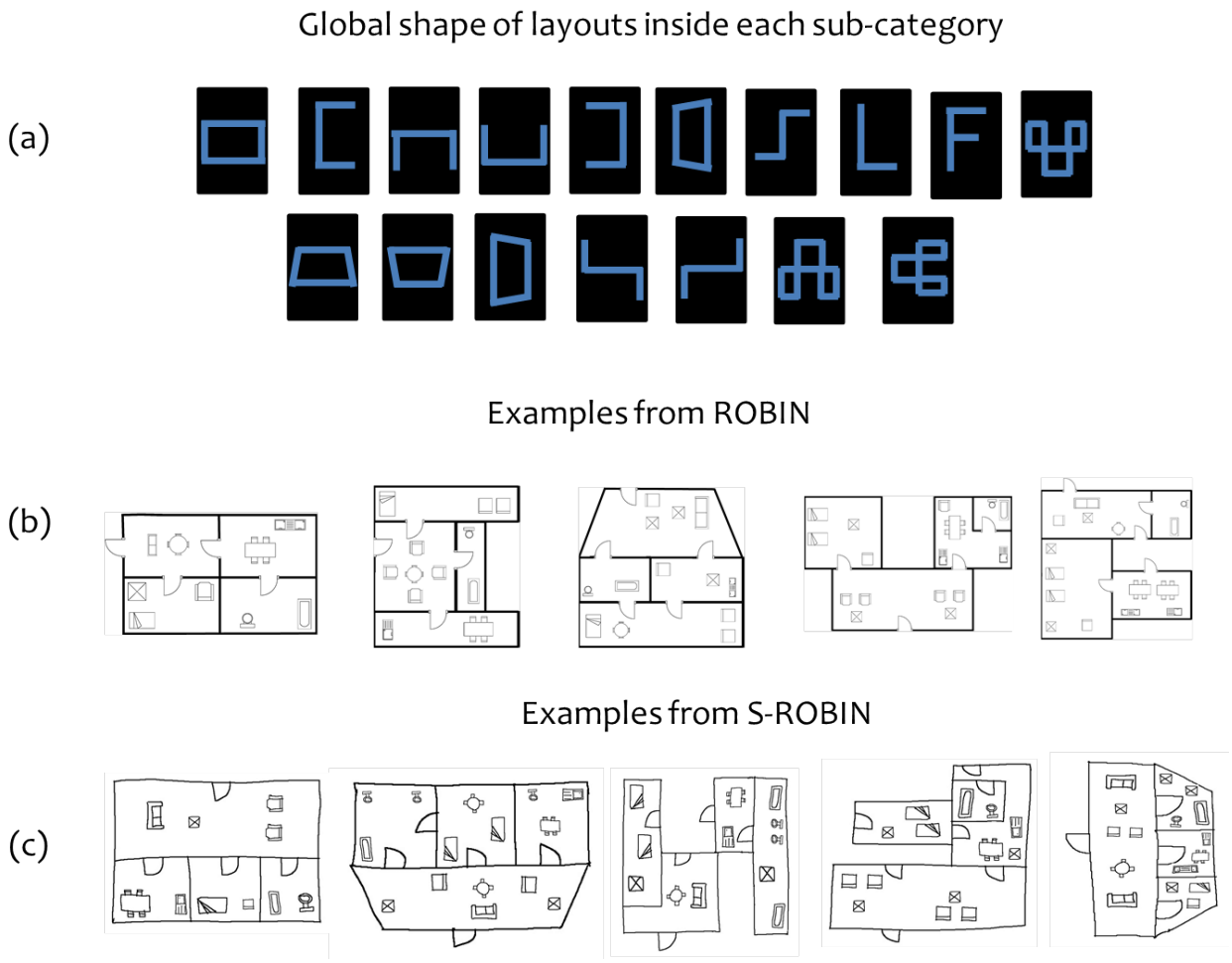
Figure A.3. : 4-Room layouts in the ROBIN and S-ROBIN dataset.

A.4 PLACEMENT OF FURNITURE

Furnitures or decors are inseparable part of a floor plan. To make the floor plans of ROBIN realistic, decor symbols as proposed by Delalandre *et al.* [2010a] were utilized. As shown in Fig. A.1, there are 12 different floor plan symbols. In each floor plan a subset of the entire symbol set was chosen and placed inside every room to give a definition to the room. As the furnitures are indicators of the purpose of the rooms, therefore, while designing, sufficient care has been taken to ensure that a house contains a i) kitchen, ii) bedroom, iv) washroom and a v) lobby. Moreover,

A.3 房屋内部的连通性

在平面图中，门被布置在指定位置，以确保各房间之间的连通性。每扇门的朝向均经过特别考量，以保证室内人员流动方向的合理性。对于每个平面图，至少有一扇门设置在外墙上，且向室内开启，作为住宅的出入口。窗户则合理布置于平面图的墙体上，以兼顾通风与美观，从而使平面图更加贴近实际。图A.3(b)展示了满足出入口要求的弧形门符号在平面图中的应用。



图A.3：ROBIN和S-ROBIN数据集中4个房间的布局。

A.4 家具的摆放

家具或装饰是平面图不可或缺的组成部分。为使ROBIN的平面图更加逼真，采用了Delalandre *et al.* [2010a] 提出的装饰符号。如图A.1所示，共有12种不同的平面图符号。在每一张平面图中，均从全部符号集中选取一个子集，并将其布置于各房间内，以明确各房间的功能属性。由于家具是房间用途的重要指示物，因此在设计过程中特别注意确保每栋住宅包含：i) 厨房、ii) 卧室、iii) 洗手间以及iv) 门厅。此外，

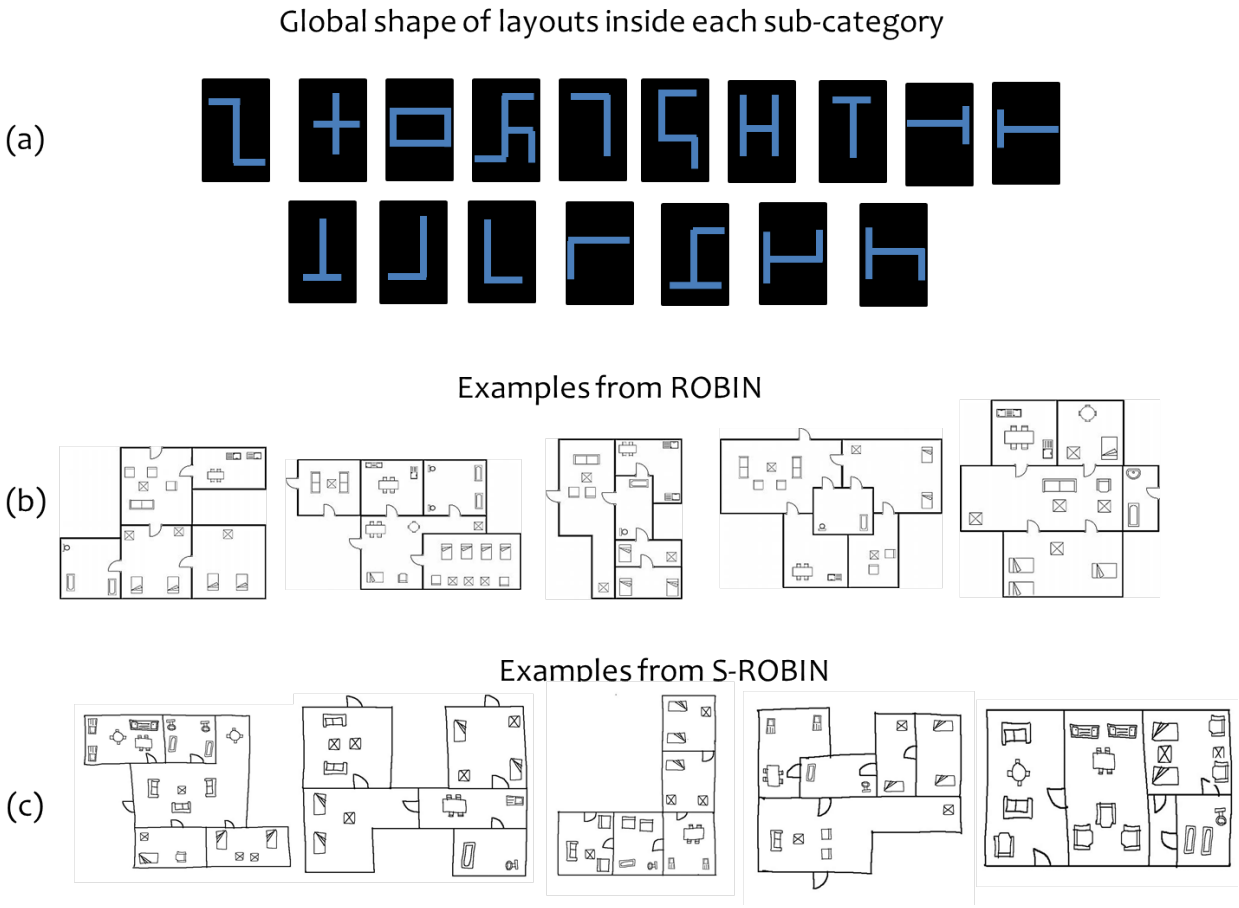
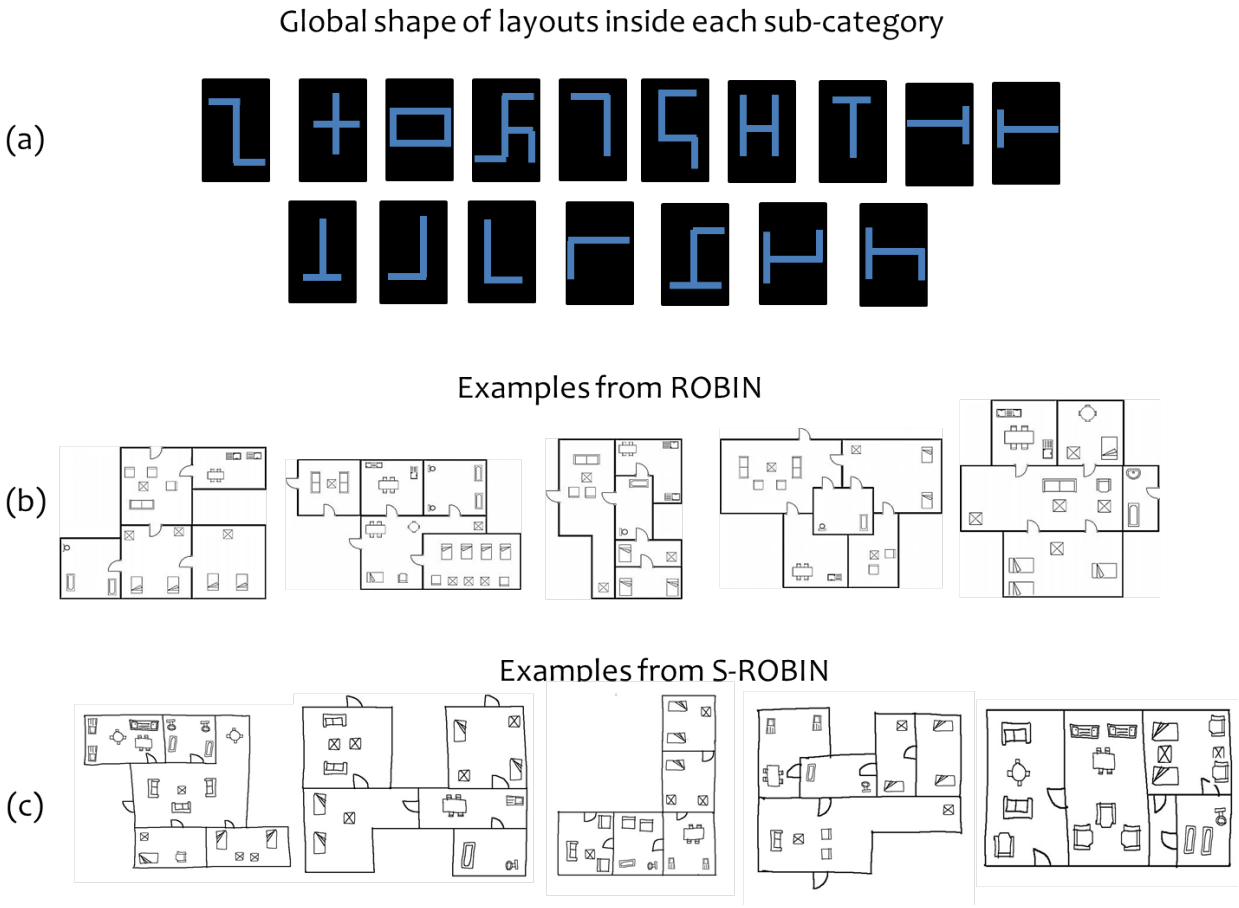


Figure A.4. : 5-Room layouts in the ROBIN and S-ROBIN dataset.

special consideration was given to the fact that the furniture is not placed in an arbitrary manner. For example, furniture should not be placed in a way that it blocks the door. Also, sufficient area should be left vacant after the furniture placement for navigation purposes.

A.5 DATASET CREATION PROCESS

ROBIN dataset was created on similar lines as the SESYD dataset [Delalandre *et al.*, 2010a]. Basic room structure of the floor plan was built using rectangular shape in Microsoft powerpoint. Thickness of each wall was kept uniform to be around 40 pixels. The decor symbols were inserted inside the rooms from the symbol library and door symbols were carefully placed on the room walls. On the other hand, to capture hand drawn sketched floor plans in S-ROBIN, volunteers were asked to draw the floor plans on a digital platform using a Wacom tablet. Wacom tablet is an interactive tool, where the user can draw on a board like tablet using a pen stylus conveniently and the resultant floor plan sketch is captured on the Microsoft paint software by adjusting the thickness of the strokes of the pen stylus. Since, offline mode is being used, the hand drawn floor plans are stored as an image and used for experimentation.



图A.4：ROBIN和S-ROBIN数据集中5个房间的布局。

特别考虑到了家具并非随意摆放这一事实。例如，家具不应以遮挡门的方式进行布置；此外，在家具布置完成后，还应留出足够的空地以供通行。

A.5 数据集创建过程

ROBIN数据集的构建方式与SESYD数据集[Delalandre等，2010a]类似。在Microsoft PowerPoint中，采用矩形形状构建了平面图的基本房间结构，并将各墙体的宽度统一设置为约40像素。装饰符号从符号库中插入到各个房间内，门符号则被精确地放置于房间墙面上。另一方面，为了采集手绘草图形式的S-ROBIN平面图，我们要求志愿者使用Wacom数位板在数字化平台上进行绘制。Wacom数位板是一种交互式工具，用户可借助笔式触控笔在类似画板的设备上便捷作画，所生成的平面图草图通过调整笔式触控笔笔触的粗细，在Microsoft Paint软件中予以记录。由于采用离线模式，手绘平面图以图像形式存储，并用于实验研究。

A.6 STATISTICS OF THE DATASET

Both the datasets contain 510 floor plans divided into 3 categories of layouts each with 17 sub categories. The unique characteristic of the datasets is that they are designed keeping in mind the need of a potential buyer. Every prospective buyer wants to have certain amenities and functionalities in their house. Thus, in the dataset, there are three broad categories, which are different from each other in terms of the number and type of rooms present in a floor plan. The broad categories are layouts with 3-rooms, 4-rooms and 5-rooms each which differ in the global shape of layout across their sub-categories. Each broad category is further classified into 17 sub-categories depending upon the global layout shape of the floor plan. Standard notations as mentioned by Delalandre *et al.* [2007], are followed for all the floor plans so that the datasets can be used for other floor plan analysis tasks as well, and not only for retrieval. The dataset has 12 different categories of furnitures inside each layout. These datasets help in better visualization of the floor plans and aid in efficient capturing of various high-level features while fine-grained retrieval. Illustrations of the various sub-categories in the floor plans and how their shapes are modeled are shown in Fig. A.2, Fig. A.3 and Fig. A.4. The size of the floor plans in ROBIN vary from 825×708 to 2916×1845 whereas the resolution in S-ROBIN dataset ranges from 591×517 pixels to 1483×884.

A.7 CONCLUSION

The rationale behind the creation of ROBIN and S-ROBIN dataset was to create a floor plan repository that can be used for research purposes by the Document Analysis and Retrieval (DAR) community. Both ROBIN and S-ROBIN dataset are sufficient for carrying out tasks like, retrieval, semantic segmentation, door to door navigation, etc. The entire dataset along with the annotations can be downloaded from [S-ROBIN, 2018].

A.6 数据集的统计信息

两个数据集均包含510张平面图，每组分为3个布局大类，每个大类又细分为17个小类。这些数据集的独特之处在于，其设计充分考虑了潜在购房者的实际需求：每位购房者都希望住宅具备特定的配套设施与功能。因此，数据集中设置了三类主要布局，它们在平面图中房间的数量与类型上各不相同。这三类分别为三居室、四居室和五居室布局，且各小类在整体形态上也存在差异。每一类布局根据其整体形态进一步划分为17个小类。为便于数据集不仅用于检索，还可应用于其他平面图分析任务，所有平面图均采用Delalandre于2007年提出的标准化标注方法。每个布局中还包含12种不同类型的家具。这些数据集有助于更直观地呈现平面图，并在细粒度检索过程中有效提取各类高层特征。平面图中各小类的具体示例及其形状建模方式分别见图A.2、图A.3和图A.4。ROBIN数据集中平面图的尺寸范围为825×708至2916×1845像素，而S-ROBIN数据集的分辨率则介于591×517像素至1483×884像素之间。

A.7 结论

创建ROBIN和S-ROBIN数据集的初衷是构建一个可用于文档分析与检索（DAR）研究领域的平面图资源库。ROBIN和S-ROBIN数据集均足以支持诸如检索、语义分割、门到门导航等任务的开展。整个数据集及其标注可从[S-ROBIN, 2018]下载。

Annexure B

GUI for floor plan analysis and retrieval

In this Chapter, a software interface for the task of content-based floor plan retrieval is discussed. The software works in the following manner:

1. The user chooses a sample floor plan image and uploads it to the server.
2. The retrieval engine extracts features from the query sample and matches with all the samples of the database.
3. The retrieved samples are rank ordered and shown to the user.

The software has a Graphical User Interface (GUI) with which the user can interact with the system and retrieve similar floor plans from the database. The software has two operational modes, which are (i) detailed mode, and (ii) direct mode. In the detailed mode, depending on the users' choice the intermediate steps of floor plan retrieval such as, segmentation, graph creation with nodes as segmented rooms and adjacencies as relations between the rooms, and finally retrieval based on the features extracted from the floor plans are shown in this GUI. On the other hand, in the direct mode the intermediate steps are not shown, and only the final result is displayed.

B.1 SYSTEM ARCHITECTURE

The web-based interface, which runs in a client server mode, has been developed to facilitate the framework for semantic retrieval of floor plans. The proposed system runs on Windows 7/XP and requires a few seconds to retrieve the rank-ordered results from the repository of floor plans. Figure B.1 shows the overall architecture for the framework. Developed under a client-server paradigm, the retrieval framework has two types of users to access the application, namely normal user and the administrator. A normal user is allowed to browse the system and perform queries, select whether he/she wants to get results of all the intermediate steps in the framework or only the final retrieval results, place a request to add a new floor plan to the repository, etc. An administrator on the other hand has all the access of the functionalities provided to a normal user, and in addition has the privilege to see the repository updation requests sent by the normal user and add/remove user accounts etc. The types of user access are controlled by a backend user database maintained using Microsoft Access. The retrieval portal is developed using HTML5, JavaScript, CSS. Apache webserver hosts the webpages, while Tomcat Java Server hosts Java servlets for receiving the query provided by the user.

附件B

GUI for floor plan analysis and retrieval

在本章中，讨论了一种用于基于内容的建筑平面图检索任务的软件界面。该软件的工作流程如下:

1. 用户选择一张示例平面图并将其上传至服务器。

2. 检索引擎从查询样本中提取特征，并与数据库中的所有样本进行匹配。

3. 检索到的样本按排序顺序展示给用户。

该软件配备了一个图形用户界面（GUI），用户可通过该界面与系统交互，并从数据库中检索相似的平面布局。软件提供两种运行模式：（i）详细模式和（ii）直接模式。在详细模式下，根据用户的选择，平面布局检索过程中的各中间步骤——包括分割、以分割后的房间为节点、以房间之间的邻接关系为边构建图，以及基于从平面布局中提取的特征进行最终检索——均会在该GUI中予以展示。而在直接模式下，则不显示这些中间步骤，仅呈现最终结果。

B.1 系统架构

基于Web的界面采用客户端—服务器模式构建，旨在为楼层平面图的语义检索框架提供便利。所提出的系统运行于Windows 7/XP平台，从楼层平面图知识库中检索并输出按相关性排序的结果仅需数秒。图B.1展示了该框架的整体架构。在客户端—服务器架构下，检索框架支持两类用户访问：普通用户和管理员。普通用户可浏览系统、执行查询，选择获取框架中所有中间步骤的结果或仅获取最终检索结果，并可提交将新楼层平面图添加至知识库的请求等。而管理员除具备普通用户的所有功能权限外，还拥有查看普通用户提交的知识库更新请求、增删用户账户等特权。用户的访问权限由后端采用Microsoft Access维护的用户数据库进行控制。检索门户使用HTML5、JavaScript和CSS开发，Apache Web服务器负责托管网页，Tomcat Java服务器则用于运行Java Servlet，以接收用户提交的查询请求。

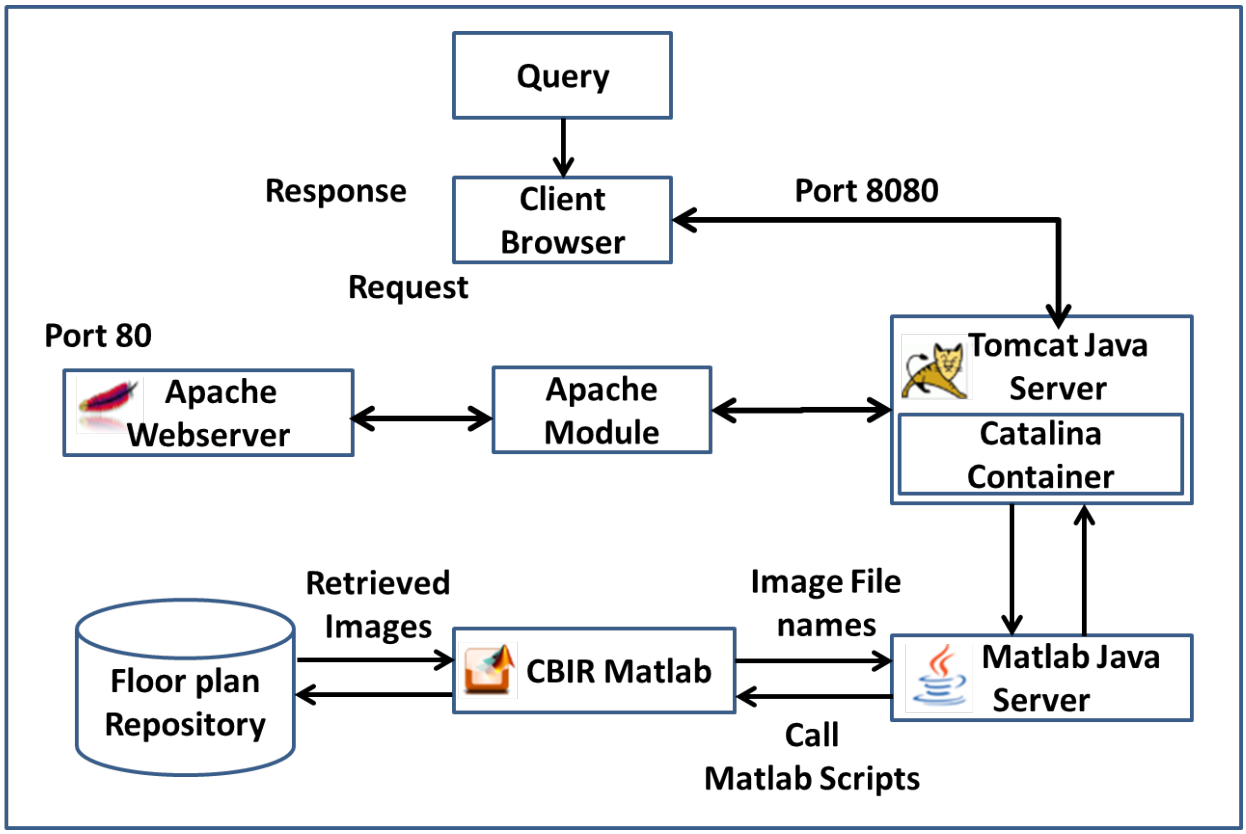


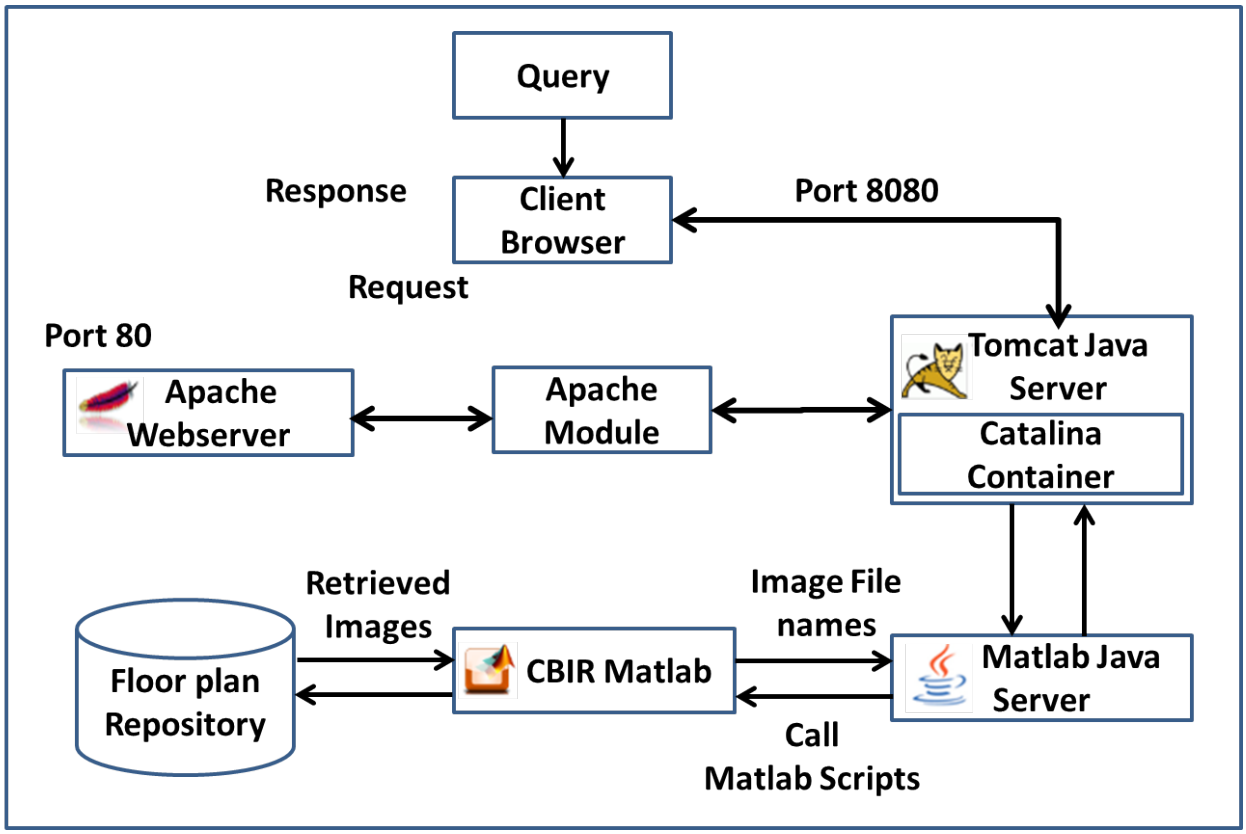
Figure B.1. : Architecture of the developed software for floor plan retrieval.

B.2 SOFTWARE FUNCTIONALITY

Figure B.2 depicts the sequence diagram of the developed software. There are five primary objects. They are: (i) user, (ii) host computer, (iii) Graphical User interface (GUI), (iv) Matlab workspace, and (v) the floor plan database. The user initiates the interaction between the software and the outside world by selecting a query floor plan image from the host computer. Through the GUI the query image is submitted to the Matlab workspace. All the necessary modules are developed in Matlab as .m files. The feature extraction, segmentation, matching and rank ordered list is done in the Matlab workspace. Based on the rank ordered list, floor plans are retrieved from the database and shown to the user. The software also has a functionality to allow the user to choose the number of floor plan samples to be displayed. Depending upon the users' choice (the value of k), the appropriate number of floor plans are shown to the user. In the next section, screen-shots of the key components of the software and their description is presented.

B.3 SCREEN-SHOTS OF THE FLOOR PLAN RETRIEVAL FRAMEWORK

Initially, when the user launches the application he/she comes across the basic screen prompting the user to select a floor plan from the repository to query into the retrieval framework as shown in Fig. B.3. As shown in Fig. B.3, there are three place holders to host the three different type of images. It can be observed that the "Search" button at the bottom of the interface is disabled at the beginning to ensure that the search is not made on null objects.



图B.1：用于楼层平面图检索的已开发软件架构。

B.2 软件功能

图B.2展示了所开发软件的时序图。系统包含五个主要对象，分别为：(i) 用户、(ii) 主计算机、(iii) 图形用户界面（GUI）、(iv) MATLAB 工作区，以及 (v) 平面布局数据库。用户通过从主计算机中选择一张查询平面图图像，启动软件与外部世界的交互。经由图形用户界面，该查询图像被提交至 MATLAB 工作区。所有必要的模块均以 .m 文件的形式在 MATLAB 中实现。特征提取、分割、匹配及排序后的结果列表均在 MATLAB 工作区内完成。根据排序后的结果列表，系统从数据库中检索相应的平面布局并呈现给用户。此外，该软件还具备一项功能，允许用户选择显示的平面图样本数量。根据用户的设定值 k，系统将相应数量的平面图展示给用户。下一节将给出软件关键组件的屏幕截图及其说明。

B.3 平面图检索框架的屏幕截图

最初，当用户启动应用程序时，会进入主界面，系统提示用户从资源库中选择一张楼层平面图，以供检索框架进行查询，如图B.3所示。如图B.3所示，界面上设有三个占位区域，用于显示三种不同类型的图像。值得注意的是，界面底部的“搜索”按钮在初始状态下处于禁用状态，以确保不会对空对象执行搜索操作。

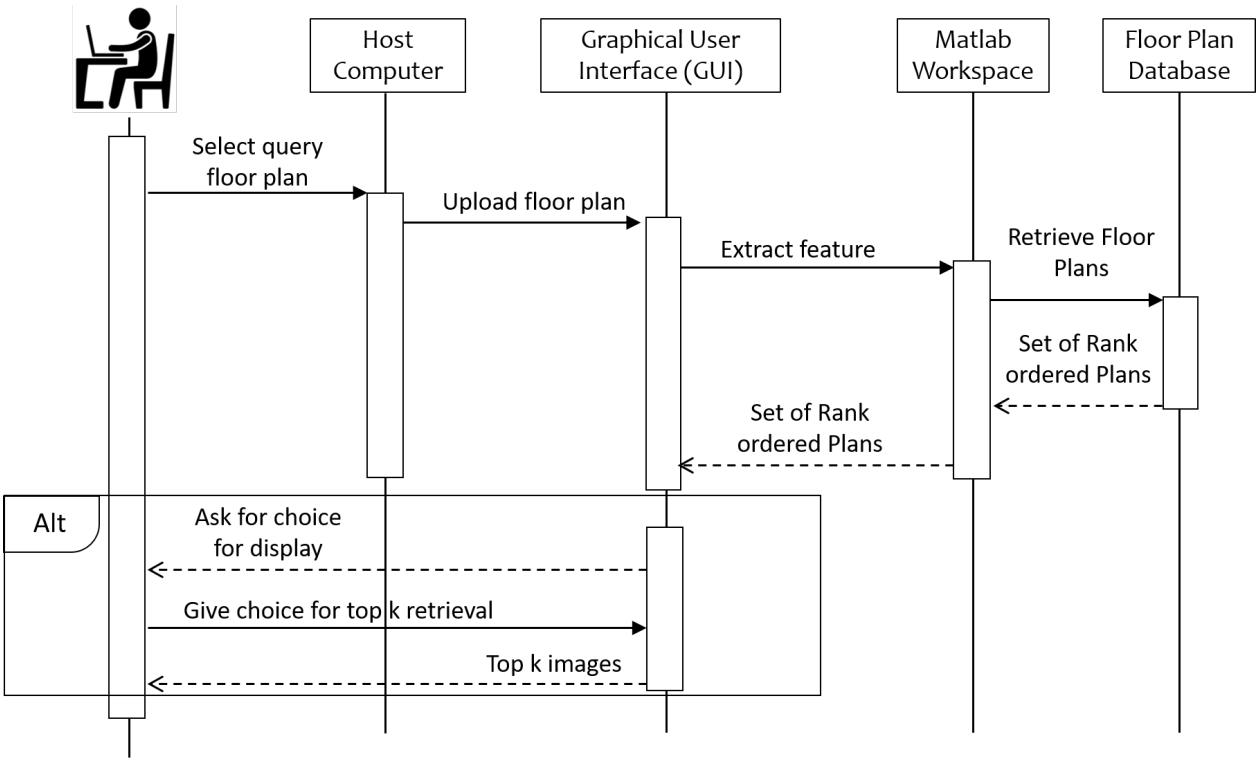


Figure B.2. : Sequence diagram depicting the overall flow of the software.

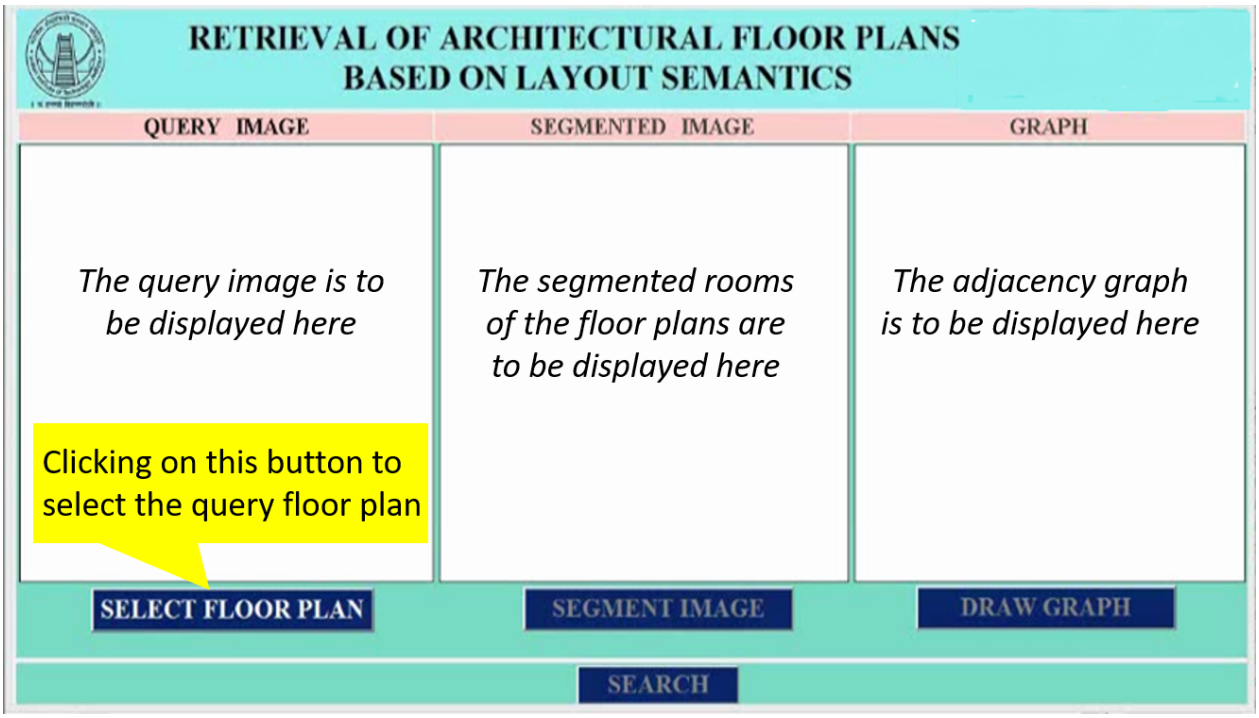
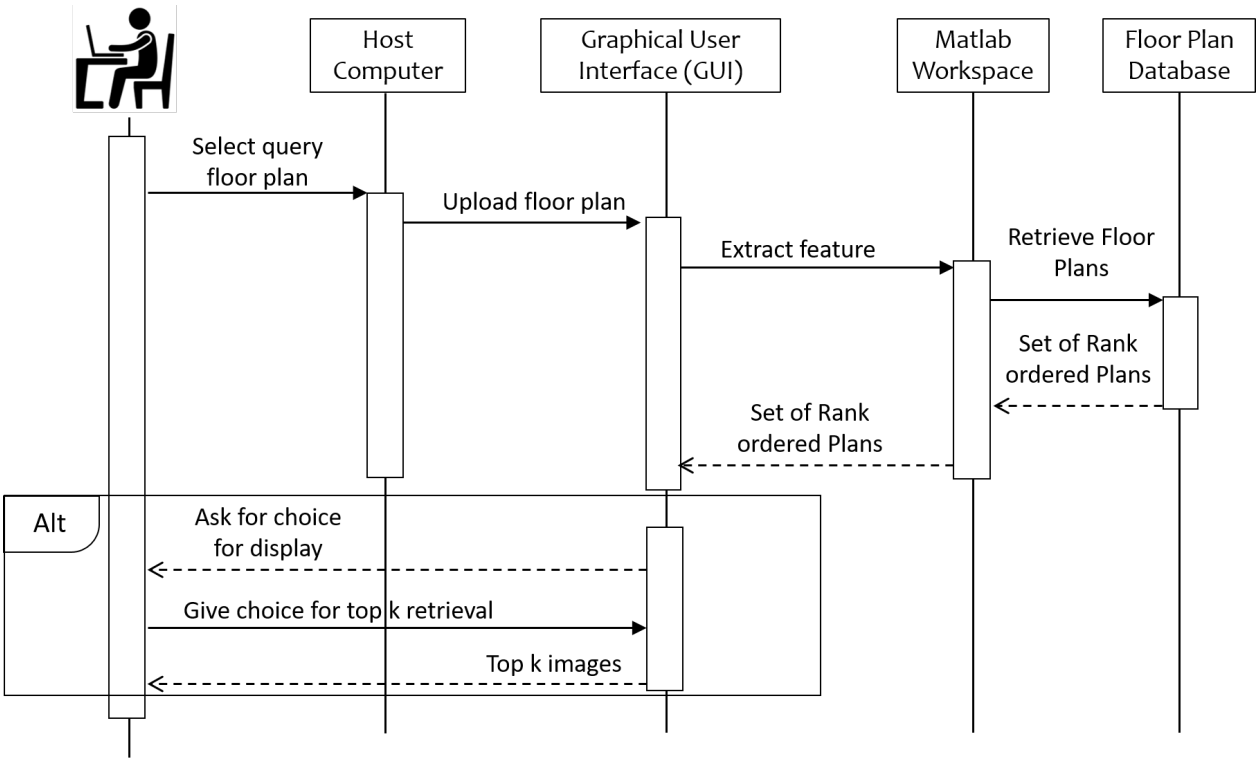
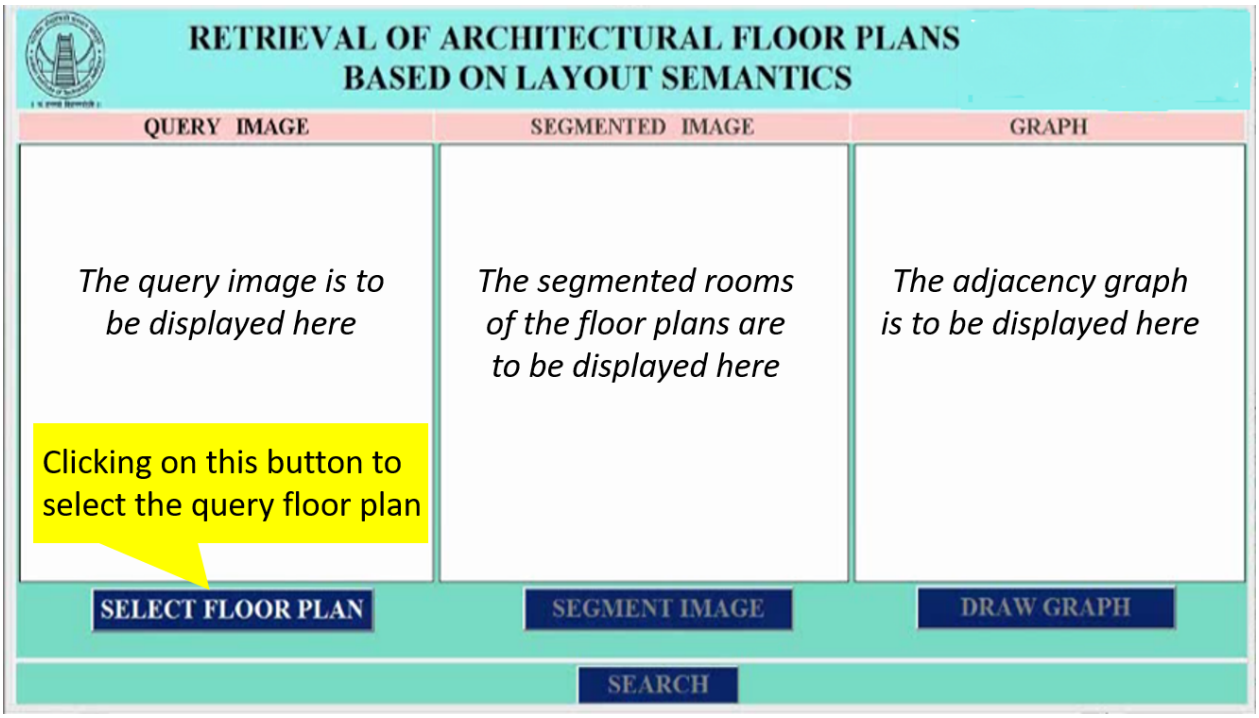


Figure B.3. : Screen prompting the user to select a floor plan of his/her choice.



图B.2：描述软件整体流程的时序图。



图B.3：屏幕提示用户选择其所需的楼层平面图。

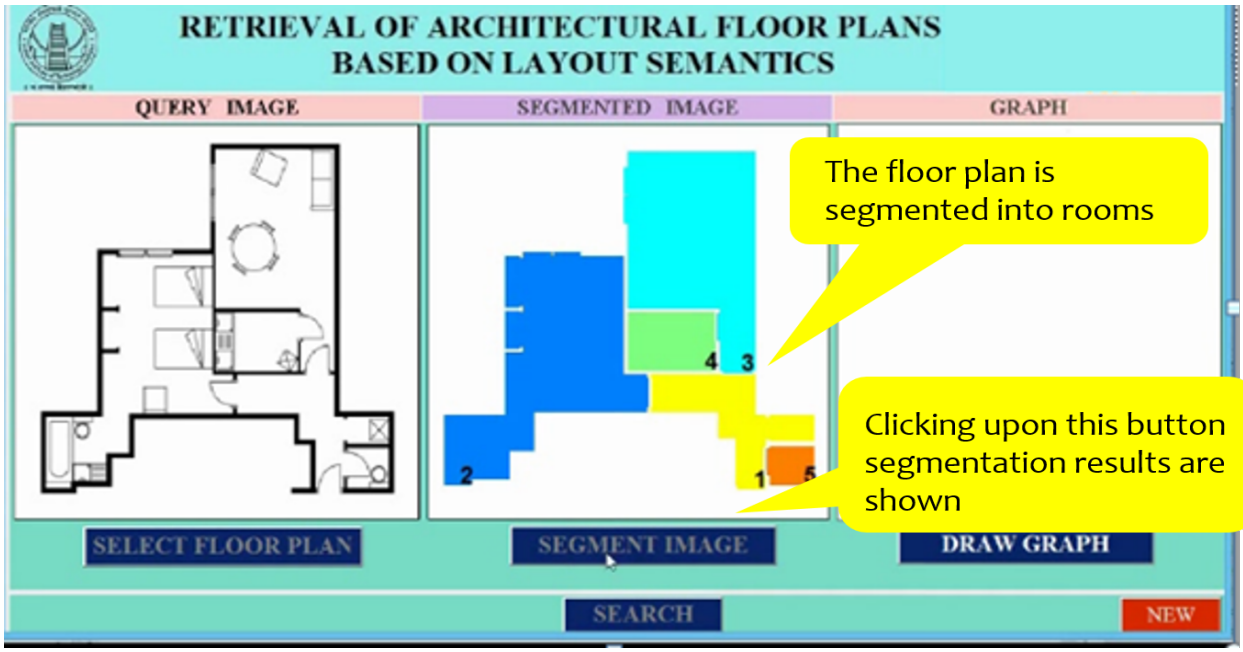


Figure B.4. : Segmentation of the selected plan into labeled rooms

Once the floor plan is chosen and shown in the appropriate place holder the user has the option to choose one of the two modes, i.e. direct or detailed mode. If the user chooses the detailed mode then the intermediate results will be shown in the appropriate place holders. Under the detailed mode, the next task is to segment the floor plan into meaningful entities like rooms and separating it from the decor elements present inside the rooms to further derive the relations between them. Figure B.4 depicts such a scenario, where the intermediate result is shown. Since

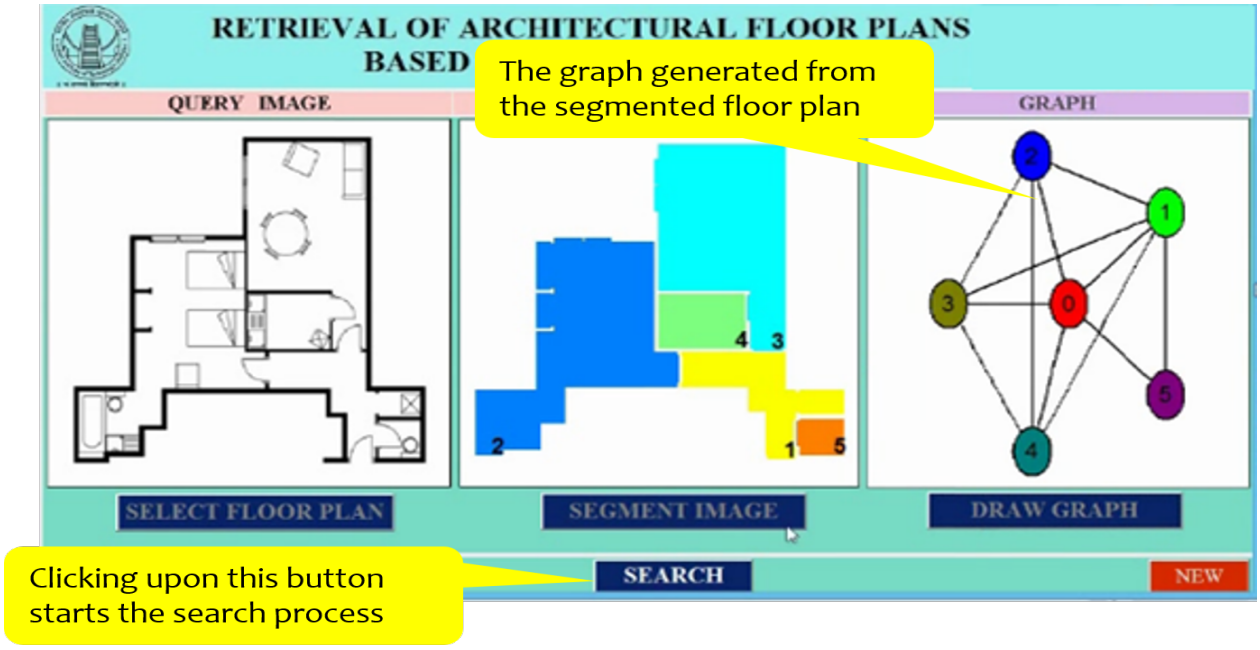
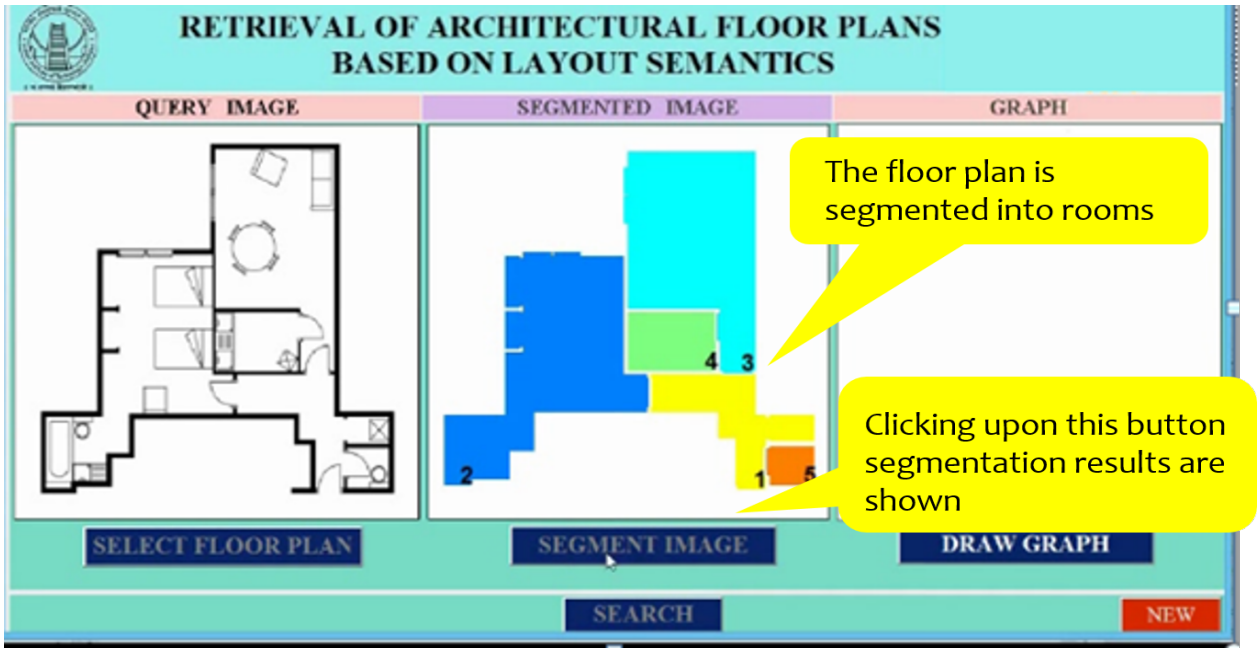
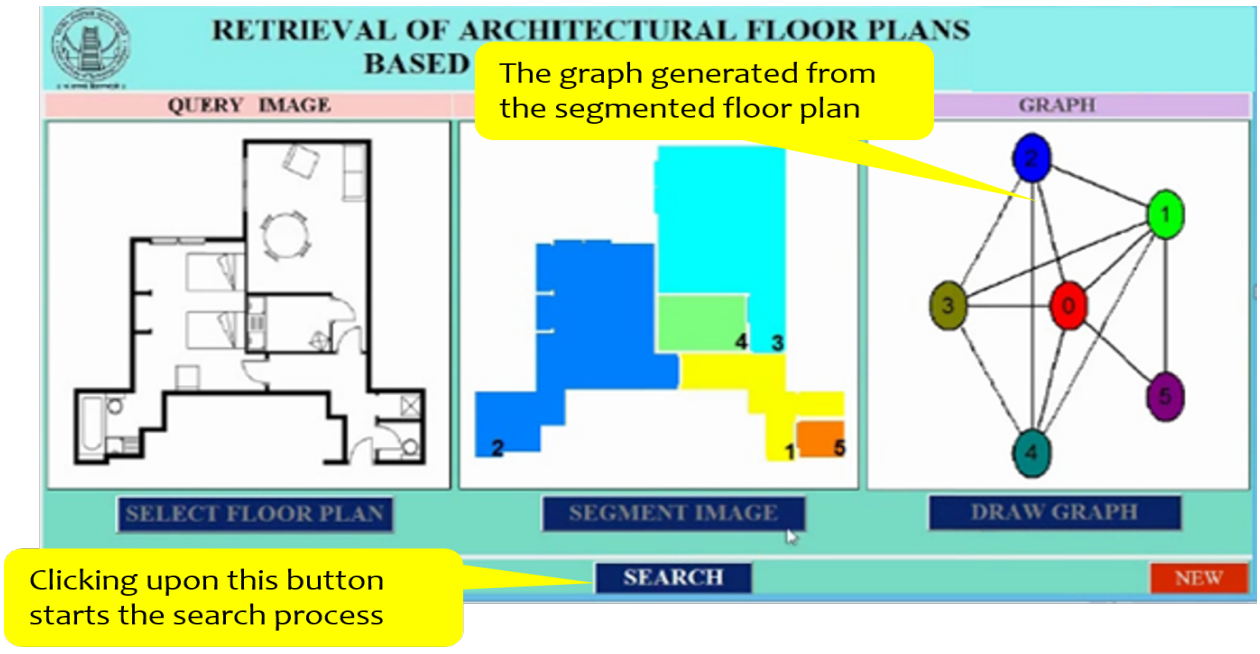


Figure B.5. : Graph embedding of the room level structure of the floor plan.



图B.4：将所选平面划分为已标注的房间

一旦选定平面图并将其显示在相应的占位符中，用户即可选择两种模式之一：直接模式或详细模式。若用户选择详细模式，则中间结果将显示在相应的占位符中。在详细模式下，下一步任务是将平面图分割为具有语义意义的实体（如房间），并将其与房间内的装饰元素相分离，以进一步推导它们之间的关系。图B.4展示了此类场景，其中显示了中间结果。由于



图B.5：楼层平面图的房间层级结构的图嵌入表示。

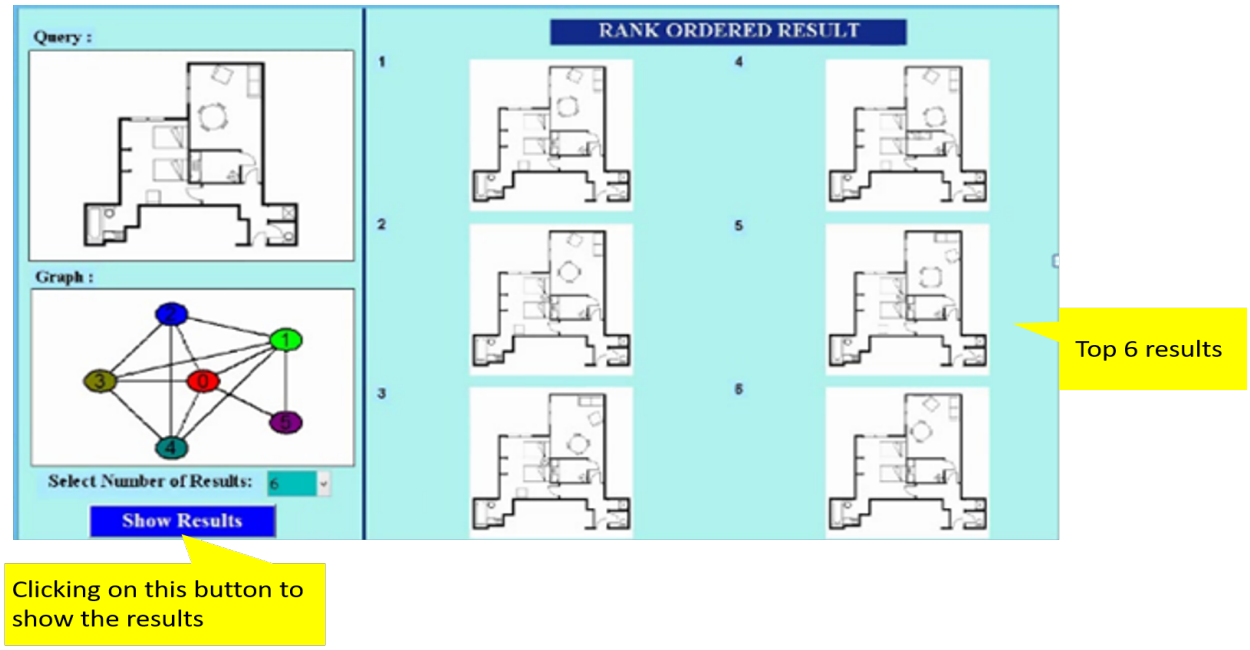


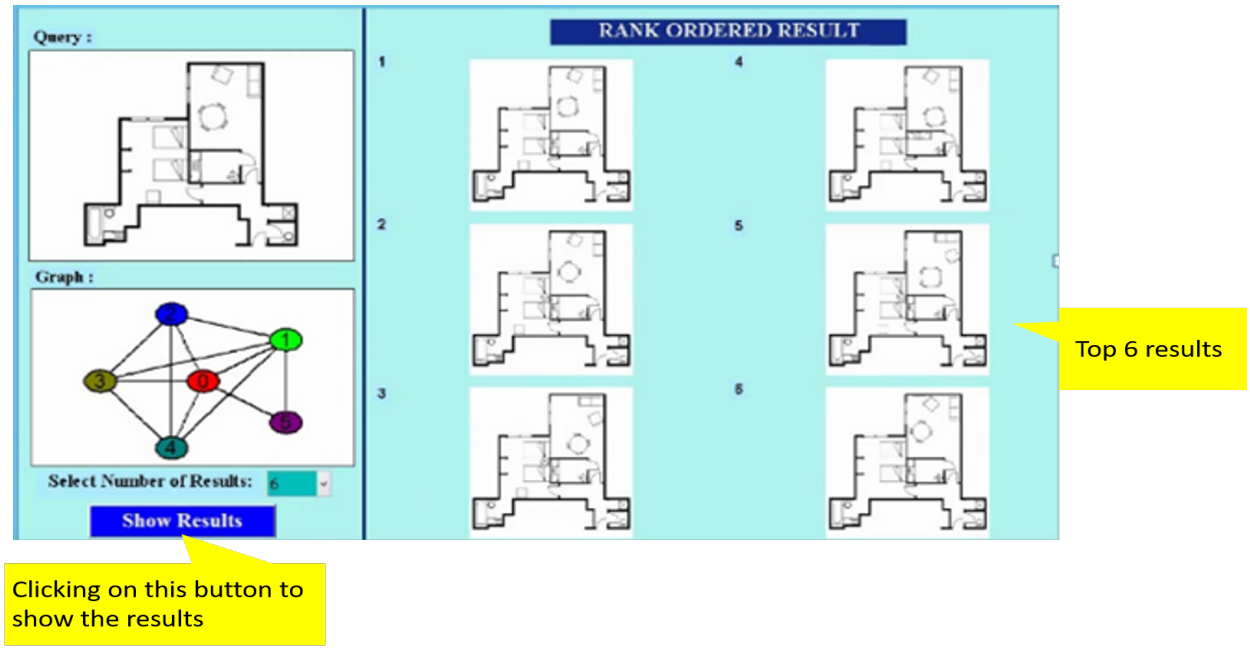
Figure B.6. : Selecting how many results to be displayed while retrieval.

the user has chosen detailed mode, the "Search" button is still disabled as one more step, i.e. the graph creation process is pending.

The next important user experience is shown in Fig. B.5. After the segmented floor plan is obtained with numbered labeling on each room, the next task is to obtain a relationship between these rooms. Therefore, adjacency is used as a parameter to define relationships between them. A graph is created where nodes represent rooms and edges represent which rooms are adjacent to each other. As shown in Fig. B.5, the "Search" button is now activated as the software is ready to do the matching between the query feature and the database features to show the retrieved result.

After the required graph is obtained, spectral analysis of the adjacency matrix is done in the back-end for the query floor plan as well as the database floor plans. The floor plans with similar adjacency are clustered together in the spectral space and thus, the GUI redirects to the second page for retrieval of similar floor plans as shown in Fig. B.6. As shown in Fig. B.6, there are three place holders in the interface. On the left hand side top the query floor plan is shown, while the corresponding graph is shown at the bottom. Just below the graph, there is a drop down list using which the user can choose the value of the number of floor plans to show as result. Upon the selection of number of results to be displayed, the rank ordered retrieval results are displayed as shown in Fig. B.7.

Figure B.7 depicts the final result for top 6 results. As seen in the results the retrieved results are numbered as per their position in the rank ordered list. After the results are displayed, the user can always change the number of results to be displayed. The user needs to choose an appropriate number from the drop-down list and then click the "'Show Results" button. The number of floor plans shown as results are updated accordingly.



图B.6：在检索时选择显示的结果数量。

当用户已选择详细模式时，“搜索”按钮仍处于禁用状态，因为图谱生成过程尚未完成。

下一个重要的用户体验如图B.5所示。在对平面图进行分割并为每个房间赋予编号标签后，下一步任务是确定这些房间之间的关系。因此，采用邻接关系作为参数来定义它们之间的关联。构建一个图结构，其中节点代表各个房间，边则表示哪些房间彼此相邻。如图B.5所示，“搜索”按钮现已激活，表明软件已准备好在查询特征与数据库特征之间进行匹配，以展示检索结果。

在获得所需图结构后，后端会对查询楼层平面图及数据库中的各楼层平面图的邻接矩阵进行谱分析。在谱空间中，具有相似邻接关系的楼层平面图会被聚类在一起，因此图形用户界面会跳转至第二页，用于检索相似的楼层平面图，如图B.6所示。如图B.6所示，界面上设有三个占位区域：左侧上方显示查询楼层平面图，下方则显示相应的图结构；紧邻图结构下方设有一个下拉列表，用户可通过该列表选择要显示的结果数量。选定显示结果的数量后，按排序顺序排列的检索结果将如图B.7所示呈现。

图B.7展示了前6条检索结果的最终呈现效果。由结果可见，检索结果按其在排序列表中的位置进行编号。结果显示后，用户可随时调整待显示的结果数量。用户需从下拉列表中选择一个合适的数值，然后单击“显示结果”按钮，此时作为结果展示的平面图数量将相应更新。

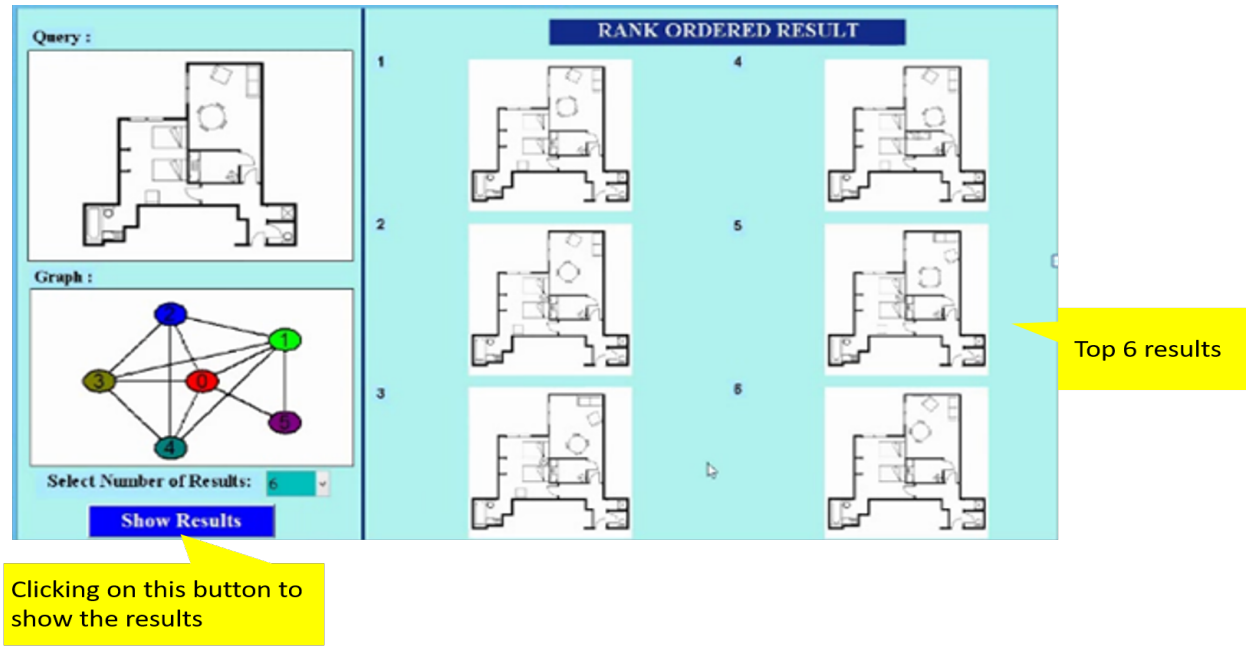
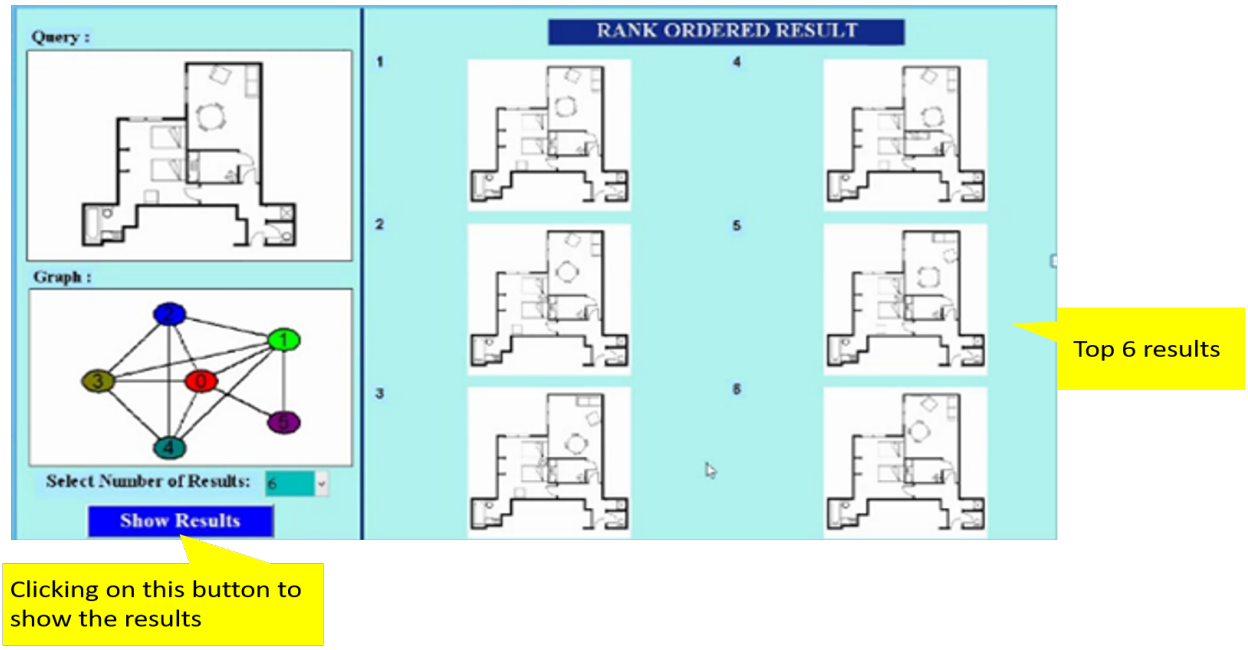


Figure B.7. : User's required floor plans on display as per their similarity from the query floor plan.

B.4 CONCLUSION

This is an initial version of a platform developed for content based floor plan retrieval. At present, the software runs on system with Matlab installed in it or a system that has a Matlab plugin installed. An extension could be to have a mobile version of the same that can be used easily without any specific software prerequisites.



图B.7：根据与查询平面图的相似度，显示用户所需的各楼层平面图。

B.4 结论

这是一个用于基于内容的平面布局检索平台的初始版本。目前，该软件可在已安装Matlab的系统或已集成Matlab插件的系统上运行。未来可进一步开发其移动版本，使其无需特定软件即可便捷使用。

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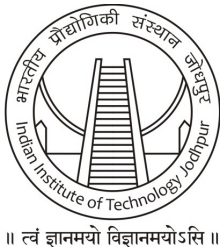
Content Based Analysis and Retrieval of Architectural Floor plans

A Thesis submitted by
Divya Sharma

in partial fulfillment of the requirements for the award of the degree of
Doctor of Philosophy

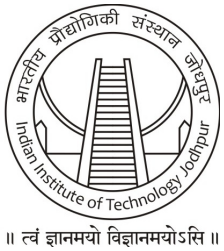


Indian Institute of Technology Jodhpur
Department of Computer Science and Engineering
August 2018

基于内容的建筑平面图分析与检索

迪维娅·夏尔马
提交的学位论文

为部分满足哲学博士学位授予要求



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August 2018

7
Conclusions

Efficient retrieval of similar floor plan images is a critical component of early design phase of construction as it is the primary reason of delay in the design/ conceptualization phase. Traditional CBIR approach is a straight forward and intuitive solution to this problem. The individual techniques proposed in literature, provide an incomplete answer to the question of representing a floor plan in a particular way to facilitate efficient search. Most of the proposed approaches in the literature are tailored for natural images, and some of them for text based document images. However, none of the known results are sufficient to tackle the problem of a unified framework that incorporates multiple modalities of query (sketch and image) for floor plan retrieval. In this chapter the contributions presented during each approach are summarized and the conclusions in the form of strengths and weaknesses of each technique are discussed. Finally, several working lines to be considered in the future are discussed.

In this thesis, it is shown that both the structural and semantic features are important for retrieval of similar floor plan images. It is also argued that hand crafted features are performing better that the learned features (using convolutional neural networks) in case of fine grained retrieval. However, for global similarity (i.e. similarity based on the overall shape of the floor plan), deep features are superior. This thesis also highlights that when sketch is used as a query mode, the traditional techniques (including deep features) fail to scale to this domain shift. Additional processing components need to be incorporated in the framework to tackle such a scenario.

The proposed system can also find application in scenarios where buyers are looking for specific features in house designs, while presenting their requirements to the architect. It can also be deployed in online real estate platforms, where the user can be directed to conveniently query their required floor plans using query as image or sketch and retrieve their required homes from the repository. The novelty of the contribution is that floor plans have been in the past used as graphical documents for analysis and symbol spotting purposes but retrieval based on significant content derived from a floor plan has not been researched. Extensive experimental evaluation on real-world datasets shows that combination of structural and semantic approaches for analysis of floor plans and also employing a combination of traditional and deep learning approaches for feature extraction add value to the overall retrieval accuracy. Both qualitative and quantitative comparisons were made with recent techniques for feature extraction to exhibit the superior performances of the proposed frameworks.

7.1 CONTRIBUTIONS

The key contributions of this thesis are:

- In Chapter 3, a novel approach to perform floor plan retrieval incorporating both symbol spotting and room level analysis is proposed by taking query by example as an image. Inclusion of the room decor arrangement during retrieval is a novel idea that can find application

7
Conclusions

高效检索相似的平面图是建筑施工早期设计阶段的关键环节，因为它是导致设计/概念化阶段延误的主要原因。传统的基于内容的图像检索（CBIR）方法为此问题提供了一种直观且直接的解决方案。然而，文献中提出的各项具体技术仅能部分回答如何以特定方式表示平面图以促进高效检索这一问题。大多数现有方法针对自然图像设计，少数则适用于基于文本的文档图像。然而，迄今为止尚无任何研究成果能够充分解决构建一个统一框架的问题，该框架需同时支持多种查询模态（草图与图像）以实现平面图检索。在本章中，我们将对各方法所作出的贡献进行总结，并以优缺点的形式讨论每种技术的特点。最后，还将探讨未来可进一步开展的研究方向。

本论文表明，结构特征和语义特征对于相似楼层平面图的检索均具有重要意义。同时指出，在细粒度检索任务中，手工设计的特征相较于基于卷积神经网络学习得到的特征表现更优。然而，对于全局相似性（即基于楼层平面整体形状的相似性），深度特征则更具优势。此外，本论文还强调，当以草图作为查询模式时，传统方法（包括基于深度特征的方法）难以适应这一领域迁移问题，因此需要在框架中引入额外的处理模块以应对此类场景。

所提出的系统还可应用于以下场景：购房者在向建筑师提出需求时，希望寻找具备特定功能特征的住宅设计方案。此外，该系统也可部署于在线房地产平台，用户可通过图像或草图进行查询，从而便捷地从数据库中检索出符合其需求的户型方案。本研究的创新之处在于：以往户型图主要作为用于分析与符号识别的图形文档，而基于从户型图中提取的关键内容进行检索的研究尚属空白。在真实数据集上开展的大量实验评估表明，将结构化方法与语义方法相结合以分析户型图，并同时采用传统方法与深度学习方法进行特征提取，能够有效提升整体检索精度。通过与现有特征提取技术的定性和定量对比，进一步验证了所提框架的优越性能。

7.1 贡献

本论文的主要贡献如下：

- 在第3章中，提出了一种新颖的方法，通过将示例查询作为图像，结合符号识别与房间级分析来实现平面图检索。在检索过程中纳入房间的装饰布局是一项创新性思路，具有潜在的应用价值。

in scenarios where the user has specific choices in terms of the fully furnished floor plans. Furthermore, a spectral embedding approach to represent graphs obtained from the layout is proposed in the baseline retrieval framework proposed in Chapter 3. This technique helped in reducing the computation time taken during the graph matching stage to a considerable level.

- In Chapter 4, a deep learning framework is proposed to extract semantic features from floor plans and those features are used for matching and retrieval of similar floor plan images. Novelty lies in using convolutional neural networks for the task of feature extraction in floor plans which prove to increase the retrieval performance as compared to hand-crafted feature extraction strategies.

- In Chapter 5, a novel framework for fine grained retrieval of floor plans using features like area, number, type of furniture and also adjacencies between the rooms is proposed. Inclusion of the high level features during retrieval is a novel idea that can find application in scenarios where the user has specific choices in terms of the fully furnished floor plans. Also, weighted feature fusion is proposed which helps in weighing features when a user is querying and the user is fixated over certain features to have a mandatory presence while retrieval. This framework has an edge over the deep learning framework in a respect that priority to certain features can be given while retrieval which in not otherwise possible in a deep learning network where globally the features are extracted.

- In Chapter 6, the introduction of sketch as a query mode while retrieval is proposed. Sketch based query mode is very intuitive because a user can conveniently draw his/her requirements on an interface as if they are drawing with pen and paper. Sketch based querying helps in capturing the content better. Two deep learning approaches, one with the usage of Cyclic GANs and the other with the usage of autoencoders are applied for domain adaptation purposes aiding to sketch to image retrieval. The performance of both the systems were obtained to be impressive while retrieval on benchmark datasets.

- Two datasets namely, ROBIN with floor plan images and S-ROBIN with sketched floor plans have also been proposed to benefit the research community to carry out research in the area of floor plan analysis and retrieval. These two datasets help in overcoming the drawbacks that the existing publicly available datasets had in terms of less variation in samples as well as inadequate number of samples to carry out deep learning and retrieval tasks.

7.2 DISCUSSION ON THE RELATIVE PERFORMANCE OF THE TECHNIQUES PROPOSED

In this section we discuss the comparative quantitative performance of the techniques mentioned in Chapters 3, 4, 5 and 6 of the thesis.

In Chapter 3, handcrafted topological features were proposed to be extracted from floor plan images. The Mean Average Precision (MAP) value obtained during retrieval on the ROBIN dataset was observed to be 0.247 in case only room layout was considered while retrieval, on the other hand incorporating both room layout and room decor features made retrieval quite specific. This was due to the fact that the layout similarity in a category of the ROBIN dataset was higher than the combined layout and decor similarity. Thus, the system gave a MAP value of 0.232 in this case. We observed that although, topology and room decor features help in distinguishing two floor plans and retrieving similar ones, but inclusively are not a good enough measure to analyse a floor plan completely. It is very difficult to curate handcrafted features powerful enough to extract

在用户对全装修平面布局方案具有特定选择权的情境下，此外，在第3章提出的基线检索框架中，还提出了一种基于谱嵌入的图表示方法。该技术显著降低了图匹配阶段的计算时间。

- 在第4章中，提出了一种深度学习框架，用于从平面图中提取语义特征，并将这些特征应用于相似平面图图像的匹配与检索。其创新之处在于，首次将卷积神经网络应用于平面图的特征提取任务，实验结果表明，该方法相较于手工设计的特征提取策略能够显著提升检索性能。

- 在第5章中，提出了一种新颖的框架，用于基于面积、家具数量与类型以及房间之间的邻接关系等特征实现建筑平面图的细粒度检索。在检索过程中引入高层语义特征是一种创新思路，可在用户对全配式建筑平面图有特定偏好时得到应用。此外，还提出了加权特征融合方法，该方法允许用户在查询时对各特征进行权重分配，并可指定某些特征在检索结果中必须出现。与深度学习框架相比，该框架的优势在于能够在检索过程中对特定特征赋予优先级，而深度学习网络通常仅进行全局特征提取，无法实现这一功能。

- 在第6章中，提出了一种以草图作为查询模式的检索方法。基于草图的查询模式具有极强的直观性，用户可以在界面上像使用纸笔作画一样便捷地绘制其需求，从而更有效地捕捉内容特征。为支持从草图到图像的检索任务，本文分别采用了两种深度学习方法：一种基于循环生成对抗网络（Cyclic GANs），另一种基于自编码器，用于实现领域适应。在基准数据集上的实验结果表明，这两种系统的检索性能均十分可观。

- 此外，还提出了两个数据集——包含平面图图像的ROBIN数据集和包含手绘平面图的S-ROBIN数据集，以促进学术界在平面图分析与检索领域的研究。这两个数据集有助于克服现有公开数据集在样本多样性不足以及样本数量不足以支持深度学习与检索任务等方面的局限性。

7.2 关于所提出技术相对性能的讨论

在本节中，我们讨论了论文第3章至第6章所提及各技术的定量性能对比。

在第3章中，我们提出从平面布局图中提取手工设计的拓扑特征。实验结果表明，在仅考虑房间布局的情况下，该方法在ROBIN数据集上的平均精度均值（MAP）为0.247；而当同时引入房间布局与室内装饰特征时，检索结果的针对性显著提高。这一现象的原因在于，ROBIN数据集中同一类别内的布局相似度高于布局与装饰特征的综合相似度，因此在后一种情况下，系统的MAP值降至0.232。我们观察到，尽管拓扑特征与室内装饰特征有助于区分不同的平面布局并实现相似布局的检索，但二者综合起来仍不足以对平面布局进行完整分析。要构建能够有效提取特征的手工设计特征，其难度极大。

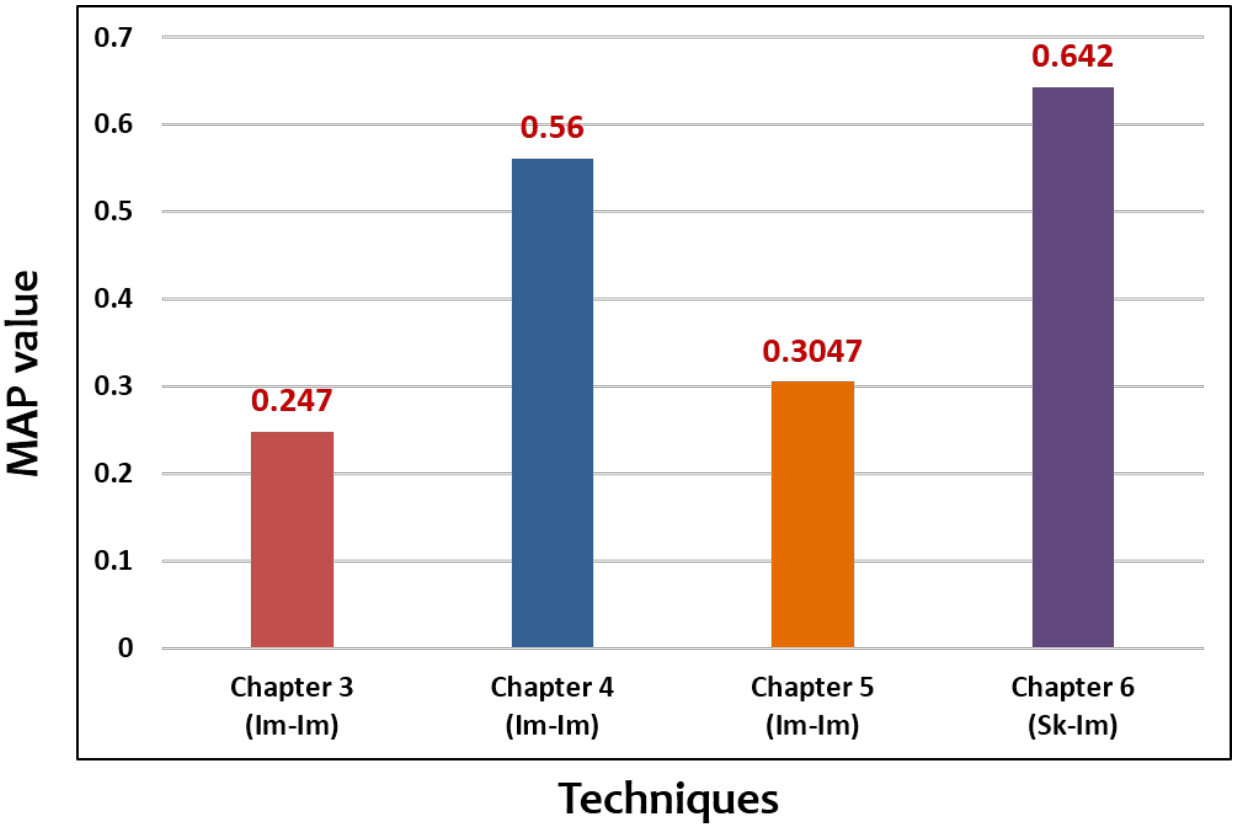


Figure 7.1. : Comparison of MAP values obtained by techniques mentioned in Chapters 3,4,5 and 6.

all the details from a floor plan. Therefore, we explored deep neural networks to extract structural and semantic information from a floor plan.

In Chapter 4, a deep learning framework using CNNs is proposed to extract semantic features from floor plan images and those features are used for matching and retrieval of similar floor plan images. As compared to Chapter 3, the deep learning framework, was able to enhance the retrieval performance considerably. We observed a MAP value of 0.56 on the ROBIN dataset. However, we observed that deep learning is convenient and bypasses the need of curating features manually, the disadvantage is that the layers in the deep learning framework extract features implicitly. This implicit feature extraction at each layer does not give the user the ability to give priorities to certain aspects or features while retrieval.

Therefore, in Chapter 5, high level semantic analysis of floor plans and their subsequent retrieval is proposed. Additionally, a provision to adjust the weights of each feature while retrieval is given in the proposed technique. Such a provision can prove to be more practical to users while looking for similar floor plans sufficing their specific preferences. We observed a MAP value of 0.3047 on the ROBIN dataset using this technique. Although, the MAP value obtained here is less than what was obtained using the deep learning framework for image retrieval, the advantages however are that, weighted feature fusion helps in emphasizing on certain features when a user is querying and has specific feature preferences while retrieval.

In Chapter 6, the querying mode changes and sketch based querying is proposed. Such a query makes the system more convenient for searching from the user's end. The conjunction of

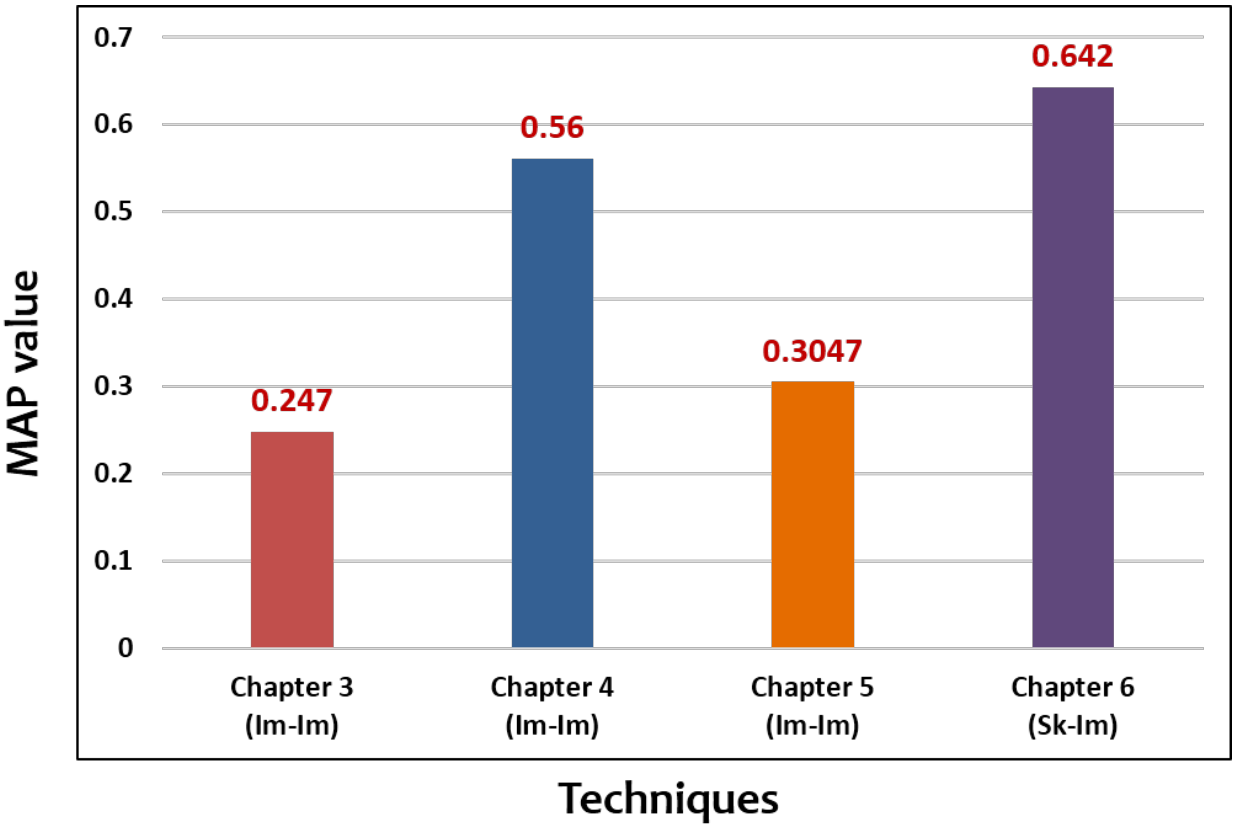


图7.1：比较第3章至第6章中所述各技术所获得的MAP值。

从平面图中获取所有细节。因此，我们探索了利用深度神经网络从平面图中提取结构信息和语义信息的方法。

在第4章中，提出了一种基于卷积神经网络（CNN）的深度学习框架，用于从平面图图像中提取语义特征，并利用这些特征实现相似平面图的匹配与检索。与第3章相比，该深度学习框架显著提升了检索性能，在ROBIN数据集上获得了0.56的平均精度均值（MAP）。然而，尽管深度学习具有便捷性并免去了人工特征工程的繁琐，其缺点在于：深度学习框架中的各层以隐式方式提取特征，这种逐层的隐式特征提取使得用户在检索时无法对特定方面或特征赋予优先级。

因此，在第5章中，提出了对平面图进行高层次语义分析并实现其检索的方法。此外，所提出的算法还提供了在检索过程中调整各特征权重的机制。这一机制在用户根据自身特定偏好搜索相似平面图时更具实用性。采用该方法在ROBIN数据集上测得的平均精度均值（MAP）为0.3047。尽管该MAP值低于使用深度学习框架进行图像检索时所获得的结果，但其优势在于：加权特征融合能够在用户进行查询且对某些特征有明确偏好时，突出相应特征的重要性。

在第6章中，查询模式发生了变化，并提出了基于草图的查询方法。这种查询方式使系统在用户端的检索操作更加便捷。此外，

CNN, Cyclic GAN and autoencoders was used in this particular framework for domain adaptation. The MAP value obtained in this case was impressive at a value of 0.642 for ROBIN dataset.

Figure 7.1 presents a bar graph comparing the MAP values obtained by each technique. Im-Im represents, image to image retrieval and Sk-Im represents, sketch to image retrieval as proposed in Chapter 6.

7.3 SCOPE OF FUTURE WORK

The work presented in this thesis, definitely opens the scope of a chain of tasks in many diverse applications, for researchers to solve in future. Some of them are given below.

- In the future, content based retrieval taking a hybrid scheme of querying can be researched. Here, the idea can be that a user can draw/scribble over a floor plan image and in real-time the query modifies according to the addition or removal of the components done by the user in the floor plan image. Such a system can be more interactive to use. Such multimodal inputs would make the process of query representation very convenient to the end-user, and would provide a rich set of specifications to further accurately retrieve similar floor plan samples.

- Future research should concentrate on capturing of better and usable content from floor plans to help in real world scenarios. Researchers may explore on deducing high-level features, which for example can be in the form of accessibility of each room, in terms of shortest path of a room to other rooms or to the exit in a floor plan. This scenario can find application in maintaining safety situations in big architectural buildings and retrieving floor plans abiding by the safety standards set by a regulatory body.

- The main issue with floor plan analysis research is the dearth of publicly available datasets as architects do not prefer to share their designs in the open. Therefore, a good future scope can be to develop a larger dedicated repository of real-world floor plans to foster research in the area of analysis and retrieval in floor plans.

- Acquiring the floor plan images may not always be in digital form. The alternate mode could be through high-resolution scanning. However, scanned documents at times are corrupted with transformations like shear, rotation, etc. Also, during digitization some noise might get added or existing noise in the document might get enhanced. Retrieval of floor plan images under such an uncontrolled situation forms a challenging research problem.

- In Chapter 6, Fig. 6.7, we already propose, a unified multimodal system which can classify query into sketch or image and accordingly perform image to image or sketch to image retrieval. Future research can also explore the combination of feature extraction techniques such as, hand-crafted techniques and deep learning techniques for retrieval. For example, in chapter 4, as we observe that deep learning fails to explicitly give preference to a feature while retrieval which was possible through the technique mentioned in chapter 5. Therefore, a suggested modification can be exploring an approach of incorporating feature importance into neural network learning. The weights connected to an important feature can be biased to have an absolute value larger than the weights connected to the other features.

在该框架中，分别采用了卷积神经网络（CNN）、循环生成对抗网络（Cyclic GAN）以及自编码器进行领域自适应。在此情形下，针对ROBIN数据集所获得的平均精度均值（MAP）达到了令人瞩目的0.642。

图7.1为柱状图，比较了各方法所获得的平均精度均值（MAP）值。其中，“Im-Im”表示图像到图像的检索，“Sk-Im”表示如第6章所提出的草图到图像的检索。

7.3 未来工作的范围

本论文所开展的工作无疑为研究人员在未来解决诸多不同应用领域中的系列任务开辟了新的研究空间，其中部分任务列举如下。

- 未来，可以研究一种基于内容的混合查询检索方案。其核心思想是：用户可在平面图图像上进行绘制或涂鸦，系统能够实时根据用户在该图像中对构件的添加或删除操作动态调整查询条件。此类系统将具有更高的交互性。多模态输入方式将极大简化最终用户的查询表达过程，并提供丰富的规格描述，从而更准确地检索出相似的平面图样本。

- 未来的研究应着重于从平面图中提取更优质、更具实用价值的信息，以支持实际应用场景。研究人员可进一步探索高层特征的推断方法，例如，通过计算某一房间到其他房间或到出口的最短路径来表征各房间的可达性。此类方法在大型建筑的安全管理以及符合监管机构制定的安全标准的平面图检索等方面具有潜在的应用价值。

- 平面图分析研究的主要问题在于公开可用的数据集匮乏，因为建筑师通常不愿将其设计成果对外共享。因此，未来的一个重要方向是构建一个规模更大的、专门用于平面图分析与检索研究的真实世界平面图数据库。

- 获取平面图图像时，其形式未必始终为数字化。另一种方式是通过高分辨率扫描。然而，扫描得到的文档有时会因剪切、旋转等变换而出现损坏；此外，在数字化过程中还可能引入噪声，或使文档中原有的噪声被放大。在如此不受控的情况下进行平面图图像的检索，构成了一个极具挑战性的研究问题。

- 在第6章的图6.7中，我们已经提出了一种统一的多模态系统，该系统能够将查询分类为草图或图像，并据此分别执行图像到图像检索或草图到图像检索。未来的研究还可以探索将手工特征提取技术与深度学习技术相结合用于检索。例如，在第4章中，我们观察到深度学习在检索过程中无法显式地对某一特征给予优先考虑，而这种优先性通过第5章所提及的技术是可以实现的。因此，一种可行的改进方案是探索一种将特征重要性融入神经网络学习的方法。与重要特征相连的权重可以被赋予较大的绝对值，以区别于与其他特征相连的权重。

Annexure A
Floor plan datasets ROBIN and S-ROBIN

Dataset plays a pivotal role in any image processing problem. Availability of a public dataset sometimes makes the life of a researcher easy. However, this is true for only generic problems, and not so for problems which are of some specific nature. Floor plan image analysis is such a problem where a large collection of samples is hardly available. For this reason two public datasets have already been proposed by researchers. They are: (i) SESYD [Delalandre *et al.*, 2010b] and (ii) CVC-FP [de las Heras *et al.*, 2015]. The former has 10 classes of floor plans, where there are 100 samples/class. The latter has 122 scanned floor plan documents divided into 4 categories based on the origin and style. The SESYD dataset is inappropriate for application to a retrieval task due to very less difference between a pair in a class of floor plan images. Also, in the CVC-FP dataset, the samples are insufficient in number for the task of floor plan retrieval.

To train and evaluate the contributions presented in this thesis, two floor plan databases, named Repository Of BuildIng floor plaNs (ROBIN) and Sketched-Repository Of BuildIng floor plaNs (S-ROBIN) are created. S-ROBIN are the sketch replicas of the ROBIN dataset containing floor plan images. These datasets have significantly aided in the content analysis and retrieval of similar floor plans as proposed in this thesis. The primary motivation while creating the ROBIN dataset was to have: i) sufficient number of sample to execute the experiments, and ii) variation (inter and intra class) among the samples. While searching for a dream house, the customer generally has the follwing sequence of requirements in his/her mind; i) the number of rooms inside the house, ii) the global shape of the house, iii) what is the pattern in which the rooms are connected to each other, and iv) if a user is looking for a well furnished home. then which furniture should be placed in the house and in which arrangement. In the following subsections, each of the points mentioned above are elaborated in detail.

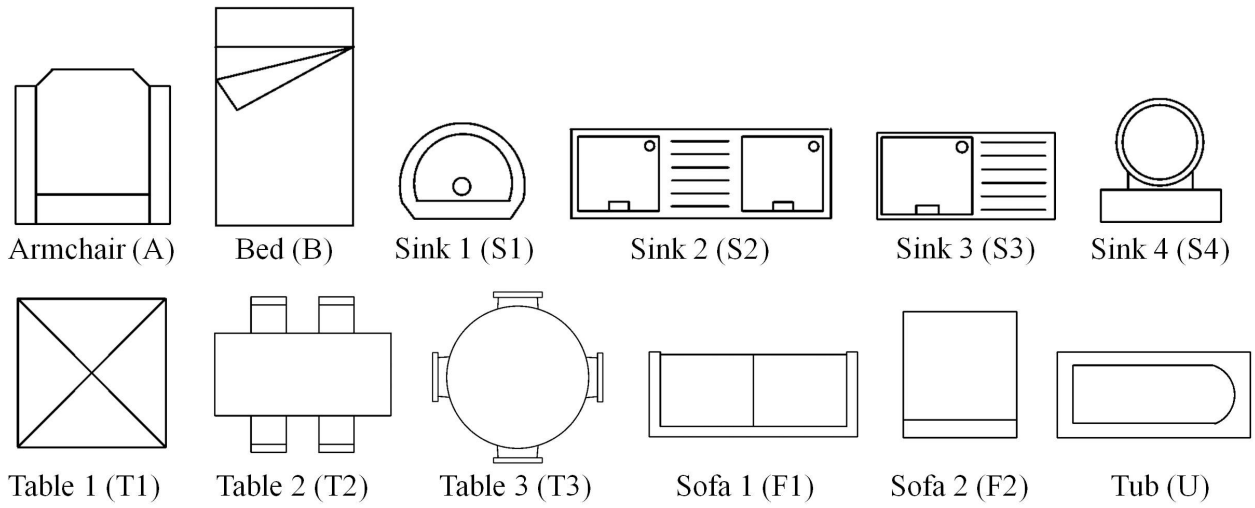


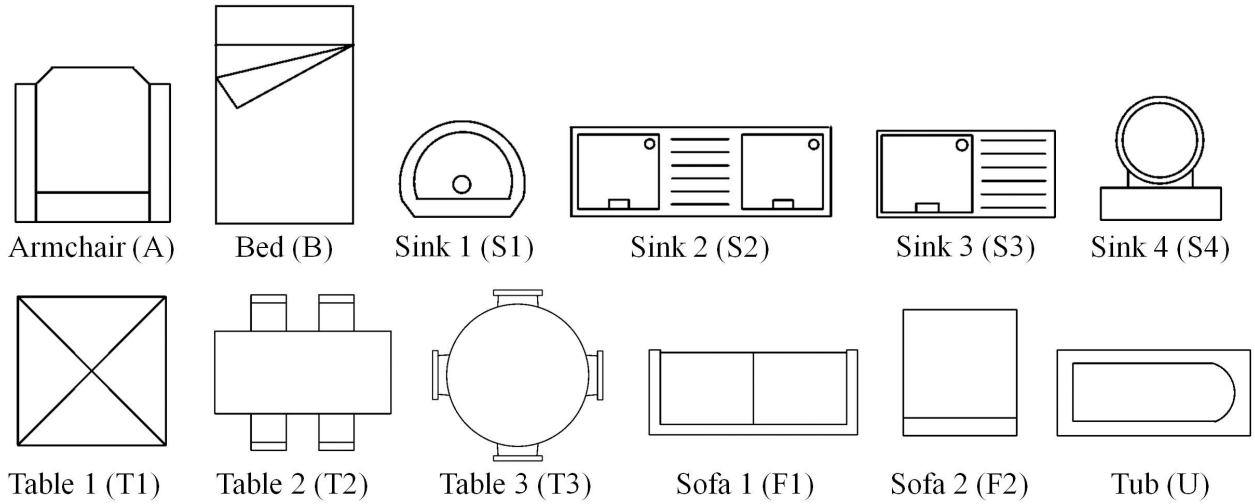
Figure A.1. : Symbol set used as furniture in ROBIN and S-ROBIN dataset.

附件A

Floor plan datasets ROBIN and S-ROBIN

数据集在任何图像处理问题中都起着至关重要的作用。公开数据集的可用性有时能够简化研究人员的工作。然而，这仅适用于通用问题，而对于某些特定类型的问题则不然。楼层平面图图像分析便属于此类问题，其中可供使用的样本数量极为有限。正因如此，研究者已提出了两个公开数据集，分别为：(i) SESYD [Delalandre *et al.*, 2010b] 和 (ii) CVC-FP [de las Heras *et al.*, 2015]。前者包含10类楼层平面图，每类有100个样本；后者则包含122份扫描得到的楼层平面图文档，按来源与风格划分为4个类别。由于同一类别中的楼层平面图样本之间差异甚微，SESYD数据集并不适于检索任务。此外，在CVC-FP数据集中，样本数量亦不足以支持楼层平面图检索任务。

为训练并评估本论文所提出的各项方法，我们构建了两个平面图数据库，分别命名为“建筑平面图资源库”（Repository Of BuildIng floor plaNs，简称ROBIN）和“草图版建筑平面图资源库”（Sketched-Repository Of BuildIng floor plaNs，简称S-ROBIN）。其中，S-ROBIN是ROBIN数据集的草图复制品，包含平面图图像。这些数据集在本论文所提出的相似平面图内容分析与检索任务中发挥了重要作用。创建ROBIN数据集的主要动机在于：i）确保实验所需的样本数量充足；ii）使样本在类间及类内均具有足够的多样性。在寻找理想住宅时，客户通常会在脑海中形成如下需求序列：i）房屋内部的房间数量；ii）房屋的整体形态；iii）各房间之间的连接模式；以及iv）若用户寻求精装修住宅，则还需明确应选用何种家具及其摆放布局。在接下来的小节中，上述各要点将予以详细阐述。



图A.1：在ROBIN和S-ROBIN数据集中用作家具的符号集。

A.1 NUMBER OF ROOMS

In ROBIN there are three categories of floor plans, depicting the three broad classes of houses. There are houses with 3, 4 and 5 rooms. The purpose and the subsequent label of each room varies depending on the various decors present in them. To be consistent with the existing floor plan datasets, the furniture symbols used in ROBIN dataset are similar to the ones used in SESYD dataset [Delalandre *et al.*, 2010a] as shown in Fig. A.1.

A.2 GLOBAL SHAPE OF THE HOUSE

The second important criteria kept in mind while designing the dataset was the global or overall shape of the layout. To design the floor plans the rooms were arranged either in the horizontal or the vertical direction with respect to each other. While designing, due consideration was given to the fact that, there is no arbitrary shaped floor plan image and also that curvilinear structures in the layout walls are avoided. Moreover, to increase the variability among the samples, difference in arrangement of rooms across 3, 4 and 5 room layout categories was also introduced. Figure A.2 (a), (b) and (c) depict the example of variation in global shapes across a layout category.

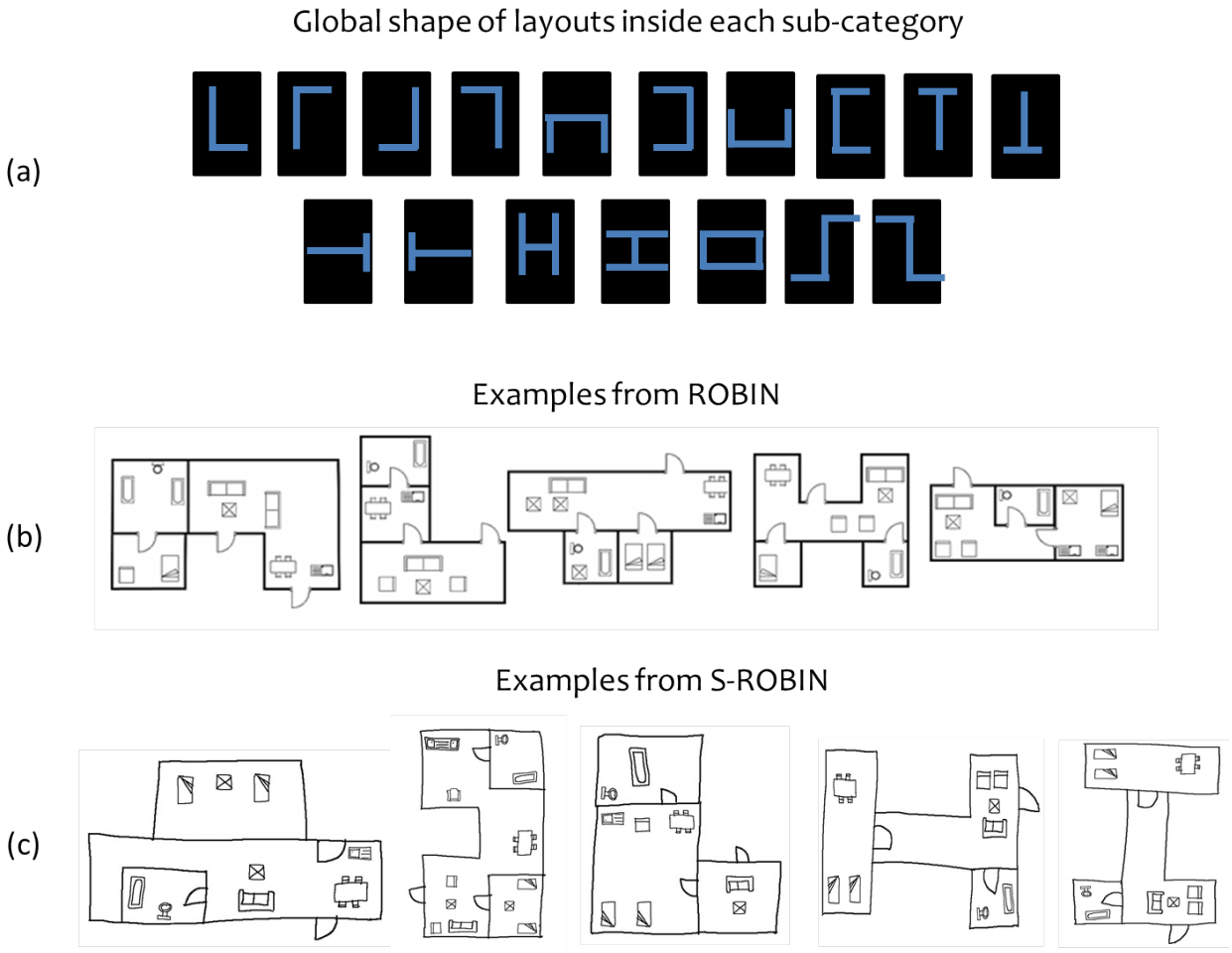


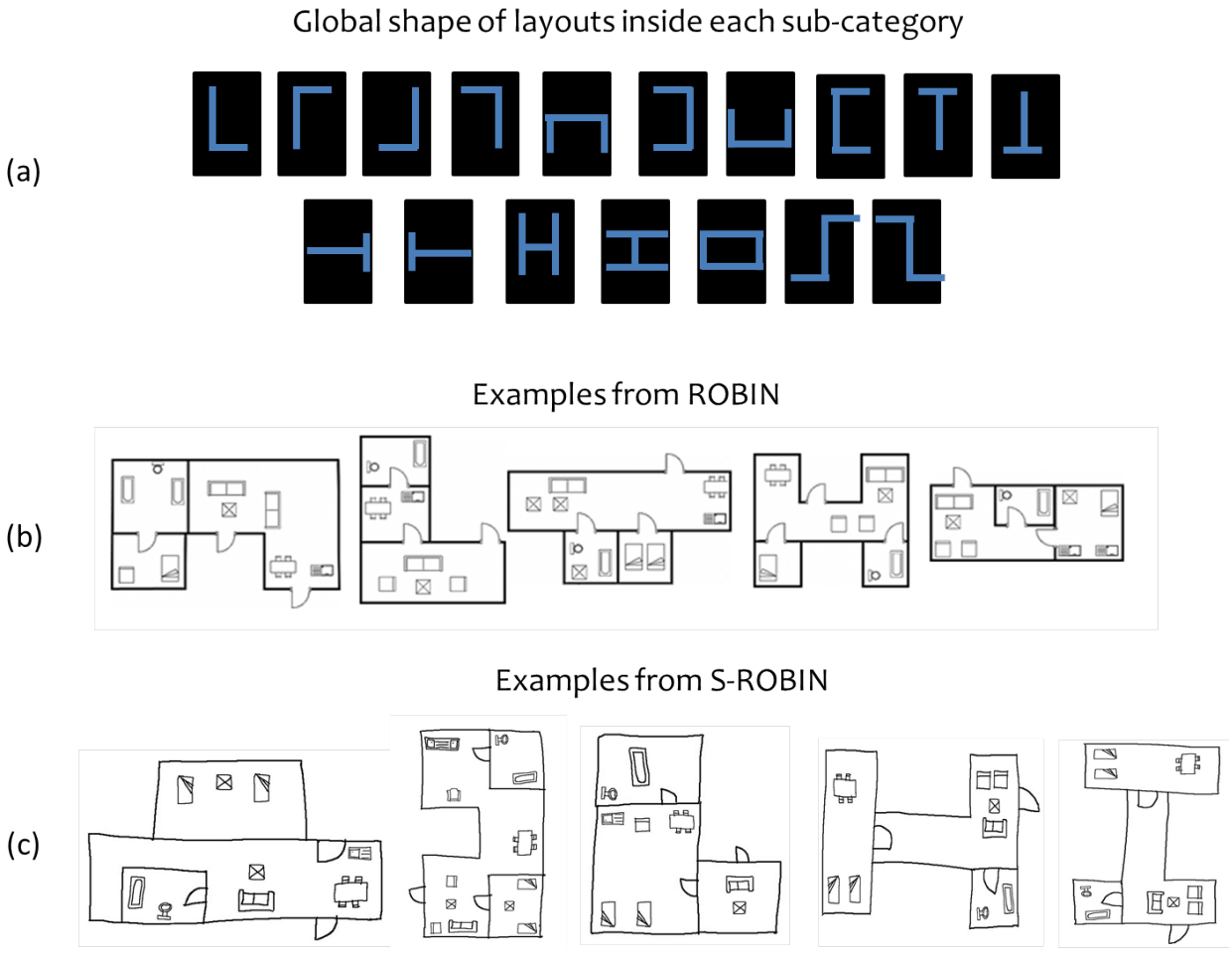
Figure A.2. : 3-Room layouts in the ROBIN and S-ROBIN dataset.

A.1 房间数量

在ROBIN数据集中，共有三类平面布局，分别对应三种主要的住宅类型。这些住宅分别包含3间、4间和5间房间。各房间的功能及其后续标注会根据其内部所配置的不同装饰而有所差异。为与现有平面布局数据集保持一致，ROBIN数据集中使用的家具符号与SESYD数据集[Delalandre, 2010a]中的符号相似，如图A.1所示。

A.2 房屋的整体形状

在设计数据集时，所遵循的第二个重要原则是布局的整体形态。在绘制平面图时，各房间之间均按水平或垂直方向进行排列。设计过程中充分考虑了以下两点：一是不存在任意形状的平面图；二是避免在墙体布局中出现曲线结构。此外，为提高样本的多样性，还在3室、4室和5室这三类布局中引入了不同的房间布置方式。图A.2（a）、（b）和（c）分别展示了同一布局类别中整体形态的差异示例。



图A.2：ROBIN和S-ROBIN数据集中3个房间的布局。

A.3 CONNECTIVITY WITHIN A HOUSE

Doors are placed at the desired places in the floor plans to ensure connectivity between the rooms. Orientation of each door is given special consideration to ensure proper direction of movement within the house. For every floor plan, there is atleast one door lying on the outer wall of the floor plan that opens inside the house and provides entry/exit for the house. Windows are appropriately placed on the walls of the floor plan to ensure proper ventilation and aesthetics. This makes the floor plan more realistic. Figure A.3 (b) can be observed for the arc-type door symbols satisfying entry-exit criteria in the floor plans.

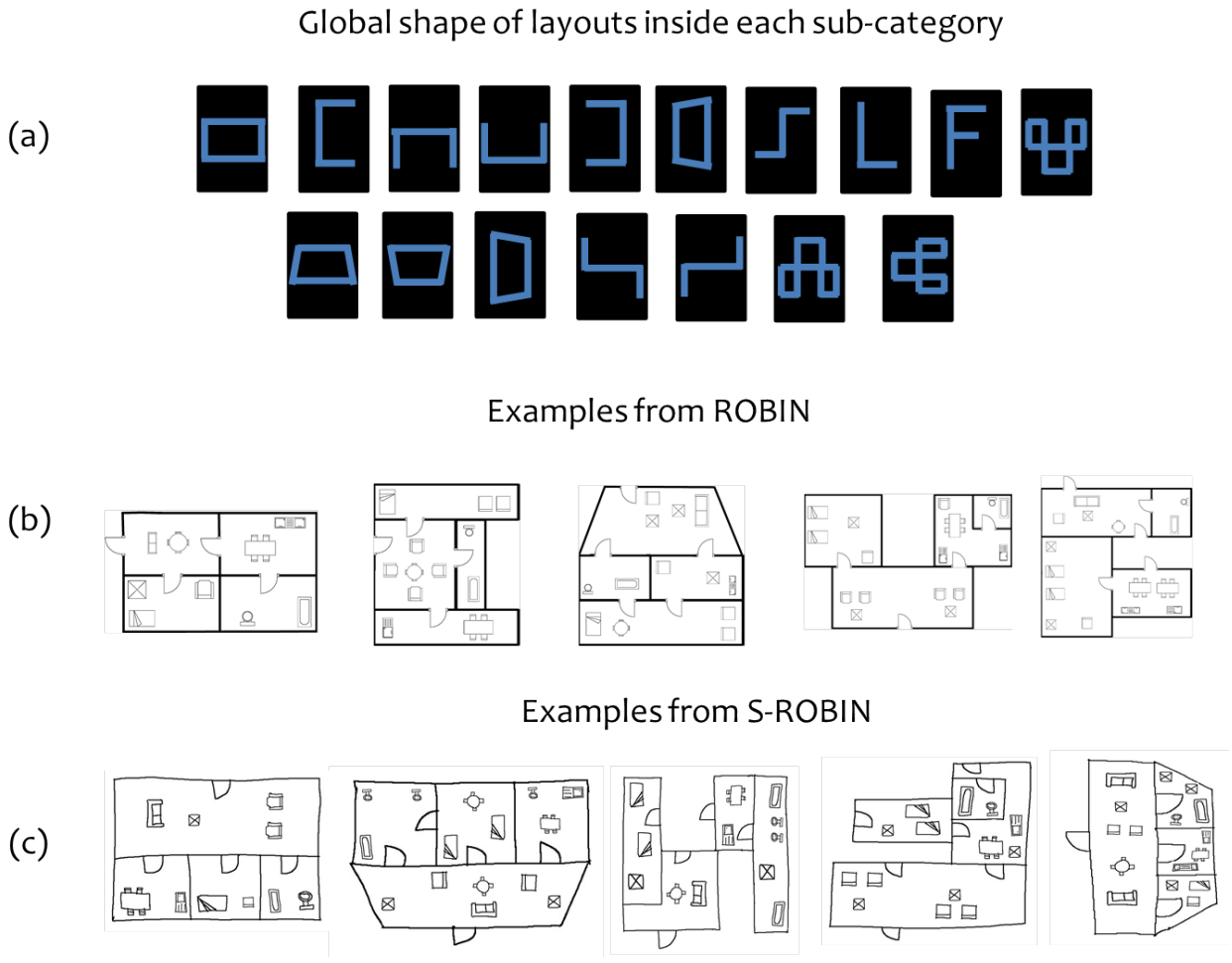


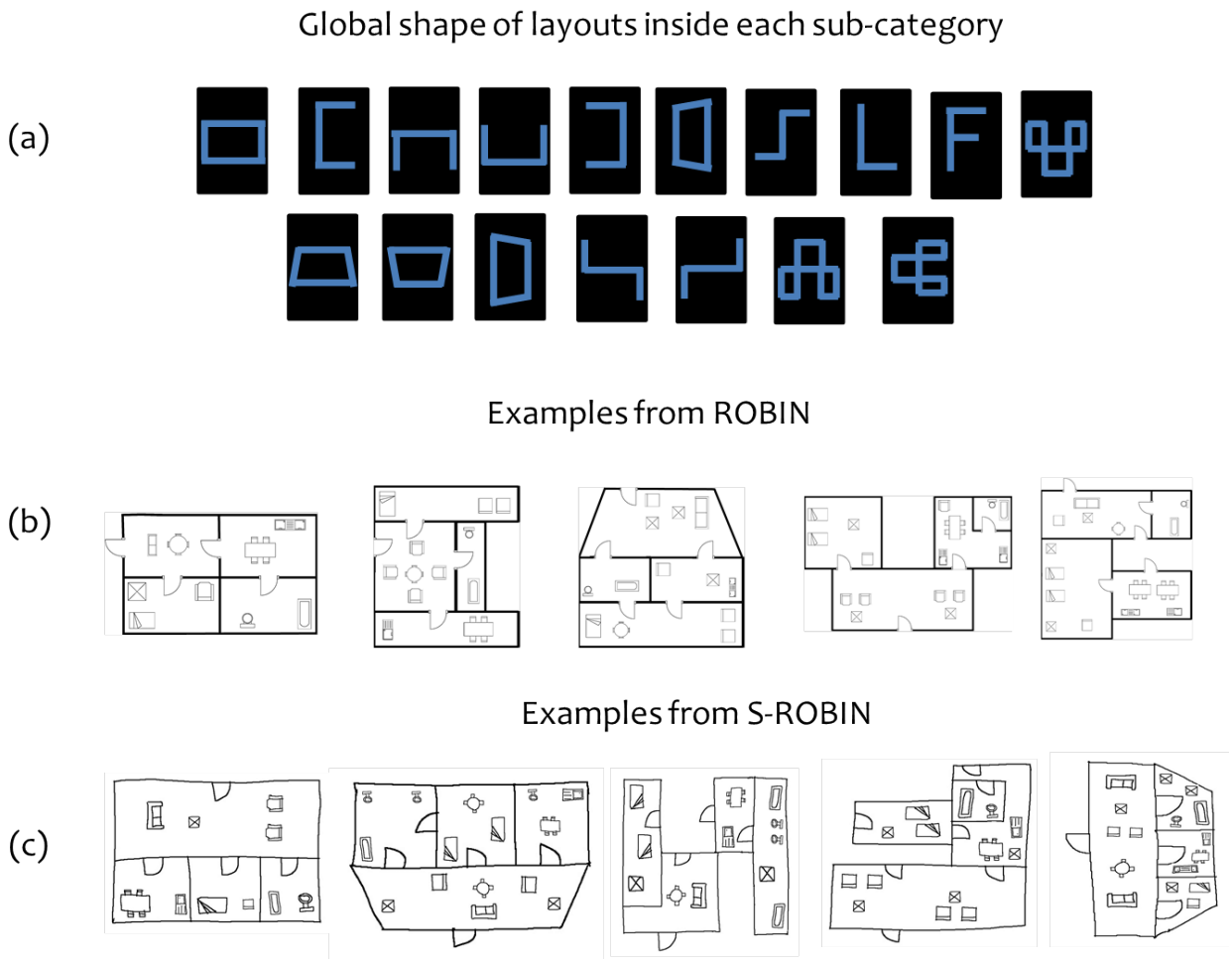
Figure A.3. : 4-Room layouts in the ROBIN and S-ROBIN dataset.

A.4 PLACEMENT OF FURNITURE

Furnitures or decors are inseparable part of a floor plan. To make the floor plans of ROBIN realistic, decor symbols as proposed by Delalandre *et al.* [2010a] were utilized. As shown in Fig. A.1, there are 12 different floor plan symbols. In each floor plan a subset of the entire symbol set was chosen and placed inside every room to give a definition to the room. As the furnitures are indicators of the purpose of the rooms, therefore, while designing, sufficient care has been taken to ensure that a house contains a i) kitchen, ii) bedroom, iv) washroom and a v) lobby. Moreover,

A.3 房屋内部的连通性

在平面图中，门被布置在指定位置，以确保各房间之间的连通性。每扇门的朝向均经过特别考量，以保证室内人员流动方向的合理性。对于每个平面图，至少有一扇门设置在外墙上，且向室内开启，作为住宅的出入口。窗户则合理布置于平面图的墙体上，以兼顾通风与美观，从而使平面图更加贴近实际。图A.3(b)展示了满足出入口要求的弧形门符号在平面图中的应用。



图A.3：ROBIN和S-ROBIN数据集中4个房间的布局。

A.4 家具的摆放

家具或装饰是平面图不可或缺的组成部分。为使ROBIN的平面图更加逼真，采用了Delalandre *et al.* [2010a] 提出的装饰符号。如图A.1所示，共有12种不同的平面图符号。在每一张平面图中，均从全部符号集中选取一个子集，并将其布置于各房间内，以明确各房间的功能属性。由于家具是房间用途的重要指示物，因此在设计过程中特别注意确保每栋住宅包含：i) 厨房、ii) 卧室、iii) 洗手间以及iv) 门厅。此外，

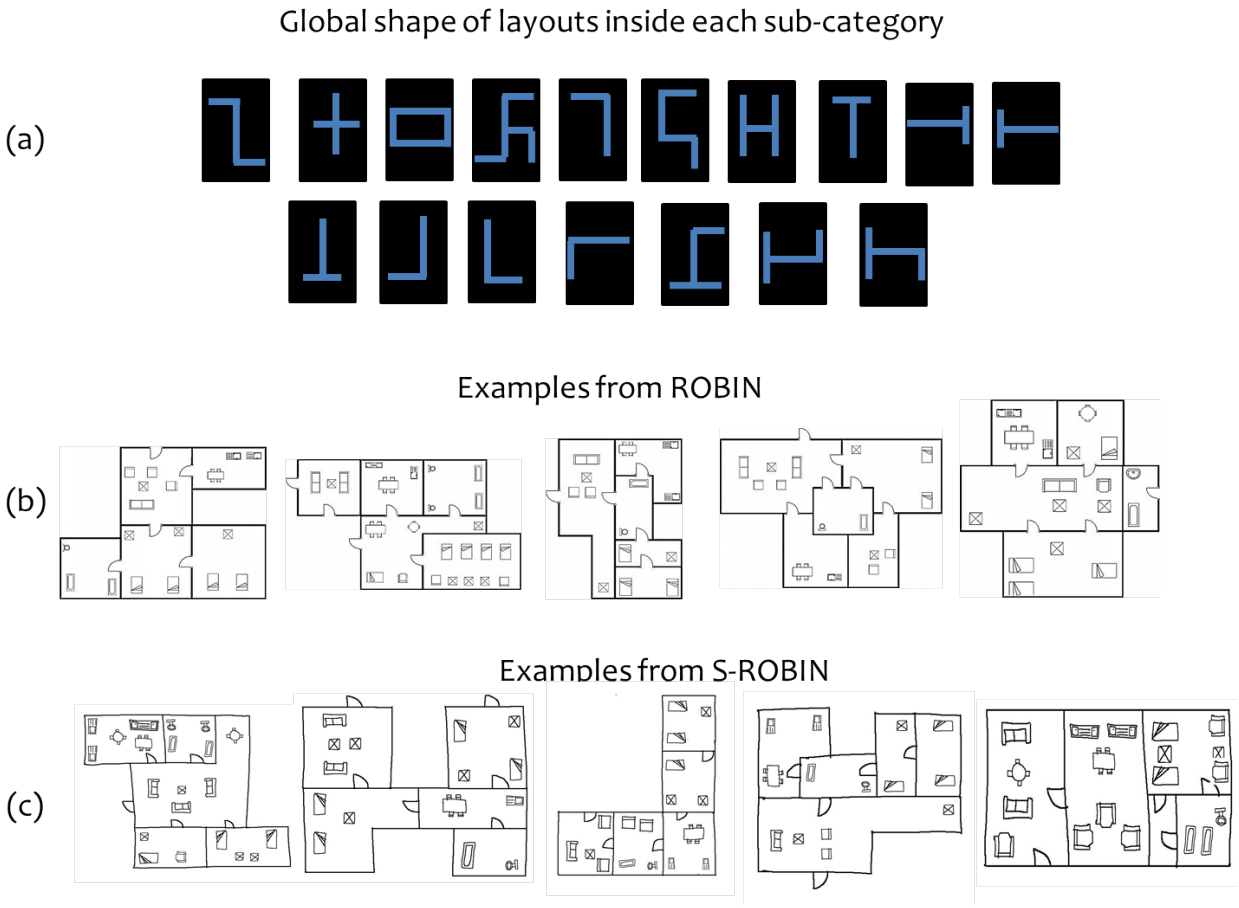
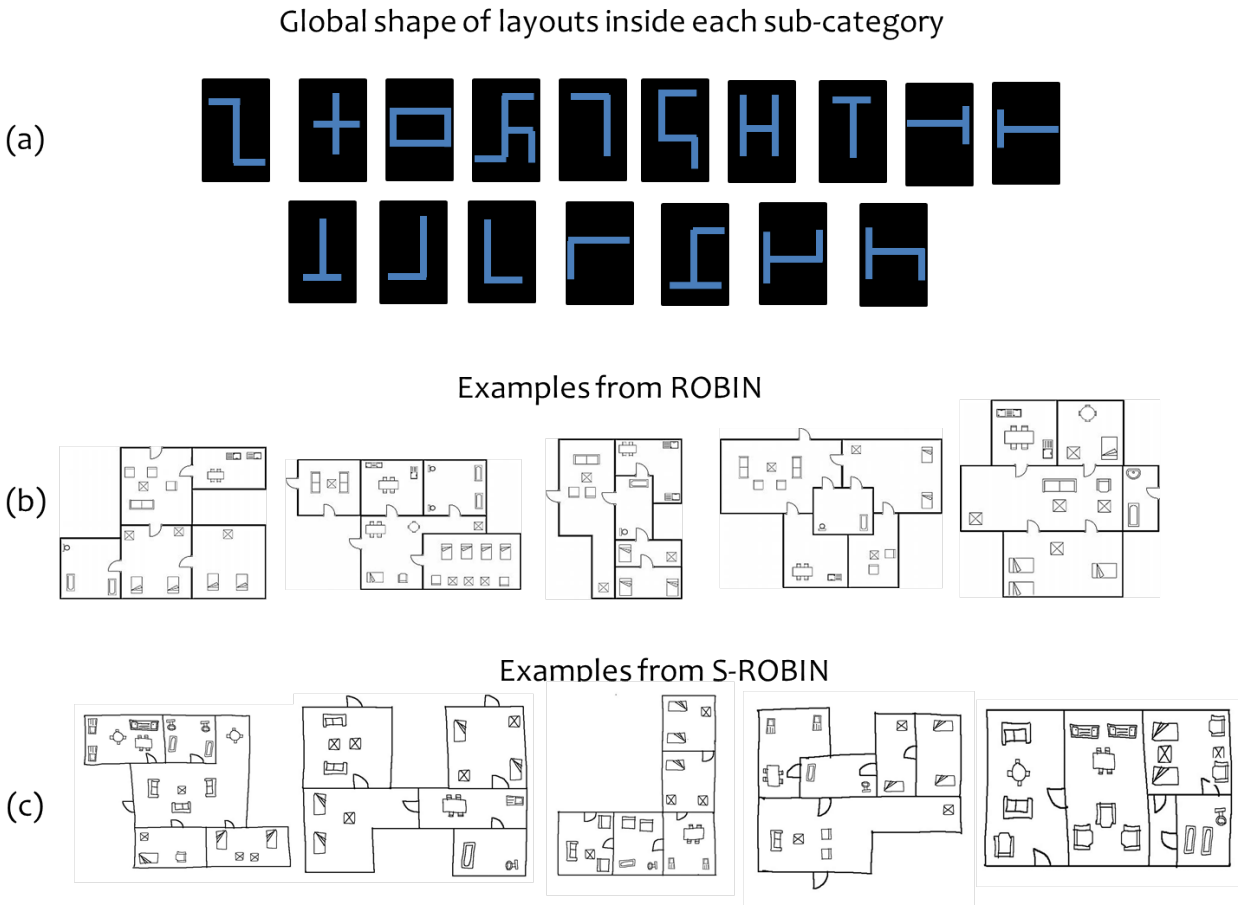


Figure A.4. : 5-Room layouts in the ROBIN and S-ROBIN dataset.

special consideration was given to the fact that the furniture is not placed in an arbitrary manner. For example, furniture should not be placed in a way that it blocks the door. Also, sufficient area should be left vacant after the furniture placement for navigation purposes.

A.5 DATASET CREATION PROCESS

ROBIN dataset was created on similar lines as the SESYD dataset [Delalandre *et al.*, 2010a]. Basic room structure of the floor plan was built using rectangular shape in Microsoft powerpoint. Thickness of each wall was kept uniform to be around 40 pixels. The decor symbols were inserted inside the rooms from the symbol library and door symbols were carefully placed on the room walls. On the other hand, to capture hand drawn sketched floor plans in S-ROBIN, volunteers were asked to draw the floor plans on a digital platform using a Wacom tablet. Wacom tablet is an interactive tool, where the user can draw on a board like tablet using a pen stylus conveniently and the resultant floor plan sketch is captured on the Microsoft paint software by adjusting the thickness of the strokes of the pen stylus. Since, offline mode is being used, the hand drawn floor plans are stored as an image and used for experimentation.



图A.4：ROBIN和S-ROBIN数据集中5个房间的布局。

特别考虑到了家具并非随意摆放这一事实。例如，家具不应以遮挡门的方式进行布置；此外，在家具布置完成后，还应留出足够的空地以供通行。

A.5 数据集创建过程

ROBIN数据集的构建方式与SESYD数据集[Delalandre等，2010a]类似。在Microsoft PowerPoint中，采用矩形形状构建了平面图的基本房间结构，并将各墙体的宽度统一设置为约40像素。装饰符号从符号库中插入到各个房间内，门符号则被精确地放置于房间墙面上。另一方面，为了采集手绘草图形式的S-ROBIN平面图，我们要求志愿者使用Wacom数位板在数字化平台上进行绘制。Wacom数位板是一种交互式工具，用户可借助笔式触控笔在类似画板的设备上便捷作画，所生成的平面图草图通过调整笔式触控笔笔触的粗细，在Microsoft Paint软件中予以记录。由于采用离线模式，手绘平面图以图像形式存储，并用于实验研究。

A.6 STATISTICS OF THE DATASET

Both the datasets contain 510 floor plans divided into 3 categories of layouts each with 17 sub categories. The unique characteristic of the datasets is that they are designed keeping in mind the need of a potential buyer. Every prospective buyer wants to have certain amenities and functionalities in their house. Thus, in the dataset, there are three broad categories, which are different from each other in terms of the number and type of rooms present in a floor plan. The broad categories are layouts with 3-rooms, 4-rooms and 5-rooms each which differ in the global shape of layout across their sub-categories. Each broad category is further classified into 17 sub-categories depending upon the global layout shape of the floor plan. Standard notations as mentioned by Delalandre *et al.* [2007], are followed for all the floor plans so that the datasets can be used for other floor plan analysis tasks as well, and not only for retrieval. The dataset has 12 different categories of furnitures inside each layout. These datasets help in better visualization of the floor plans and aid in efficient capturing of various high-level features while fine-grained retrieval. Illustrations of the various sub-categories in the floor plans and how their shapes are modeled are shown in Fig. A.2, Fig. A.3 and Fig. A.4. The size of the floor plans in ROBIN vary from 825×708 to 2916×1845 whereas the resolution in S-ROBIN dataset ranges from 591×517 pixels to 1483×884.

A.7 CONCLUSION

The rationale behind the creation of ROBIN and S-ROBIN dataset was to create a floor plan repository that can be used for research purposes by the Document Analysis and Retrieval (DAR) community. Both ROBIN and S-ROBIN dataset are sufficient for carrying out tasks like, retrieval, semantic segmentation, door to door navigation, etc. The entire dataset along with the annotations can be downloaded from [S-ROBIN, 2018].

A.6 数据集的统计信息

两个数据集均包含510张平面图，每组分为3个布局大类，每个大类又细分为17个小类。这些数据集的独特之处在于，其设计充分考虑了潜在购房者的实际需求：每位购房者都希望住宅具备特定的配套设施与功能。因此，数据集中设置了三类主要布局，它们在平面图中房间的数量与类型上各不相同。这三类分别为三居室、四居室和五居室布局，且各小类在整体形态上也存在差异。每一类布局根据其整体形态进一步划分为17个小类。为便于数据集不仅用于检索，还可应用于其他平面图分析任务，所有平面图均采用Delalandre于2007年提出的标准化标注方法。每个布局中还包含12种不同类型的家具。这些数据集有助于更直观地呈现平面图，并在细粒度检索过程中有效提取各类高层特征。平面图中各小类的具体示例及其形状建模方式分别见图A.2、图A.3和图A.4。ROBIN数据集中平面图的尺寸范围为825×708至2916×1845像素，而S-ROBIN数据集的分辨率则介于591×517像素至1483×884像素之间。

A.7 结论

创建ROBIN和S-ROBIN数据集的初衷是构建一个可用于文档分析与检索（DAR）研究领域的平面图资源库。ROBIN和S-ROBIN数据集均足以支持诸如检索、语义分割、门到门导航等任务的开展。整个数据集及其标注可从[S-ROBIN, 2018]下载。